

(19) **United States**

(12) **Patent Application Publication**
DE CROZALS et al.

(10) **Pub. No.: US 2025/0283132 A1**
(43) **Pub. Date: Sep. 11, 2025**

(54) **METHODS AND KITS FOR ENZYMATIC
SYNTHESIS OF G4-PRONE
POLYNUCLEOTIDES**

(30) **Foreign Application Priority Data**

Apr. 2, 2021 (EP) EP21305428

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Publication Classification

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(51) **Int. Cl.**
C12P 19/34 (2006.01)
C12N 9/12 (2006.01)
(52) **U.S. Cl.**
CPC **C12P 19/34** (2013.01); **C12N 9/1264**
(2013.01); **C12Y 207/07031** (2013.01)

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(57) **ABSTRACT**

The present invention is directed to methods, compositions and kits for template-free enzymatic synthesis of polynucleotides having sequences capable of forming G-quadruplex (G4) structures. In accordance with the invention elongation reactions affected by G4 formation are carried out in the presence of polyC oligonucleotides, such as polyC initiators, that inhibit or prevent formation of either intra-strand or inter-strand G4 structures.

Specification includes a Sequence Listing.

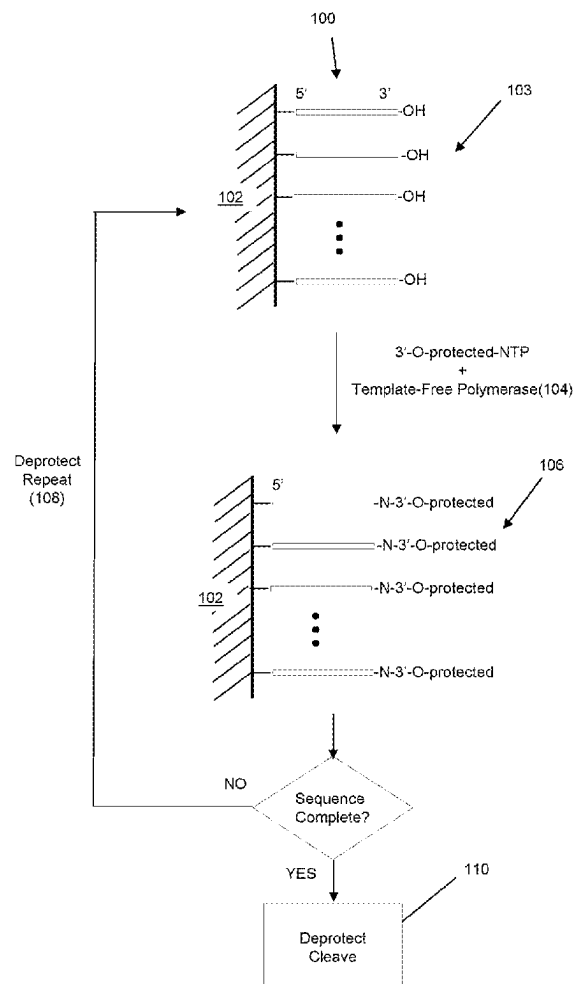
(21) Appl. No.: **18/285,014**

(22) PCT Filed: **Apr. 1, 2022**

(86) PCT No.: **PCT/EP2022/058814**

§ 371 (c)(1),

(2) Date: **Sep. 29, 2023**



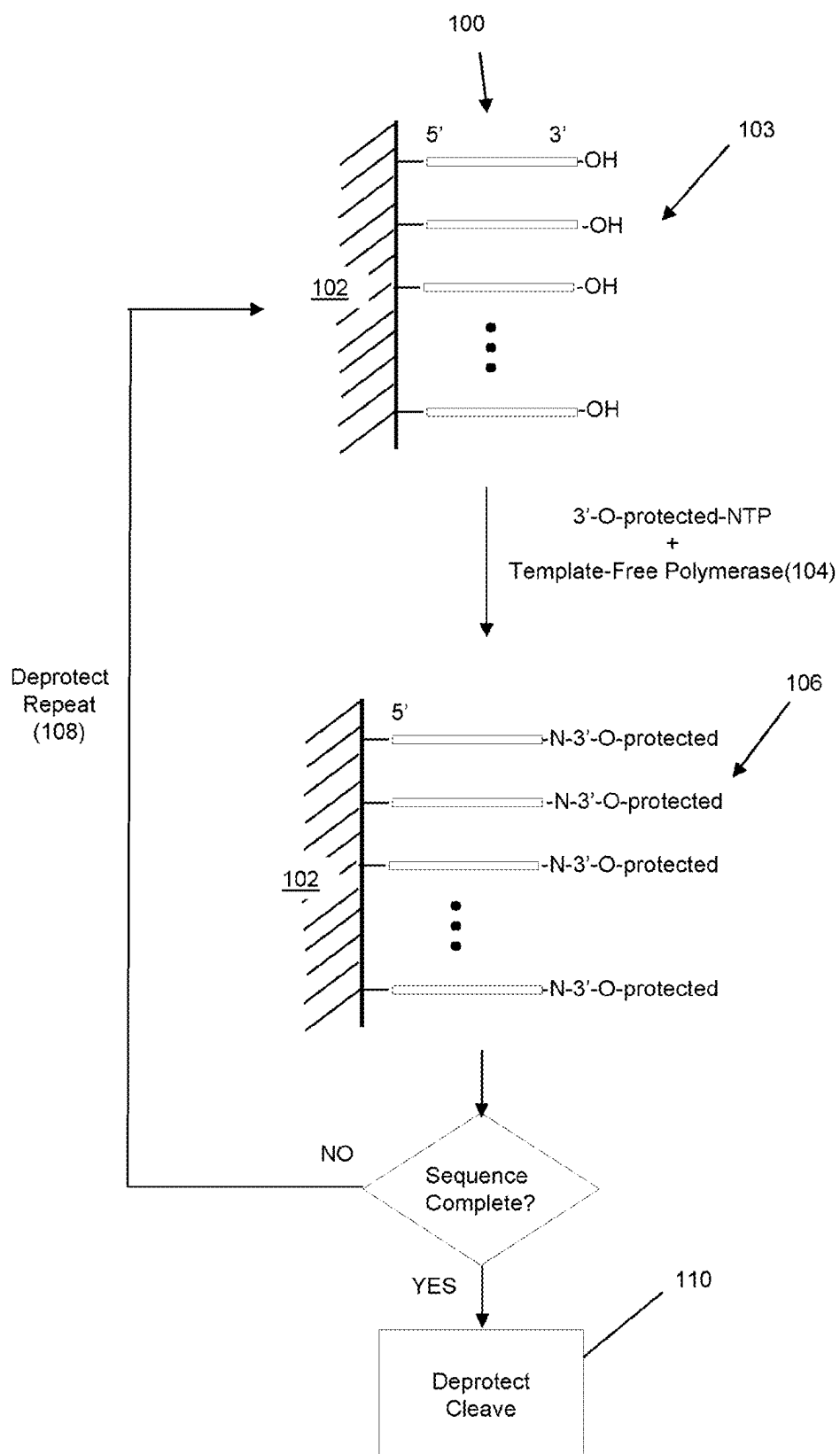


Fig. 1

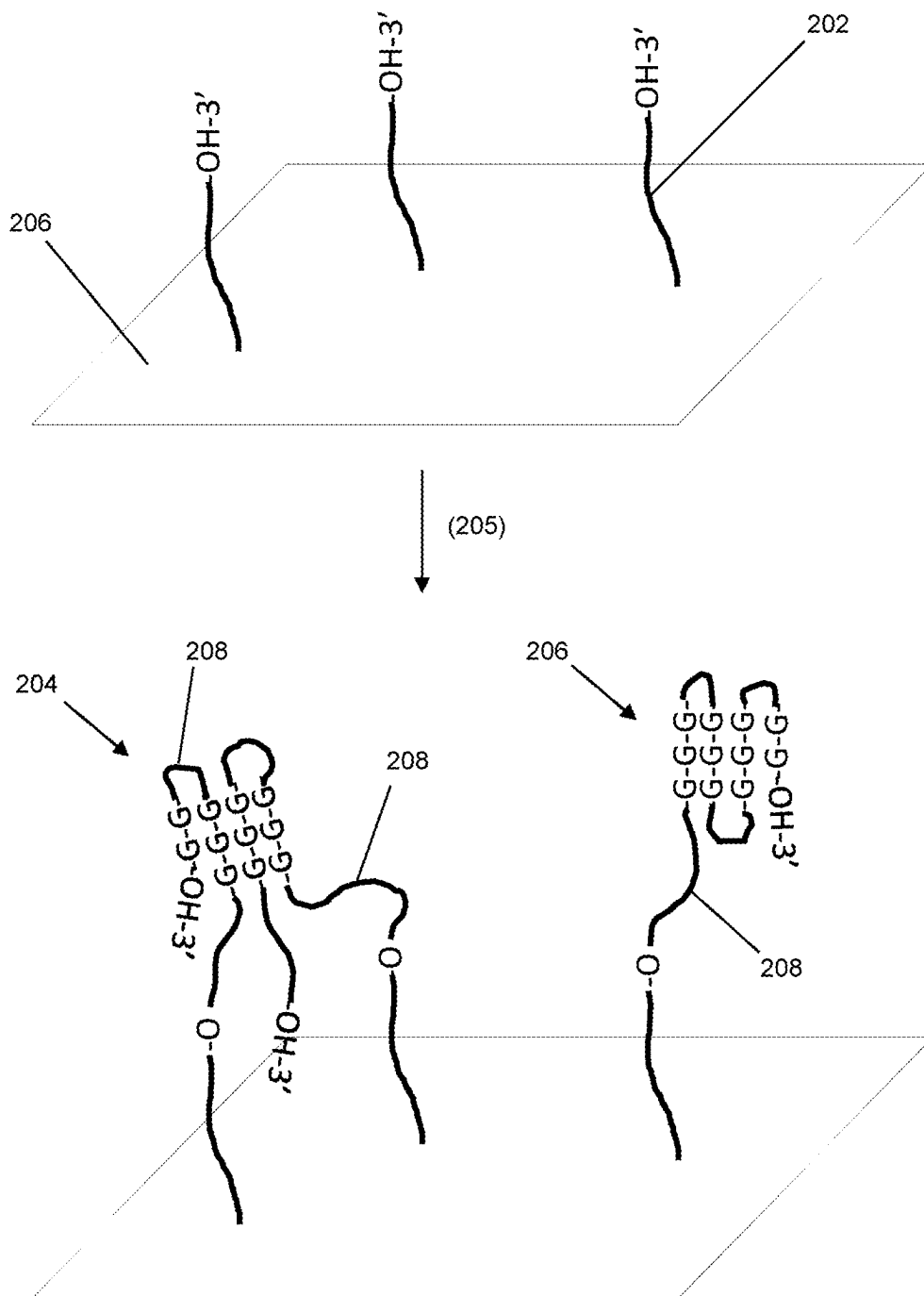


Fig. 2A

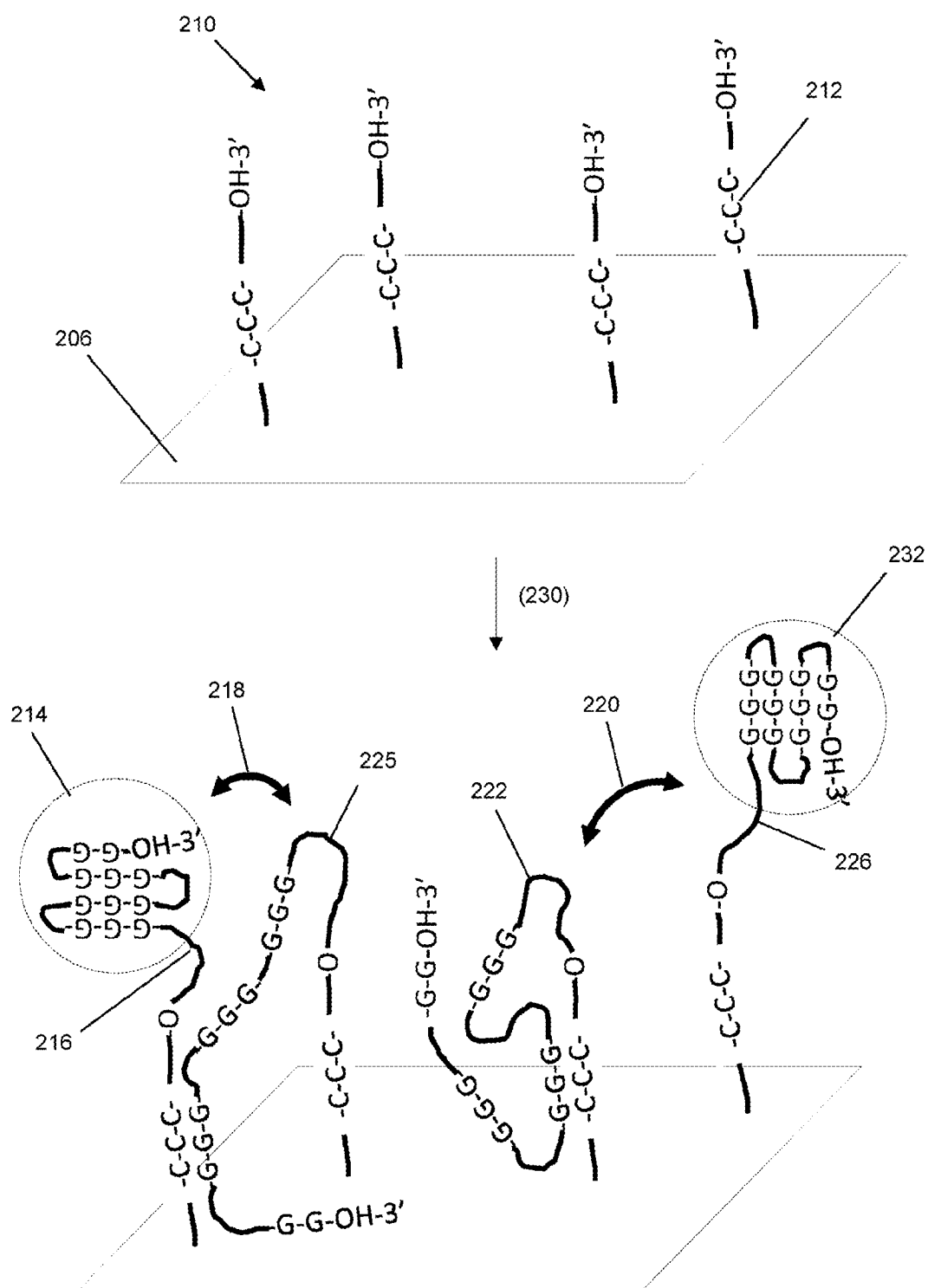


Fig. 2B

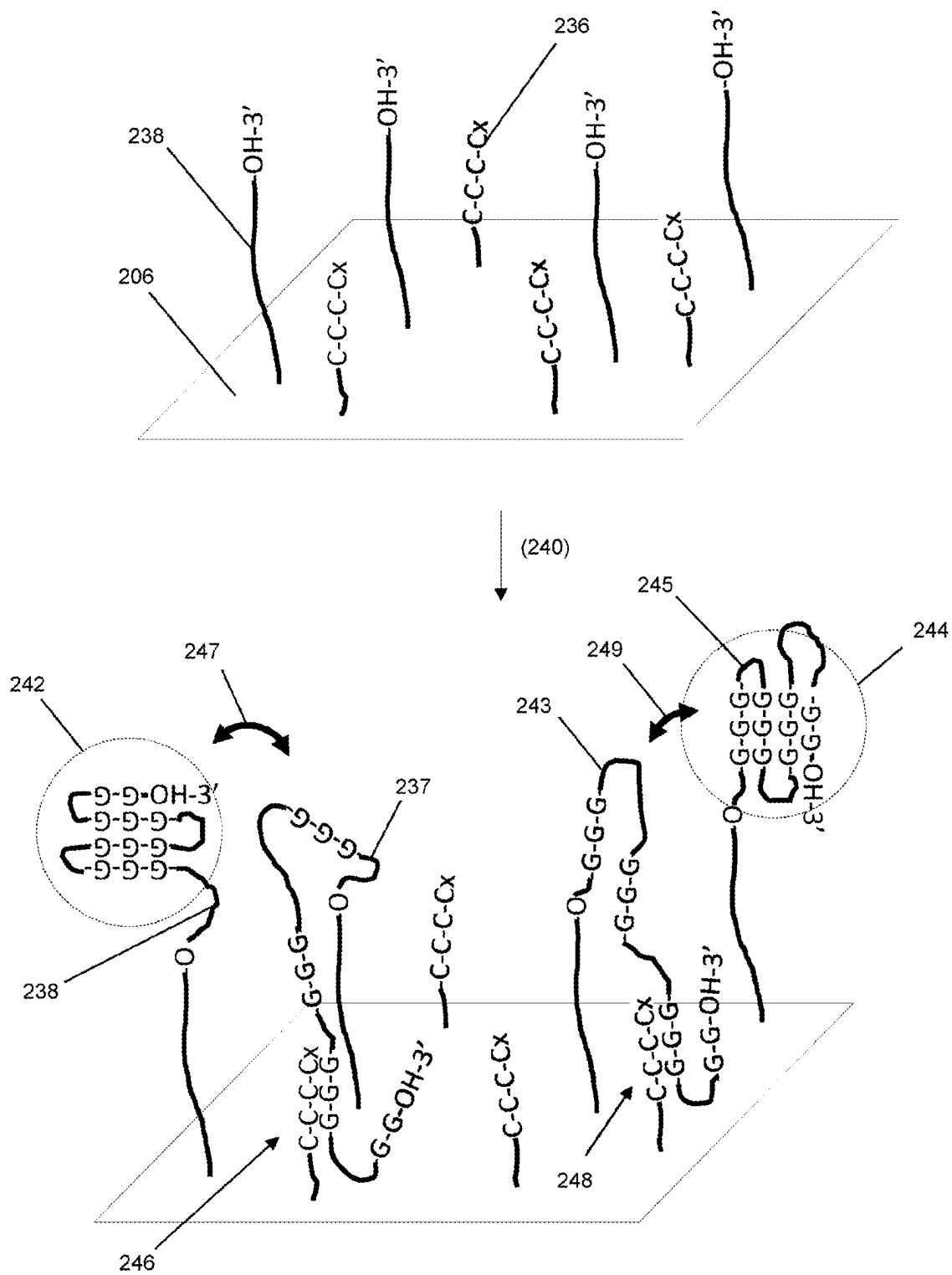


Fig. 2C

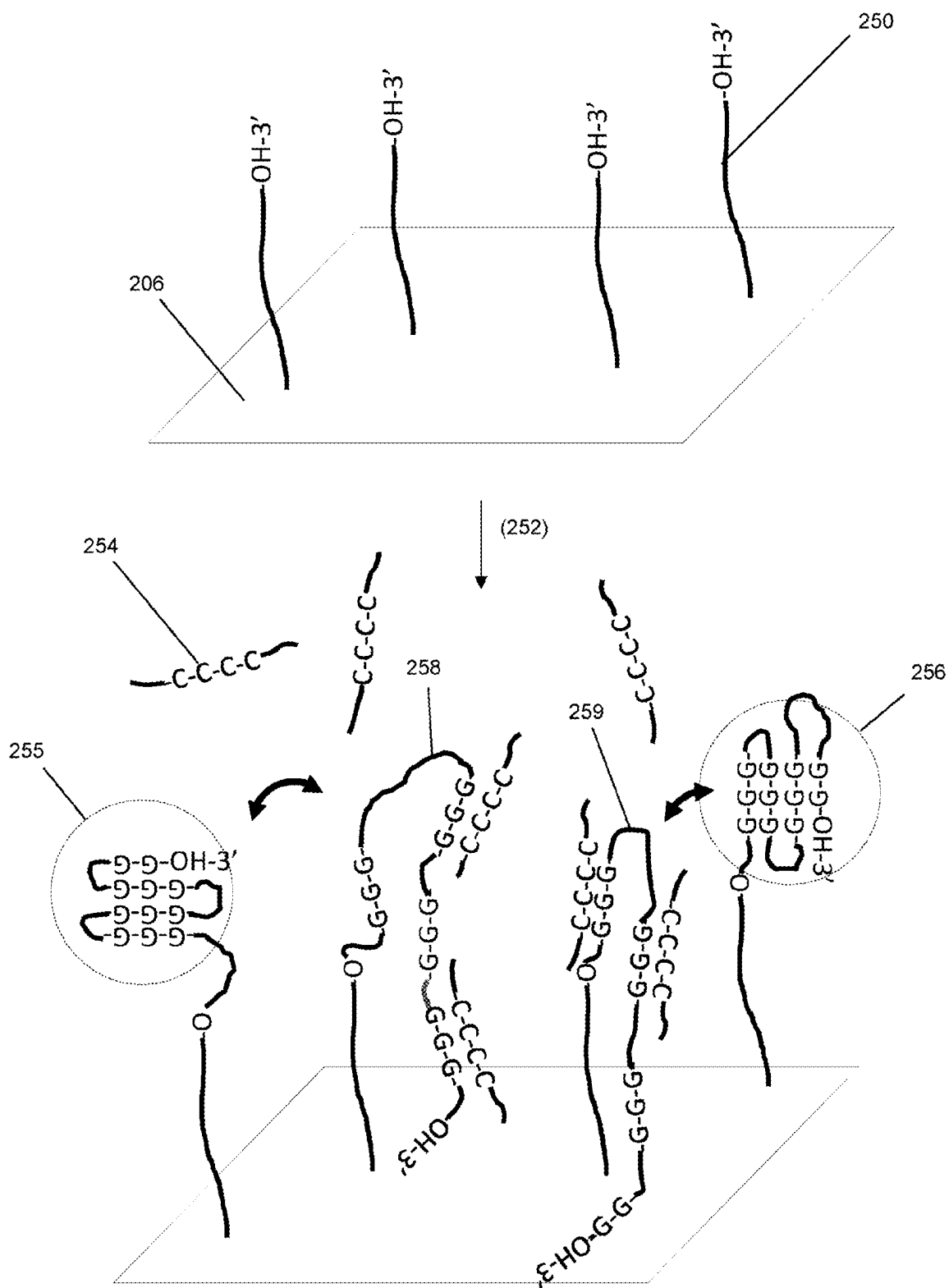


Fig. 2D

METHODS AND KITS FOR ENZYMATIC SYNTHESIS OF G4-PRONE POLYNUCLEOTIDES

BACKGROUND

[0001] Interest in enzymatic approaches to polynucleotide synthesis has recently increased not only because of increased demand for synthetic polynucleotides in many areas, such as synthetic biology, CRISPR-Cas9 applications, and high-throughput sequencing, but also because of the limitations of chemical approaches to polynucleotide synthesis, such as upper limits on product length and the use and the need to dispose of organic solvents, Jensen et al, *Biochemistry*, 57: 1821-1832 (2018). Enzymatic synthesis is attractive because of its specificity and efficiency and its requiring only mild aqueous-compatible reagents and reaction conditions.

[0002] Currently, most enzymatic approaches for both DNA and RNA synthesis employ template-free polymerases to repeatedly add 3'-O-blocked nucleoside triphosphates to a single stranded initiator or an elongated strand attached to a support followed by deblocking until a polynucleotide of the desired sequence is obtained, e.g. Hiatt and Rose, International patent publication WO96/07669. The inventors have discovered that template-free polymerases, such as, terminal deoxynucleotidyltransferases (TdT), do not efficiently couple nucleotides to elongated strands at sequences that are capable of forming G-quadruplexes (G4s). G4 structures are commonly occurring secondary structures important for their involvement in a variety natural processes, e.g. Kwok et al, *Trends in Biotechnology*, 35(10): 997-1013 (2017); Murat et al, *Curr. Opin. Genet. Devel.*, 25: 22-29 (2014); and the like. Thus, the state of the art of enzymatic synthesis currently is deficient in its capability to synthesize polynucleotides prone to forming G4 structures.

[0003] In view of the interest in extending the application of template-free enzymatic synthesis of polynucleotides, the field would be advanced if methods were available to increase the efficiency and yield of target polynucleotide containing G4 structures.

SUMMARY OF THE INVENTION

[0004] The present invention is directed to methods and kits for template-free enzymatic synthesis of either DNA or RNA polynucleotides that employ agents in a synthesis reaction for disrupting the formation of G4 structures. In one aspect, such agents are polycytidylate or polydeoxycytidylate oligonucleotides (collectively referred to herein as "polyC oligonucleotides"). In some embodiments, polyC oligonucleotides are components of initiators and/or synthesis supports such that they are capable of interacting with G-rich domains of polynucleotides being synthesized. In other embodiments, polyC oligonucleotides are free in solution during selected coupling or elongation steps during synthesis.

[0005] In some embodiments, the invention is directed to methods of synthesizing a polynucleotide having a predetermined sequence capable of forming a G4 structure, wherein the method comprises the steps of: (a) providing, attached to a synthesis support, initiators each with a free 3'-hydroxyl; (b) repeating in a reaction mixture including the synthesis support, until the polynucleotide is formed, cycles of (i) contacting under elongation conditions the initiators or

elongated fragments having free 3'-O-hydroxyls with a 3'-O-blocked nucleoside triphosphate and a template-independent polymerase so that the initiators or elongated fragments are elongated by incorporation of a 3'-O-blocked nucleoside triphosphate to form 3'-O-blocked elongated fragments, and (ii) deblocking the elongated fragments to form elongated fragments having free 3'-hydroxyls, wherein the reaction mixture for elongating the initiators or elongated fragments comprise polyC oligonucleotides capable of forming duplexes with regions of the polynucleotide. In some embodiments, initiators each comprise a segment consisting of a polyC oligonucleotide. In other embodiments, a synthesis support is provided having polyC oligonucleotides attached thereto, eventually in addition to initiators. In still other embodiments, polyC oligonucleotides are in solution as a component of an elongation reaction mixture for selected elongation cycles in which G4 structure formation has a high likelihood of occurrence. In some embodiments, such selected elongation cycles may be determined by a conventional G4 prediction algorithm.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 diagrammatically illustrates a method of template-free enzymatic synthesis of a polynucleotide.

[0007] FIGS. 2A-2D diagrammatically illustrate various embodiments for including in a reaction mixture polyC oligonucleotides.

DETAILED DESCRIPTION OF THE INVENTION

[0008] The general principles of the invention are disclosed in more detail herein particularly by way of examples, such as those shown in the drawings and described in detail. It should be understood, however, that the intention is not to limit the invention to the particular embodiments described. The invention is amenable to various modifications and alternative forms, specifics of which are shown for several embodiments. The intention is to cover all modifications, equivalents, and alternatives falling within the principles and scope of the invention.

[0009] The practice of the present invention may employ, unless otherwise indicated, conventional techniques and descriptions of organic chemistry, molecular biology (including recombinant techniques), cell biology, and biochemistry, which are within the skill of the art. Such conventional techniques may include, but are not limited to, preparation and use of synthetic peptides, synthetic polynucleotides, monoclonal antibodies, nucleic acid cloning, amplification, sequencing and analysis, and related techniques. Protocols for such conventional techniques can be found in product literature from manufacturers and in standard laboratory manuals, such as *Genome Analysis: A Laboratory Manual Series* (Vols. I-IV); *PCR Primer: A Laboratory Manual*; and *Molecular Cloning: A Laboratory Manual* (all from Cold Spring Harbor Laboratory Press); Lutz and Bornscheuer, Editors, *Protein Engineering Handbook* (Wiley-VCH, 2009); Hermanson, *Bioconjugate Techniques*, Second Edition (Academic Press, 2008); and like references.

[0010] The invention is directed to improvements to template-free enzymatic synthesis of polynucleotides, especially DNA or RNA, which permit higher yields of long polynucleotides by providing synthesis conditions that suppress or disrupt the formation of G-quadruplex (or G4)

secondary structures in growing chains. Without the intention of being limited to a particular theory or hypothesis, it is believed that the formation of G4 structures limits access to synthesis reagents, such as template-free polymerases, thereby inhibiting chain extension, or elongation, and thereby increasing the variability of product length. In part, the invention is based on a recognition and appreciation that the negative effects of such secondary structures on product yield can be mitigated or suppressed by providing elongation (or extension or coupling) conditions that include agents, particularly polyC oligonucleotides, which disrupt the formation of G4 structures. In particular, it is believed that such disruption occurs by providing alternative stable configurations, e.g. duplexes with polyC regions, that a growing strand having G-rich regions can occupy. In view of the above, in some embodiments, initiators of the invention may comprise polyG oligonucleotides wherein an alternative stable configuration may be a G4 structure between the polyG oligonucleotide in the initiator and G-rich sequences of the polynucleotide being synthesized. Such polyG oligonucleotides may have a length in the range of from 2 to 20 guanylates or deoxyguanylates.

[0011] G-quadruplex structures are common in nature and may be predicted using available algorithms, e.g. Lombardi et al, *Nucleic Acids Research*, 48(1): 1-15 (2020), and like references. $G_{3+N_{1-7}G_{3+N_{1-7}G_{3+N_{1-7}G_{3+}}$ is a common G4 motif, where "N" is any nucleotide and "3+" means 3 or more G's in a row. As used herein, the term "G4-prone polynucleotide" means a polynucleotide having a nucleotide sequence that can form a G4 structure under elongation reaction conditions. In some embodiments, "G4-prone polynucleotide" means a polynucleotide having a nucleotide sequence that a conventional G4 prediction algorithm indicates as likely to form a G4 structure, e.g. Burge et al, *Nucleic Acids Research*, 34(19): 5402-5415 (2006); Huppert et al, *Nucleic Acids Research*, 33(9): 2908-2916 (2005); Kwok et al, *Trends in Biochemistry*, 35(10): 997-1013 (2017); Lombardi et al (cited above); Murat et al, *Curr. Opin. Genetic & Development*, 25: 22-29 (2014); Todd et al, *Nucleic Acids Research*, 33(9): 2901-2907 (2005); Bedrat et al, *Nucleic Acids Research*, 44(4): 1746-1759 (2016); and the like.

[0012] PolyC oligonucleotides of the invention may have lengths of 2 or more nucleotides. In some embodiments, polyC oligonucleotides of the invention have lengths in the range of from 2 to 60 nucleotides, or from 2 to 50 nucleotides, or from 2 to 40 nucleotides, or from 2 to 30 nucleotides, or from 2 to 20 nucleotides. In other embodiments, polyC oligonucleotides of the invention have lengths in the range of from 6 to 60 nucleotides, or from 6 to 50 nucleotides, or from 6 to 40 nucleotides, or from 6 to 30 nucleotides, or from 6 to 20 nucleotides. In some embodiments, initiators of the invention have a length in the range of from 6 to 50 nucleotides and polyC oligonucleotides make up fifty percent or more of the initiator sequence. In some embodiments, polyC oligonucleotides of the invention have lengths, concentrations and/or configurations (i.e. segment of initiator, independently attached to same solid support as initiator, or in free solution) of sufficient magnitude to increase the purity of the synthesized polynucleotides by 20 percent or more as compared to an equivalent synthesis using a polyT initiator.

[0013] In some embodiments, a plurality of polyC oligonucleotides may be present in a larger oligonucleotide, such

as an initiator, which is a component of an elongation reaction mixture. For example, an initiator may have a segment comprising several polyC oligonucleotides separated by a non-C nucleotide, e.g. -CCCTCCCTCCCT- (SEQ ID NO: 159), -CCCTCCCTCCCCCT- (SEQ ID NO: 160), or the like. In some embodiments, such composites of polyC oligonucleotides may be selected for ease of manufacturing.

[0014] Whenever initiators (described more fully below) comprise polyC oligonucleotides, the polyC oligonucleotides may comprise all or a portion of the initiators. In different embodiments, polyC oligonucleotides may be provided in any or all of the following configurations: (i) as part of initiators, (ii) as part of oligonucleotides attached to the same synthesis support as initiators, and (iii) as part of oligonucleotides in free solution. In each case the part of an oligonucleotide or initiator that is polyC may be the entire oligonucleotide or initiator. In regard to (ii), such oligonucleotides may be attached to a synthesis support by either a 5' end or a 3' end. In some embodiments, whenever an oligonucleotide of (ii) is attached by a 5' end its 3' end is capped so that nucleotides are not attached to it during coupling or elongation steps. Likewise, in regard to (iii), polyC oligonucleotides in free solution have their 3' ends capped so that nucleotides are not attached to them during coupling or elongation steps.

[0015] FIGS. 2A-2D illustrate various aspects of the invention. FIG. 2A illustrates synthesis support (200) with oligonucleotide initiators (202) attached by their 5' ends to support (200). After synthesis (205) of polynucleotides (208) containing polyG segments, either inter-strand (204) G4 structures may form or intra-strand (206) G4 structures may form to inhibit further extension of the polynucleotides. In some cases (e.g. 206) inhibition does not occur until the synthesis of the final G-rich segment of the polynucleotides takes place, which allows G4 formation. Thus, in some embodiments, free solution polyC oligonucleotides need be introduced into coupling reactions only in selected coupling steps which may be determined for particular polynucleotides using G4 prediction algorithms such as described in Lombardi et al (cited above). FIG. 2B illustrates embodiment (i) of the previous paragraph in which polyC oligonucleotides are components of initiators. In the upper panel of FIG. 2B initiators (210) are attached to synthesis support (206). Each initiator contains a polyC oligonucleotide segment (212). After synthesis up to where an intra-strand G4 structure starts to form, the bottom panel of FIG. 2B shows three possible configurations of the polynucleotides in their interactions with themselves and each other. Strands (216) and (226) illustrate at (214) and (232), respectively, G4 structures at their 3' ends, which inhibit a template-free polymerase, such as a TdT, from participating in an extension reaction. Strand (222) illustrates a polynucleotide in which one of its polyG segments forms an intra-strand duplex with the polyC component of the initiator, which thereby inhibits the formation of a G4 structure at its 3' end. Strands (216) and (225) illustrate the formation of an inter-strand duplex between the polyC component of the initiator of strand (216) and one of the polyG segments of strand (225), thereby disrupting the formation of a G4 structure at the end of strand (225). As mentioned above, it is believed that because of the formation of the G-C duplexes and the transitions (e.g. (218) and (220)) of the strands between duplex states and G4 states, template-free polymerases have an opportunity to interact with free 3'-hy-

droxyls of the growing chains for catalyzing a coupling reaction that otherwise would occur with much lower efficiency.

[0016] FIG. 2C illustrates an embodiment in which polyC oligonucleotides (236) are provided as separate oligonucleotides attached to the same synthesis support (206) as initiators (238). PolyC oligonucleotides may be attached by either their 5' ends or their 3' ends; however, if attached by their 5' ends, preferably the 3' ends are capped (indicated in the figure as an "x") so that they are not extended during extension steps. The initiators and polyC oligonucleotides may be attached using conventional attachment chemistries. Typically both would have reactive moieties (e.g. amines) on their attachment ends and would be reacted with complementary moieties on the synthesis support in relative concentrations selected so that the density of polyC oligonucleotides were high enough to permit inter-strand duplexes to form. The bottom panel of FIG. 2C shows the interaction of polyC and polyG regions after synthesis (240) up to the point where intra-strand G4 structures, e.g. (242) and (244), can form. In this embodiment, only inter-strand C-G duplexes, e.g. (246) and (248), between strands being synthesized and the polyC oligonucleotides. As above, the polyC oligonucleotides in proximity to the synthesized strands (237, 238, 243, 245) permit transitions, e.g. (247) and (249), between duplex states, (246) and (248) and G4 states (242) and (244), respectively, which allow more efficient extension reactions to take place.

[0017] FIG. 2D illustrates an embodiment in which polyC oligonucleotides are provided in solution as a component of a reaction mixture. In the upper panel of FIG. 2D, synthesis support (206) is illustrated with initiators (250) and no polyC oligonucleotides. After synthesis (252) of target polynucleotides up to the point in which intra-strand G4 structures begin to form, e.g. (255) and (256), a plurality of extension steps may be performed in reaction mixtures that contain polyC oligonucleotides (254). As shown for stands (258) and (259) the polyC oligonucleotides in solution form duplexes with polyG segments that otherwise would contribute to G4 structures.

Template-Free Enzymatic Synthesis of DNA

[0018] Generally, methods of template-free (or equivalently, "template-independent") enzymatic DNA synthesis or RNA synthesis comprise repeated cycles of steps, such as illustrated in FIG. 1, in which a predetermined nucleotide is coupled to an initiator or growing chain in each cycle. The general elements of template-free enzymatic synthesis of polynucleotides is described in the following references: Ybert et al, International patent publication WO/2015/159023; Ybert et al, International patent publication WO/2017/216472; Hyman, U.S. Pat. No. 5,436,143; Hiatt et al, U.S. Pat. No. 5,763,594; Jensen et al, *Biochemistry*, 57: 1821-1832 (2018); Mathews et al, *Organic & Biomolecular Chemistry*, DOI: 0.1039/c6ob01371f (2016); Schmitz et al, *Organic Lett.*, 1(11): 1729-1731 (1999).

[0019] Initiator polynucleotides (100) are provided, for example, attached to solid support (120), which have free 3'-hydroxyl groups (130). To the initiator polynucleotides (100) (or elongated initiator polynucleotides in subsequent cycles) are added a 3'-O-protected-dNTP or 3'-O-protected-rNTP and a template-free polymerase, such as a TdT or variant thereof usually for DNA synthesis (e.g. Ybert et al, WO/2017/216472; Champion et al, WO2019/135007) or a

polyA polymerase (PAP) or polyU polymerase (PUP) or variant thereof usually for RNA synthesis (e.g. Heinisch et al, WO2021/018919) under conditions (140) effective for the enzymatic incorporation of the 3'-O-protected-NTP onto the 3' end of the initiator polynucleotides (100) (or elongated initiator polynucleotides). This reaction produces elongated initiator polynucleotides whose 3'-hydroxyls are protected (106). If the elongated sequence is not complete, then another cycle of addition is implemented (108). If the elongated initiator polynucleotide contains a completed sequence, then the 3'-O-protection group may be removed, or deprotected, and the desired sequence may be cleaved from the original initiator polynucleotide (110). Such cleavage may be carried out using any of a variety of single strand cleavage techniques, for example, by inserting a cleavable nucleotide at a predetermined location within the original initiator polynucleotide. An exemplary cleavable nucleotide may be a uracil nucleotide which is cleaved by uracil DNA glycosylase. If the elongated initiator polynucleotide does not contain a completed sequence, then the 3'-O-protection groups are removed to expose free 3'-hydroxyls (103) and the elongated initiator polynucleotides are subjected to another cycle of nucleotide addition and deprotection.

[0020] As used herein, the terms "protected" and "blocked" in reference to specified groups, such as, a 3'-hydroxyls of a nucleotide or a nucleoside, are used interchangeably and are intended to mean a moiety is attached covalently to the specified group that prevents a chemical change to the group during a chemical or enzymatic process. Whenever the specified group is a 3'-hydroxyl of a nucleoside triphosphate, or an extended fragment (or "extension intermediate") in which a 3'-protected (or blocked)-nucleoside triphosphate has been incorporated, the prevented chemical change is a further, or subsequent, extension of the extended fragment (or "extension intermediate") by an enzymatic coupling reaction.

[0021] As used herein, an "initiator" (or equivalent terms, such as, "initiating fragment," "initiator nucleic acid," "initiator oligonucleotide," or the like) refers to a short oligonucleotide sequence with a free 3'-hydroxyl at its end, which can be further elongated by a template-free polymerase, such as TdT. In one embodiment, the initiating fragment is a DNA initiating fragment. In an alternative embodiment, the initiating fragment is an RNA initiating fragment. In some embodiments, an initiating fragment possesses between 3 and 100 nucleotides, in particular between 3 and 20 nucleotides, which may be all or partially polyC. In some embodiments, the initiating fragment is single-stranded. In alternative embodiments, the initiating fragment may be double-stranded. In some embodiments, an initiator oligonucleotide may be attached to a synthesis support by its 5' end; and in other embodiments, an initiator oligonucleotide may be attached indirectly to a synthesis support by forming a duplex with a complementary oligonucleotide that is directly attached to the synthesis support, e.g. through a covalent bond. In some embodiments a synthesis support is a solid support which may be a discrete region of a planar solid, or may be a bead.

[0022] In some embodiments, an initiator may comprise a non-nucleic acid compound having a free hydroxyl to which a TdT may couple a 3'-O-protected dNTP, e.g. Baiga, U.S. patent publications US2019/0078065 and US2019/0078126.

[0023] Synthesis supports to which PolyC-containing initiators are attached may comprise polymers, porous or

non-porous solids, including beads or microspheres, planar surfaces, such as a glass slide, membrane, or the like. In some embodiments, a solid support, or synthesis support, may comprise magnetic beads, particle-based resins, such as agarose, or the like.

[0024] Synthesis supports include, but are not limited to, soluble supports, such as, polymer supports, including polyethylene glycol (PEG) supports, dendrimer supports and the like; non-swellable solid supports, such as, polystyrene particles, Dynabeads, and the like; swellable solid supports, such as resins or gels, including agarose. Synthesis supports may also form part of reaction chambers, such as, the filter membrane of a filter plate. Guidance for selecting soluble supports is found in references Bonora et al, *Nucleic Acids Research*, 21(5): 1213-1217 (1993); Dickerson et al, *Chem. Rev.* 102: 3325-3344 (2002); Fishman et al, *J. Org. Chem.*, 68: 9843-9846 (2003); Gavert et al, *Chem. Rev.* 97: 489-509 (1997); Shchepinov et al, *Nucleic Acids Research*, 25(22): 4447-4454 (1997); and like references. Guidance for selecting solid supports is found in Brown et al, *Synlett* 1998(8): 817-827; Maeta et al, U.S. Pat. No. 9,045,573; Beaucage and Iyer, *Tetrahedron*, 48(12): 2223-2311 (1992); and the like. Guidance for attaching oligonucleotides to solid supports is found in Arndt-Jovin et al, *Eur. J. Biochem.*, 54: 411-418 (1975); Ghosh et al, *Nucleic Acids Research*, 15(13): 5353-5372 (1987); Integrated DNA Technologies, "Strategies for attaching oligonucleotides to solid supports," 2014 (v6); Gokmen et al, *Progress in Polymer Science* 37: 365-405 (2012); and like references.

[0025] In some embodiments, the solid-phase support will typically be comprised of porous beads or particles in the form of a resin or gel. Numerous materials are suitable as solid-phase supports for the synthesis of polynucleotides. As used herein, the term "particle" includes, without limitation, a "microparticle" or "nanoparticle" or "bead" or "microbead" or "microsphere." Particles or beads useful in the invention include, for example, beads measuring 1 to 300 microns in diameter, or 20 to 300 microns in diameter, or 30 to 300 microns in diameter, or beads measuring larger than 300 microns in diameter. A particle comprising polyC-containing initiators can be made of glass, plastic, polystyrene, resin, gel, agarose, sepharose, and/or other suitable materials. Of particular interest are porous resin particles or beads, such as, agarose beads. Exemplary agarose particles include Sepharose™ beads. In some embodiments, cyanogen bromide-activated 4% crosslinked agarose beads having diameters in the range of 40-165 μ m may be derivatized with polyC-containing initiators for use with methods of the invention. In other embodiments, cyanogen bromide-activated 6% crosslinked agarose beads having diameters in the range of 200-300 μ m may be used with methods of the invention. In the latter two embodiments, polyC-containing oligonucleotide initiators having a 5'-aminolinker may be coupled to the Sepharose™ beads for use with the invention. Other desirable linkers for agarose beads include thiol and epoxy linkers.

[0026] In some embodiments, a porous resin support derivatized with polyC-containing initiators has average pore diameters of at least 10 nm, or at least 20 nm, or at least 50 nm. In other embodiments, such porous resin support has an average pore diameter in the range of from 10 nm to 500 nm, or in the range of from 50 nm to 500 nm.

[0027] In some embodiments, polyC-containing initiators are attached to planar supports for massively parallel syn-

thesis of oligonucleotides, e.g. via inkjet delivery of reagents, such as described by Horgan et al, International patent publication WO2020/020608, which is incorporated herein by reference. In some embodiments such planar supports comprise a uniform coating of polyC-containing initiators with protected 3'-hydroxyls, wherein, for example, discrete reaction sites may be defined by delivering deprotection solution to discrete locations. In other embodiments, such planar supports comprise an array of discrete reaction sites each containing polyC-containing initiators, which, for example, may be formed on a substrate by photolithographic methods of Brennan, U.S. Pat. No. 5,474,796; Peck et al, U.S. patent Ser. No. 10/384,189; Indermuhle et al, U.S. patent Ser. No. 10/669,304; Fixe et al, *Materials Research Society Symposium Proceedings*, Volume 723, *Molecularly Imprinted Materials—Sensors and Other Devices*, Symposia (San Francisco, California on Apr. 2-5, 2002); or like references.

[0028] After synthesis is completed polynucleotides with the desired nucleotide sequence may be released from initiators and the solid supports by cleavage. A wide variety of cleavable linkages or cleavable nucleotides may be used for this purpose. In some embodiments, cleaving the desired polynucleotide leaves a natural free 5'-hydroxyl on a cleaved strand; however, in alternative embodiments, a cleaving step may leave a moiety, e.g. a 5'-phosphate, that may be removed in a subsequent step, e.g. by phosphatase treatment. Cleaving steps may be carried out chemically, thermally, enzymatically or by photochemical methods. In some embodiments, cleavable nucleotides may be nucleotide analogs such as deoxyuridine or 8-oxo-deoxyguanosine that are recognized by specific glycosylases (e.g. uracil deoxyglycosylase followed by endonuclease VIII, and 8-oxoguanine DNA glycosylase, respectively). In some embodiments, cleavage may be accomplished by providing initiators with a deoxyinosine as the penultimate 3' nucleotide, which may be cleaved by endonuclease V at the 3' end of the initiator leaving a 5'-phosphate on the released polynucleotide. Further methods for cleaving single stranded polynucleotides are disclosed in the following references, which are incorporated by reference: U.S. Pat. Nos. 5,739,386, 5,700,642 and 5,830,655; and U.S. Patent Publication Nos. 2003/0186226 and 2004/0106728; and in Urdea and Horn, U.S. Pat. No. 5,367,066.

[0029] In some embodiments, cleavage by glycosylases and/or endonucleases may require a double stranded DNA substrate.

[0030] Returning to FIG. 1, in some embodiments, an ordered sequence of nucleotides are coupled to an initiator nucleic acid using a template-free polymerase, such as TdT, in the presence of 3'-O-protected NTPs in each synthesis step. In some embodiments, the method of synthesizing an oligonucleotide comprises the steps of (a) providing an initiator having a free 3'-hydroxyl (**100**); (b) reacting (**104**) under extension conditions the initiator or an extension intermediate having a free 3'-hydroxyl with a template-free polymerase in the presence of a 3'-O-protected nucleoside triphosphate to produce a 3'-O-protected extension intermediate (**106**); (c) deprotecting the extension intermediate to produce an extension intermediate with a free 3'-hydroxyl (**108**); and (d) repeating steps (b) and (c) (**110**) until the polynucleotide is synthesized. (Sometimes the terms "extension intermediate" and "elongation fragment" are used interchangeably). In some embodiments, an initiator is provided

as an oligonucleotide attached to a solid support, e.g. by its 5' end. The above method may also include a washing step after each reaction, or extension, step, as well as after each de-protecting step. For example, the step of reacting may include a sub-step of removing unincorporated nucleoside triphosphates, e.g. by washing, after a predetermined incubation period, or reaction time. Such predetermined incubation periods or reaction times typically may be a few seconds, e.g. 30 sec, to several minutes, e.g. 30 min.

[0031] When the sequence of polynucleotides on a synthesis support includes reverse complementary subsequences, secondary intra-molecular or cross-molecular structures may be created by the formation of hydrogen bonds between the reverse complementary regions. In some embodiments, base protecting moieties for exocyclic amines are selected so that hydrogens of the protected nitrogen cannot participate in hydrogen bonding, thereby preventing the formation of such secondary structures. That is, base protecting moieties may be employed to prevent the formation of hydrogen bonds, such as are formed in normal base pairing, for example, between nucleosides A and T and between G and C. At the end of a synthesis, the base protecting moieties may be removed and the polynucleotide product may be cleaved from the solid support, for example, by cleaving it from its initiator.

[0032] In addition to providing 3'-O-blocked NTP monomers with base protection groups, elongation reactions may be performed at higher temperatures using thermal stable template-free polymerases. For example, a thermal stable template-free polymerase having activity above 40° C. may be employed; or, in some embodiments, a thermal stable template-free polymerase having activity in the range of from 40-85° C. may be employed; or, in some embodiments, a thermal stable template-free polymerase having activity in the range of from 40-65° C. may be employed.

[0033] In some embodiments, elongation conditions may include adding solvents to an elongation reaction mixture that inhibit hydrogen bonding or base stacking. Such solvents include water miscible solvents with low dielectric constants, such as dimethyl sulfoxide (DMSO), methanol, and the like. Likewise, in some embodiments, elongation conditions may include the provision of chaotropic agents that include, but are not limited to, n-butanol, ethanol, guanidinium chloride, lithium perchlorate, lithium acetate, magnesium chloride, phenol, 2-propanol, sodium dodecyl sulfate, thiourea, urea, and the like. In some embodiments, elongation conditions include the presence of a secondary-structure-suppressing amount of DMSO. In some embodiments, elongation conditions may include the provision of DNA binding proteins that inhibit the formation of secondary structures, wherein such proteins include, but are not limited to, single-stranded binding proteins, helicases, DNA glycolases, and the like.

[0034] 3'-O-blocked dNTPs without base protection may be purchased from commercial vendors or synthesized using published techniques, e.g. U.S. Pat. No. 7,057,026; Guo et al, Proc. Natl. Acad. Sci., 105(27): 9145-9150 (2008); Benner, U.S. Pat. Nos. 7,544,794 and 8,212,020; International patent publications WO2004/005667, WO91/06678; Canard et al, Gene (cited herein); Metzker et al, Nucleic Acids Research, 22: 4259-4267 (1994); Meng et al, J. Org. Chem., 14: 3248-3252 (3006); U.S. patent publication 2005/037991. 3'-O-blocked dNTPs with base protection may be synthesized as described below.

[0035] When base-protected dNTPs are employed the method of FIG. 1 may further include a step (e) removing base protecting moieties, which in the case of acyl or amidine protection groups may (for example) include treating with concentrated ammonia.

[0036] The above method may also include one or more capping steps in addition to washing steps after the reacting, or extending, step. A first capping step may cap, or render inert to further extensions, unreacted 3'-OH groups on partially synthesized polynucleotides. Such capping step is usually implemented after a coupling step, and whenever a capping compound is used, it is selected to be unreactive with protection groups of the monomer just coupled to the growing strands. In some embodiments, such capping steps may be implemented by coupling (for example, by a second enzymatic coupling step) a capping compound that renders the partially synthesized polynucleotide incapable of further couplings, e.g. with TdT. Such capping compounds may be a dideoxynucleoside triphosphate. In other embodiments, non-extended strands with free 3'-hydroxyls may be degraded by treating them with a 3'-exonuclease activity, e.g. Exo I. For example, see Hyman, U.S. Pat. No. 5,436, 143. Likewise, in some embodiments, strands that fail to be deblocked may be treated to either remove the strand or render it inert to further extensions. A second capping step may be implemented after a deprotection step, to render the affected strands inert from any subsequent coupling or deprotection any 3'-O protection, or blocking groups. Capping compounds of such second capping step are selected so that they do not react with free 3'-hydroxyls that may be present. In some embodiments, such second capping compound may be a conjugate of an aldehyde group and a hydrophobic group. The latter group permits separation based on hydrophobicity, e.g. Andrus, U.S. Pat. No. 5,047, 524.

[0037] Exemplary reaction conditions for an elongation step (also sometimes referred to as an extension step or a coupling step) comprise the following: 2.0-20. μ M purified TdT; 125-600 μ M 3'-O-blocked dNTP (e.g. 3'-O—NH₂-blocked dNTP); about 10 to about 500 mM potassium cacodylate buffer (pH between 6.5 and 7.5) and from about 0.01 to about 10 mM of a divalent cation (e.g. CoCl₂ or MnCl₂), where the elongation reaction may be carried out in a 50 μ L reaction volume, at a temperature within the range RT to 45° C., for 3-5 minutes. In embodiments, in which the 3'-O-blocked dNTPs are 3'-O—NH₂-blocked dNTPs, reaction conditions for a deblocking step may comprise the following: 700-1500 mM NaNO₂; 500-1000 mM sodium acetate (adjusted with acetic acid to pH in the range of 4.8-6.5), where the deblocking reaction may be carried out in a 50 μ L volume, at a temperature within the range of RT to 45° C. for 30 seconds to several minutes. Washes may be performed with the cacodylate buffer without the components of the coupling reaction (e.g. enzyme, monomer, divalent cations).

[0038] Depending on particular applications, the steps of deblocking and/or cleaving may include a variety of chemical or physical conditions, e.g. light, heat, pH, presence of specific reagents, such as enzymes, which are able to cleave a specified chemical bond. Guidance in selecting 3'-O-blocking groups and corresponding de-blocking conditions may be found in the following references, which are incorporated by reference: Benner, U.S. Pat. Nos. 7,544,794 and 8,212,020; 5,808,045; 8,808,988; International patent pub-

lication WO91/06678; and references cited below. In some embodiments, the cleaving agent (also sometimes referred to as a de-blocking reagent or agent) is a chemical cleaving agent, such as, for example, dithiothreitol (DTT). In alternative embodiments, a cleaving agent may be an enzymatic cleaving agent, such as, for example, a phosphatase, which may cleave a 3'-phosphate blocking group. It will be understood by the person skilled in the art that the selection of deblocking agent depends on the type of 3'-nucleotide blocking group used, whether one or multiple blocking groups are being used, whether initiators are attached to living cells or organisms or to solid supports, and the like, that necessitate mild treatment. For example, a phosphine, such as tris(2-carboxyethyl)phosphine (TCEP) can be used to cleave a 3'-azidomethyl groups, palladium complexes can be used to cleave a 3'-allyl groups, or sodium nitrite can be used to cleave a 3'-amino group. In particular embodiments, the cleaving reaction involves TCEP, a palladium complex or sodium nitrite.

[0039] As noted above, in some embodiments it is desirable to employ two or more blocking groups that may be removed using orthogonal de-blocking conditions. The following exemplary pairs of blocking groups may be used in parallel synthesis embodiments. It is understood that other blocking group pairs, or groups containing more than two, may be available for use in these embodiments of the invention.

3'-O-NH ₂	3'-O-azidomethyl
3'-O-NH ₂	3'-O-allyl, 3'-O-propargyl
3'-O-NH ₂	3'-O-phosphate
3'-O-azidomethyl	3'-O-allyl, 3'-O-propargyl
3'-O-azidomethyl	3'-O-phosphate
3'-O-allyl, 3'-O-propargyl	3'-O-phosphate

[0040] Synthesizing oligonucleotides on living cells requires mild deblocking, or deprotection, conditions, that is, conditions that do not disrupt cellular membranes, denature proteins, interfere with key cellular functions, or the like. In some embodiments, deprotection conditions are within a range of physiological conditions compatible with cell survival. In such embodiments, enzymatic deprotection is desirable because it may be carried out under physiological conditions. In some embodiments specific enzymatically removable blocking groups are associated with specific enzymes for their removal. For example, ester- or acyl-based blocking groups may be removed with an esterase, such as acetylase, or like enzyme, and a phosphate blocking group may be removed with a 3' phosphatase, such as T4 polynucleotide kinase. By way of example, 3'-O-phosphates may be removed by treatment with a solution of 100 mM Tris-HCl (pH 6.5) 10 mM MgCl₂, 5 mM 2-mercaptoethanol, and one Unit T4 polynucleotide kinase. The reaction proceeds for one minute at a temperature of 37° C.

[0041] In some embodiments, 3'-O blocking groups include 3'-O-azidomethyl, 3'-O—NH₂, 3'-O-allyl. In some embodiments, blocking group include 3'-O-methyl, 3'-O-(2-nitrobenzyl), 3'-O-allyl, 3'-O-amine, 3'-O-azidomethyl, 3'-O-tert-butoxy ethoxy, 3'-O-(2-cyanoethyl), 3'-O-nitro, and 3'-O-propargyl. In other embodiments, the 3'-blocked nucleotide triphosphate is blocked by either a 3'-O-azidomethyl or a 3'-O—NH₂. Synthesis and use of such 3'-blocked nucleoside triphosphates are disclosed in the following references: U.S. Pat. Nos. 9,410,197; 8,808,988; 6,664,097;

5,744,595; 7,544,794; 8,034,923; 8,212,020; 10,472,383; Guo et al, Proc. Natl. Acad. Sci., 105(27): 9145-9150 (2008); and like references.

[0042] Depending on particular applications, the steps of deblocking and/or cleaving may include a variety of chemical or physical conditions, e.g. light, heat, pH, presence of specific reagents, such as enzymes, which are able to cleave a specified chemical bond. Guidance in selecting 3'-O-blocking groups and corresponding de-blocking conditions may be found in references, such as Wuts, Green's Protection Groups in Organic Chemistry, 5th Edition (Wiley 2014). In some embodiments, the cleaving agent (also sometimes referred to as a de-blocking reagent or agent) is a chemical cleaving agent, such as, for example, dithiothreitol (DTT). In alternative embodiments, a cleaving agent may be an enzymatic cleaving agent, such as, for example, a phosphatase, which may cleave a 3'-phosphate blocking group. It will be understood by the person skilled in the art that the selection of deblocking agent depends on the type of 3'-nucleotide blocking group used, whether one or multiple blocking groups are being used, whether initiators are attached to living cells or organisms or to solid supports, and the like, that necessitate mild treatment. For example, a phosphine, such as tris(2-carboxyethyl)phosphine (TCEP) can be used to cleave a 3'-O-azidomethyl group, palladium complexes can be used to cleave 3'-O-allyl group and 3'-O-propargyl group, or sodium nitrite can be used to cleave a 3'-O-amino group.

Template-Free Enzymatic Synthesis of RNA

[0043] Methods of the invention comprise the enzymatic synthesis of RNA. In some embodiments, such methods comprise the steps described in FIG. 1 using as a template-free polymerase a poly(A) polymerase (PAP) or a poly(U) polymerase. In some embodiments, PAPs and/or PUPs are used to synthesize a polyribonucleic acid using 3'-O-reversibly protected-rNTP precursors, wherein a single PUP or PAP variant may be employed for coupling all ribonucleoside triphosphate monomers, or in alternative embodiments. In some embodiments, different PUPs and PAPs may be employed for coupling different kinds ribonucleoside triphosphate monomers in the synthesis of a particular RNA. Likewise, in other embodiments, PAPs and/or PUPs may be used to synthesize a polydeoxyribonucleic acid using 3'-O-reversibly protected-dNTP precursors, wherein a single PUP or PAP is employed for coupling all deoxyribonucleoside triphosphate (dNTP) monomers, or in an alternative embodiment, wherein different PUP and PAP polymerases may be employed for coupling different kinds of deoxyribonucleoside triphosphate monomers. In some embodiments for RNA synthesis, the same 3'-O-reversible protecting groups described above for deoxyribonucleotides may also be used with ribonucleotide monomers. In some embodiments for RNA synthesis, said 3'-O-blocked nucleoside triphosphate is a 3'-O-azidomethyl-ribonucleoside triphosphate. In some embodiments, methods may employ PAP and/or PUP variants that have been modified by genetic engineering to improve efficiency of coupling 3'-O-blocked-ribonucleoside triphosphates and 3'-O-blocked-2'-deoxyribonucleoside triphosphates to growing polynucleotide chains in a synthesis, for example, as described below.

[0044] In some embodiments, the method of synthesizing an oligoribonucleotide of a predetermined sequence comprises the steps of (a) providing an initiator having a free

3'-hydroxyl; (b) reacting under elongation conditions the initiator or an elongation fragment having a free 3'-hydroxyl with a PAP or a PUP in the presence of a 3'-O-blocked ribonucleoside triphosphate to produce a 3'-O-blocked elongation fragment; (c) deblocking the elongation fragment to produce an elongation fragment with a free 3'-hydroxyl; and (d) repeating steps (b) and (c) until the polyribonucleotide of the predetermined sequence is synthesized, wherein the reaction mixture for elongating the initiators or elongated fragments comprise polyC oligonucleotides capable of forming duplexes with the polynucleotide. In some embodiments, initiators each comprise a segment consisting of a polyC oligonucleotide. In other embodiments, a synthesis support is provided having polyC oligonucleotides attached thereto, eventually in addition to initiators. In still other embodiments, polyC oligonucleotides are provided in solution as a component of an elongation reaction mixture for selected elongation cycles in which G4 structure formation has a high likelihood of occurrence.

[0045] In some embodiments, as noted above, an initiator is provided as an oligonucleotide attached to a solid support, e.g. by its 5' end. The above method may also include washing steps after the reaction, or extension, step, as well as after the de-blocking step. For example, the step of reacting may include a sub-step of removing unincorporated ribonucleoside triphosphates, e.g. by washing, after a predetermined incubation period, or reaction time. Such predetermined incubation periods or reaction times may be a few seconds, e.g. 30 sec, to several minutes, e.g. 30 min.

[0046] The above method may also include capping step (s) as well as washing steps after the reacting, or extending, step, as well as after the deblocking step. As mentioned above, in some embodiments, capping steps may be included in which non-extended free 3'-hydroxyls are reacted with compounds that prevents any further extensions of the capped strand. In some embodiments, such compound may be a dideoxynucleoside triphosphate. In other embodiments, non-extended strands with free 3'-hydroxyls may be degraded by treating them with a 3'-exoribonuclease activity, e.g. RNase R (Epicentre). Likewise, in some embodiments, strands that fail to be deblocked may be treated to either remove the strand or render it inert to further extensions.

[0047] Exemplary reaction conditions for an extension or elongation step using PAP or PUP comprise the following: Reaction conditions 1 (for primer+AM-rATP): 250 uM AM-rATP, 0.1 uM ATT0488-(rA)5, 1 uM PAP, 1x ATP buffer (20 mM Tris-HCl, 0.6 mM MnCl₂, 0.02 mM EDTA, 0.1% BSA, 10% glycerol, 100 mM imidazole, pH 7-8), 37 C, 30 min. Reaction condition 2 (for primer+AM-rGTP): 250 uM rGTP, 0.1 uM ATT0488-(rA)5, 1 uM PAP, 1x GTP buffer (0.6 mM MnCl₂, 0.1% BSA, 10 mM imidazole, pH 6), 37 C, 30 min. In the foregoing, "AM-rNTP" refers to 3'-O-azidomethyl-ribonucleoside triphosphate.

Template-Free Polymerases for Polynucleotide Synthesis

[0048] A variety of different template-free polymerases are available for use in methods of the invention. Template-free polymerases include, but are not limited to, polX family polymerases (including DNA polymerases β , γ and μ), poly(A) polymerases (PAPs), poly(U) polymerases (PUPs), DNA polymerase θ , and the like, for example, described in the following references: Ybert et al, International patent publication WO2017/216472; Champion et al, U.S. patent Ser. No. 10/435,676; Champion et al, International patent publication WO2020/099451; Heinisch et al, International patent publication WO2021/018919. In particular, terminal deoxynucleotidyltransferases (TdT) and variants thereof are useful in template-free DNA synthesis and PAPs and PUPs and variants thereof are useful in template-free RNA synthesis.

[0049] In some embodiments, TdT variants are employed with the invention which display increased incorporation activity with respect to 3'-O-modified nucleoside triphosphates. For example, such TdT variants may be produced using techniques described in Champion et al, U.S. patent Ser. No. 10/435,676, which is incorporated herein by reference. In some embodiments, a TdT variant is employed having an amino acid sequence at least 80 percent identical to a TdT having an amino acid sequence of any of SEQ ID NOs 7 through 20, inclusive, and 24 through 39, inclusive, and one or more of the substitutions listed in Table 1, wherein the TdT variant (i) is capable of synthesizing a nucleic acid fragment without a template and (ii) is capable of incorporating a 3'-O-modified nucleotide onto a free 3'-hydroxyl of a nucleic acid fragment. In some embodiments, the above TdT variants include a substitution at every position listed in Table 1. In some embodiments, the above percent identity value is at least 85 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 90 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 95 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 97 percent identity; in some embodiments, the above percent identity value is at least 98 percent identity; in some embodiments, the above percent identity value is at least 99 percent identity. As used herein, the percent identity values used to compare a reference sequence to a variant sequence do not include the expressly specified amino acid positions containing substitutions of the variant sequence; that is, the percent identity relationship is between sequences of a reference protein and sequences of a variant protein outside of the expressly specified positions containing substitutions in the variant. Thus, for example, if the reference sequence and the variant sequence each comprised 100 amino acids and the variant sequence had mutations at positions 25 and 81, then the percent homology would be in regard to sequences 1-24, 26-80 and 82-100.

TABLE 1

SEQ ID NO		Substitutions					
Animal							
1	Mouse	M192R/Q	C302G/R	R336L/N	R454P/N/A/V	E457N/L/T/S/K	
7	Mouse	M63R/Q	C173G/R	R207L/N	R325P/N/A/V	E328N/L/T/S/K	
8	Bovine	M63R/Q	C173G/R	R207L/N	R324P/N/A/V	E327N/L/T/S/K	

TABLE 1-continued

SEQ ID NO	Animal	Substitutions					
9	Human	M63R/Q	C173G/R	R207L/N	R324P/N/A/V	E327N/L/T/S/K	
10	Chicken	—	C172G/R	R206L/N	R320P/N/A/V	—	
11	Possum	M63R/Q	C173G/R	R207L/N	R331P/N/A/V	E334N/L/T/S/K	
12	Shrew	M63R/Q	C173G/R	R207L/N	—	E328N/L/T/S/K	
13	Python	—	C174G/R	R208L/N	R331P/N/A/V	E334N/L/T/S/K	
14	Canine	M73R/Q	C173G/R	R207L/N	R325P/N/A/V	E328N/L/T/S/K	
15	Mole	M64R/Q	C174G/R	R208L/N	—	E329N/L/T/S/K	
16	Pika	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	
17	Hedgehog	M63R/Q	C173G/R	R207L/N	R328P/N/A/V	E331N/L/T/S/K	
18	Tree shrew	—	C173G/R	R207L/N	R325P/N/A/V	E328N/L/T/S/K	
19	Platypus	M63R/Q	C182G/R	R216L/N	R338P/N/A/V	E341N/L/T/S/K	
20	Jerboa	M66R/Q	C176G/R	R210L/N	R328P/N/A/V	E331N/L/T/S/K	
24	Canary	—	C170G/R	R204L/N	R326P/N/A/V	E329N/L/T/S/K	
25	Neopelma	—	C158G/R	R192L/N	R314P/N/A/V	E317N/L/T/S/K	
26	Alligator	—	—	R205L/N	R327P/N/A/V	E330N/L/T/S/K	
27	Xenopus	—	—	R205L/N	R324P/N/A/V	E327N/L/T/S/K	
28	Tiger snake	—	—	R205L/N	R327P/N/A/V	E330N/L/T/S/K	
29	Brown trout	—	—	R192L/N	R311P/N/A/V	E314N/L/T/S/K	
30	Electric eel	—	—	R205L/N	R321P/N/A/V	E325N/L/T/S/K	
31	Walking fish	—	—	R205L/N	R322P/N/A/V	E325N/L/T/S/K	
32	Guppy	—	—	R205L/N	R322P/N/A/V	E325N/L/T/S/K	
33	Rat	M48R/Q	C158G/R	R192L/N	R310P/N/A/V	E313N/L/T/S/K	
34	Rat	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	
35	Colobus monkey	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	
36	Pig	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	
37	Tiger	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	
38	Water buffalo	M48R/Q	C158G/R	R192L/N	R310P/N/A/V	E313N/L/T/S/K	
39	Marmot	M61R/Q	C171G/R	R205L/N	R323P/N/A/V	E326N/L/T/S/K	

[0050] In some embodiments, a TdT variant of the invention is derived from a TdT comprising an amino acid sequence at least 80 percent identical to an amino acid sequence selected from SEQ ID NOs 40 through 75, inclusive, and one or more of the substitutions listed in Table 1, wherein the TdT variant (i) is capable of synthesizing a nucleic acid fragment without a template and (ii) is capable of incorporating a 3'-O-modified nucleotide onto a free 3'-hydroxyl of a nucleic acid fragment. In some embodiments, the above TdT variants include a substitution at every position listed in Table 2. In some embodiments, the above percent identity value is at least 85 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 90 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 95 percent identity with the indicated SEQ ID NOs; in some embodiments, the above percent identity value is at least 97 percent identity; in some embodiments, the above percent identity value is at least 98 percent identity; in some embodiments, the above percent identity value is at least 99 percent identity. As above, the

percent identity values used to compare a reference sequence to a variant sequence do not include the expressly specified amino acid positions containing substitutions of the variant sequence; that is, the percent identity relationship is between sequences of a reference protein and sequences of a variant protein outside of the expressly specified positions containing substitutions in the variant.

[0051] TdT variants of SEQ ID NOs 40 through 54, inclusive, 56, 59, 61, 63, 65, 67, 69, 70, 73 and 74 includes substitutions at one or more of the indicated amino acid positions as listed in Table 2 in addition to a stabilizing substitution of the glutamine at position 4 (or a functionally equivalent position). In other embodiments, TdT variants of the invention are derived from natural TdTs such as those listed in Table 2 with a substitution at every one of the indicated amino acid positions in addition to the stabilizing substitution of the glutamine at position 4. In some embodiments, such stabilizing amino acid substituted for glutamine is selected from the group consisting of E, S, D and N. In other embodiments, the stabilizing amino acid is E.

TABLE 2

SEQ ID NO	Animal	Substitutions					
1	Mouse	M192R/Q	C302G/R	R336L/N	R454P/N/A/V	E457N/L/T/S/K	
40	Mouse	M44R/Q	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K	
41	Bovine	M44R/Q	C154G/R	R188L/N	R305P/N/A/V	E308N/L/T/S/K	
42	Human	M44R/Q	C154G/R	R188L/N	R305P/N/A/V	E308N/L/T/S/K	
43	Chicken	—	C154G/R	R188L/N	R302P/N/A/V	—	
44	Possum	M44R/Q	C154G/R	R188L/N	R312P/N/A/V	E315N/L/T/S/K	
45	Shrew	M44R/Q	C154G/R	R188L/N	—	E309N/L/T/S/K	
46	Canine	M44R/Q	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K	
47	Mole	M44R/Q	C154G/R	R188L/N	—	E309N/L/T/S/K	
48	Pika	M44R/Q	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K	

TABLE 2-continued

SEQ ID NO	Animal	Substitutions				
49	Hedgehog	M44R/Q	C154G/R	R188L/N	R309P/N/A/V	E312N/L/T/S/K
50	Tree shrew	—	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K
51	Platypus	M44R/Q	C163G/R	R197L/N	R319P/N/A/V	E322N/L/T/S/K
52	Canary	—	C153G/R	R187L/N	R309P/N/A/V	—
53	Neopelma	—	C154G/R	R188L/N	R310P/N/A/V	E311N/L/T/S/K
54	Alligator	—	—	R188L/N	R310P/N/A/V	E313N/L/T/S/K
56	Xenopus	—	—	R188L/N	R307P/N/A/V	E310N/L/T/S/K
59	Brown Trout	—	—	R188L/N	—	E310N/L/T/S/K
61	Electric eel	—	—	R188L/N	—	—
63	Walking fish	—	—	R188L/N	R305P/N/A/V	E308N/L/T/S/K
65	Guppy	—	—	R188L/N	R305P/N/A/V	E308N/L/T/S/K
67	Rat	—	—	R188L/N	R306P/N/A/V	E309N/L/T/S/K
69	Piliocolobus	—	—	R188L/N	R306P/N/A/V	E309N/L/T/S/K
70	Pig	M44R/Q	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K
73	Water buffalo	M44R/Q	C154G/R	R188L/N	R305P/N/A/V	E308N/L/T/S/K
74	Marmot	M44R/Q	C154G/R	R188L/N	R306P/N/A/V	E309N/L/T/S/K

[0052] In some embodiments, further TdT variants for use with methods of the invention include one or more of the substitutions of methionine, cysteine, arginine (first position), arginine (second position) or glutamic acid, as shown in Table 2.

[0053] In some embodiments, a TdT variant comprising an amino acid sequence at least ninety percent identical to an amino acid sequence of SEQ ID NOs 55, 57, 58, 60, 62, 64, 66, 68, 71, 72, and 75 through 112, inclusive, may also be used with the present invention.

[0054] In regard to TdT variants of SEQ ID NOs 7 through 112, in some embodiments, a 3'-O-modified nucleotide may comprise a 3'-O—NH₂-nucleoside triphosphate, a 3'-O-azidomethyl-nucleoside triphosphate, a 3'-O-allyl-nucleoside triphosphate, a 3'-O-(2-nitrobenzyl)-nucleoside triphosphate, or a 3'-O-propargyl-nucleoside triphosphate.

[0055] A wide variety of PAPs may be used with the method of the invention, including PAP variants that have been engineered for improved characteristics, such as, higher incorporation rates of 3'-O-protected-rNTPs (including for particular protection groups, such as, 3'-O-azidomethyl), greater stability and shelf life, thermostability, solubility, and the like. In particular, a yeast PAP with a mutation at M310 (SEQ ID NO: 1), or a functionally equivalent residue in other PAPs, such as PAPs from various different species, shows improved incorporation of 3'-O-protected rNTPs with respect to a wildtype PAP. In some embodiments, a yeast PAP variant of the invention has an amino acid sequence of SEQ ID NO: 1 except for a substitution at M310. In some embodiments, such substitution is selected from M310F/Y/V/E/T. In particular, substitutions M310F/Y allow the incorporation of 3'-O-amino-rATPs and substitutions M310V/E/T improve the rate of incorporation of 3'-O-protected-rGTPs. In other embodiments, a yeast PAP variant of the invention has an amino acid sequence with at least 90 percent identity of SEQ ID NO: 1 except for a substitution at M310.

[0056] PAP variants for use with the invention include those listed in Table 3 below. In some embodiments PAP variants of the invention comprise at least a substitution at the second position indicated in Table 3. In other embodiments, embodiments of PAP variants of the invention comprise at least a substitution at the first position indicated in Table 3.

TABLE 3

PAP Variants: Positions of Substitutions			
SEQ ID NO	Organism	First Position	Second Position
113	yeast	V234	M310
114	<i>Myceliophthora</i>	V240	M318
115	<i>Thielavia</i>	V240	M318
116	<i>Pyronema</i>	I237	M316
117	<i>Tilletia</i>	V232	M309
118	<i>Clathrospora</i>	V240	M316
119	<i>Drechslerella</i>	V196	M272
120	<i>Magnaportheopsis</i>	V240	M316
121	<i>Cryptococcus</i>	V229	M307
122	<i>Golovinomyces</i>	V236	M313
123	<i>Hortaea</i>	V236	M312
124	<i>Valsa</i>	V241	M317
125	<i>Wallemia</i>	V233	M316
126	<i>Xylaria</i>	V240	M316
127	<i>Chaetomium</i>	V240	M312
128	<i>Lachancea</i>	V234	M310
129	<i>Schizosaccharomyces</i>	V233	M309
130	<i>Exophiala</i>	V237	M317
131	<i>Scedosporium</i>	V238	M314
132	<i>Trichoderma</i>	V231	M307
133	<i>Aspergillus</i>	V239	M315
134	<i>Sodiomyces</i>	V240	M316
135	<i>Neohortaea</i>	V235	M311

[0057] In some embodiments, a substitution at a first position as indicated in Table 3 is A or G (thus, for example, for SEQ ID NO: 113, the substitution may be written V234A/G). In some embodiments, a substitution at a second position as indicated in Table 3 is F, Y, V, E, or T (thus, for example, for SEQ ID NO: 113, the substitution may be written M310F/Y/V/E/T).

[0058] In some embodiments, a PAP variant of the invention has one or more of the substitutions of Table 3 and a percent identity value of at least 80 percent identity with the indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 90 percent identity with the indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 95 percent identity with the indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 97 percent identity; in some embodiments, the above percent identity value is at least 98 percent identity; in some embodiments, the above percent identity value is at least 99 percent identity.

[0059] In some embodiments, a thermostable PAP is employed so that the method may be practiced at a temperature that reduces or eliminates the formation of secondary structures in the RNA or DNA being synthesized. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PAP is higher than 40° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PAP is higher than 50° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PAP is between 40° C. and 85° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PAP is between 50° C. and 85° C.

[0060] As with PAPs, a wide variety of PUPs may be used with the method of the invention, including PUP variants that have been engineered for improved characteristics, such as, higher incorporation rates of 3'-O-protected-rNTPs (including for particular protection groups, such as, 3'-O-azidomethyl), greater stability and shelf life, thermostability, solubility, and the like. PUP variants for use with the invention include those listed in Table 4 below. In some embodiments PUP variants of the invention comprise at least a substitution at the first position indicated in Table 4. In other embodiments, embodiments of PUP variants of the invention comprise at least a substitution at the second position indicated in Table 4.

TABLE 4

PUP Variants: Positions of Substitutions			
SEQ ID NO	Organism	First Position	Second Position
136	<i>S. pombe</i>	Y212	H336
137	<i>T. brucei</i>	Y189	L303
138	<i>S. pombe</i>	Y184	H308
139	<i>T. boudieri</i>	Y227	H364
140	<i>D. stenobrocha</i>	Y478	H613
141	<i>Phytomonas</i>	Y192	L306
142	<i>B. saltans</i>	Y186	L326
143	<i>A. deanei</i>	Y243	L392
144	<i>P. lactucaedebilis</i>	Y196	H330
145	<i>S. culicis</i>	Y253	L392
146	<i>B. meristosporus</i>	Y284	H408
147	<i>N. californiae</i>	Y182	H310
148	<i>Perkinsela</i>	Y187	L394
149	<i>S. complicate</i>	Y203	H331
150	<i>S. ochraceum</i>	Y224	F349
151	<i>G. androsaceus</i>	Y204	Y332
152	<i>T. equiperdum</i>	Y337	L473
153	<i>M. conica</i>	Y296	H431
154	<i>P. murina</i>	Y291	H423
155	<i>S. japonicus</i>	Y218	H340
156	<i>A. nigricans</i>	Y366	H509

[0061] In some embodiments, a substitution at a first position as indicated in Table 4 is A or G (thus, for example, for SEQ ID NO: 136, the substitution may be written Y212A/G). In some embodiments, a substitution at a second position as indicated in Table 4 is F, Y, V, E, or T (thus, for example, for SEQ ID NO: 4, the substitution may be written H336F/Y/V/E/T)

[0062] In some embodiments, a PUP variant of the invention has one or more of the substitutions of Table 4 and a percent identity value of at least 80 percent identity with the indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 90 percent identity with the

indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 95 percent identity with the indicated SEQ ID NO; in some embodiments, the above percent identity value is at least 97 percent identity; in some embodiments, the above percent identity value is at least 98 percent identity; in some embodiments, the above percent identity value is at least 99 percent identity.

[0063] In some embodiments, a thermostable PUP is employed so that the method may be practiced at a temperature that reduces or eliminates the formation of secondary structures in the RNA or DNA being synthesized. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PUP is higher than 40° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PUP is higher than 50° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PUP is between 40° C. and 85° C. In some embodiments, the temperature range within which the highest incorporation rate occurs for the thermostable PUP is between 50° C. and 85° C.

[0064] TdT, PAP and PUP variants for use with the invention each comprise an amino acid sequence having a percent sequence identity with a specified SEQ ID NO, subject to the presence of indicated substitutions. In some embodiments, the number and type of sequence differences between a variant of the invention described in this manner and the specified SEQ ID NO may be due to substitutions, deletion and/or insertions, and the amino acids substituted, deleted and/or inserted may comprise any amino acid. In some embodiments, such deletions, substitutions and/or insertions comprise only naturally occurring amino acids. In some embodiments, substitutions comprise only conservative, or synonymous, amino acid changes, as described in Grantham, Science, 185: 862-864 (1974). That is, a substitution of an amino acid can occur only among members of its set of synonymous amino acids. In some embodiments, sets of synonymous amino acids that may be employed are set forth in Table 5A.

TABLE 5A

Synonymous Sets of Amino Acids I	
Amino Acid	Synonymous Set
Ser	Ser, Thr, Gly, Asn
Arg	Arg, Gln, Lys, Glu, His
Leu	Ile, Phe, Tyr, Met, Val, Leu
Pro	Gly, Ala, Thr, Pro
Thr	Pro, Ser, Ala, Gly, His, Gln, Thr
Ala	Gly, Thr, Pro, Ala
Val	Met, Tyr, Phe, Ile, Leu, Val
Gly	Gly, Ala, Thr, Pro, Ser
Ile	Met, Tyr, Phe, Val, Leu, Ile
Phe	Trp, Met, Tyr, Ile, Val, Leu, Phe
Tyr	Trp, Met, Phe, Ile, Val, Leu, Tyr
Cys	Cys, Ser, Thr
His	His, Glu, Lys, Gln, Thr, Arg
Gln	Gln, Glu, Lys, Asn, His, Thr, Arg
Asn	Asn, Gln, Asp, Ser
Lys	Lys, Glu, Gln, His, Arg
Asp	Asp, Glu, Asn
Glu	Glu, Asp, Lys, Asn, Gln, His, Arg
Met	Met, Phe, Ile, Val, Leu
Trp	Trp

[0065] In some embodiments, sets of synonymous amino acids that may be employed are set forth in Table 5B.

TABLE 5B

Synonymous Sets of Amino Acids II	
Amino Acid	Synonymous Set
Ser	Ser
Arg	Arg, Lys, His
Leu	Ile, Phe, Met, Leu
Pro	Ala, Pro
Thr	Thr
Ala	Pro, Ala
Val	Met, Ile Val
Gly	Gly
Ile	Met, Phe, Val, Leu, Ile
Phe	Met, Tyr, Ile, Leu, Phe
Tyr	Trp, Met
Cys	Cys, Ser
His	His, Gln, Arg
Gln	Gln, Glu, His
Asn	Asn, Asp
Lys	Lys, Arg
Asp	Asp, Asn
Glu	Glu, Gln
Met	Met, Phe, Ile, Val, Leu
Trp	Trp

[0066] TdT, PAP and PUP variants for use with the invention are produced by conventional biotechnology techniques and may include an affinity tag for purification, which may be attached to the N-terminus, C-terminus or at an interior position of the template-free polymerase. In some embodiments, affinity tags are cleaved before the template-free polymerase is used. In other embodiments, affinity tags are not cleaved before use. In some embodiments, a peptide affinity tag is inserted into a loop 2 region of a TdT variant. An exemplary N-terminal His-tag for use with TdT variants of the invention is MASSHHHHHHSSGSEN-LYFQTGSSG- (SEQ ID NO: 6)). Guidance for selecting a peptide affinity tag is described in the following references: Terpe, Appl. Microbiol. Biotechnol., 60: 523-533 (2003); Arnau et al, Protein Expression and Purification, 48: 1-13 (2006); Kimple et al, Curr. Protoc. Protein Sci., 73: Unit-9.9 (2015); Kimple et al, U.S. Pat. No. 7,309,575; Lichty et al, Protein Expression and Purification, 41: 98-105 (2005); and the like. Guidance for selecting a peptide affinity tag is described in the following references: Terpe, Appl. Microbiol. Biotechnol., 60: 523-533 (2003); Arnau et al, Protein Expression and Purification, 48: 1-13 (2006); Kimple et al, Curr. Protoc. Protein Sci., 73: Unit-9.9 (2015); Kimple et al, U.S. Pat. No. 7,309,575; Lichty et al, Protein Expression and Purification, 41: 98-105 (2005); and the like.

Measurement of Nucleotide Incorporation Activity

[0067] The efficiency of nucleotide incorporation by variants used with the invention may be measured by an extension, or elongation, assay, e.g. as described in Boule et al (cited below); Bentolila et al (cited below); and Hiatt et al, U.S. Pat. No. 5,808,045, the latter of which is incorporated herein by reference. Briefly, in one form of such an assay, a fluorescently labeled oligonucleotide having a free 3'-hydroxyl is reacted with a template-free polymerase, such as a TdT, under extension conditions for a predetermined duration in the presence of a reversibly blocked nucleoside triphosphate, after which the extension reaction is stopped

and the amounts of extension products and unextended oligonucleotide are quantified after separation by gel electrophoresis. By such assays, the incorporation efficiency of a variant template-free polymerase may be readily compared to the efficiencies of other variants or to that of wild type or reference polymerases. In some embodiments, a measure of template-free polymerase efficiency may be a ratio (given as a percentage) of amount of extended product using the variant template-free polymerase over the amount of extended product using wild type template-free polymerase, or reference polymerase, in an equivalent assay.

[0068] In some embodiments, the following particular extension assay may be used to measure incorporation efficiencies of TdTs: The primer used is the following:

(SEQ ID NO: 5)

5' -AAAAAAAAAAAAAGGGG-3'

The primer has also an ATTO fluorescent dye on the 5' extremity. Representative modified nucleotides used (noted as dNTP in Table 6) include 3'-O-amino-2',3'-dideoxynucleotides-5'-triphosphates (—ONH₂, Firebird Biosciences), such as 3'-O-amino-2',3'-dideoxyadenosine-5'-triphosphate. For each different variant tested, one tube is used for the reaction. The reagents are added to the tube, starting from water, and then in the order of Table 6. After 30 min at 37° C. the reaction is stopped by addition of formamide (Sigma).

TABLE 6

Extension Activity Assay Reagents		
Reagent	Concentration	Volume
H ₂ O	—	12 µL
Activity buffer	10x	2 µL
dNTP	250 µM	2 µL
Purified enzyme	20 µM	2 µL
Fluorescent primer	500 nM	2 µL

The Activity buffer comprises, for example, TdT reaction buffer (available from New England Biolabs) supplemented with CoCl₂.

[0069] The product of the assay is analyzed by conventional polyacrylamide gel electrophoresis. For example, products of the above assay may be analyzed in a 16 percent polyacrylamide denaturing gel (Bio-Rad). Gels are made just before the analysis by pouring polyacrylamide inside glass plates and let it polymerize. The gel inside the glass plates is mounted on an adapted tank filed with TBE buffer (Sigma) for the electrophoresis step. The samples to be analyzed are loaded on the top of the gel. A voltage of 500 to 2,000V is applied between the top and bottom of the gel for 3 to 6 h at room temperature. After separation, gel fluorescence is scanned using, for example, a Typhoon scanner (GE Life Sciences). The gel image is analyzed using ImageJ software (imagej.nih.gov/ij/), or its equivalent, to calculate the percentage of incorporation of the modified nucleotides.

[0070] The elongation efficiency of a template-free polymerase may also be measured in the following hairpin completion assay. In such assay, a test polynucleotide is provided with a free 3' hydroxyl such that under reaction conditions it is substantially only single stranded, but that upon extension with a polymerase, such as a TdT variant, it forms a stable hairpin structure comprising a single stranded

loop and a double stranded stem. This allows the detection of an extension of the 3' end by the presence of the double stranded polynucleotide. The double stranded structure may be detected in a variety of ways including, but not limited to, (i) fluorescent dyes that preferentially fluoresce upon intercalation into the double stranded structure, (ii) fluorescent resonance energy transfer (FRET) between an acceptor (or donor) on the extended polynucleotide and a donor (or acceptor) on an oligonucleotide that forms a triplex with the newly formed hairpin stem, (iii) FRET acceptors and donors that are both attached to the test polynucleotide and that are brought into FRET proximity upon formation of a hairpin, or the like. In some embodiments, a stem portion of a test polynucleotide after extension by a single nucleotide is in the range of 4 to 6 basepairs in length; in other embodiments, such stem portion is 4 to 5 basepairs in length; and in still other embodiments, such stem portion is 4 basepairs in length. In some embodiments, a test polynucleotide has a length in the range of from 10 to 20 nucleotides; in other embodiments, a test polynucleotide has a length in the range of from 12 to 15 nucleotides. In some embodiments, it is advantageous or convenient to extend the test polynucleotide with a nucleotide that maximizes the difference between the melting temperatures of the stem without extension and the stem with extension; thus, in some embodiments, a test polynucleotide is extended with a dC or dG (and accordingly the test polynucleotide is selected to have an appropriate complementary nucleotide for stem formation).

[0071] Exemplary test polynucleotides for hairpin completion assays include p875 (5'-CAGTTAAAACT) (SEQ ID NO: 2) which is completed by extending with a dGTP; p876 (5'-GAGTTAAAACT) (SEQ ID NO: 3) which is completed by extending with a dCTP; and p877 (5'-CAGCAAGGCT) (SEQ ID NO: 4) which is completed by extending with a dGTP. Exemplary reaction conditions for such test polynucleotides may comprise: 2.5-5 μ M of test polynucleotide, 1:4000 dilution of GelRed® (intercalating dye from Biotium, Inc., Fremont, CA), 200 mM Cacodylate KOH pH 6.8, 1 mM CoCl₂, 0-20% of DMSO and 3'-ONH₂ dGTP and TdT at desired concentrations. Completion of the hairpin may be monitored by an increase in fluorescence of GelRed® dye using a conventional fluorimeter, such as a TECAN reader at a reaction temperature of 28-38° C., using an excitation filter set to 360 nm and an emission filter set to 635 nm.

Kits

[0072] The invention includes a variety of kits for practicing methods of the invention. In one aspect, kits of the invention comprise a synthesis support having attached thereto, eventually by a 5' end, an initiator comprising a polyC oligonucleotide. In some embodiments, such synthesis support is a solid support. In further embodiments, such solid support may comprise particles, which may be porous particles or nonporous particles. Nonporous particles may, for example, comprise magnetic beads. In other embodiments, such particles may comprise porous particles, such as resins or gels. In some embodiments, such resins comprise an agarose resin. In some of the above embodiments, initiators attached to a solid support each comprise one or more polyC oligonucleotides each with length in the range of from 2 to 30 nucleotides. In some embodiments, such solid support is a population of microparticles, especially

nonporous microparticles. In other embodiments, such solid support is a population of porous microparticles. In some embodiments such porous microparticles are agarose microparticles. In some embodiments, such solid support is a planar support, such as a glass slide. In some embodiments, such planar support has a uniform coating of initiators containing one or more polyC oligonucleotides. In other embodiments, such planar support has an array of discrete reaction sites each comprising a coating of initiators containing one or more polyC oligonucleotides. In some embodiments, kits of the invention further include one or more template-free polymerase variants in a formulation, or in formulations, if provided separately, suitable for carrying out template-free enzymatic polynucleotide synthesis as described herein. Such kits may also include synthesis buffers for each template-free polymerase variant that provide reaction conditions for optimizing the template-free addition or incorporation of a 3'-O-protected dNTP to a growing strand. In embodiments for synthesizing DNA, a template-free polymerase is a TdT variant. In embodiments for synthesizing RNA, a template-free polymerase is a PAP and/or PUP variant.

[0073] In some embodiments, kits of the invention may comprise a solid support having attached thereto by a 5' end an initiator comprising a polyC oligonucleotide and separate polyC oligonucleotides attached to the same solid support. In additional embodiments, such kits may comprise polyC oligonucleotides in a solution.

[0074] In some embodiments, kits of the invention further include 3'-O-reversibly protected dNTPs. In such embodiments, the 3'-O-reversibly protected dNTPs may comprise 3'-O-amino-dNTPs or 3'-O-azidomethyl-dNTPs. In further embodiments, kits may include one or more of the following items, either separately or together with the above-mentioned items: (i) deprotection or de-blocking reagents for carrying out a deprotecting or deblocking step as described herein, (ii) solid supports with initiators attached thereto, (iii) cleavage reagents for releasing completed polynucleotides from solid supports, (iv) wash reagents or buffers for removing unreacted 3'-O-reversibly protected dNTPs at the end of an enzymatic addition or coupling step, and (v) post-synthesis processing reagents, such as purification columns, desalting reagents, eluting reagents, and the like.

[0075] In regard to items (ii) and (iii) above, certain initiators and cleavage reagents go together. For example, an initiator comprising an inosine cleavable nucleotide may come with an endonuclease V cleavage reagent; an initiator comprising a nitrobenzyl photocleavable linker may come with a suitable light source for cleaving the photocleavable linker; an initiator comprising a uracil may come with a uracil DNA glycosylase cleavage reagent; and the like.

Example

Synthesis of G4-Forming Polynucleotides

With and Without PolyC Initiators

[0076] In this example, eight polydideoxyribonucleotides having G4-prone sequences are synthesized with and without PolyC initiators substantially following the exemplary synthesis protocol described above. Solid supports are CNBr-activated 45 μ m agarose beads having either polyC initiator (-CCCCCCCCCCCCCTdT-3' (SEQ ID NO: 157)) attached via a C15 linker or non-polyC initiator

(-TTTTTTTTTTdIT-3' (SEQ ID NO: 158)) attached via a C15 linker. The template-free polymerase is TdT variant M77 (SEQ ID NO: 106) having N-terminal affinity tag (SEQ ID NO: 6). After synthesis, polynucleotide product is cleaved from the solid supports using EndoV endonuclease, following the protocol described in Creton, International patent publication WO/2020/165137. Cleaved synthesis products are analyzed by capillary electrophoresis to determine the purity of the desired polynucleotide and samples of product are sequenced to assess deletion, substitution and insertion errors. By both measures the use of polyC-containing initiators showed significantly improved yields of the desired products. The purity data indicates that use of the polyC initiator increased the purity of the G4-prone polynucleotide products by an average percentage of 20 percent or greater. The following tables compare error rates of various types in the sequences sampled from the polynucleotides synthesized with and without polyC-containing initiators.

Error Rates in Average Percentages Using Non-PolyC Initiators

	A	C	G	T
Substitutions	0.08	0.66	0.17	0.04
Insertions	0.16	0.10	0.13	0.17
Deletions	1.11	0.62	0.64	0.52

Error Rates in Average Percentages Using PolyC Initiators

	A	C	G	T
Substitutions	0.08	0.21	0.09	0.03
Insertions	0.06	0.05	0.09	0.08
Deletions	0.94	0.28	0.37	0.30

Definitions

[0077] Unless otherwise specifically defined herein, terms and symbols of nucleic acid chemistry, biochemistry, genetics, and molecular biology used herein follow those of standard treatises and texts in the field, e.g. Kornberg and Baker, DNA Replication, Second Edition (W.H. Freeman, New York, 1992); Lehninger, Biochemistry, Second Edition (Worth Publishers, New York, 1975); Strachan and Read, Human Molecular Genetics, Second Edition (Wiley-Liss, New York, 1999).

[0078] “Functionally equivalent” in reference to amino acid positions in two or more different TdTs means (i) the amino acids at the respective positions play the same functional role in an activity of the TdTs, and (ii) the amino acids occur at homologous amino acid positions in the amino acid sequences of the respective TdTs. It is possible to identify positionally equivalent or homologous amino acid residues in the amino acid sequences of two or more different TdTs on the basis of sequence alignment and/or molecular modelling. In some embodiments, functionally equivalent amino acid positions belong to inefficiency motifs that are conserved among the amino acid sequences of TdTs of evolutionarily related species, e.g. genus, families, or the like. Examples of such conserved inefficiency motifs are

described in Motea et al, Biochim. Biophys. Acta. 1804(5): 1151-1166 (2010); Delarue et al, EMBO J., 21: 427-439 (2002); and like references.

[0079] “Kit” refers to any delivery system, such as a package, for delivering materials or reagents for carrying out a method implemented by a system or apparatus of the invention. In some embodiments, consumables materials or reagents are delivered to a user of a system or apparatus of the invention in a package referred to herein as a “kit.” In the context of the invention, such delivery systems include, usually packaging methods and materials that allow for the storage, transport, or delivery of materials, such as, synthesis supports, oligonucleotides, 3'-O-protected-dNTPs, and the like. For example, kits may include one or more enclosures (e.g., boxes) containing solid supports with polyC initiators attached and/or supporting materials. Such contents may be delivered to the intended recipient together or separately. For example, a first container may contain solid supports with polyC initiators attached, while a second or more containers contain a 3'-O-protected-deoxynucleoside triphosphates, a template-free polymerase, for example, a specific TdT variant, and appropriate buffers.

[0080] “Mutant” or “variant,” which are used interchangeably, refer to polypeptides derived from a natural or reference TdT polypeptide described herein, and comprising a modification or an alteration, i.e., a substitution, insertion, and/or deletion, at one or more positions. Variants may be obtained by various techniques well known in the art. In particular, examples of techniques for altering the DNA sequence encoding the wild-type protein, include, but are not limited to, site-directed mutagenesis, random mutagenesis, sequence shuffling and synthetic oligonucleotide construction. Mutagenesis activities consist in deleting, inserting or substituting one or several amino-acids in the sequence of a protein or in the case of the invention of a polymerase. The following terminology is used to designate a substitution: L238A denotes that amino acid residue (Leucine, L) at position 238 of a reference, or wild type, sequence is changed to an Alanine (A). A132V/I/M denotes that amino acid residue (Alanine, A) at position 132 of the parent sequence is substituted by one of the following amino acids: Valine (V), Isoleucine (I), or Methionine (M). The substitution can be a conservative or non-conservative substitution. Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine, asparagine and threonine), hydrophobic amino acids (methionine, leucine, isoleucine, cysteine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine and serine).

[0081] “Polynucleotide” or “oligonucleotide” are used interchangeably and each mean a linear polymer of nucleotide monomers or analogs thereof. Monomers making up polynucleotides and oligonucleotides are capable of specifically binding to a natural polynucleotide by way of a regular pattern of monomer-to-monomer interactions, such as Watson-Crick type of base pairing, base stacking, Hoogsteen or reverse Hoogsteen types of base pairing, or the like. Such monomers and their internucleosidic linkages may be naturally occurring or may be analogs thereof, e.g. naturally occurring or non-naturally occurring analogs. Non-naturally occurring analogs may include PNAs, phosphorothioate internucleosidic linkages, bases containing linking groups

permitting the attachment of labels, such as fluorophores, or haptens, and the like. Whenever the use of an oligonucleotide or polynucleotide requires enzymatic processing, such as extension by a polymerase, ligation by a ligase, or the like, one of ordinary skill would understand that oligonucleotides or polynucleotides in those instances would not contain certain analogs of internucleosidic linkages, sugar moieties, or bases at any or some positions. Polynucleotides typically range in size from a few monomeric units, e.g. 5-40, when they are usually referred to as "oligonucleotides," to several thousand monomeric units. Whenever a polynucleotide or oligonucleotide is represented by a sequence of letters (upper or lower case), such as "ATGCCTG," it will be understood that the nucleotides are in 5'→3' order from left to right and that "A" denotes deoxyadenosine, "C" denotes deoxycytidine, "G" denotes deoxyguanosine, and "T" denotes thymidine, "I" denotes deoxyinosine, "U" denotes uridine, unless otherwise indicated or obvious from context. Unless otherwise noted the terminology and atom numbering conventions will follow those disclosed in Strachan and Read, *Human Molecular Genetics 2* (Wiley-Liss, New York, 1999). Usually polynucleotides comprise the four natural nucleosides (e.g. deoxyadenosine, deoxycytidine, deoxyguanosine, deoxythymidine for DNA or their ribose counterparts for RNA) linked by phosphodiester linkages; however, they may also comprise non-natural nucleotide analogs, e.g. including modified bases, sugars, or internucleosidic linkages. It is clear to those skilled in the art that where an enzyme has specific oligonucleotide or polynucleotide substrate requirements for activity, e.g. single stranded DNA, RNA/DNA duplex, or the like, then selection of appropriate composition for the oligonucleotide or polynucleotide substrates is well within the knowledge of one of ordinary skill, especially with guidance from treatises, such as Sambrook et al, *Molecular Cloning, Second Edition* (Cold Spring Harbor Laboratory, New York, 1989), and like references. Likewise, the oligonucleotide and polynucleotide may refer to either a single stranded form or a double stranded form (i.e. duplexes of an oligonucleotide or polynucleotide and its respective complement). It will be clear to one of ordinary skill which form or whether both forms are intended from the context of the terms usage.

[0082] "Primer" means an oligonucleotide, either natural or synthetic that is capable, upon forming a duplex with a polynucleotide template, of acting as a point of initiation of nucleic acid synthesis and being extended from its 3' end along the template so that an extended duplex is formed. Extension of a primer is usually carried out with a nucleic acid polymerase, such as a DNA or RNA polymerase. The sequence of nucleotides added in the extension process is determined by the sequence of the template polynucleotide. Usually primers are extended by a DNA polymerase. Primers usually have a length in the range of from 14 to 40 nucleotides, or in the range of from 18 to 36 nucleotides. Primers are employed in a variety of nucleic acid amplification reactions, for example, linear amplification reactions using a single primer, or polymerase chain reactions, employing two or more primers. Guidance for selecting the lengths and sequences of primers for particular applications is well known to those of ordinary skill in the art, as evidenced by the following references that are incorporated by reference: Dieffenbach, editor, *PCR Primer: A Laboratory Manual*, 2nd Edition (Cold Spring Harbor Press, New York, 2003).

[0083] "Sequence identity" refers to the number (or fraction, usually expressed as a percentage) of matches (e.g., identical amino acid residues) between two sequences, such as two polypeptide sequences or two polynucleotide sequences. The sequence identity is determined by comparing the sequences when aligned so as to maximize overlap and identity while minimizing sequence gaps. In particular, sequence identity may be determined using any of a number of mathematical global or local alignment algorithms, depending on the length of the two sequences. Sequences of similar lengths are preferably aligned using a global alignment algorithm (e.g. Needleman and Wunsch algorithm; Needleman and Wunsch, 1970) which aligns the sequences optimally over the entire length, while sequences of substantially different lengths are preferably aligned using a local alignment algorithm (e.g. Smith and Waterman algorithm (Smith and Waterman, 1981) or Altschul algorithm (Altschul et al., 1997; Altschul et al., 2005)). Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software available on internet web sites such as <http://blast.ncbi.nlm.nih.gov/or> <http://www.ebi.ac.uk/Tools/emboss/>. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithm needed to achieve maximal alignment over the full length of the sequences being compared. For purposes herein, % amino acid sequence identity values refer to values generated using the pair wise sequence alignment program EMBOSS Needle, that creates an optimal global alignment of two sequences using the Needleman-Wunsch algorithm, wherein all search parameters are set to default values, i.e. Scoring matrix=BLOSUM62, Gap open=10, Gap extend=0.5, End gap penalty=false, End gap open=10 and End gap extend=0.5.

[0084] "Substitution" means that an amino acid residue is replaced by another amino acid residue. Preferably, the term "substitution" refers to the replacement of an amino acid residue by another selected from the naturally-occurring standard 20 amino acid residues, rare naturally-occurring amino acid residues (e.g. hydroxyproline, hydroxylysine, aliohydroxylysine, 6-N-methyllysine, N-ethylglycine, N-methylglycine, N-ethylasparagine, allo-isoleucine, N-methylisoleucine, N-methylvaline, pyroglutamine, aminobutyric acid, ornithine, norleucine, norvaline), and non-naturally occurring amino acid residue, often made synthetically, (e.g. cyclohexyl-alanine). Preferably, the term "substitution" refers to the replacement of an amino acid residue by another selected from the naturally-occurring standard 20 amino acid residues. The sign "+" indicates a combination of substitutions. The amino acids are herein represented by their one-letter or three-letters code according to the following nomenclature: A: alanine (Ala); C: cysteine (Cys); D: aspartic acid (Asp); E: glutamic acid (Glu); F: phenylalanine (Phe); G: glycine (Gly); H: histidine (His); I: isoleucine (Ile); K: lysine (Lys); L: leucine (Leu); M: methionine (Met); N: asparagine (Asn); P: proline (Pro); Q: glutamine (Gln); R: arginine (Arg); S: serine (Ser); T: threonine (Thr); V: valine (Val); W: tryptophan (Trp) and Y: tyrosine (Tyr). In the present document, the following terminology is used to designate a substitution: L238A denotes that amino acid residue (Leucine, L) at position 238 of the

parent sequence is changed to an Alanine (A). A132V/I/M denotes that amino acid residue (Alanine, A) at position 132 of the parent sequence is substituted by one of the following amino acids: Valine (V), Isoleucine (I), or Methionine (M). The substitution can be a conservative or non-conservative substitution. Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine, asparagine and threonine), hydrophobic amino acids (methionine, leucine, isoleucine, cysteine and valine), aromatic amino acids (phe-

nylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine and serine).

[0085] This disclosure is not intended to be limited to the scope of the particular forms set forth, but is intended to cover alternatives, modifications, and equivalents of the variations described herein. Further, the scope of the disclosure fully encompasses other variations that may become obvious to those skilled in the art in view of this disclosure. The scope of the present invention is limited only by the appended claims.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 160

<210> SEQ ID NO 1

<211> LENGTH: 510

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 1

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1          5          10          15
Arg Gln Leu Gly Thr Pro Val Ala Ser Thr Pro Tyr Asp Ile Arg Phe
20          25          30
Arg Asp Leu Val Leu Phe Ile Leu Glu Lys Lys Met Gly Thr Thr Arg
35          40          45
Arg Ala Phe Leu Met Glu Leu Ala Arg Arg Lys Gly Phe Arg Val Glu
50          55          60
Asn Glu Leu Ser Asp Ser Val Thr His Ile Val Ala Glu Asn Asn Ser
65          70          75          80
Gly Ser Asp Val Leu Glu Trp Leu Gln Leu Gln Asn Ile Lys Ala Ser
85          90          95
Ser Glu Leu Glu Leu Leu Asp Ile Ser Trp Leu Ile Glu Cys Met Gly
100         105         110
Ala Gly Lys Pro Val Glu Met Met Gly Arg His Gln Leu Val Val Asn
115         120         125
Arg Asn Ser Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala
130         135         140
Pro Ala Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr
145         150         155         160
Leu Asn Asn Tyr Asn Gln Leu Phe Thr Asp Ala Leu Asp Ile Leu Ala
165         170         175
Glu Asn Asp Glu Leu Arg Glu Asn Glu Gly Ser Cys Leu Ala Phe Met
180         185         190
Arg Ala Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met
195         200         205
Lys Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Ser Ile
210         215         220
Ile Glu Gly Ile Ile Glu Asp Gly Glu Ser Ser Glu Ala Lys Ala Val
225         230         235         240
Leu Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe
245         250         255
Gly Val Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg
260         265         270

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Thr Leu Ser Lys Ile Gln Ser Asp Lys Ser Leu Arg Phe Thr Gln Met
 275 280 285
 Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Asn
 290 295 300
 Arg Pro Glu Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val
 305 310 315 320
 Thr Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Arg
 325 330 335
 Gly Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu
 340 345 350
 Ala Thr Glu Asp Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe
 355 360 365
 Trp Lys Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr
 370 375 380
 Phe Glu Lys Phe Lys Gln Pro Ser Arg Lys Val Asp Ala Leu Asp His
 385 390 395 400
 Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu Asp His Gly Arg Val His
 405 410 415
 Ser Glu Lys Ser Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg
 420 425 430
 Val Asp Leu Val Met Cys Pro Tyr Asp Arg Arg Ala Phe Ala Leu Leu
 435 440 445
 Gly Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala
 450 455 460
 Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Arg
 465 470 475 480
 Thr Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala
 485 490 495
 His Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 500 505 510

<210> SEQ ID NO 2
 <211> LENGTH: 12
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Test Polynucleotide p875

<400> SEQUENCE: 2

cagttaaaaa ct

12

<210> SEQ ID NO 3
 <211> LENGTH: 11
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Test Polynucleotide p876

<400> SEQUENCE: 3

gagttaaaac t

11

<210> SEQ ID NO 4
 <211> LENGTH: 10
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Test Polynucleotide p877

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<400> SEQUENCE: 4

cagcaaggct 10

<210> SEQ ID NO 5

<211> LENGTH: 18

<212> TYPE: DNA

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

<400> SEQUENCE: 5

aaaaaaaaa aaaagggg 18

<210> SEQ ID NO 6

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: affinity tag for purifying TdT variants

<400> SEQUENCE: 6

Met Ala Ser Ser His His His His His Ser Ser Gly Ser Glu Asn
1 5 10 15Leu Tyr Phe Gln Thr Gly Ser Ser Gly
20 25

<210> SEQ ID NO 7

<211> LENGTH: 381

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 7

Asn Ser Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro
1 5 10 15Ala Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu
20 25 30Asn Asn Tyr Asn Gln Leu Phe Thr Asp Ala Leu Asp Ile Leu Ala Glu
35 40 45Asn Asp Glu Leu Arg Glu Asn Glu Gly Ser Cys Leu Ala Phe Met Arg
50 55 60Ala Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys
65 70 75 80Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Ser Ile Ile
85 90 95Glu Gly Ile Ile Glu Asp Gly Glu Ser Ser Glu Ala Lys Ala Val Leu
100 105 110Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
115 120 125Val Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr
130 135 140Leu Ser Lys Ile Gln Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln
145 150 155 160Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Asn Arg
165 170 175Pro Glu Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Thr
180 185 190

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Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Arg Gly
 195 200 205
 Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala
 210 215 220
 Thr Glu Asp Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe Trp
 225 230 235 240
 Lys Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe
 245 250 255
 Glu Lys Phe Lys Gln Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe
 260 265 270
 Gln Lys Cys Phe Leu Ile Leu Lys Leu Asp His Gly Arg Val His Ser
 275 280 285
 Glu Lys Ser Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg Val
 290 295 300
 Asp Leu Val Met Cys Pro Tyr Asp Arg Arg Ala Phe Ala Leu Leu Gly
 305 310 315 320
 Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr
 325 330 335
 His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Arg Thr
 340 345 350
 Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His
 355 360 365
 Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 370 375 380

<210> SEQ ID NO 8
 <211> LENGTH: 380
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Bovine

<400> SEQUENCE: 8

Asp Tyr Ser Ala Thr Pro Asn Pro Gly Phe Gln Lys Thr Pro Pro Leu
 1 5 10 15
 Ala Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Lys Thr Thr Leu
 20 25 30
 Asn Asn Tyr Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu
 35 40 45
 Asn Ser Glu Phe Lys Glu Asn Glu Val Ser Tyr Val Thr Phe Met Arg
 50 55 60
 Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys
 65 70 75 80
 Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Ile Ile
 85 90 95
 Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu
 100 105 110
 Asn Asp Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
 115 120 125
 Val Gly Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Ser
 130 135 140
 Leu Ser Lys Ile Met Ser Asp Lys Thr Leu Lys Phe Thr Lys Met Gln
 145 150 155 160

<400> SEQUENCE: 9

Asp 1	Tyr	Ser	Asp	Ser 5	Thr	Asn	Pro	Gly	Pro 10	Pro	Lys	Thr	Pro	Pro 15	Ile
Ala	Val	Gln	Lys 20	Ile	Ser	Gln	Tyr	Ala 25	Cys	Gln	Arg	Arg	Thr 30	Thr	Leu
Asn	Asn	Cys 35	Asn	Gln	Ile	Phe	Thr 40	Asp	Ala	Phe	Asp	Ile 45	Leu	Ala	Glu
Asn	Cys 50	Glu	Phe	Arg	Glu	Asn 55	Glu	Asp	Ser	Cys	Val 60	Thr	Phe	Met	Arg
Ala 65	Ala	Ser	Val	Leu 70	Lys	Ser	Leu	Pro	Phe	Thr 75	Ile	Ile	Ser	Met	Lys 80
Asp	Thr	Glu	Gly	Ile 85	Pro	Cys	Leu	Gly	Ser 90	Lys	Val	Lys	Gly	Ile 95	Ile
Glu	Glu	Ile	Ile 100	Glu	Asp	Gly	Glu	Ser 105	Ser	Glu	Val	Lys	Ala 110	Val	Leu
Asn	Asp	Glu	Arg 115	Tyr	Gln	Ser	Phe	Lys 120	Leu	Phe	Thr	Ser 125	Val	Phe	Gly
Val	Gly	Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr

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130					135					140					
Leu	Ser	Lys	Val	Arg	Ser	Asp	Lys	Ser	Leu	Lys	Phe	Thr	Arg	Met	Gln
145					150					155					160
Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg
			165						170					175	
Ala	Glu	Ala	Glu	Ala	Val	Ser	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala
			180					185					190		
Phe	Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly
	195						200					205			
Lys	Lys	Met	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser
	210					215					220				
Thr	Glu	Asp	Glu	Glu	Gln	Leu	Leu	Gln	Lys	Val	Met	Asn	Leu	Trp	Glu
225					230					235					240
Lys	Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu
			245						250					255	
Lys	Leu	Arg	Leu	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln
			260					265					270		
Lys	Cys	Phe	Leu	Ile	Phe	Lys	Leu	Pro	Arg	Gln	Arg	Val	Asp	Ser	Asp
		275					280					285			
Gln	Ser	Ser	Trp	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp
	290					295					300				
Leu	Val	Leu	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp
305					310					315					320
Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His
				325					330					335	
Glu	Arg	Lys	Met	Ile	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys
			340					345					350		
Arg	Ile	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu
		355					360					365			
Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala				
	370					375					380				

<210> SEQ ID NO 10

<211> LENGTH: 376

<212> TYPE: PRT

<213> ORGANISM: Gallus domesticus

<400> SEQUENCE: 10

Gln	Tyr	Pro	Thr	Leu	Lys	Thr	Pro	Glu	Ser	Glu	Val	Ser	Ser	Phe	Thr
1				5					10					15	
Ala	Ser	Lys	Val	Ser	Gln	Tyr	Ser	Cys	Gln	Arg	Lys	Thr	Thr	Leu	Asn
			20					25					30		
Asn	Cys	Asn	Lys	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Met	Ala	Glu	Asn
		35					40					45			
Tyr	Glu	Phe	Lys	Glu	Asn	Glu	Ile	Phe	Cys	Leu	Glu	Phe	Leu	Arg	Ala
	50				55						60				
Ala	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Val	Thr	Arg	Met	Lys	Asp
65					70					75				80	
Ile	Gln	Gly	Leu	Pro	Cys	Met	Gly	Asp	Arg	Val	Arg	Asp	Val	Ile	Glu
			85					90						95	
Glu	Ile	Ile	Glu	Glu	Gly	Glu	Ser	Ser	Arg	Ala	Lys	Asp	Val	Leu	Asn
			100					105						110	

-continued

Asp	Glu	Arg	Tyr	Lys	Ser	Phe	Lys	Glu	Phe	Thr	Ser	Val	Phe	Gly	Val
		115					120					125			
Gly	Val	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Leu	Arg	Thr	Val
	130					135					140				
Glu	Glu	Val	Lys	Ala	Asp	Lys	Thr	Leu	Lys	Leu	Ser	Lys	Met	Gln	Arg
145					150					155					160
Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Ser	Lys	Ala
			165					170						175	
Glu	Ala	Asp	Ala	Val	Ser	Ser	Ile	Val	Lys	Asn	Thr	Val	Cys	Thr	Phe
		180						185					190		
Leu	Pro	Asp	Ala	Leu	Val	Thr	Ile	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys
		195					200					205			
Lys	Ile	Gly	His	Asp	Ile	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Gln	Arg
	210					215					220				
Glu	Asp	Asp	Glu	Leu	Leu	His	Lys	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Ile
225					230					235					240
Ile	Glu	Ser	Thr	Phe	Val	Lys	Glu	Gln	Ile	Pro	Ser	Arg	His	Val	Asp
			245						250					255	
Ala	Met	Asp	His	Phe	Gln	Lys	Cys	Phe	Ala	Ile	Leu	Lys	Leu	Tyr	Gln
			260					265					270		
Pro	Arg	Val	Asp	Asn	Ser	Ser	Tyr	Asn	Met	Ser	Lys	Lys	Cys	Asp	Met
		275					280					285			
Ala	Glu	Val	Lys	Asp	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Ile	Thr
	290					295					300				
Pro	Phe	Glu	Gln	Tyr	Ala	Tyr	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser	Arg
305					310					315					320
Gln	Phe	Gly	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys	Met
			325						330					335	
Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Arg	Lys	Arg	Val	Phe	Leu
			340					345					350		
Lys	Ala	Gly	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr
		355					360					365			
Val	Glu	Pro	Trp	Glu	Arg	Asn	Ala								
	370					375									

<210> SEQ ID NO 11

<211> LENGTH: 387

<212> TYPE: PRT

<213> ORGANISM: Didelphis marsupialis

<400> SEQUENCE: 11

Ser	Ala	Asn	Pro	Asp	Pro	Thr	Ala	Gly	Thr	Leu	Asn	Ile	Leu	Pro	Pro
1			5						10					15	
Thr	Thr	Lys	Thr	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Ile
			20					25					30		
Asn	Asn	His	Asn	Gln	Arg	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Lys
		35					40					45			
Asn	Tyr	Glu	Phe	Lys	Glu	Asn	Asp	Asp	Thr	Cys	Leu	Thr	Phe	Met	Arg
	50					55					60				
Ala	Ile	Ser	Val	Leu	Lys	Cys	Leu	Pro	Phe	Glu	Val	Val	Ser	Leu	Lys
65					70					75				80	
Asp	Thr	Glu	Gly	Leu	Pro	Trp	Ile	Gly	Asp	Glu	Val	Lys	Gly	Ile	Met
			85					90						95	

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Glu Glu Ile Ile Glu Asp Gly Glu Ser Leu Glu Val Gln Ala Val Leu
 100 105 110
 Asn Asp Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
 115 120 125
 Val Gly Leu Lys Thr Ala Asp Lys Trp Tyr Arg Met Gly Phe Arg Thr
 130 135 140
 Leu Asn Lys Ile Arg Ser Asp Lys Thr Leu Lys Leu Thr Lys Met Gln
 145 150 155 160
 Lys Ala Gly Leu Cys Tyr Tyr Glu Asp Leu Ile Asp Cys Val Ser Lys
 165 170 175
 Ala Glu Ala Asp Ala Val Ser Leu Leu Val Gln Asp Ala Val Trp Thr
 180 185 190
 Phe Leu Pro Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly
 195 200 205
 Lys Glu Phe Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ala
 210 215 220
 Glu Lys Glu Gln Glu Asp Gln Leu Leu Gln Lys Val Thr Asn Leu Trp
 225 230 235 240
 Lys Lys Gln Gly Leu Leu Leu Tyr Cys Asp Leu Ile Glu Ser Thr Phe
 245 250 255
 Glu Asp Leu Lys Leu Pro Ser Arg Lys Ile Asp Ala Leu Asp His Phe
 260 265 270
 Gln Lys Cys Phe Leu Ile Leu Lys Leu Tyr His His Lys Glu Asp Lys
 275 280 285
 Arg Lys Trp Glu Met Pro Thr Gly Ser Asn Glu Ser Glu Ala Lys Ser
 290 295 300
 Trp Lys Ala Ile Arg Val Asp Leu Val Val Cys Pro Tyr Asp Arg Tyr
 305 310 315 320
 Ala Phe Ala Leu Leu Gly Trp Ser Gly Ser Arg Gln Phe Glu Arg Asp
 325 330 335
 Leu Arg Arg Tyr Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His
 340 345 350
 Ala Leu Tyr Asp Lys Thr Lys Lys Ile Phe Leu Lys Ala Lys Ser Glu
 355 360 365
 Glu Glu Ile Phe Ala His Leu Gly Leu Glu Tyr Ile Gln Pro Ser Glu
 370 375 380
 Arg Asn Ala
 385

<210> SEQ ID NO 12
 <211> LENGTH: 381
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: shrew

<400> SEQUENCE: 12

Asp Cys Pro Ala Ser His Asp Ser Ser Pro Gln Lys Thr Glu Ser Ala
 1 5 10 15
 Ala Val Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu
 20 25 30
 Asn Asn His Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu
 35 40 45

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Asn Cys Glu Phe Arg Glu Asn Glu Gly Ser Tyr Val Thr Tyr Met Arg
 50                               55                               60

Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Ser Ile Ile Ser Met Lys
65                               70                               75                               80

Asp Thr Glu Gly Ile Pro Cys Leu Ala Asp Lys Val Lys Cys Val Ile
                               85                               90                               95

Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu
                               100                               105                               110

Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
                               115                               120                               125

Val Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Leu Gly Phe Arg Thr
                               130                               135                               140

Leu Ser Gly Ile Met Asn Asp Lys Thr Leu Lys Leu Thr His Met Gln
145                               150                               155                               160

Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg
                               165                               170                               175

Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala
                               180                               185                               190

Phe Leu Pro Asp Ala Ile Val Thr Met Thr Gly Gly Phe Arg Arg Gly
                               195                               200                               205

Lys Lys Val Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala
210                               215                               220

Thr Glu Glu Gln Glu Gln Gln Leu Leu His Lys Val Ile Thr Phe Trp
225                               230                               235                               240

Glu Lys Glu Gly Leu Leu Leu Tyr Cys Asp Leu Tyr Glu Ser Thr Phe
                               245                               250                               255

Glu Lys Leu Lys Met Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe
                               260                               265                               270

Gln Lys Cys Phe Leu Ile Leu Lys Leu His Arg Glu Cys Val Asp Asp
                               275                               280                               285

Gly Thr Ser Ser Gln Leu Gln Gly Lys Thr Trp Lys Ala Ile Arg Val
290                               295                               300

Asp Leu Val Val Cys Pro Tyr Glu Cys Arg Ala Phe Ala Leu Leu Gly
305                               310                               315                               320

Trp Thr Gly Ser Pro Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr
                               325                               330                               335

His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr
                               340                               345                               350

Lys Arg Lys Phe Leu Ser Ala Asp Ser Glu Glu Asp Ile Phe Ala His
                               355                               360                               365

Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
370                               375                               380

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<210> SEQ ID NO 13
<211> LENGTH: 387
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: python

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<400> SEQUENCE: 13

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Glu Lys Tyr Gln Leu Pro Glu Asp Glu Asp Arg Ser Val Thr Ser Asp
1              5              10              15

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Leu Asp Arg Asp Ser Ile Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr
 20 25 30
 Leu Lys Asn Tyr Asn Gln Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala
 35 40 45
 Glu Asn Tyr Glu Phe Asn Glu Asn Lys Gly Phe Cys Thr Ala Phe Arg
 50 55 60
 Arg Ala Ala Ser Val Leu Lys Cys Leu Pro Phe Thr Ile Val Gln Val
 65 70 75 80
 His Asp Ile Glu Gly Val Pro Trp Met Gly Lys Gln Val Lys Gly Ile
 85 90 95
 Ile Glu Asp Ile Ile Glu Glu Gly Glu Ser Ser Lys Val Lys Ala Val
 100 105 110
 Leu Asp Asn Glu Asn Tyr Arg Ser Val Lys Leu Phe Thr Ser Val Phe
 115 120 125
 Gly Val Gly Leu Lys Thr Ser Asp Lys Trp Tyr Arg Met Gly Leu Arg
 130 135 140
 Thr Leu Glu Glu Val Lys Arg Asp Lys Asn Leu Lys Leu Thr Arg Met
 145 150 155 160
 Gln Lys Ala Gly Phe Leu His Tyr Asp Asp Leu Thr Ser Cys Val Ser
 165 170 175
 Lys Ala Glu Ala Asp Ala Ala Ser Leu Ile Val Gln Asp Val Val Trp
 180 185 190
 Lys Ile Val Pro Asn Ala Ile Val Thr Ile Ala Gly Gly Phe Arg Arg
 195 200 205
 Gly Lys Gln Thr Gly His Asp Val Asp Phe Leu Ile Thr Val Pro Gly
 210 215 220
 Ser Lys Gln Glu Glu Glu Glu Leu Leu His Thr Val Ile Asp Ile Trp
 225 230 235 240
 Lys Lys Gln Glu Leu Leu Leu Tyr Tyr Asp Leu Ile Glu Ser Thr Phe
 245 250 255
 Glu Asp Thr Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe
 260 265 270
 Gln Lys Cys Phe Ala Ile Leu Lys Val His Lys Glu Arg Glu Asp Lys
 275 280 285
 Gly Asn Ser Ile Arg Ser Lys Ala Phe Ser Glu Glu Glu Ile Lys Asp
 290 295 300
 Trp Lys Ala Ile Arg Val Asp Leu Val Val Val Pro Phe Glu Gln Tyr
 305 310 315 320
 Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser Thr Gln Phe Glu Arg Asp
 325 330 335
 Leu Arg Arg Tyr Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His
 340 345 350
 Ala Leu Tyr Asp Lys Thr Lys Lys Ile Phe Leu Asn Ala Ala Ser Glu
 355 360 365
 Glu Glu Ile Phe Ala His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu
 370 375 380
 Arg Asn Ala
 385

<210> SEQ ID NO 14

<211> LENGTH: 381

-continued

<212> TYPE: PRT

<213> ORGANISM: Canis familiaris

<400> SEQUENCE: 14

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Asp Tyr Thr Ala Ser Pro Asn Pro Glu Leu Gln Lys Thr Leu Pro Val
1      5      10      15
Ala Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu
20      25      30
Asn Asn Tyr Asn Asn Val Phe Thr Asp Ala Phe Glu Val Leu Ala Glu
35      40      45
Asn Tyr Glu Phe Arg Glu Asn Glu Val Phe Ser Leu Thr Phe Met Arg
50      55      60
Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys
65      70      75      80
Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Gln Val Lys Cys Ile Ile
85      90      95
Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu
100     105     110
Asn Asp Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
115     120     125
Val Gly Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr
130     135     140
Leu Ser Lys Ile Lys Ser Asp Lys Ser Leu Lys Phe Thr Pro Met Gln
145     150     155     160
Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg
165     170     175
Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Gly Ala
180     185     190
Phe Leu Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly
195     200     205
Lys Lys Met Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser
210     215     220
Thr Asp Glu Asp Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp
225     230     235     240
Glu Arg Lys Gly Leu Leu Leu Tyr Cys Asp Leu Val Glu Ser Thr Phe
245     250     255
Glu Lys Leu Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe
260     265     270
Gln Lys Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Gly
275     280     285
Gly Lys Cys Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val
290     295     300
Asp Leu Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly
305     310     315     320
Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Ser
325     330     335
His Glu Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr
340     345     350
Lys Lys Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His
355     360     365
Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
370     375     380

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<210> SEQ ID NO 15
<211> LENGTH: 382
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: mole

<400> SEQUENCE: 15

Gly Asp Cys Pro Ala Ser His Asp Ser Ser Pro Gln Lys Thr Glu Ser
1      5      10      15

Ala Ala Val Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr
20      25      30

Leu Asn Asn His Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala
35      40      45

Glu Asn Cys Glu Phe Arg Glu Asn Glu Gly Ser Tyr Val Thr Tyr Met
50      55      60

Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Ser Ile Ile Ser Met
65      70      75      80

Lys Asp Thr Glu Gly Ile Pro Cys Leu Ala Asp Lys Val Lys Cys Val
85      90      95

Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val
100     105     110

Leu Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe
115     120     125

Gly Val Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Leu Gly Phe Arg
130     135     140

Thr Leu Ser Gly Ile Met Asn Asp Lys Thr Leu Lys Leu Thr His Met
145     150     155     160

Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr
165     170     175

Arg Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp
180     185     190

Ala Phe Leu Pro Asp Ala Ile Val Thr Met Thr Gly Gly Phe Arg Arg
195     200     205

Gly Lys Lys Val Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu
210     215     220

Ala Thr Glu Glu Gln Glu Gln Leu Leu His Lys Val Ile Thr Phe
225     230     235     240

Trp Glu Lys Glu Gly Leu Leu Leu Tyr Cys Asp Leu Tyr Glu Ser Thr
245     250     255

Phe Glu Lys Leu Lys Met Pro Ser Arg Lys Val Asp Ala Leu Asp His
260     265     270

Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu His Arg Glu Cys Val Asp
275     280     285

Asp Gly Thr Ser Ser Gln Leu Gln Gly Lys Thr Trp Lys Ala Ile Arg
290     295     300

Val Asp Leu Val Val Cys Pro Tyr Glu Cys Arg Ala Phe Ala Leu Leu
305     310     315     320

Gly Trp Thr Gly Ser Pro Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala
325     330     335

Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys
340     345     350

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Thr Lys Arg Lys Phe Leu Ser Ala Asp Ser Glu Glu Asp Ile Phe Ala
355 360 365

His Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
370 375 380

<210> SEQ ID NO 16

<211> LENGTH: 379

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: artificial protein

<400> SEQUENCE: 16

Glu Tyr Ser Ala Asn Pro Ser Pro Gly Pro Gln Ala Thr Pro Ala Val
1 5 10 15

Tyr Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
20 25 30

His Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr
35 40 45

Glu Phe Lys Glu Asn Glu Gly Cys Tyr Val Thr Tyr Met Arg Ala Ala
50 55 60

Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Val Ser Met Lys Asp Thr
65 70 75 80

Glu Gly Ile Pro Cys Leu Glu Asp Lys Val Lys Ser Ile Met Glu Glu
85 90 95

Ile Ile Glu Glu Gly Glu Ser Ser Glu Val Lys Ala Val Leu Ser Asp
100 105 110

Glu Arg Tyr Gln Cys Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115 120 125

Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Ser Leu Ser
130 135 140

Asn Ile Arg Leu Asp Lys Ser Leu Lys Phe Thr Gln Met Gln Lys Ala
145 150 155 160

Gly Phe Arg Tyr Tyr Glu Asp Ile Val Ser Cys Val Thr Arg Ala Glu
165 170 175

Ala Glu Ala Val Asp Val Leu Val Asn Glu Ala Val Arg Ala Phe Leu
180 185 190

Pro Asp Ala Phe Ile Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys
195 200 205

Ile Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Leu Thr Glu
210 215 220

Glu Asp Glu Gln Gln Leu Leu His Lys Val Met Asn Leu Trp Glu Lys
225 230 235 240

Lys Gly Leu Leu Leu Tyr His Asp Leu Val Glu Ser Thr Phe Glu Lys
245 250 255

Leu Lys Gln Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys
260 265 270

Cys Phe Leu Ile Phe Lys Leu Tyr His Glu Arg Val Gly Gly Asp Arg
275 280 285

Cys Arg Gln Pro Glu Gly Lys Asp Trp Lys Ala Ile Arg Val Asp Leu
290 295 300

Val Met Cys Pro Tyr Glu Cys His Ala Phe Ala Leu Leu Gly Trp Thr
305 310 315 320

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Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Ser His Glu
325 330 335
Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
340 345 350
Val Phe Leu Gln Ala Glu Asn Glu Glu Glu Ile Phe Ala His Leu Gly
355 360 365
Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
370 375

<210> SEQ ID NO 17
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Erinaceus europaeus

<400> SEQUENCE: 17

Asp Ala Ser Phe Gly Ser Asn Pro Gly Ser Gln Asn Thr Pro Pro Leu
1 5 10 15
Ala Ile Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Ser Leu
20 25 30
Asn Asn Cys Asn His Ile Phe Thr Asp Ala Leu Asp Ile Leu Ala Glu
35 40 45
Asn His Glu Phe Arg Glu Asn Glu Val Ser Cys Val Ala Phe Met Arg
50 55 60
Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys
65 70 75 80
Asp Thr Lys Gly Ile Pro Cys Leu Gly Asp Lys Ala Lys Cys Val Ile
85 90 95
Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Ile Leu
100 105 110
Asn Asp Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly
115 120 125
Val Gly Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr
130 135 140
Leu Asn Lys Ile Met Ser Asp Lys Thr Leu Lys Leu Thr Arg Met Gln
145 150 155 160
Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Ala Lys
165 170 175
Ala Glu Ala Asp Ala Val Ser Val Leu Val Gln Glu Ala Val Trp Ala
180 185 190
Phe Leu Pro Asp Ala Met Val Thr Met Thr Gly Gly Phe Arg Arg Gly
195 200 205
Lys Lys Leu Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ala
210 215 220
Thr Glu Glu Glu Glu Gln Gln Leu Leu Pro Lys Val Ile Asn Phe Trp
225 230 235 240
Glu Arg Lys Gly Leu Leu Tyr His Asp Leu Val Glu Ser Thr Phe
245 250 255
Glu Lys Leu Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe
260 265 270
Gln Lys Cys Phe Leu Ile Leu Lys Leu His Leu Gln His Val Asn Gly
275 280 285
Val Gly Asn Ser Lys Thr Gly Gln Gln Glu Gly Lys Asn Trp Lys Ala

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290	295	300
Ile Arg Val Asp Leu Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala		
305	310	315 320
Leu Leu Gly Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg		
	325	330 335
Phe Ala Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr		
	340	345 350
Asp Lys Thr Lys Arg Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile		
	355	360 365
Phe Ala His Leu Gly Leu Asp Tyr Ile Asp Pro Trp Glu Arg Asn Ala		
	370	375 380
<210> SEQ ID NO 18		
<211> LENGTH: 381		
<212> TYPE: PRT		
<213> ORGANISM: Unknown		
<220> FEATURE:		
<223> OTHER INFORMATION: tree shrew		
<400> SEQUENCE: 18		
Asp His Ser Thr Ser Pro Ser Pro Gly Pro Gln Lys Thr Pro Ala Leu		
1	5	10 15
Ala Val Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu		
	20	25 30
Asn Asn Cys Asn Arg Val Phe Thr Asp Ala Phe Glu Thr Leu Ala Glu		
	35	40 45
Asn Tyr Glu Phe Arg Glu Asn Glu Asp Ser Ser Val Ile Phe Leu Arg		
	50	55 60
Ala Ala Ser Val Leu Arg Ser Leu Pro Phe Thr Ile Thr Ser Met Arg		
	65	70 75 80
Asp Thr Glu Gly Leu Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile		
	85	90 95
Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Asn Ala Val Leu		
	100	105 110
Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly		
	115	120 125
Val Gly Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr		
	130	135 140
Leu Ser Arg Val Arg Ser Asp Lys Ser Leu His Leu Thr Arg Met Gln		
	145	150 155 160
Gln Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Ala Ser Cys Val Thr Arg		
	165	170 175
Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Gly Ala		
	180	185 190
Phe Leu Pro Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly		
	195	200 205
Lys Lys Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser		
	210	215 220
Thr Glu Glu Lys Glu Glu Leu Leu Gln Lys Val Leu Asn Leu Trp		
	225	230 235 240
Glu Lys Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe		
	245	250 255
Glu Lys Leu Lys Thr Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe		

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260					265					270					
Pro	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Gly
	275						280					285			
Asp	Lys	Pro	Ser	Gln	Gln	Glu	Gly	Lys	Ser	Trp	Lys	Ala	Ile	Arg	Val
	290					295					300				
Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Arg	His	Ala	Phe	Ala	Leu	Leu	Gly
	305					310					315				320
Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr
			325						330					335	
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
			340					345					350		
Lys	Arg	Val	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Asp	Ile	Phe	Ala	His
		355					360					365			
Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala			
	370					375					380				

<210> SEQ ID NO 19

<211> LENGTH: 394

<212> TYPE: PRT

<213> ORGANISM: Ornithorhynchus anatinus

<400> SEQUENCE: 19

Leu	Thr	Asn	Ser	Ala	Pro	Ile	Asn	Cys	Met	Thr	Glu	Thr	Pro	Ser	Leu
1				5					10					15	
Ala	Thr	Lys	Gln	Val	Ser	Gln	Tyr	Ala	Cys	Glu	Arg	Arg	Thr	Thr	Leu
			20					25					30		
Asn	Asn	Cys	Asn	Gln	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Lys
		35					40					45			
Asp	Phe	Glu	Phe	Arg	Glu	Asn	Glu	Gly	Ile	Cys	Leu	Ala	Phe	Met	Arg
	50					55					60				
Ala	Ile	Ser	Val	Leu	Lys	Cys	Leu	Pro	Phe	Thr	Ile	Val	Arg	Met	Lys
	65				70						75			80	
Asp	Ile	Glu	Gly	Val	Pro	Trp	Leu	Gly	Asp	Gln	Val	Lys	Ser	Ile	Ile
			85					90						95	
Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Ser	Val	Lys	Ala	Val	Leu
			100					105					110		
Asn	Asp	Glu	Arg	Tyr	Arg	Ser	Phe	Gln	Leu	Phe	Asn	Ser	Val	Phe	Glu
		115					120					125			
Val	Gly	Leu	Thr	Asp	Asn	Gly	Glu	Asn	Gly	Ile	Ala	Arg	Gly	Phe	Gln
	130					135					140				
Thr	Leu	Asn	Glu	Val	Ile	Thr	Asp	Glu	Asn	Ile	Ser	Leu	Thr	Lys	Thr
	145				150						155			160	
Thr	Leu	Ser	Thr	Ser	Leu	Trp	Asn	Tyr	Leu	Pro	Gly	Phe	Leu	Tyr	Tyr
			165					170						175	
Glu	Asp	Leu	Val	Ser	Cys	Val	Ala	Lys	Glu	Glu	Ala	Asp	Ala	Val	Tyr
		180						185					190		
Leu	Ile	Val	Lys	Glu	Ala	Val	Arg	Ala	Phe	Leu	Pro	Glu	Ala	Leu	Val
		195					200					205			
Thr	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Ile	Gly	His	Asp	Val
	210					215					220				
Asp	Phe	Leu	Ile	Ser	Asp	Pro	Glu	Ser	Gly	Gln	Asp	Glu	Gln	Leu	Leu
	225				230						235				240

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Pro Asn Ile Ile Lys Leu Trp Glu Lys Gln Glu Leu Leu Leu Tyr Tyr
    245                250                255

Asp Leu Val Glu Ser Thr Phe Glu Lys Thr Lys Ile Pro Ser Arg Lys
    260                265                270

Val Asp Ala Met Asp His Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu
    275                280                285

His His Gln Lys Val Asp Ser Gly Arg Tyr Lys Pro Pro Pro Glu Ser
    290                295                300

Lys Asn His Glu Ala Lys Asn Trp Lys Ala Ile Arg Val Asp Leu Val
    305                310                315                320

Met Cys Pro Phe Glu Gln Tyr Ala Tyr Ala Leu Leu Gly Trp Thr Gly
    325                330                335

Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Lys
    340                345                350

Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Ile
    355                360                365

Phe Leu Lys Ala Glu Ser Glu Glu Asp Ile Phe Thr His Leu Gly Leu
    370                375                380

Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
    385                390

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<210> SEQ ID NO 20

<211> LENGTH: 384

<212> TYPE: PRT

<213> ORGANISM: *Jaculus blanfordi*

<400> SEQUENCE: 20

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Ser Ser Glu Leu Glu Leu Leu Asp Val Ser Trp Leu Ile Glu Cys Met
1      5      10      15

Gly Ala Gly Lys Pro Val Glu Met Thr Gly Arg His Gln Leu Val Lys
20     25     30

Gln Thr Phe Cys Leu Pro Gly Phe Ile Leu Gln Asp Ala Phe Asp Ile
35     40     45

Leu Ala Glu Asn Cys Glu Phe Arg Glu Asn Glu Ala Ser Cys Val Glu
50     55     60

Phe Met Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Ile
65     70     75     80

Ser Val Lys Asp Thr Glu Gly Ile Pro Trp Leu Gly Gly Lys Val Lys
85     90     95

Cys Val Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys
100    105    110

Ala Leu Leu Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser
115    120    125

Val Phe Gly Val Gly Leu Lys Thr Ala Glu Arg Trp Phe Arg Met Gly
130    135    140

Phe Arg Thr Leu Ser Thr Val Lys Leu Asp Lys Ser Leu Thr Phe Thr
145    150    155    160

Arg Met Gln Lys Ala Gly Phe Leu His Tyr Glu Asp Leu Val Ser Cys
165    170    175

Val Thr Arg Ala Glu Ala Glu Ala Val Ser Val Leu Val Gln Gln Ala
180    185    190

Val Val Ala Phe Leu Pro Asp Ala Leu Val Ser Met Thr Gly Gly Phe
195    200    205

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Arg	Arg	Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser
210						215					220				
Pro	Glu	Ala	Thr	Glu	Glu	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr
225				230						235					240
Asn	Phe	Trp	Glu	Gln	Lys	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	His	Val	Glu
			245						250					255	
Ser	Thr	Phe	Glu	Lys	Cys	Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu
			260					265					270		
Asp	His	Phe	Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	Tyr	Arg	Glu	Arg
			275				280					285			
Val	Asp	Ser	Val	Lys	Ser	Ser	Gln	Gln	Glu	Gly	Lys	Gly	Trp	Lys	Ala
	290					295					300				
Ile	Arg	Val	Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Cys	Arg	Ala	Phe	Ala
305				310						315					320
Leu	Leu	Gly	Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg
				325					330					335	
Tyr	Ala	Thr	His	Glu	Arg	Lys	Met	Arg	Leu	Asp	Asn	His	Ala	Leu	Tyr
			340					345					350		
Asp	Lys	Thr	Lys	Arg	Val	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile
			355				360					365			
Phe	Ala	His	Leu	Gly	Leu	Glu	Tyr	Ile	Glu	Pro	Leu	Glu	Arg	Asn	Ala
	370					375					380				

<210> SEQ ID NO 21

<211> LENGTH: 364

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Puma

<400> SEQUENCE: 21

Val	Val	Lys	Lys	Ile	Pro	Leu	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu
1			5					10					15		
Asn	Asn	Phe	Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Val	Leu	Ala	Glu
			20				25					30			
Asn	Tyr	Glu	Phe	Lys	Glu	Asn	Glu	Ile	Ser	Ser	Ala	Thr	Phe	Met	Arg
	35					40					45				
Ala	Ala	Ser	Val	Leu	Lys	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Lys	Asp
	50				55					60					
Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Cys	Val	Ile	Glu
65				70					75					80	
Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn
			85					90						95	
Asp	Glu	Arg	Tyr	Gln	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val
			100					105					110		
Gly	Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu
		115					120					125			
Ser	Lys	Ile	Lys	Ser	Asp	Lys	Thr	Leu	Lys	Phe	Thr	Gln	Met	Gln	Lys
	130					135					140				
Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg	Ala
145					150					155					160
Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe
			165						170					175	

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Leu Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys
 180 185 190
 Lys Ile Gly His Asp Val Asp Phe Leu Ile Thr Ile Pro Gly Ser Thr
 195 200 205
 Asp Glu Glu Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Gln
 210 215 220
 Arg Lys Glu Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu
 225 230 235 240
 Lys Leu Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln
 245 250 255
 Lys Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Gly
 260 265 270
 Lys Cys Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp
 275 280 285
 Leu Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp
 290 295 300
 Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His
 305 310 315 320
 Glu Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys
 325 330 335
 Lys Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu
 340 345 350
 Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 22
 <211> LENGTH: 362
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: reptilian
 <400> SEQUENCE: 22

Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr Asn Lys Lys Tyr
 1 5 10 15
 Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr Glu Met Arg Glu Asn
 20 25 30
 Ala Gly Ala Cys Leu Ala Phe Arg Arg Ala Ala Ser Val Leu Lys Phe
 35 40 45
 Leu Pro Phe Ala Ile Val Arg Ser Asn Asp Ile Glu Gly Leu Pro Trp
 50 55 60
 Met Gly Glu Gln Val Lys Gly Ile Ile Glu Asp Ile Leu Glu Glu Gly
 65 70 75 80
 Gln Ser Pro Lys Val Glu Ala Val Leu Asn Asn Glu Lys Tyr Arg Ser
 85 90 95
 Phe Lys Leu Phe Thr Ser Val Phe Gly Val Ala Leu Lys Thr Ser Glu
 100 105 110
 Lys Trp Phe Met Met Gly Leu Arg Asn Leu Glu Asp Val Lys Leu Asn
 115 120 125
 Gln Asn Leu Gln Leu Thr Arg Met Gln Lys Ala Gly Leu Gln His Tyr
 130 135 140
 Glu Asp Leu Ile Ser Tyr Val Ser Lys Ala Glu Ala Asp Ser Thr Ser
 145 150 155 160

-continued

Leu Met Val Lys Asp Thr Val Trp Lys Phe Ser Pro Ser Ala Leu Val
 165 170 175
 Thr Leu Thr Gly Gly Phe Arg Arg Gly Lys Lys Met Gly His Asp Val
 180 185 190
 Asp Phe Leu Ile Thr Val Pro Gly Ser Arg Pro Asn Glu Glu Leu Leu
 195 200 205
 His Leu Val Ile Asp Cys Trp Lys Lys Gln Gly Leu Leu Leu Tyr Tyr
 210 215 220
 Asp Leu Ile Glu Ser Thr Phe Glu Lys Ser Lys Leu Pro Ser Gln Arg
 225 230 235 240
 Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe Ala Ile Leu Lys Leu
 245 250 255
 His Lys Glu Arg Val Asn Gln Gly Thr Ser Leu Pro Pro Val Ala Ser
 260 265 270
 Thr Val Glu Glu Ile Lys Asp Trp Lys Ala Ile Arg Val Asp Leu Val
 275 280 285
 Val Ser Pro Phe Glu Gln His Ala Phe Ala Leu Leu Gly Trp Thr Gly
 290 295 300
 Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Lys
 305 310 315 320
 Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Ile
 325 330 335
 Phe Leu Ser Ala Ser Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu
 340 345 350
 Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 23
 <211> LENGTH: 364
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: shrew

<400> SEQUENCE: 23

Val Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn
 1 5 10 15
 Asn His Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn
 20 25 30
 Cys Glu Phe Arg Glu Asn Glu Gly Ser Tyr Val Thr Tyr Met Arg Ala
 35 40 45
 Ala Ser Val Leu Lys Ser Leu Pro Phe Ser Ile Ile Ser Met Lys Asp
 50 55 60
 Thr Glu Gly Ile Pro Cys Leu Ala Asp Lys Val Lys Cys Val Ile Glu
 65 70 75 80
 Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn
 85 90 95
 Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val
 100 105 110
 Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Leu Gly Phe Arg Thr Leu
 115 120 125
 Ser Gly Ile Met Asn Asp Lys Thr Leu Lys Leu Thr His Met Gln Lys
 130 135 140

-continued

Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala
 145 150 155 160

Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe
 165 170 175

Leu Pro Asp Ala Ile Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys
 180 185 190

Lys Val Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr
 195 200 205

Glu Glu Gln Glu Gln Gln Leu Leu His Lys Val Ile Thr Phe Trp Glu
 210 215 220

Lys Glu Gly Leu Leu Leu Tyr Cys Asp Leu Tyr Glu Ser Thr Phe Glu
 225 230 235 240

Lys Leu Lys Met Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln
 245 250 255

Lys Cys Phe Leu Ile Leu Lys Leu His Arg Glu Cys Val Asp Asp Gly
 260 265 270

Thr Ser Ser Gln Leu Gln Gly Lys Thr Trp Lys Ala Ile Arg Val Asp
 275 280 285

Leu Val Val Cys Pro Tyr Glu Cys Arg Ala Phe Ala Leu Leu Gly Trp
 290 295 300

Thr Gly Ser Pro Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His
 305 310 315 320

Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys
 325 330 335

Arg Lys Phe Leu Ser Ala Asp Ser Glu Glu Asp Ile Phe Ala His Leu
 340 345 350

Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 24

<211> LENGTH: 382

<212> TYPE: PRT

<213> ORGANISM: *Serinus canaria*

<400> SEQUENCE: 24

Ser Pro Pro Leu Asn Thr Pro Glu Leu Glu Met Pro Ser Phe Ile Ala
 1 5 10 15

Thr Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr Leu Asn Asn
 20 25 30

Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Val Met Ala Glu Asn Tyr
 35 40 45

Glu Phe Lys Glu Asn Glu Ile Phe Cys Leu Glu Phe Leu Arg Ala Ala
 50 55 60

Ser Leu Leu Lys Ser Leu Pro Phe Ser Val Thr Arg Met Lys Asp Ile
 65 70 75 80

Gln Gly Leu Pro Cys Met Gly Asp Gln Val Arg Asp Val Ile Glu Ile
 85 90 95

Ile Glu Glu Gly Glu Ser Ser Arg Val Lys Glu Val Leu Asn Asp Glu
 100 105 110

Arg Tyr Lys Ala Phe Lys Gln Phe Thr Ser Val Phe Gly Val Gly Val
 115 120 125

Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Val Gly Glu

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130	135	140
Val Lys Ala Asp Lys Thr Leu Lys Leu Ser Lys Met Gln Lys Ala Gly		
145	150	155 160
Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Ser Lys Ala Glu Ala		
	165	170 175
Asp Ala Val Ser Leu Ile Val Lys Asn Thr Val Cys Thr Phe Leu Pro		
	180	185 190
Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Asn Ile		
	195	200 205
Gly His Asp Ile Asp Phe Leu Ile Thr Asn Pro Gly Pro Arg Glu Asp		
	210	215 220
Asp Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly Leu		
	225	230 235 240
Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Val Lys Glu Gln Leu		
	245	250 255
Pro Ser Arg Lys Ile Asp Ala Met Asp His Phe Gln Lys Cys Phe Ala		
	260	265 270
Ile Leu Lys Leu Tyr Gln Pro Arg Val Asp Asn Ser Thr Cys Asn Thr		
	275	280 285
Ser Lys Lys Leu Glu Met Ala Glu Val Lys Asp Trp Lys Ala Ile Arg		
	290	295 300
Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ser Tyr Ala Leu Leu		
	305	310 315 320
Gly Trp Thr Gly Ser Arg Gln Phe Gly Arg Asp Leu Arg Arg Tyr Ala		
	325	330 335
Ala His Glu Arg Arg Met Ile Leu Asp Asn His Gly Leu Tyr Asp Arg		
	340	345 350
Thr Lys Arg Ile Phe Leu Lys Ala Gly Ser Glu Glu Glu Ile Phe Ala		
	355	360 365
His Leu Gly Leu Asp Tyr Val Glu Pro Trp Glu Arg Asn Ala		
	370	375 380

<210> SEQ ID NO 25

<211> LENGTH: 370

<212> TYPE: PRT

<213> ORGANISM: Neopelma chrysocephalum

<400> SEQUENCE: 25

Phe Ile Ala Arg Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr		
1	5	10 15
Leu Asn Asn Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Met Ala		
	20	25 30
Glu Asn Tyr Glu Phe Lys Glu Asn Glu Ile Phe Cys Leu Glu Phe Leu		
	35	40 45
Arg Ala Ala Ser Leu Leu Lys Tyr Leu Pro Phe Pro Val Thr Arg Met		
	50	55 60
Lys Asp Ile Gln Gly Leu Pro Cys Ile Gly Asp Gln Val Arg Asp Val		
	65	70 75 80
Ile Glu Gly Ile Ile Glu Glu Gly Glu Ser Ser Arg Val Lys Glu Val		
	85	90 95
Leu Asn Asp Glu Arg Tyr Lys Ala Phe Lys Gln Phe Thr Ser Val Phe		
	100	105 110

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<210> SEQ ID NO 26
<211> LENGTH: 383
<212> TYPE: PRT
<213> ORGANISM: Alligator sinensis

<400> SEQUENCE: 26
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Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro	Ser	Val
1				5					10					15	
Asp	Lys	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn
			20					25					30		
Tyr	Asn	Lys	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Cys
	35						40					45			
Glu	Phe	Arg	Glu	Asn	Arg	Leu	Gly	Cys	Leu	Glu	Phe	Leu	Arg	Ala	Ala
	50					55					60				
Ser	Val	Leu	Lys	Phe	Leu	Pro	Phe	Pro	Ile	Val	Lys	Met	Lys	Asn	Ile
65				70					75					80	
Glu	Gly	Leu	Pro	Cys	Met	Gly	Asp	Lys	Val	Lys	Cys	Val	Ile	Glu	Glu
				85				90					95		

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Ile Leu Glu Glu Gly Glu Ser Cys Gln Ala Lys Glu Ile Leu Asn Asp
      100                      105                      110

Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
      115                      120                      125

Leu Lys Thr Thr Glu Lys Trp Tyr Arg Met Gly Phe Arg Thr Leu Glu
      130                      135                      140

Glu Val Lys Ala Glu Lys Thr Leu Lys Leu Ser Arg Met Gln Ile Ala
      145                      150                      155                      160

Gly Phe Leu His Tyr Glu Asp Ile Ile Ser Tyr Val Ser Lys Ala Glu
      165                      170                      175

Ala Asp Ala Val Ser Leu Leu Ile Lys Asp Thr Val Cys Met Phe Leu
      180                      185                      190

Pro Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Lys
      195                      200                      205

Thr Gly His Asp Val Asp Phe Leu Ile Thr Asn Pro Gly Pro Glu Glu
      210                      215                      220

Glu Lys Glu Leu Leu His Lys Val Val Asp Leu Trp Glu Lys Gln Gly
      225                      230                      235                      240

Leu Leu Leu Tyr Tyr Asp Val Ile Glu Ser Thr Phe Glu Lys Glu Lys
      245                      250                      255

Arg Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe
      260                      265                      270

Ala Ile Leu Lys Leu His Gln Gln Arg Arg Gly Asn Ser Asn Ser Asn
      275                      280                      285

Ile Ser Lys Glu Ser Asp Lys Ala Glu Val Lys Asp Trp Lys Ala Ile
      290                      295                      300

Arg Val Asp Leu Val Ile Ser Pro Phe Glu Gln Tyr Ala Tyr Ala Leu
      305                      310                      315                      320

Leu Gly Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr
      325                      330                      335

Ala Ser Arg Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp
      340                      345                      350

Lys Thr Lys Arg Thr Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe
      355                      360                      365

Ala His Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
      370                      375                      380

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<210> SEQ ID NO 27

<211> LENGTH: 380

<212> TYPE: PRT

<213> ORGANISM: *Xenopus laevis*

<400> SEQUENCE: 27

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Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Glu
1      5      10      15

Val Lys Val Ser Gln Tyr Ala Cys Gln Arg Cys Thr Thr Leu Gln Asp
      20      25      30

Thr Asn Arg Ile Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu His Phe
      35      40      45

Glu Phe Cys Glu Asn Lys Gly Arg Thr Val Ala Phe Leu Arg Ala Ser
      50      55      60

Ser Leu Ile Lys Ser Leu Pro Phe Pro Ile Thr Ala Met Lys Glu Leu

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65	70	75	80
Glu Gly Leu Pro Trp	Leu Gly Asp Gln Met Lys Gly Ile Ile Glu Glu		
	85	90	95
Ile Leu Glu Glu Gly Lys Ser Tyr Lys Val Leu Glu Val Met Asn Glu			
	100	105	110
Glu Arg Tyr Lys Ser Phe Lys Gln Phe Thr Ser Val Phe Gly Val Gly			
	115	120	125
Leu Lys Thr Ser Asp Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu			
	130	135	140
Glu Ile Lys Asn Glu Lys Glu Leu Lys Leu Thr Lys Met Gln Lys Cys			
	145	150	155
Gly Leu Leu Tyr Tyr Glu Asp Ile Thr Ser Tyr Val Ser Arg Ala Glu			
	165	170	175
Ala Glu Thr Thr Glu Gln Leu Ile Lys Ser Ile Val Trp Lys Phe Val			
	180	185	190
Pro Asp Ala Ile Val Thr Leu Thr Gly Gly Phe Arg Arg Gly Lys Lys			
	195	200	205
Lys Gly His Asp Val Asp Ile Leu Ile Thr Cys Ala Arg Lys Gly Lys			
	210	215	220
Glu Lys Asn Ile Leu His Asn Thr Met Ser Val Leu Lys Asn Arg Gly			
	225	230	235
Leu Leu Leu Phe Tyr Asn Ile Ile Glu Ser Thr Phe Asp Glu Thr Lys			
	245	250	255
Leu Pro Ser Arg His Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe			
	260	265	270
Thr Ile Leu Lys Leu Pro Lys Arg Gln Met Asp Ile Gly Asn Ile Ile			
	275	280	285
Asp Pro His Glu Cys Glu Arg Lys Asn Trp Lys Ala Val Arg Leu Asp			
	290	295	300
Leu Val Ile Thr Pro Tyr Glu Gln Tyr Pro Tyr Ala Leu Leu Gly Trp			
	305	310	315
Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His			
	325	330	335
Glu Lys Arg Met Met Leu Asp Asn His Gly Leu Tyr Asp Lys Thr Lys			
	340	345	350
Asn Asn Phe Leu Lys Ala Asn Asn Glu Glu Asp Ile Phe Lys Gln Leu			
	355	360	365
Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala			
	370	375	380

<210> SEQ ID NO 28

<211> LENGTH: 383

<212> TYPE: PRT

<213> ORGANISM: *Notechis scutatus*

<400> SEQUENCE: 28

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val			
1	5	10	15
Asp Ser Ile Ser Pro Tyr Ala Cys Gln Arg Arg Thr Thr Leu Lys Asn			
	20	25	30
Tyr Asn Gln Lys Phe Thr Asp Ala Phe Glu Ile Leu Val Glu Asn Tyr			
	35	40	45

-continued

Glu	Phe	Asn	Glu	Asn	Lys	Glu	Phe	Cys	Leu	Ala	Phe	Gly	Arg	Ala	Ala
50						55					60				
Ser	Leu	Leu	Lys	Cys	Leu	Pro	Phe	Thr	Val	Val	Arg	Val	Asn	Asp	Ile
65					70					75					80
Glu	Gly	Leu	Pro	Trp	Met	Gly	Lys	Gln	Val	Arg	Glu	Ile	Ile	Glu	Asp
				85					90					95	
Ile	Leu	Glu	Glu	Gly	Glu	Ser	Ser	Lys	Val	Lys	Ala	Val	Leu	Asn	Asp
			100					105					110		
Glu	Asn	Tyr	Arg	Ser	Ile	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
		115					120					125			
Leu	Arg	Thr	Ser	Glu	Lys	Trp	Tyr	Arg	Met	Gly	Leu	Arg	Thr	Leu	Glu
	130					135					140				
Glu	Val	Lys	Cys	Asn	Lys	Asn	Leu	Thr	Leu	Thr	Arg	Met	Gln	Lys	Ala
145					150					155					160
Gly	Phe	Phe	Tyr	Tyr	Asp	Asp	Leu	Ile	Ser	Ser	Val	Ser	Lys	Ala	Glu
				165					170					175	
Ala	Asp	Ala	Ala	Thr	Gln	Ile	Val	Gln	Asp	Thr	Val	Trp	Lys	Ile	Leu
			180					185					190		
Pro	Asn	Ala	Val	Val	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Gln
		195					200					205			
Thr	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Val	Pro	Gly	Ser	Arg	Gln
	210					215					220				
Glu	Glu	Glu	Leu	Leu	His	Pro	Val	Ile	Asp	Ile	Trp	Lys	Lys	Gln	Glu
225					230					235					240
Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Ile	Glu	Ser	Thr	Phe	Glu	Asn	Thr	Lys
				245					250					255	
Leu	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe
			260					265					270		
Ala	Ile	Leu	Lys	Val	His	Lys	Glu	Arg	Val	Asn	Lys	Gly	Ser	Ala	Val
		275					280					285			
Gln	Ser	Asn	Val	Phe	Ala	Glu	Glu	Gly	Thr	Lys	Asp	Trp	Lys	Ala	Ile
	290					295					300				
Arg	Val	Asp	Leu	Val	Val	Thr	Pro	Phe	Gln	His	Tyr	Ala	Phe	Ala	Leu
305					310					315					320
Leu	Gly	Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr
				325					330					335	
Ala	Thr	Gln	Glu	Lys	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp
			340					345					350		
Lys	Thr	Lys	Lys	Ile	Phe	Leu	Ser	Ala	Ala	Asn	Glu	Glu	Glu	Ile	Phe
		355					360					365			
Ser	His	Leu	Gly	Leu	Asp	Tyr	Leu	Glu	Pro	Trp	Glu	Arg	Asn	Ala	
	370					375					380				

<210> SEQ ID NO 29

<211> LENGTH: 367

<212> TYPE: PRT

<213> ORGANISM: Salmo truttal

<400> SEQUENCE: 29

Thr	Ala	Ala	Ala	Asn	Val	Ser	Gln	Tyr	Ala	Cys	Leu	Arg	Arg	Thr	Thr
1				5					10					15	

Thr	Glu	Asn	His	Asn	Lys	Ile	Phe	Thr	Asp	Val	Leu	Glu	Glu	Leu	Ala
		20						25					30		

-continued

Glu Asn Ser Glu Phe Asn Glu Ser Lys Gly Pro Cys Leu Ala Phe Arg
 35 40 45
 Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Ser Ala Val His Cys Leu
 50 55 60
 Gly Ala Ile Gln Gly Leu Pro Cys Leu Gly Glu His Thr Lys Ala Val
 65 70 75 80
 Met Glu Glu Ile Leu Ile Phe Gly Arg Ser Phe Lys Val Glu Glu Val
 85 90 95
 Gln Ser Asp Glu Arg Tyr Gln Ala Leu Lys Leu Phe Thr Ser Val Phe
 100 105 110
 Gly Val Gly Pro Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg
 115 120 125
 Ser Leu Lys Glu Ile Leu Ala Glu Pro Asn Ile Gln Leu Asn Arg Met
 130 135 140
 Gln Arg Ala Gly Phe Leu Tyr Tyr Arg Asp Ile Ser Lys Ala Val Ser
 145 150 155 160
 Lys Ala Glu Ala Lys Ala Leu Ser Ser Ile Ile Glu Glu Thr Ala His
 165 170 175
 Trp Ile Ala Pro Asp Ser Ile Leu Ala Leu Thr Gly Gly Phe Arg Arg
 180 185 190
 Gly Lys Glu Tyr Gly His Asp Val Asp Phe Leu Leu Thr Met Pro Glu
 195 200 205
 Met Gly Lys Glu Glu Gly Leu Leu Leu Arg Val Ile Asp Arg Leu Arg
 210 215 220
 Asp Gln Gly Ile Leu Leu Tyr Cys Glu His Gln Asp Ser Thr Phe Asp
 225 230 235 240
 Met Ser Lys Leu Pro Ser His Arg Phe Glu Ala Met Asp His Phe Glu
 245 250 255
 Lys Cys Phe Leu Ile Leu Arg Leu Glu Glu Gly Gln Val Glu Gly Asp
 260 265 270
 Gly Gly Leu Gln Lys Asp Pro Gly Glu Ser Arg Gly Trp Arg Ala Val
 275 280 285
 Arg Val Asp Leu Val Ala Pro Pro Val Asp Arg Tyr Ala Phe Val Leu
 290 295 300
 Leu Gly Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Phe
 305 310 315 320
 Ala Ser Lys Glu Arg Gly Met Cys Leu Asp Asn His Ala Leu Tyr Asp
 325 330 335
 Lys Thr Lys Lys Leu Phe Leu Pro Ala Thr Ser Glu Glu Asp Ile Phe
 340 345 350
 Ala His Leu Gly Leu Glu Tyr Val Glu Pro Trp Gln Arg Asn Ala
 355 360 365

<210> SEQ ID NO 30

<211> LENGTH: 377

<212> TYPE: PRT

<213> ORGANISM: Electrophorus electricus

<400> SEQUENCE: 30

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Leu
 1 5 10 15
 Pro Thr Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asp Asn

-continued

20					25					30					
His	Asn	Lys	Val	Phe	Thr	Asp	Ala	Leu	Glu	Val	Leu	Ile	Glu	Asn	Tyr
	35						40					45			
Glu	Phe	Ser	Asp	Asn	Lys	Gly	Ala	Cys	Val	Gly	Phe	Arg	Arg	Ala	Ala
	50					55					60				
Ser	Val	Leu	Lys	Ser	Leu	Pro	Lys	Pro	Leu	Arg	Cys	Leu	Lys	Asp	Met
	65					70					75				80
Glu	Gly	Leu	Pro	Cys	Leu	Gly	Asp	Asp	Thr	Lys	Ala	Ile	Ile	Glu	Glu
				85					90					95	
Ile	Tyr	Glu	Cys	Gly	Ser	Ser	Ser	Arg	Val	Glu	Asn	Ile	Leu	Ser	Asp
		100						105					110		
Glu	Lys	Tyr	Gln	Thr	Leu	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
		115						120					125		
Pro	Lys	Thr	Gly	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ala	Leu	Glu
		130				135					140				
Gln	Val	His	Ser	Glu	Pro	Ser	Ile	Gln	Leu	Asn	Lys	Met	Gln	Ala	Ala
	145					150					155				160
Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Ile	Ser	Lys	Pro	Val	Ser	Arg	Ala	Glu
				165					170					175	
Ala	Lys	Ala	Val	Gly	Cys	Ile	Ile	Glu	Glu	Val	Ala	Ser	Cys	Phe	Ser
			180					185					190		
Ser	Ser	Val	Thr	Ile	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Glu
		195					200						205		
Phe	Gly	His	Asp	Val	Asp	Phe	Leu	Leu	Ser	Ile	Pro	Glu	Pro	Gly	Lys
	210					215					220				
Glu	Asp	Gly	Leu	Leu	Pro	Ala	Val	Ile	Asp	Arg	Leu	Arg	Lys	Gln	Gly
	225					230					235				240
Ile	Leu	Leu	Tyr	Ser	Asp	Leu	Gln	Glu	Ser	Thr	Leu	Gln	Gln	Trp	Lys
				245					250					255	
Arg	Pro	Ser	Arg	Cys	Phe	Asp	Ser	Met	Asp	His	Phe	Gln	Lys	Cys	Phe
			260					265					270		
Leu	Ile	Val	Lys	Leu	Trp	Thr	Arg	Leu	Val	Glu	Gly	His	Arg	Glu	Asp
		275					280						285		
Pro	Ser	Ser	Gln	Arg	Asp	Trp	Lys	Ala	Val	Arg	Val	Asp	Leu	Val	Val
	290					295					300				
Pro	Pro	Val	Asp	Cys	Tyr	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Ser	Gly	Ser
	305					310					315				320
Thr	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Arg	Leu	Glu	Arg	Arg
				325					330					335	
Met	Leu	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Thr	Asn	Thr	Phe
			340					345					350		
Leu	Gln	Ala	Lys	Thr	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp
		355					360						365		
Tyr	Ile	Glu	Pro	Trp	Gln	Arg	Asn	Ala							
	370					375									

<210> SEQ ID NO 31

<211> LENGTH: 378

<212> TYPE: PRT

<213> ORGANISM: Anabas testudineus

<400> SEQUENCE: 31

-continued

Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro	Ser	Val
1				5					10					15	
Ala	Thr	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Thr	Glu	Asn
			20					25					30		
Asn	Asn	Lys	Ile	Leu	Thr	Asp	Ala	Phe	Glu	Val	Leu	Ala	Glu	Ser	Tyr
		35					40					45			
Glu	Leu	Asn	Gln	Leu	Glu	Gly	Pro	Cys	Leu	Ala	Phe	Arg	Arg	Ala	Ala
	50					55					60				
Ser	Val	Leu	Lys	Ser	Leu	Pro	Trp	Ala	Val	Gln	Cys	Leu	Gly	Ala	Thr
65					70					75					80
Gln	Gly	Leu	Pro	Cys	Leu	Gly	Glu	His	Thr	Lys	Ala	Leu	Ile	Glu	Glu
			85						90					95	
Ile	Leu	Gln	Tyr	Gly	His	Ser	Phe	Glu	Val	Glu	Lys	Ile	Leu	Ser	Asp
			100					105					110		
Glu	Arg	Tyr	Gln	Thr	Leu	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
		115						120					125		
Pro	Lys	Thr	Ala	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ser	Phe	Ser
	130					135					140				
Asp	Ile	Leu	Ala	Glu	Pro	Ser	Ile	Gln	Leu	Asn	Arg	Met	Gln	Gln	Ser
145					150					155					160
Gly	Phe	Leu	His	Tyr	Gly	Asp	Ile	Ser	Arg	Ala	Val	Ser	Lys	Ala	Glu
			165						170					175	
Ala	Arg	Ala	Leu	Gly	Asn	Ile	Ile	Asp	Glu	Ala	Val	His	Ala	Ile	Thr
			180					185					190		
Pro	Asp	Gly	Ile	Leu	Ala	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Glu
		195					200					205			
Phe	Gly	His	Asp	Val	Asp	Phe	Ile	Val	Thr	Thr	Pro	Glu	Gln	Gly	Lys
	210					215					220				
Glu	Glu	Thr	Leu	Leu	Pro	Asn	Ile	Ile	Asp	Arg	Leu	Lys	Glu	Gln	Gly
225					230					235					240
Ile	Leu	Leu	Tyr	Ser	Asp	Tyr	Gln	Thr	Ser	Thr	Phe	Asp	Ile	Ser	Lys
			245						250					255	
Leu	Pro	Ser	His	Lys	Phe	Glu	Ala	Met	Asp	His	Phe	Ala	Lys	Cys	Phe
			260					265					270		
Leu	Ile	Leu	Arg	Leu	Glu	Gly	Ser	Leu	Val	Asp	Arg	Gly	Leu	Asn	Ser
		275					280					285			
Thr	Glu	Gly	Asp	Ser	Arg	Gly	Trp	Arg	Ala	Val	Arg	Val	Asp	Leu	Val
	290					295					300				
Ser	Pro	Pro	Met	Glu	Arg	Tyr	Ala	Tyr	Ala	Leu	Leu	Gly	Trp	Thr	Gly
305					310					315					320
Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Arg	Leu	Glu	Gln
			325						330					335	
His	Met	Leu	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Lys	Glu
			340					345					350		
Phe	Leu	Ala	Ala	Thr	Thr	Glu	Arg	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu
		355					360					365			
Glu	Tyr	Ile	Glu	Pro	Trp	Gln	Arg	Asn	Ala						
	370					375									

<210> SEQ ID NO 32

<211> LENGTH: 378

<212> TYPE: PRT

-continued

<213> ORGANISM: *Poecilia reticulata*

<400> SEQUENCE: 32

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Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val
1          5          10          15
Asp Lys Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Val Glu Asn
20          25          30
Asn Asn Arg Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Asn Tyr
35          40          45
Glu Phe Asn Glu Ile Glu Gly Arg Cys Leu Ala Phe Arg Arg Ala Ala
50          55          60
Ser Val Leu Lys Ser Leu Pro Trp Ala Val Arg Ser Val Gly Ala Thr
65          70          75          80
Leu Asp Leu Pro Cys Leu Gly Glu His Thr Thr Ala Val Met Lys Glu
85          90          95
Ile Leu Gln Tyr Gly Arg Ser Phe Glu Val Glu Lys Ile Leu Ser Asp
100         105         110
Glu Arg Cys Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115         120         125
Pro Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Phe Ser
130         135         140
Asp Val Leu Ala Gln Pro Gly Ile His Leu Asn Arg Met Gln Gln Ser
145         150         155         160
Gly Phe Leu His Tyr Gly Asp Ile Ser Arg Ala Val Ser Lys Ala Glu
165         170         175
Ala Arg Ala Val Gly Asn Ile Ile Asp Glu Ala Val His Val Ile Thr
180         185         190
Pro Asn Ala Ile Leu Ala Leu Thr Gly Gly Phe Arg Arg Gly Lys Asp
195         200         205
Phe Gly His Asp Val Asp Phe Ile Val Thr Thr Thr Glu Leu Gly Lys
210         215         220
Glu Lys Asn Leu Leu Ile Ser Val Ile Glu Ser Leu Lys Lys Gln Gly
225         230         235         240
Leu Leu Leu Phe Ser Asp Tyr Gln Ala Ser Thr Phe Asp Ile Ser Lys
245         250         255
Leu Pro Ser His Arg Phe Glu Ala Met Asp His Phe Ala Lys Cys Phe
260         265         270
Leu Ile Leu Arg Leu Glu Gly Ser Arg Val Glu Gly Gly Leu Gln Arg
275         280         285
Ala Gln Ala Asp Gly Arg Gly Trp Arg Ala Val Arg Val Asp Leu Val
290         295         300
Ser Pro Pro Ala Asp Arg Phe Ala Phe Thr Met Leu Gly Trp Thr Gly
305         310         315         320
Ser Arg Met Phe Glu Arg Asp Leu Arg Arg Phe Ala Arg Leu Glu Arg
325         330         335
Gln Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Glu
340         345         350
Phe Leu Thr Ala Ala Thr Glu Lys Asp Ile Phe Asp His Leu Gly Leu
355         360         365
Glu Tyr Ile Glu Pro Trp Gln Arg Asn Ala
370         375

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<210> SEQ ID NO 33
<211> LENGTH: 366
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 33

Pro Leu Met Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr
1      5      10      15
Leu Asn Asn His Asn Gln Leu Phe Thr Asp Ala Phe Asp Ile Leu Ala
      20      25      30
Glu Asn Tyr Glu Phe Arg Glu Asn Glu Val Ser Cys Leu Pro Phe Met
      35      40      45
Arg Ala Ala Ser Val Leu Lys Ser Leu Ser Phe Pro Ile Val Ser Met
      50      55      60
Lys Asp Ile Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val
      65      70      75      80
Ile Glu Gly Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val
      85      90      95
Leu Asn Asp Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe
      100     105     110
Gly Val Gly Leu Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg
      115     120     125
Thr Leu Ser Lys Ile Lys Ser Asp Lys Ser Leu Arg Phe Thr His Met
      130     135     140
Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Asn
      145     150     155     160
Arg Ala Glu Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val
      165     170     175
Ala Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Arg
      180     185     190
Gly Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu
      195     200     205
Ala Thr Glu Glu Glu Glu Gln Gln Leu Leu His Lys Val Thr Asn Phe
      210     215     220
Trp Arg Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr
      225     230     235     240
Phe Glu Lys Phe Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His
      245     250     255
Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu His Arg Gly Leu Val Arg
      260     265     270
Ser Glu Glu Ser Gly Gln Gln Glu Gly Lys Asp Trp Lys Ala Ile Arg
      275     280     285
Val Asp Leu Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu
      290     295     300
Gly Trp Thr Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala
      305     310     315     320
Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys
      325     330     335
Thr Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala
      340     345     350
His Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
      355     360     365

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<210> SEQ ID NO 34
<211> LENGTH: 379
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 34

Ser Leu Ser Pro Val Pro Gly Ser Gln Thr Val Pro Pro Pro Leu Met
1          5          10          15

Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
20          25          30

His Asn Gln Leu Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Tyr
35          40          45

Glu Phe Arg Glu Asn Glu Val Ser Cys Leu Pro Phe Met Arg Ala Ala
50          55          60

Ser Val Leu Lys Ser Leu Ser Phe Pro Ile Val Ser Met Lys Asp Ile
65          70          75          80

Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile Glu Gly
85          90          95

Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp
100         105         110

Glu Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115         120         125

Leu Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser
130         135         140

Lys Ile Lys Ser Asp Lys Ser Leu Arg Phe Thr His Met Gln Lys Ala
145         150         155         160

Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Asn Arg Ala Glu
165         170         175

Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Ala Phe Leu
180         185         190

Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Met
195         200         205

Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr Glu
210         215         220

Glu Glu Glu Gln Gln Leu Leu His Lys Val Thr Asn Phe Trp Arg Gln
225         230         235         240

Gln Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Glu Lys
245         250         255

Phe Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys
260         265         270

Cys Phe Leu Ile Leu Lys Leu His Arg Gly Leu Val Arg Ser Glu Glu
275         280         285

Ser Gly Gln Gln Glu Gly Lys Asp Trp Lys Ala Ile Arg Val Asp Leu
290         295         300

Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr
305         310         315         320

Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
325         330         335

Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
340         345         350

Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly

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355	360	365
Leu Asp Tyr Ile Glu Pro Trp	Glu Arg Asn Ala	
370	375	

<210> SEQ ID NO 35
 <211> LENGTH: 379
 <212> TYPE: PRT
 <213> ORGANISM: Piliocolobus tephrosceles
 <400> SEQUENCE: 35

Ser Asp Ser Thr Asn Pro Gly Leu Pro Lys Thr Pro Pro Thr Ala Ile	
1	15
Gln Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn	
20	30
Cys Asn Gln Ile Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Cys	
35	45
Glu Phe Arg Glu Asn Glu Asp Ser Cys Val Thr Phe Met Arg Ala Ala	
50	60
Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr	
65	80
Glu Gly Ile Pro Cys Leu Gly Ser Lys Val Lys Cys Ile Ile Glu Glu	
85	95
Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp	
100	110
Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly	
115	125
Leu Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser	
130	140
Lys Val Arg Ser Asp Glu Ser Leu Lys Phe Thr Arg Met Gln Arg Ala	
145	160
Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu	
165	175
Ala Glu Ala Val Ser Ala Leu Val Lys Glu Ala Val Trp Ala Phe Leu	
180	190
Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys	
195	205
Met Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu	
210	220
Asp Glu Glu Gln Gln Leu Leu Gln Lys Val Met Asn Leu Trp Glu Lys	
225	240
Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys	
245	255
Leu Arg Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys	
260	270
Cys Phe Leu Ile Phe Lys Leu Pro Leu Gln Arg Val Asp Ser Asp Gln	
275	285
Ser Ser Trp Gln Gly Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu	
290	300
Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr	
305	320
Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu	
325	335

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Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
      340              345              350

Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly
      355              360              365

Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
      370              375

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<210> SEQ ID NO 36
<211> LENGTH: 379
<212> TYPE: PRT
<213> ORGANISM: Sus scrofa

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<400> SEQUENCE: 36

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Ser Ala Ser Pro Ser Pro Gly Ser Gln Asn Thr Leu Pro Pro Ala Val
1      5      10      15

Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
20     25     30

Cys Asn His Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Asn Tyr
35     40     45

Glu Phe Arg Glu Asn Glu Thr Phe Cys Leu Ala Phe Met Arg Ala Ala
50     55     60

Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr
65     70     75     80

Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile Glu Glu
85     90     95

Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp
100    105    110

Glu Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115    120    125

Leu Lys Thr Ser Glu Arg Trp Phe Arg Met Gly Phe Arg Ser Leu Ser
130    135    140

Lys Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met Gln Lys Ala
145    150    155    160

Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu
165    170    175

Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Gln Ala Phe Leu
180    185    190

Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys
195    200    205

Met Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Asp
210    215    220

Asp Glu Glu Gln Gln Leu Leu Pro Lys Val Val Asn Leu Trp Glu Arg
225    230    235    240

Glu Gly Leu Leu Leu Tyr Cys Asp Leu Val Glu Ser Thr Leu Glu Lys
245    250    255

Ser Lys Leu Pro Ser Arg Asn Val Asp Ala Leu Asp His Phe Gln Lys
260    265    270

Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Gly Met
275    280    285

Ser Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu
290    295    300

Val Met Cys Pro Tyr Glu Leu Arg Ala Phe Ala Leu Leu Gly Trp Thr
305    310    315    320

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<210> SEQ ID NO 37
<211> LENGTH: 379
<212> TYPE: PRT
<213> ORGANISM: Panthera tigris altaica
```

Tyr 1	Ser	Ala	Ser	Pro 5	Asn	Pro	Glu	Leu	Gln 10	Lys	Thr	Pro	Pro	Leu 15	Val
Lys	Lys	Ile	Pro 20	Leu	Tyr	Ala	Cys	Gln 25	Arg	Arg	Thr	Thr	Leu 30	Asn	Asn
Phe	Asn	His	Val	Phe	Thr	Asp	Ala 40	Phe	Glu	Val	Leu	Ala 45	Glu	Asn	Tyr
Glu	Phe	Lys	Glu	Asn	Glu	Val	Ser 55	Ser	Ala	Thr	Phe	Met 60	Arg	Ala	Ala
Ser 65	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Lys	Asp	Thr 80
Glu	Gly	Ile	Pro	Cys 85	Leu	Gly	Asp	Lys	Val 90	Lys	Cys	Val	Ile	Glu 95	Glu
Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp
Glu	Arg	Tyr	Gln	Ser	Phe	Lys	Leu 120	Phe	Thr	Ser	Val	Phe 125	Gly	Val	Gly
Leu	Lys 130	Thr	Ser	Glu	Lys	Trp 135	Phe	Arg	Met	Gly	Phe 140	Arg	Thr	Leu	Ser
Lys 145	Ile	Lys	Ser	Asp	Lys 150	Thr	Leu	Lys	Phe	Thr 155	Gln	Met	Gln	Lys	Ala
Gly	Phe	Leu	Tyr	Tyr 165	Glu	Asp	Leu	Val	Ser 170	Cys	Val	Thr	Arg	Ala	Glu
Ala	Glu	Ala	Val	Gly 180	Val	Leu	Val	Lys 185	Glu	Ala	Val	Trp	Ala 190	Phe	Leu
Pro	Asp	Ala 195	Phe	Val	Thr	Met	Thr 200	Gly	Gly	Phe	Arg	Arg 205	Gly	Lys	Lys
Ile 210	Gly	His	Asp	Val	Asp	Phe 215	Leu	Ile	Thr	Ser	Pro 220	Gly	Ser	Thr	Asp
Glu 225	Glu	Glu	Glu	Glu	Leu 230	Leu	Pro	Lys	Val	Ile 235	Asn	Leu	Trp	Gln	Arg
Lys	Glu	Leu	Leu	Leu 245	Tyr	Tyr	Asp	Leu	Val 250	Glu	Ser	Thr	Phe	Glu 255	Lys
Leu	Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp 265	Ala	Leu	Asp	His	Phe	Gln	Lys
Cys	Phe	Leu	Ile	Leu	Lys	Leu	His 280	His	Gln	Arg	Val	Asp 285	Ser	Gly	Lys
Cys	Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu

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<210> SEQ ID NO 38
<211> LENGTH: 365
<212> TYPE: PRT
<213> ORGANISM: Bubalus bubalis
```

Leu	Ala	Val	Lys	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Lys	Thr	Thr
1			5						10					15	
Leu	Asn	Asn	Tyr	Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala
			20					25					30		
Glu	Asn	Ser	Glu	Phe	Lys	Glu	Asn	Glu	Val	Ser	Tyr	Val	Thr	Phe	Met
		35					40					45			
Arg	Ala	Ala	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met
	50					55					60				
Lys	Asp	Thr	Gln	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Cys	Val
65					70					75				80	
Ile	Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val
			85						90					95	
Leu	Asn	Asp	Glu	Arg	Tyr	Gln	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe
			100					105					110		
Gly	Val	Gly	Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg
		115					120					125			
Ser	Leu	Ser	Lys	Ile	Thr	Ser	Asp	Lys	Thr	Leu	Lys	Phe	Thr	Lys	Met
	130					135					140				
Gln	Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr
145					150					155					160
Arg	Ala	Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp
			165						170					175	
Ala	Phe	Leu	Pro	Asp	Ala	Phe	Ile	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg
			180					185					190		
Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly
		195					200					205			
Ser	Ala	Glu	Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp
	210					215					220				
Glu	Lys	Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe
225				230						235					240
Glu	Lys	Phe	Lys	Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe
			245						250					255	
Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Ser
		260						265					270		

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Gly	Arg	Ser	Asn	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val
		275					280					285			
Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly
	290					295					300				
Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Ile	Arg	Arg	Tyr	Ala	Thr
305					310					315					320
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
			325					330						335	
Lys	Arg	Val	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His
			340					345					350		
Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala			
	355						360					365			

<210> SEQ ID NO 39

<211> LENGTH: 379

<212> TYPE: PRT

<213> ORGANISM: Marmota flaviventris

<400> SEQUENCE: 39

Pro	Asp	His	Ser	Asn	Ser	Asp	Pro	Gln	Arg	Ile	Pro	Pro	Pro	Ala	Val
1				5					10					15	
Gln	Thr	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn
		20						25					30		
Cys	Asn	Arg	Val	Phe	Thr	Asp	Ala	Phe	Asp	Val	Leu	Ala	Glu	Asn	Tyr
	35						40					45			
Glu	Phe	Arg	Glu	Asn	Glu	Ser	Cys	Ser	Val	Val	Phe	Met	Arg	Ala	Ala
	50					55					60				
Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Arg	Asp	Leu
65					70					75				80	
Glu	Gly	Ile	Pro	Cys	Leu	Glu	Gly	Lys	Ala	Lys	Ser	Ile	Ile	Glu	Glu
			85						90					95	
Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp
		100						105					110		
Glu	Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
		115					120					125			
Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Ser
	130					135					140				
Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Lys	Phe	Thr	His	Met	Gln	Lys	Ala
145					150					155				160	
Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg	Ala	Glu
			165						170					175	
Ala	Glu	Ala	Val	Ser	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu
			180					185					190		
Pro	Asp	Ala	Phe	Ile	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Asn
	195						200					205			
Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Ala	Glu	Ala	Thr	Glu
	210					215					220				
Glu	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asn	Leu	Trp	Glu	Lys
225					230					235				240	
Lys	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys
			245						250					255	
Leu	Lys	Thr	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys
			260					265					270		

<400> SEQUENCE: 40

Lys 1	Ile	Ser	Gln	Tyr 5	Ala	Cys	Gln	Arg	10	Thr	Thr	Leu	Asn	Asn	Tyr
Asn	Gln	Leu	Phe 20	Thr	Asp	Ala	Leu	Asp 25	Ile	Leu	Ala	Glu	Asn 30	Asp	Glu
Leu	Arg	Glu 35	Asn	Glu	Gly	Ser 40	Cys	Leu	Ala	Phe	Met 45	Arg	Ala	Ser	Ser
Val 50	Leu	Lys	Ser	Leu	Pro	Phe 55	Pro	Ile	Thr	Ser	Met 60	Lys	Asp	Thr	Glu
Gly 65	Ile	Pro	Cys	Leu	Gly 70	Asp	Lys	Val	Lys	Ser 75	Ile	Ile	Glu	Gly	Ile 80
Ile	Glu	Asp	Gly 85	Glu	Ser	Ser	Glu	Ala	Lys 90	Ala	Val	Leu	Asn	Asp	Glu
Arg	Tyr	Lys 100	Ser	Phe	Lys	Leu	Phe	Thr 105	Ser	Val	Phe	Gly 110	Val	Gly	Leu
Lys	Thr 115	Ala	Glu	Lys	Trp	Phe 120	Arg	Met	Gly	Phe	Arg 125	Thr	Leu	Ser	Lys
Ile 130	Gln	Ser	Asp	Lys	Ser	Leu 135	Arg	Phe	Thr	Gln	Met 140	Gln	Lys	Ala	Gly
Phe 145	Leu	Tyr	Tyr	Glu 150	Asp	Leu	Val	Ser	Cys 155	Val	Asn	Arg	Pro	Glu	Ala 160
Glu	Ala	Val	Ser 165	Met	Leu	Val	Lys	Glu	Ala 170	Val	Val	Thr	Phe	Leu	Pro 175
Asp	Ala	Leu 180	Val	Thr	Met	Thr	Gly	Gly 185	Phe	Arg	Arg	Gly 190	Lys	Met	Thr
Gly 195	His	Asp	Val	Asp	Phe	Leu 200	Ile	Thr	Ser	Pro	Glu 205	Ala	Thr	Glu	Asp
Glu 210	Glu	Gln	Gln	Leu	Leu	His 215	Lys	Val	Thr	Asp	Phe 220	Trp	Lys	Gln	Gln
Gly 225	Leu	Leu	Leu	Tyr 230	Cys	Asp	Ile	Leu	Glu	Ser 235	Thr	Phe	Glu	Lys	Phe 240
Lys	Gln	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys

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245				250				255							
Phe	Leu	Ile	Leu	Lys	Leu	Asp	His	Gly	Arg	Val	His	Ser	Glu	Lys	Ser
			260						265				270		
Gly	Gln	Gln	Glu	Gly	Lys	Gly	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val
			275				280						285		
Met	Cys	Pro	Tyr	Asp	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
			290				295						300		
Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg
			305				310				315				320
Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Arg	Thr	Lys	Arg	Val
			325								330				335
Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu
			340												350
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
			355						360						

<210> SEQ ID NO 41

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Bovine

<400> SEQUENCE: 41

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Lys	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Ser	Glu
			20					25					30		
Phe	Lys	Glu	Asn	Glu	Val	Ser	Tyr	Val	Thr	Phe	Met	Arg	Ala	Ala	Ser
			35					40				45			
Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Lys	Asp	Thr	Glu
			50			55					60				
Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Cys	Ile	Ile	Glu	Glu	Ile
			65		70					75				80	
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	Glu
			85					90						95	
Arg	Tyr	Gln	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
			100					105					110		
Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Ser	Leu	Ser	Lys
			115				120					125			
Ile	Met	Ser	Asp	Lys	Thr	Leu	Lys	Phe	Thr	Lys	Met	Gln	Lys	Ala	Gly
			130			135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg	Ala	Glu	Ala
			145		150					155				160	
Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro
			165					170						175	
Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Ile
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Ala	Glu	Asp
			195				200					205			
Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys	Lys	Gly
			210			215					220				
Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Phe	Lys

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225	230	235	240
Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys Cys Phe	245	250	255
Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Ser Lys Ser Asn	260	265	270
Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val Met	275	280	285
Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser	290	295	300
Arg Gln Phe Glu Arg Asp Ile Arg Arg Tyr Ala Thr His Glu Arg Lys	305	310	315
Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val Phe	325	330	335
Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp	340	345	350
Tyr Ile Glu Pro Trp Glu Arg Asn Ala	355	360	

<210> SEQ ID NO 42

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Cys	1	5	10	15
Asn Gln Ile Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Cys Glu	20	25	30	
Phe Arg Glu Asn Glu Asp Ser Cys Val Thr Phe Met Arg Ala Ala Ser	35	40	45	
Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Glu	50	55	60	
Gly Ile Pro Cys Leu Gly Ser Lys Val Lys Gly Ile Ile Glu Glu Ile	65	70	75	80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu	85	90	95	
Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu	100	105	110	
Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser Lys	115	120	125	
Val Arg Ser Asp Lys Ser Leu Lys Phe Thr Arg Met Gln Lys Ala Gly	130	135	140	
Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu Ala	145	150	155	160
Glu Ala Val Ser Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro	165	170	175	
Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys Met	180	185	190	
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu Asp	195	200	205	
Glu Glu Gln Leu Leu Gln Lys Val Met Asn Leu Trp Glu Lys Lys Gly	210	215	220	

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Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Leu Arg
225                230                235                240

Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe
                245                250                255

Leu Ile Phe Lys Leu Pro Arg Gln Arg Val Asp Ser Asp Gln Ser Ser
                260                265                270

Trp Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val Leu
                275                280                285

Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser
290                295                300

Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg Lys
305                310                315                320

Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Ile Phe
                325                330                335

Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp
                340                345                350

Tyr Ile Glu Pro Trp Glu Arg Asn Ala
355                360

<210> SEQ ID NO 43
<211> LENGTH: 358
<212> TYPE: PRT
<213> ORGANISM: Gallus domesticus

<400> SEQUENCE: 43

Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr Leu Asn Asn Cys
1          5          10          15

Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Met Ala Glu Asn Tyr Glu
20          25          30

Phe Lys Glu Asn Glu Ile Phe Cys Leu Glu Phe Leu Arg Ala Ala Ser
35          40          45

Val Leu Lys Ser Leu Pro Phe Pro Val Thr Arg Met Lys Asp Ile Gln
50          55          60

Gly Leu Pro Cys Met Gly Asp Arg Val Arg Asp Val Ile Glu Glu Ile
65          70          75          80

Ile Glu Glu Gly Glu Ser Ser Arg Ala Lys Asp Val Leu Asn Asp Glu
85          90          95

Arg Tyr Lys Ser Phe Lys Glu Phe Thr Ser Val Phe Gly Val Gly Val
100         105         110

Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Leu Arg Thr Val Glu Glu
115         120         125

Val Lys Ala Asp Lys Thr Leu Lys Leu Ser Lys Met Gln Arg Ala Gly
130         135         140

Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Ser Lys Ala Glu Ala
145         150         155         160

Asp Ala Val Ser Ser Ile Val Lys Asn Thr Val Cys Thr Phe Leu Pro
165         170         175

Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Lys Ile
180         185         190

Gly His Asp Ile Asp Phe Leu Ile Thr Ser Pro Gly Gln Arg Glu Asp
195         200         205

Asp Glu Leu Leu His Lys Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu
210         215         220

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Ser Thr Phe Val Lys Glu Gln Ile Pro Ser Arg His Val Asp Ala Met
 225 230 235 240
 Asp His Phe Gln Lys Cys Phe Ala Ile Leu Lys Leu Tyr Gln Pro Arg
 245 250 255
 Val Asp Asn Ser Ser Tyr Asn Met Ser Lys Lys Cys Asp Met Ala Glu
 260 265 270
 Val Lys Asp Trp Lys Ala Ile Arg Val Asp Leu Val Ile Thr Pro Phe
 275 280 285
 Glu Gln Tyr Ala Tyr Ala Leu Leu Gly Trp Thr Gly Ser Arg Gln Phe
 290 295 300
 Gly Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg Lys Met Met Leu
 305 310 315 320
 Asp Asn His Ala Leu Tyr Asp Lys Arg Lys Arg Val Phe Leu Lys Ala
 325 330 335
 Gly Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp Tyr Val Glu
 340 345 350
 Pro Trp Glu Arg Asn Ala
 355

<210> SEQ ID NO 44

<211> LENGTH: 368

<212> TYPE: PRT

<213> ORGANISM: Didelphis marsupialis

<400> SEQUENCE: 44

Thr Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Ile Asn Asn His
 1 5 10 15
 Asn Gln Arg Phe Thr Asp Ala Phe Glu Ile Leu Ala Lys Asn Tyr Glu
 20 25 30
 Phe Lys Glu Asn Asp Asp Thr Cys Leu Thr Phe Met Arg Ala Ile Ser
 35 40 45
 Val Leu Lys Cys Leu Pro Phe Glu Val Val Ser Leu Lys Asp Thr Glu
 50 55 60
 Gly Leu Pro Trp Ile Gly Asp Glu Val Lys Gly Ile Met Glu Glu Ile
 65 70 75 80
 Ile Glu Asp Gly Glu Ser Leu Glu Val Gln Ala Val Leu Asn Asp Glu
 85 90 95
 Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110
 Lys Thr Ala Asp Lys Trp Tyr Arg Met Gly Phe Arg Thr Leu Asn Lys
 115 120 125
 Ile Arg Ser Asp Lys Thr Leu Lys Leu Thr Lys Met Gln Lys Ala Gly
 130 135 140
 Leu Cys Tyr Tyr Glu Asp Leu Ile Asp Cys Val Ser Lys Ala Glu Ala
 145 150 155 160
 Asp Ala Val Ser Leu Leu Val Gln Asp Ala Val Trp Thr Phe Leu Pro
 165 170 175
 Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Glu Phe
 180 185 190
 Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ala Glu Lys Glu
 195 200 205
 Gln Glu Asp Gln Leu Leu Gln Lys Val Thr Asn Leu Trp Lys Lys Gln

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210	215	220
Gly Leu Leu Leu Tyr Cys Asp Leu Ile Glu Ser Thr Phe Glu Asp Leu		
225	230	235 240
Lys Leu Pro Ser Arg Lys Ile Asp Ala Leu Asp His Phe Gln Lys Cys		
	245	250 255
Phe Leu Ile Leu Lys Leu Tyr His His Lys Glu Asp Lys Arg Lys Trp		
	260	265 270
Glu Met Pro Thr Gly Ser Asn Glu Ser Glu Ala Lys Ser Trp Lys Ala		
	275	280 285
Ile Arg Val Asp Leu Val Val Cys Pro Tyr Asp Arg Tyr Ala Phe Ala		
	290	295 300
Leu Leu Gly Trp Ser Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg		
305	310	315 320
Tyr Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr		
	325	330 335
Asp Lys Thr Lys Lys Ile Phe Leu Lys Ala Lys Ser Glu Glu Glu Ile		
	340	345 350
Phe Ala His Leu Gly Leu Glu Tyr Ile Gln Pro Ser Glu Arg Asn Ala		
	355	360 365

<210> SEQ ID NO 45
 <211> LENGTH: 362
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: shrew

<400> SEQUENCE: 45

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn His
1 5 10 15
Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Cys Glu
20 25 30
Phe Arg Glu Asn Glu Gly Ser Tyr Val Thr Tyr Met Arg Ala Ala Ser
35 40 45
Val Leu Lys Ser Leu Pro Phe Ser Ile Ile Ser Met Lys Asp Thr Glu
50 55 60
Gly Ile Pro Cys Leu Ala Asp Lys Val Lys Cys Val Ile Glu Glu Ile
65 70 75 80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
85 90 95
Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
100 105 110
Lys Thr Ala Glu Lys Trp Phe Arg Leu Gly Phe Arg Thr Leu Ser Gly
115 120 125
Ile Met Asn Asp Lys Thr Leu Lys Leu Thr His Met Gln Lys Ala Gly
130 135 140
Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu Ala
145 150 155 160
Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro
165 170 175
Asp Ala Ile Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys Val
180 185 190
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr Glu Glu

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195					200					205					
Gln	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Ile	Thr	Phe	Trp	Glu	Lys	Glu
210						215					220				
Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Leu	Tyr	Glu	Ser	Thr	Phe	Glu	Lys	Leu
225					230					235					240
Lys	Met	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys
				245					250					255	
Phe	Leu	Ile	Leu	Lys	Leu	His	Arg	Glu	Cys	Val	Asp	Asp	Gly	Thr	Ser
			260					265					270		
Ser	Gln	Leu	Gln	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val
			275					280					285		
Val	Cys	Pro	Tyr	Glu	Cys	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
	290					295					300				
Ser	Pro	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg
305						310					315				320
Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Lys
			325						330					335	
Phe	Leu	Ser	Ala	Asp	Ser	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu
			340					345					350		
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
	355					360									

<210> SEQ ID NO 46

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Canis familiaris

<400> SEQUENCE: 46

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	Asn	Val	Phe	Thr	Asp	Ala	Phe	Glu	Val	Leu	Ala	Glu	Asn	Tyr	Glu
			20					25					30		
Phe	Arg	Glu	Asn	Glu	Val	Phe	Ser	Leu	Thr	Phe	Met	Arg	Ala	Ala	Ser
		35					40					45			
Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Lys	Asp	Thr	Glu
	50					55					60				
Gly	Ile	Pro	Cys	Leu	Gly	Asp	Gln	Val	Lys	Cys	Ile	Ile	Glu	Glu	Ile
65				70						75				80	
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	Glu
			85					90					95		
Arg	Tyr	Gln	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
		100						105					110		
Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Ser	Lys
		115					120					125			
Ile	Lys	Ser	Asp	Lys	Ser	Leu	Lys	Phe	Thr	Pro	Met	Gln	Lys	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg	Ala	Glu	Ala
145					150					155				160	
Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Gly	Ala	Phe	Leu	Pro
			165					170						175	
Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Met
	180						185						190		

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Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Thr	Asp	Glu
	195						200					205			
Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Arg	Lys
	210					215					220				
Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Leu
225					230					235					240
Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys
			245					250						255	
Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Gly	Gly	Lys	Cys
		260						265					270		
Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val
	275						280					285			
Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
	290					295					300				
Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Ser	His	Glu	Arg
305					310					315					320
Lys	Met	Ile	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Lys	Ile
			325						330					335	
Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu
			340					345					350		
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
	355						360								

<210> SEQ ID NO 47

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: mole

<400> SEQUENCE: 47

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	His
1				5					10					15	
Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Cys	Glu
			20					25					30		
Phe	Arg	Glu	Asn	Glu	Gly	Ser	Tyr	Val	Thr	Tyr	Met	Arg	Ala	Ala	Ser
		35				40						45			
Val	Leu	Lys	Ser	Leu	Pro	Phe	Ser	Ile	Ile	Ser	Met	Lys	Asp	Thr	Glu
	50					55					60				
Gly	Ile	Pro	Cys	Leu	Ala	Asp	Lys	Val	Lys	Cys	Val	Ile	Glu	Glu	Ile
65				70						75					80
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	Glu
			85					90						95	
Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
			100					105					110		
Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Leu	Gly	Phe	Arg	Thr	Leu	Ser	Gly
		115					120					125			
Ile	Met	Asn	Asp	Lys	Thr	Leu	Lys	Leu	Thr	His	Met	Gln	Lys	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Thr	Arg	Ala	Glu	Ala
145					150					155					160
Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro
			165					170							175

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Asp	Ala	Ile	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Val
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Glu	Ala	Thr	Glu	Glu
		195					200					205			
Gln	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Ile	Thr	Phe	Trp	Glu	Lys	Glu
	210					215					220				
Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Leu	Tyr	Glu	Ser	Thr	Phe	Glu	Lys	Leu
225					230					235					240
Lys	Met	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys
				245					250					255	
Phe	Leu	Ile	Leu	Lys	Leu	His	Arg	Glu	Cys	Val	Asp	Asp	Gly	Thr	Ser
			260					265					270		
Ser	Gln	Leu	Gln	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val
		275					280					285			
Val	Cys	Pro	Tyr	Glu	Cys	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
	290					295						300			
Ser	Pro	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg
305					310					315					320
Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Lys
				325					330					335	
Phe	Leu	Ser	Ala	Asp	Ser	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu
			340					345					350		
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
		355					360								

<210> SEQ ID NO 48

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Ochotona princeps

<400> SEQUENCE: 48

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	His
1			5					10					15		
Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Tyr	Glu
		20						25				30			
Phe	Lys	Glu	Asn	Glu	Gly	Cys	Tyr	Val	Thr	Tyr	Met	Arg	Ala	Ala	Ser
		35				40					45				
Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Val	Ser	Met	Lys	Asp	Thr	Glu
	50				55					60					
Gly	Ile	Pro	Cys	Leu	Glu	Asp	Lys	Val	Lys	Ser	Ile	Met	Glu	Glu	Ile
65				70					75					80	
Ile	Glu	Glu	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Ser	Asp	Glu
			85					90					95		
Arg	Tyr	Gln	Cys	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
		100					105					110			
Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Ser	Leu	Ser	Asn
		115					120					125			
Ile	Arg	Leu	Asp	Lys	Ser	Leu	Lys	Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly
	130					135					140				
Phe	Arg	Tyr	Tyr	Glu	Asp	Ile	Val	Ser	Cys	Val	Thr	Arg	Ala	Glu	Ala
145				150						155				160	
Glu	Ala	Val	Asp	Val	Leu	Val	Asn	Glu	Ala	Val	Arg	Ala	Phe	Leu	Pro
			165					170						175	

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Asp Ala Phe Ile Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys Ile
 180 185 190
 Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Leu Thr Glu Glu
 195 200 205
 Asp Glu Gln Gln Leu Leu His Lys Val Met Asn Leu Trp Glu Lys Lys
 210 215 220
 Gly Leu Leu Leu Tyr His Asp Leu Val Glu Ser Thr Phe Glu Lys Leu
 225 230 235 240
 Lys Gln Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys Cys
 245 250 255
 Phe Leu Ile Phe Lys Leu Tyr His Glu Arg Val Gly Gly Asp Arg Cys
 260 265 270
 Arg Gln Pro Glu Gly Lys Asp Trp Lys Ala Ile Arg Val Asp Leu Val
 275 280 285
 Met Cys Pro Tyr Glu Cys His Ala Phe Ala Leu Leu Gly Trp Thr Gly
 290 295 300
 Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Ser His Glu Arg
 305 310 315 320
 Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val
 325 330 335
 Phe Leu Gln Ala Glu Asn Glu Glu Glu Ile Phe Ala His Leu Gly Leu
 340 345 350
 Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 49

<211> LENGTH: 365

<212> TYPE: PRT

<213> ORGANISM: Erinaceus europaeus

<400> SEQUENCE: 49

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Ser Leu Asn Asn Cys
 1 5 10 15
 Asn His Ile Phe Thr Asp Ala Leu Asp Ile Leu Ala Glu Asn His Glu
 20 25 30
 Phe Arg Glu Asn Glu Val Ser Cys Val Ala Phe Met Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Lys
 50 55 60
 Gly Ile Pro Cys Leu Gly Asp Lys Ala Lys Cys Val Ile Glu Glu Ile
 65 70 75 80
 Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Ile Leu Asn Asp Glu
 85 90 95
 Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110
 Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Asn Lys
 115 120 125
 Ile Met Ser Asp Lys Thr Leu Lys Leu Thr Arg Met Gln Lys Ala Gly
 130 135 140
 Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Ala Lys Ala Glu Ala
 145 150 155 160
 Asp Ala Val Ser Val Leu Val Gln Glu Ala Val Trp Ala Phe Leu Pro

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165					170					175					
Asp	Ala	Met	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Leu
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ala	Thr	Glu	Glu
		195					200					205			
Glu	Glu	Gln	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Phe	Trp	Glu	Arg	Lys
		210				215					220				
Gly	Leu	Leu	Leu	Tyr	His	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Leu
225					230					235					240
Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys
				245					250					255	
Phe	Leu	Ile	Leu	Lys	Leu	His	Leu	Gln	His	Val	Asn	Gly	Val	Gly	Asn
		260						265					270		
Ser	Lys	Thr	Gly	Gln	Gln	Glu	Gly	Lys	Asn	Trp	Lys	Ala	Ile	Arg	Val
		275					280					285			
Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly
	290					295					300				
Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Thr
305					310					315					320
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
				325					330					335	
Lys	Arg	Ile	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His
			340					345					350		
Leu	Gly	Leu	Asp	Tyr	Ile	Asp	Pro	Trp	Glu	Arg	Asn	Ala			
		355					360					365			

<210> SEQ ID NO 50

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: tree shrew

<400> SEQUENCE: 50

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Cys
1				5					10					15	
Asn	Arg	Val	Phe	Thr	Asp	Ala	Phe	Glu	Thr	Leu	Ala	Glu	Asn	Tyr	Glu
		20						25					30		
Phe	Arg	Glu	Asn	Glu	Asp	Ser	Ser	Val	Ile	Phe	Leu	Arg	Ala	Ala	Ser
		35				40						45			
Val	Leu	Arg	Ser	Leu	Pro	Phe	Thr	Ile	Thr	Ser	Met	Arg	Asp	Thr	Glu
	50					55					60				
Gly	Leu	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Cys	Val	Ile	Glu	Glu	Ile
65				70						75					80
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Asn	Ala	Val	Leu	Asn	Asp	Glu
			85					90					95		
Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
		100						105					110		
Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Ser	Arg
		115				120						125			
Val	Arg	Ser	Asp	Lys	Ser	Leu	His	Leu	Thr	Arg	Met	Gln	Gln	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Ala	Ser	Cys	Val	Thr	Arg	Ala	Glu	Ala

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145	150	155	160
Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Gly Ala Phe Leu Pro	165	170	175
Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Lys Thr	180	185	190
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu Glu	195	200	205
Lys Glu Glu Glu Leu Leu Gln Lys Val Leu Asn Leu Trp Glu Lys Lys	210	215	220
Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Leu	225	230	235
Lys Thr Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Pro Lys Cys	245	250	255
Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Gly Asp Lys Pro	260	265	270
Ser Gln Gln Glu Gly Lys Ser Trp Lys Ala Ile Arg Val Asp Leu Val	275	280	285
Met Cys Pro Tyr Glu Arg His Ala Phe Ala Leu Leu Gly Trp Thr Gly	290	295	300
Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg	305	310	315
Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val	325	330	335
Phe Leu Lys Ala Glu Ser Glu Glu Asp Ile Phe Ala His Leu Gly Leu	340	345	350
Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala	355	360	

<210> SEQ ID NO 51

<211> LENGTH: 375

<212> TYPE: PRT

<213> ORGANISM: Ornithorhynchus anatinus

<400> SEQUENCE: 51

Gln Val Ser Gln Tyr Ala Cys Glu Arg Arg Thr Thr Leu Asn Asn Cys	1	5	10	15
Asn Gln Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Lys Asp Phe Glu	20	25	30	
Phe Arg Glu Asn Glu Gly Ile Cys Leu Ala Phe Met Arg Ala Ile Ser	35	40	45	
Val Leu Lys Cys Leu Pro Phe Thr Ile Val Arg Met Lys Asp Ile Glu	50	55	60	
Gly Val Pro Trp Leu Gly Asp Gln Val Lys Ser Ile Ile Glu Glu Ile	65	70	75	80
Ile Glu Asp Gly Glu Ser Ser Ser Val Lys Ala Val Leu Asn Asp Glu	85	90	95	
Arg Tyr Arg Ser Phe Gln Leu Phe Asn Ser Val Phe Glu Val Gly Leu	100	105	110	
Thr Asp Asn Gly Glu Asn Gly Ile Ala Arg Gly Phe Gln Thr Leu Asn	115	120	125	
Glu Val Ile Thr Asp Glu Asn Ile Ser Leu Thr Lys Thr Thr Leu Ser	130	135	140	

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Thr	Ser	Leu	Trp	Asn	Tyr	Leu	Pro	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu
145				150					155						160
Val	Ser	Cys	Val	Ala	Lys	Glu	Glu	Ala	Asp	Ala	Val	Tyr	Leu	Ile	Val
			165					170						175	
Lys	Glu	Ala	Val	Arg	Ala	Phe	Leu	Pro	Glu	Ala	Leu	Val	Thr	Leu	Thr
			180					185					190		
Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu
		195					200					205			
Ile	Ser	Asp	Pro	Glu	Ser	Gly	Gln	Asp	Glu	Gln	Leu	Leu	Pro	Asn	Ile
	210					215					220				
Ile	Lys	Leu	Trp	Glu	Lys	Gln	Glu	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val
225					230					235					240
Glu	Ser	Thr	Phe	Glu	Lys	Thr	Lys	Ile	Pro	Ser	Arg	Lys	Val	Asp	Ala
			245						250					255	
Met	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln
			260					265					270		
Lys	Val	Asp	Ser	Gly	Arg	Tyr	Lys	Pro	Pro	Pro	Glu	Ser	Lys	Asn	His
		275					280					285			
Glu	Ala	Lys	Asn	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Met	Cys	Pro
	290					295					300				
Phe	Glu	Gln	Tyr	Ala	Tyr	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser	Arg	Gln
305					310					315					320
Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Lys	Lys	Met	Met
			325						330					335	
Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Lys	Ile	Phe	Leu	Lys
			340					345					350		
Ala	Glu	Ser	Glu	Glu	Asp	Ile	Phe	Thr	His	Leu	Gly	Leu	Asp	Tyr	Ile
		355					360					365			
Glu	Pro	Trp	Glu	Arg	Asn	Ala									
	370					375									

<210> SEQ ID NO 52

<211> LENGTH: 365

<212> TYPE: PRT

<213> ORGANISM: Serinus canaria

<400> SEQUENCE: 52

Lys	Val	Ser	Gln	Tyr	Ser	Cys	Gln	Arg	Lys	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	Lys	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Val	Met	Ala	Glu	Asn	Tyr	Glu
			20					25					30		
Phe	Lys	Glu	Asn	Glu	Ile	Phe	Cys	Leu	Glu	Phe	Leu	Arg	Ala	Ala	Ser
		35					40					45			
Leu	Leu	Lys	Ser	Leu	Pro	Phe	Ser	Val	Thr	Arg	Met	Lys	Asp	Ile	Gln
	50					55					60				
Gly	Leu	Pro	Cys	Met	Gly	Asp	Gln	Val	Arg	Asp	Val	Ile	Glu	Ile	Ile
65					70					75				80	
Glu	Glu	Gly	Glu	Ser	Ser	Arg	Val	Lys	Glu	Val	Leu	Asn	Asp	Glu	Arg
			85						90					95	
Tyr	Lys	Ala	Phe	Lys	Gln	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Val	Lys
		100						105					110		
Thr	Ser	Glu	Lys	Trp	Tyr	Arg	Met	Gly	Leu	Arg	Thr	Val	Gly	Glu	Val
		115					120					125			

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Lys Ala Asp Lys Thr Leu Lys Leu Ser Lys Met Gln Lys Ala Gly Phe
 130 135 140
 Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Ser Lys Ala Glu Ala Asp
 145 150 155 160
 Ala Val Ser Leu Ile Val Lys Asn Thr Val Cys Thr Phe Leu Pro Asp
 165 170 175
 Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Arg Gly Lys Asn Ile Gly
 180 185 190
 His Asp Ile Asp Phe Leu Ile Thr Asn Pro Gly Pro Arg Glu Asp Asp
 195 200 205
 Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly Leu Leu
 210 215 220
 Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Val Lys Glu Gln Leu Pro
 225 230 235 240
 Ser Arg Lys Ile Asp Ala Met Asp His Phe Gln Lys Cys Phe Ala Ile
 245 250 255
 Leu Lys Leu Tyr Gln Pro Arg Val Asp Asn Ser Thr Cys Asn Thr Ser
 260 265 270
 Lys Lys Leu Glu Met Ala Glu Val Lys Asp Trp Lys Ala Ile Arg Val
 275 280 285
 Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ser Tyr Ala Leu Leu Gly
 290 295 300
 Trp Thr Gly Ser Arg Gln Phe Gly Arg Asp Leu Arg Arg Tyr Ala Ala
 305 310 315 320
 His Glu Arg Arg Met Ile Leu Asp Asn His Gly Leu Tyr Asp Arg Thr
 325 330 335
 Lys Arg Ile Phe Leu Lys Ala Gly Ser Glu Glu Glu Ile Phe Ala His
 340 345 350
 Leu Gly Leu Asp Tyr Val Glu Pro Trp Glu Arg Asn Ala
 355 360 365

<210> SEQ ID NO 53

<211> LENGTH: 366

<212> TYPE: PRT

<213> ORGANISM: Neopelma chrysocephalum

<400> SEQUENCE: 53

Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr Leu Asn Asn Tyr
 1 5 10 15
 Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Met Ala Glu Asn Tyr Glu
 20 25 30
 Phe Lys Glu Asn Glu Ile Phe Cys Leu Glu Phe Leu Arg Ala Ala Ser
 35 40 45
 Leu Leu Lys Tyr Leu Pro Phe Pro Val Thr Arg Met Lys Asp Ile Gln
 50 55 60
 Gly Leu Pro Cys Ile Gly Asp Gln Val Arg Asp Val Ile Glu Gly Ile
 65 70 75 80
 Ile Glu Glu Gly Glu Ser Ser Arg Val Lys Glu Val Leu Asn Asp Glu
 85 90 95
 Arg Tyr Lys Ala Phe Lys Gln Phe Thr Ser Val Phe Gly Val Gly Val
 100 105 110
 Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Val Glu Glu

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115					120					125					
Leu	Lys	Ala	Asp	Lys	Thr	Leu	Lys	Leu	Ser	Lys	Met	Gln	Lys	Ala	Gly
130						135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Cys	Val	Ser	Lys	Ala	Glu	Ala
145					150					155					160
Asp	Ala	Val	Thr	Leu	Ile	Val	Lys	Asn	Thr	Val	Ser	Thr	Phe	Leu	Pro
				165					170						175
Asp	Ala	Leu	Val	Thr	Ile	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Met
			180					185					190		
Gly	His	Asp	Ile	Asp	Phe	Leu	Ile	Thr	Asn	Pro	Gly	Pro	Arg	Glu	Asp
		195					200					205			
Asp	Glu	Leu	Leu	His	Lys	Val	Val	Asp	Leu	Trp	Lys	Lys	Gln	Gly	Leu
	210						215				220				
Leu	Leu	Tyr	Cys	Asp	Ile	Ile	Glu	Ser	Thr	Phe	Val	Glu	Glu	Gln	Leu
225					230					235					240
Pro	Ser	Arg	Lys	Val	Asp	Ala	Met	Asp	Asn	Phe	Gln	Lys	Cys	Phe	Thr
				245					250						255
Ile	Leu	Lys	Leu	Tyr	Gln	Pro	Gly	Val	Asp	Asn	Ser	Ser	Tyr	Asn	Met
			260					265							270
Ser	Lys	Lys	Ser	Asp	Met	Ala	Glu	Val	Lys	Asp	Trp	Lys	Ala	Ile	Arg
		275					280					285			
Val	Asp	Leu	Val	Ile	Thr	Pro	Phe	Glu	Gln	Tyr	Ala	Tyr	Ala	Leu	Leu
	290						295				300				
Gly	Trp	Thr	Gly	Ser	Arg	Glu	Phe	Gly	Arg	Asp	Leu	Arg	Arg	Tyr	Ala
305					310					315					320
Ser	His	Glu	Arg	Lys	Met	Ile	Leu	Asp	Asn	His	Gly	Leu	Tyr	Asp	Arg
				325					330						335
Arg	Lys	Arg	Ile	Phe	Leu	Lys	Ala	Gly	Ser	Glu	Glu	Glu	Ile	Phe	Ala
			340					345							350
His	Leu	Gly	Leu	Asp	Tyr	Val	Glu	Pro	Trp	Glu	Arg	Asn	Ala		
	355						360					365			

<210> SEQ ID NO 54

<211> LENGTH: 366

<212> TYPE: PRT

<213> ORGANISM: Alligator sinensis

<400> SEQUENCE: 54

Lys	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	Lys	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Cys	Glu
			20					25					30		
Phe	Arg	Glu	Asn	Arg	Leu	Gly	Cys	Leu	Glu	Phe	Leu	Arg	Ala	Ala	Ser
		35					40					45			
Val	Leu	Lys	Phe	Leu	Pro	Phe	Pro	Ile	Val	Lys	Met	Lys	Asn	Ile	Glu
	50					55					60				
Gly	Leu	Pro	Cys	Met	Gly	Asp	Lys	Val	Lys	Cys	Val	Ile	Glu	Glu	Ile
65					70					75					80
Leu	Glu	Glu	Gly	Glu	Ser	Cys	Gln	Ala	Lys	Glu	Ile	Leu	Asn	Asp	Glu
			85						90					95	
Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Leu
			100					105						110	

-continued

Lys	Thr	Thr	Glu	Lys	Trp	Tyr	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu	Glu
		115					120					125			
Val	Lys	Ala	Glu	Lys	Thr	Leu	Lys	Leu	Ser	Arg	Met	Gln	Ile	Ala	Gly
	130					135					140				
Phe	Leu	His	Tyr	Glu	Asp	Ile	Ile	Ser	Tyr	Val	Ser	Lys	Ala	Glu	Ala
145					150					155					160
Asp	Ala	Val	Ser	Leu	Leu	Ile	Lys	Asp	Thr	Val	Cys	Met	Phe	Leu	Pro
				165					170					175	
Asp	Ala	Leu	Val	Thr	Ile	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Lys	Thr
		180						185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Asn	Pro	Gly	Pro	Glu	Glu	Glu
		195					200					205			
Lys	Glu	Leu	Leu	His	Lys	Val	Val	Asp	Leu	Trp	Glu	Lys	Gln	Gly	Leu
	210					215					220				
Leu	Leu	Tyr	Tyr	Asp	Val	Ile	Glu	Ser	Thr	Phe	Glu	Lys	Glu	Lys	Arg
225					230					235					240
Pro	Ser	Arg	Lys	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe	Ala
				245					250					255	
Ile	Leu	Lys	Leu	His	Gln	Gln	Arg	Arg	Gly	Asn	Ser	Asn	Ser	Asn	Ile
			260					265					270		
Ser	Lys	Glu	Ser	Asp	Lys	Ala	Glu	Val	Lys	Asp	Trp	Lys	Ala	Ile	Arg
		275					280					285			
Val	Asp	Leu	Val	Ile	Ser	Pro	Phe	Glu	Gln	Tyr	Ala	Tyr	Ala	Leu	Leu
	290					295					300				
Gly	Trp	Thr	Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Tyr	Ala
305					310					315					320
Ser	Arg	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys
				325					330					335	
Thr	Lys	Arg	Thr	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala
			340					345					350		
His	Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala		
		355					360					365			

<210> SEQ ID NO 55

<211> LENGTH: 366

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: trunc new

<400> SEQUENCE: 55

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	Lys	Lys	Phe	Thr	Asp	Ala	Phe	Glu	Ile	Leu	Ala	Glu	Asn	Tyr	Glu
			20					25					30		
Phe	Asn	Glu	Asn	Lys	Gly	Phe	Cys	Leu	Ala	Phe	Arg	Arg	Ala	Ala	Ser
		35					40					45			
Val	Leu	Lys	Phe	Leu	Pro	Phe	Thr	Ile	Val	Arg	Met	Asn	Asp	Ile	Gln
		50				55					60				
Gly	Leu	Pro	Trp	Met	Gly	Asp	Gln	Val	Lys	Arg	Val	Ile	Glu	Glu	Ile
65					70					75					80
Leu	Glu	Glu	Gly	Glu	Ser	Ser	Lys	Val	Lys	Glu	Val	Leu	Asn	Asp	Glu
				85					90					95	

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Lys	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Arg	100	105	110
Lys	Thr	Ser	Glu	Lys	Trp	Tyr	Arg	Met	Gly	Leu	Arg	Thr	Leu	Glu	Glu	115	120	125
Val	Lys	Ala	Asp	Lys	Thr	Leu	Lys	Leu	Thr	Lys	Met	Gln	Lys	Ala	Gly	130	135	140
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Ile	Ser	Arg	Val	Ser	Lys	Ala	Glu	Ala	145	150	155
Asp	Ala	Val	Ser	Leu	Ile	Val	Lys	Asp	Thr	Val	Trp	Lys	Phe	Leu	Pro	165	170	175
Asp	Ala	Leu	Val	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Lys	Met	180	185	190
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Thr	Pro	Gly	Pro	Glu	Glu	Glu	195	200	205
Glu	Glu	Leu	Leu	His	Lys	Val	Ile	Asp	Leu	Trp	Lys	Lys	Gln	Gly	Leu	210	215	220
Leu	Leu	Tyr	Tyr	Asp	Ile	Ile	Glu	Ser	Thr	Phe	Glu	Lys	Thr	Lys	Leu	225	230	235
Pro	Ser	Arg	Thr	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe	Ala	245	250	255
Ile	Leu	Lys	Leu	His	Lys	Gln	Arg	Val	Asp	Asn	Gly	Asn	Ser	Asn	Gln	260	265	270
Ser	Lys	Glu	Phe	Asp	Lys	Glu	Glu	Thr	Lys	Asp	Trp	Lys	Ala	Ile	Arg	275	280	285
Val	Asp	Leu	Val	Ile	Thr	Pro	Phe	Glu	Gln	Tyr	Ala	Tyr	Ala	Leu	Leu	290	295	300
Gly	Trp	Thr	Gly	Ser	Ala	Phe	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	305	310	315
Thr	His	Glu	Lys	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	325	330	335
Thr	Lys	Arg	Ile	Phe	Leu	Lys	Ala	Ala	Ser	Glu	Glu	Glu	Ile	Phe	Ala	340	345	350
His	Leu	Gly	Leu	Asp	Tyr	Leu	Glu	Pro	Trp	Glu	Arg	Asn	Ala			355	360	365

<210> SEQ ID NO 56

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: *Xenopus laevis*

<400> SEQUENCE: 56

Lys	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Cys	Thr	Thr	Leu	Gln	Asp	Thr	1	5	10	15
Asn	Arg	Ile	Phe	Thr	Asp	Ala	Phe	Asp	Ile	Leu	Ala	Glu	His	Phe	Glu	20	25	30	
Phe	Cys	Glu	Asn	Lys	Gly	Arg	Thr	Val	Ala	Phe	Leu	Arg	Ala	Ser	Ser	35	40	45	
Leu	Ile	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ala	Met	Lys	Glu	Leu	Glu	50	55	60	
Gly	Leu	Pro	Trp	Leu	Gly	Asp	Gln	Met	Lys	Gly	Ile	Ile	Glu	Glu	Ile	65	70	75	80
Leu	Glu	Glu	Gly	Lys	Ser	Tyr	Lys	Val	Leu	Glu	Val	Met	Asn	Glu	Glu	85	90	95	

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Arg Tyr Lys Ser Phe Lys Gln Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110
 Lys Thr Ser Asp Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu Glu
 115 120 125
 Ile Lys Asn Glu Lys Glu Leu Lys Leu Thr Lys Met Gln Lys Cys Gly
 130 135 140
 Leu Leu Tyr Tyr Glu Asp Ile Thr Ser Tyr Val Ser Arg Ala Glu Ala
 145 150 155 160
 Glu Thr Thr Glu Gln Leu Ile Lys Ser Ile Val Trp Lys Phe Val Pro
 165 170 175
 Asp Ala Ile Val Thr Leu Thr Gly Gly Phe Arg Arg Gly Lys Lys Lys
 180 185 190
 Gly His Asp Val Asp Ile Leu Ile Thr Cys Ala Arg Lys Gly Lys Glu
 195 200 205
 Lys Asn Ile Leu His Asn Thr Met Ser Val Leu Lys Asn Arg Gly Leu
 210 215 220
 Leu Leu Phe Tyr Asn Ile Ile Glu Ser Thr Phe Asp Glu Thr Lys Leu
 225 230 235 240
 Pro Ser Arg His Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe Thr
 245 250 255
 Ile Leu Lys Leu Pro Lys Arg Gln Met Asp Ile Gly Asn Ile Ile Asp
 260 265 270
 Pro His Glu Cys Glu Arg Lys Asn Trp Lys Ala Val Arg Leu Asp Leu
 275 280 285
 Val Ile Thr Pro Tyr Glu Gln Tyr Pro Tyr Ala Leu Leu Gly Trp Thr
 290 295 300
 Gly Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 305 310 315 320
 Lys Arg Met Met Leu Asp Asn His Gly Leu Tyr Asp Lys Thr Lys Asn
 325 330 335
 Asn Phe Leu Lys Ala Asn Asn Glu Glu Asp Ile Phe Lys Gln Leu Gly
 340 345 350
 Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 57

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Xenopus sp.

<400> SEQUENCE: 57

Lys Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Gln Asp Thr
 1 5 10 15
 Asn Arg Ile Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Tyr Glu
 20 25 30
 Phe Asn Glu Asn Lys Gly Pro Cys Val Ala Phe Arg Arg Ala Ser Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Phe Pro Ile Val Ala Met Lys Asp Leu Glu
 50 55 60
 Gly Leu Pro Trp Leu Gly Asp Gln Met Lys Arg Ile Ile Glu Glu Ile
 65 70 75 80

-continued

Leu Glu Glu Gly Lys Ser Ser Lys Val Val Glu Val Met Asn Asp Glu
 85 90 95
 Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg
 100 105 110
 Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu Glu
 115 120 125
 Ile Lys Ser Asp Lys Glu Leu Lys Leu Thr Lys Met Gln Lys Ala Gly
 130 135 140
 Leu Leu Tyr Tyr Glu Asp Ile Thr Ser Ala Val Ser Lys Ala Glu Ala
 145 150 155 160
 Glu Ala Val Glu Gln Leu Ile Lys Ser Ile Val Trp Lys Phe Val Pro
 165 170 175
 Asp Ala Ile Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys Met
 180 185 190
 Gly His Asp Val Asp Ile Leu Ile Thr Thr Pro Arg Lys Gly Glu Lys
 195 200 205
 Glu Glu Ile Leu His Lys Thr Ile Asn Val Leu Lys Lys Arg Gly Leu
 210 215 220
 Leu Leu Phe Tyr Asn Ile Ile Glu Ser Thr Phe Asp Glu Thr Lys Leu
 225 230 235 240
 Pro Ser Arg His Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe Thr
 245 250 255
 Ile Leu Lys Leu Gln Lys Gln Gln Leu Asp Asn Gly Asn Ser Asn Glu
 260 265 270
 Pro Phe Glu Thr Glu Lys Lys Asn Trp Lys Ala Ile Arg Val Asp Leu
 275 280 285
 Val Ile Thr Pro Tyr Glu Gln Tyr Ala Tyr Ala Leu Leu Gly Trp Thr
 290 295 300
 Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 305 310 315 320
 Lys Arg Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Asn
 325 330 335
 Ile Phe Leu Lys Ala Asn Ser Glu Glu Asp Ile Phe Thr His Leu Gly
 340 345 350
 Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 58

<211> LENGTH: 367

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Notechis sp.

<400> SEQUENCE: 58

Ser Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr
 1 5 10 15
 Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr Glu
 20 25 30
 Phe Asn Glu Asn Lys Gly Phe Cys Leu Ala Phe Arg Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Phe Leu Pro Phe Thr Ile Val Arg Val Asn Asp Ile Glu
 50 55 60

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Gly Leu Pro Trp Met Gly Glu Gln Val Lys Arg Ile Ile Glu Asp Ile
 65 70 75 80
 Leu Glu Glu Gly Glu Ser Ser Lys Val Lys Ala Val Leu Asn Asn Glu
 85 90 95
 Asn Tyr Arg Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg
 100 105 110
 Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Leu Glu Glu
 115 120 125
 Val Lys Ala Asp Lys Asn Leu Lys Leu Thr Arg Met Gln Lys Ala Gly
 130 135 140
 Phe Leu His Tyr Glu Asp Leu Ile Ser Arg Val Ser Lys Ala Glu Ala
 145 150 155 160
 Asp Ala Ala Ser Leu Ile Val Lys Asp Thr Val Trp Lys Phe Leu Pro
 165 170 175
 Asn Ala Leu Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys Met
 180 185 190
 Gly His Asp Val Asp Phe Leu Ile Thr Val Pro Gly Ser Arg Gln Glu
 195 200 205
 Glu Glu Glu Leu Leu His Thr Val Ile Asp Leu Trp Lys Lys Gln Gly
 210 215 220
 Leu Leu Leu Tyr Tyr Asp Leu Ile Glu Ser Thr Phe Glu Lys Thr Lys
 225 230 235 240
 Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe
 245 250 255
 Ala Ile Leu Lys Leu His Lys Glu Arg Val Asp Lys Gly Asn Ser Ile
 260 265 270
 Gln Ser Lys Ala Phe Ala Glu Glu Glu Thr Lys Asp Trp Lys Ala Ile
 275 280 285
 Arg Val Asp Leu Val Val Thr Pro Phe Glu Gln Tyr Ala Phe Ala Leu
 290 295 300
 Leu Gly Trp Thr Gly Ser Pro Phe Phe Asn Arg Asp Leu Arg Arg Tyr
 305 310 315 320
 Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp
 325 330 335
 Lys Thr Lys Lys Ile Phe Leu Lys Ala Ala Ser Glu Glu Glu Ile Phe
 340 345 350
 Ala His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
 355 360 365

<210> SEQ ID NO 59

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: Salmo trutta

<400> SEQUENCE: 59

Asn Val Ser Gln Tyr Ala Cys Leu Arg Arg Thr Thr Thr Glu Asn His
 1 5 10 15
 Asn Lys Ile Phe Thr Asp Val Leu Glu Leu Ala Glu Asn Ser Glu
 20 25 30
 Phe Asn Glu Ser Lys Gly Pro Cys Leu Ala Phe Arg Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Ser Ala Val His Cys Leu Gly Ala Ile Gln

-continued

50					55					60					
Gly	Leu	Pro	Cys	Leu	Gly	Glu	His	Thr	Lys	Ala	Val	Met	Glu	Glu	Ile
65				70					75					80	
Leu	Ile	Phe	Gly	Arg	Ser	Phe	Lys	Val	Glu	Glu	Val	Gln	Ser	Asp	Glu
			85					90					95		
Arg	Tyr	Gln	Ala	Leu	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Pro
			100					105					110		
Lys	Thr	Ala	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ser	Leu	Lys	Glu
		115					120					125			
Ile	Leu	Ala	Glu	Pro	Asn	Ile	Gln	Leu	Asn	Arg	Met	Gln	Arg	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Arg	Asp	Ile	Ser	Lys	Ala	Val	Ser	Lys	Ala	Glu	Ala
145				150					155					160	
Lys	Ala	Leu	Ser	Ser	Ile	Ile	Glu	Glu	Thr	Ala	His	Trp	Ile	Ala	Pro
			165						170				175		
Asp	Ser	Ile	Leu	Ala	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Glu	Tyr
		180					185						190		
Gly	His	Asp	Val	Asp	Phe	Leu	Leu	Thr	Met	Pro	Glu	Met	Gly	Lys	Glu
		195					200					205			
Glu	Gly	Leu	Leu	Leu	Arg	Val	Ile	Asp	Arg	Leu	Arg	Asp	Gln	Gly	Ile
	210					215					220				
Leu	Leu	Tyr	Cys	Glu	His	Gln	Asp	Ser	Thr	Phe	Asp	Met	Ser	Lys	Leu
225				230					235					240	
Pro	Ser	His	Arg	Phe	Glu	Ala	Met	Asp	His	Phe	Glu	Lys	Cys	Phe	Leu
			245					250					255		
Ile	Leu	Arg	Leu	Glu	Glu	Gly	Gln	Val	Glu	Gly	Asp	Gly	Gly	Leu	Gln
		260					265						270		
Lys	Asp	Pro	Gly	Glu	Ser	Arg	Gly	Trp	Arg	Ala	Val	Arg	Val	Asp	Leu
	275						280					285			
Val	Ala	Pro	Pro	Val	Asp	Arg	Tyr	Ala	Phe	Val	Leu	Leu	Gly	Trp	Thr
	290					295					300				
Gly	Ser	Arg	Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Ser	Lys	Glu
305				310					315					320	
Arg	Gly	Met	Cys	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Lys
			325					330					335		
Leu	Phe	Leu	Pro	Ala	Thr	Ser	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly
		340					345					350			
Leu	Glu	Tyr	Val	Glu	Pro	Trp	Gln	Arg	Asn	Ala					
	355					360									

<210> SEQ ID NO 60

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: trunc new

<400> SEQUENCE: 60

Thr	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Met	Glu	Asn	His
1			5					10					15		
Asn	Lys	Ile	Phe	Thr	Asp	Ala	Phe	Glu	Val	Leu	Ala	Glu	Asn	Ser	Glu
		20					25					30			
Phe	Asn	Glu	Ser	Lys	Glu	Ser	Arg	Asp	Ala	Phe	Arg	Arg	Ala	Ala	Ser

-continued

35					40					45					
Val	Leu	Lys	Ser	Leu	Pro	Ser	Ala	Val	Arg	Ser	Leu	Gly	Asp	Thr	Glu
50						55					60				
Gly	Leu	Pro	Cys	Leu	Gly	Glu	His	Thr	Lys	Arg	Val	Ile	Glu	Glu	Ile
65					70					75					80
Leu	Glu	Tyr	Gly	Arg	Ser	Ser	Lys	Val	Glu	Asp	Ile	Leu	Ser	Asp	Glu
				85					90					95	
Arg	Tyr	Gln	Thr	Met	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Pro
				100					105					110	
Lys	Thr	Ala	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ser	Leu	Glu	Glu
				115					120					125	
Val	Arg	Ala	Glu	Pro	Ser	Ile	His	Leu	Asn	Arg	Met	Gln	Arg	Ala	Gly
				130					135					140	
Phe	Leu	Tyr	Tyr	Arg	Asp	Ile	Ser	Lys	Arg	Val	Ser	Lys	Ala	Glu	Ala
145						150					155				160
Lys	Ala	Leu	Gly	Asn	Ile	Ile	Glu	Glu	Thr	Val	His	Ala	Ile	Ala	Pro
				165					170					175	
Asp	Ala	Ile	Val	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Glu	Phe
				180					185					190	
Gly	His	Asp	Val	Asp	Phe	Ile	Leu	Thr	Thr	Pro	Glu	Met	Gly	Lys	Glu
				195					200					205	
Glu	Gly	Leu	Leu	Pro	Arg	Val	Ile	Asp	Arg	Leu	Arg	Asn	Gln	Gly	Ile
				210					215					220	
Leu	Leu	Tyr	Cys	Asp	Tyr	Gln	Glu	Ser	Thr	Phe	Asp	Met	Ser	Lys	Leu
225						230					235				240
Pro	Ser	Arg	Thr	Phe	Glu	Ala	Met	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu
				245					250					255	
Ile	Ile	Lys	Leu	Lys	Glu	Gly	Leu	Val	Glu	Gly	Glu	Gly	Gly	Leu	Gln
				260					265					270	
Ser	Asp	Pro	Gly	Asp	Arg	Arg	Gly	Trp	Arg	Ala	Val	Arg	Val	Asp	Leu
				275					280					285	
Val	Ala	Pro	Pro	Val	Asp	Arg	Tyr	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr
				290					295					300	
Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Arg	Leu	Glu
305						310					315				320
Arg	Arg	Met	Leu	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Lys
				325					330					335	
Ile	Phe	Leu	Pro	Ala	Lys	Thr	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly
				340					345					350	
Leu	Glu	Tyr	Ile	Glu	Pro	Trp	Gln	Arg	Asn	Ala					
				355					360						

<210> SEQ ID NO 61

<211> LENGTH: 360

<212> TYPE: PRT

<213> ORGANISM: Electrophorus electricus

<400> SEQUENCE: 61

Thr	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asp	Asn	His
1				5						10				15	

Asn	Lys	Val	Phe	Thr	Asp	Ala	Leu	Glu	Val	Leu	Ile	Glu	Asn	Tyr	Glu
				20				25					30		

-continued

Phe	Ser	Asp	Asn	Lys	Gly	Ala	Cys	Val	Gly	Phe	Arg	Arg	Ala	Ala	Ser
		35					40					45			
Val	Leu	Lys	Ser	Leu	Pro	Lys	Pro	Leu	Arg	Cys	Leu	Lys	Asp	Met	Glu
	50					55					60				
Gly	Leu	Pro	Cys	Leu	Gly	Asp	Asp	Thr	Lys	Ala	Ile	Ile	Glu	Glu	Ile
65					70					75					80
Tyr	Glu	Cys	Gly	Ser	Ser	Ser	Arg	Val	Glu	Asn	Ile	Leu	Ser	Asp	Glu
				85					90					95	
Lys	Tyr	Gln	Thr	Leu	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Pro
			100					105					110		
Lys	Thr	Gly	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ala	Leu	Glu	Gln
		115					120					125			
Val	His	Ser	Glu	Pro	Ser	Ile	Gln	Leu	Asn	Lys	Met	Gln	Ala	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Ile	Ser	Lys	Pro	Val	Ser	Arg	Ala	Glu	Ala
145					150					155					160
Lys	Ala	Val	Gly	Cys	Ile	Ile	Glu	Glu	Val	Ala	Ser	Cys	Phe	Ser	Ser
				165					170					175	
Ser	Val	Thr	Ile	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Arg	Gly	Lys	Glu	Phe
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Leu	Ser	Ile	Pro	Glu	Pro	Gly	Lys	Glu
		195					200					205			
Asp	Gly	Leu	Leu	Pro	Ala	Val	Ile	Asp	Arg	Leu	Arg	Lys	Gln	Gly	Ile
	210					215					220				
Leu	Leu	Tyr	Ser	Asp	Leu	Gln	Glu	Ser	Thr	Leu	Gln	Gln	Trp	Lys	Arg
225					230					235					240
Pro	Ser	Arg	Cys	Phe	Asp	Ser	Met	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu
				245					250					255	
Ile	Val	Lys	Leu	Trp	Thr	Arg	Leu	Val	Glu	Gly	His	Arg	Glu	Asp	Pro
			260					265					270		
Ser	Ser	Gln	Arg	Asp	Trp	Lys	Ala	Val	Arg	Val	Asp	Leu	Val	Val	Pro
		275					280					285			
Pro	Val	Asp	Cys	Tyr	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Ser	Gly	Ser	Thr
	290					295					300				
Gln	Phe	Glu	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Arg	Leu	Glu	Arg	Arg	Met
305					310					315					320
Leu	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Thr	Asn	Thr	Phe	Leu
				325					330					335	
Gln	Ala	Lys	Thr	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr
			340					345					350		
Ile	Glu	Pro	Trp	Gln	Arg	Asn	Ala								
		355					360								

<210> SEQ ID NO 62

<211> LENGTH: 360

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: trunc new

<400> SEQUENCE: 62

Thr	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asp	Asn	His
1				5						10					15

-continued

Asn	Lys	Ile	Phe	Thr	Asp	Ala	Leu	Glu	Val	Leu	Ala	Glu	Asn	Tyr	Glu
		20						25					30		
Phe	Ser	Asp	Ser	Lys	Glu	Ser	Arg	Asp	Ala	Phe	Arg	Arg	Ala	Ala	Ser
	35						40				45				
Val	Leu	Lys	Ser	Leu	Pro	Thr	Pro	Leu	Arg	Ser	Leu	Gln	Asp	Thr	Glu
	50					55					60				
Gly	Leu	Pro	Cys	Leu	Gly	Glu	Glu	Thr	Lys	Arg	Val	Ile	Glu	Glu	Ile
65					70					75					80
Phe	Glu	Glu	Gly	Ser	Ser	Ser	Arg	Val	Glu	Asp	Ile	Leu	Ser	Asp	Glu
			85					90					95		
Arg	Tyr	Gln	Thr	Leu	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Pro
			100					105					110		
Lys	Thr	Ala	Glu	Lys	Trp	Tyr	Arg	Arg	Gly	Leu	Arg	Ser	Leu	Glu	Gln
		115					120					125			
Val	Arg	Ser	Asp	Pro	Ser	Ile	His	Leu	Asn	Arg	Met	Gln	Thr	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Ile	Ser	Lys	Pro	Val	Ser	Lys	Ala	Glu	Ala
145					150					155					160
Lys	Ala	Leu	Gly	Asn	Ile	Ile	Glu	Glu	Thr	Ala	Ser	Ala	Phe	Ser	Pro
				165					170					175	
Ser	Val	Thr	Ile	Thr	Leu	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Glu	Phe
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Leu	Thr	Val	Pro	Glu	Pro	Gly	Lys	Glu
		195					200					205			
Asp	Gly	Leu	Leu	Pro	Ala	Val	Ile	Asp	Arg	Leu	Arg	Ser	Gln	Gly	Ile
	210					215					220				
Leu	Leu	Tyr	Ser	Asp	Phe	Gln	Glu	Ser	Thr	Phe	Asp	Leu	Ser	Lys	Leu
225					230					235					240
Pro	Ser	Arg	Arg	Phe	Glu	Ala	Met	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu
				245					250					255	
Ile	Val	Lys	Leu	Lys	Ala	Gly	Leu	Val	Glu	Gly	Glu	Gly	Ala	Asp	Pro
			260					265					270		
Gly	Asp	Arg	Arg	Asp	Trp	Arg	Ala	Val	Arg	Val	Asp	Leu	Val	Ala	Pro
	275						280					285			
Pro	Val	Asp	Arg	Tyr	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Ser	Gly	Ser	Ala
	290					295					300				
Phe	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Phe	Ala	Arg	Leu	Glu	Arg	Gly	Met
305					310					315					320
Leu	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Thr	Asn	Thr	Phe	Leu
				325					330					335	
Gln	Ala	Lys	Thr	Glu	Glu	Asp	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr
			340					345					350		
Ile	Glu	Pro	Trp	Gln	Arg	Asn	Ala								
	355					360									

<210> SEQ ID NO 63

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: Anabas testudineus

<400> SEQUENCE: 63

Thr	Val	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Thr	Glu	Asn	Asn
1				5						10				15	

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Asn Lys Ile Leu Thr Asp Ala Phe Glu Val Leu Ala Glu Ser Tyr Glu
    20                      25                      30
Leu Asn Gln Leu Glu Gly Pro Cys Leu Ala Phe Arg Arg Ala Ala Ser
    35                      40                      45
Val Leu Lys Ser Leu Pro Trp Ala Val Gln Cys Leu Gly Ala Thr Gln
    50                      55                      60
Gly Leu Pro Cys Leu Gly Glu His Thr Lys Ala Leu Ile Glu Glu Ile
    65                      70                      75                      80
Leu Gln Tyr Gly His Ser Phe Glu Val Glu Lys Ile Leu Ser Asp Glu
    85                      90                      95
Arg Tyr Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly Pro
    100                     105                     110
Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Phe Ser Asp
    115                     120                     125
Ile Leu Ala Glu Pro Ser Ile Gln Leu Asn Arg Met Gln Gln Ser Gly
    130                     135                     140
Phe Leu His Tyr Gly Asp Ile Ser Arg Ala Val Ser Lys Ala Glu Ala
    145                     150                     155                     160
Arg Ala Leu Gly Asn Ile Ile Asp Glu Ala Val His Ala Ile Thr Pro
    165                     170                     175
Asp Gly Ile Leu Ala Leu Thr Gly Gly Phe Arg Arg Gly Lys Glu Phe
    180                     185                     190
Gly His Asp Val Asp Phe Ile Val Thr Thr Pro Glu Gln Gly Lys Glu
    195                     200                     205
Glu Thr Leu Leu Pro Asn Ile Ile Asp Arg Leu Lys Glu Gln Gly Ile
    210                     215                     220
Leu Leu Tyr Ser Asp Tyr Gln Thr Ser Thr Phe Asp Ile Ser Lys Leu
    225                     230                     235                     240
Pro Ser His Lys Phe Glu Ala Met Asp His Phe Ala Lys Cys Phe Leu
    245                     250                     255
Ile Leu Arg Leu Glu Gly Ser Leu Val Asp Arg Gly Leu Asn Ser Thr
    260                     265                     270
Glu Gly Asp Ser Arg Gly Trp Arg Ala Val Arg Val Asp Leu Val Ser
    275                     280                     285
Pro Pro Met Glu Arg Tyr Ala Tyr Ala Leu Leu Gly Trp Thr Gly Ser
    290                     295                     300
Arg Gln Phe Glu Arg Asp Leu Arg Arg Phe Ala Arg Leu Glu Gln His
    305                     310                     315                     320
Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Glu Phe
    325                     330                     335
Leu Ala Ala Thr Thr Glu Arg Asp Ile Phe Ala His Leu Gly Leu Glu
    340                     345                     350
Tyr Ile Glu Pro Trp Gln Arg Asn Ala
    355                     360

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<210> SEQ ID NO 64

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: walking fish

<400> SEQUENCE: 64

-continued

Thr Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Thr Glu Asn Asn
 1 5 10 15
 Asn Lys Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Ser Tyr Glu
 20 25 30
 Phe Asn Glu Met Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Trp Ala Val Gln Ser Leu Gly Ala Thr Gln
 50 55 60
 Asp Leu Pro Cys Leu Gly Glu His Thr Lys Arg Val Met Glu Glu Ile
 65 70 75 80
 Leu Gln Tyr Gly Arg Ser Phe Glu Val Glu Lys Ile Leu Ser Asp Glu
 85 90 95
 Arg Tyr Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly Pro
 100 105 110
 Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Phe Glu Asp
 115 120 125
 Ile Arg Ala Glu Pro Ser Ile His Leu Asn Arg Met Gln Gln Ser Gly
 130 135 140
 Phe Leu His Tyr Gly Asp Ile Ser Arg Ala Val Ser Lys Ala Glu Ala
 145 150 155 160
 Arg Ala Leu Gly Asn Ile Ile Asp Glu Ala Val His Ala Ile Thr Pro
 165 170 175
 Asp Ala Ile Leu Ala Leu Thr Gly Gly Phe Arg Leu Gly Lys Glu Phe
 180 185 190
 Gly His Asp Val Asp Phe Ile Val Thr Thr Pro Glu Leu Gly Lys Glu
 195 200 205
 Glu Ser Leu Leu Pro Asn Ile Ile Asp Arg Leu Lys Lys Gln Gly Ile
 210 215 220
 Leu Leu Tyr Ser Asp Tyr Gln Ala Ser Thr Phe Asp Met Ser Lys Leu
 225 230 235 240
 Pro Ser His Arg Phe Glu Ala Met Asp His Phe Ala Lys Cys Phe Leu
 245 250 255
 Ile Leu Arg Leu Glu Gly Ser Gln Val Glu Gly Gly Leu Asn Ser Ala
 260 265 270
 Glu Gly Asp Ser Arg Gly Trp Arg Ala Val Arg Val Asp Leu Val Ser
 275 280 285
 Pro Pro Met Asp Arg Tyr Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser
 290 295 300
 Ala Phe Phe Asn Arg Asp Leu Arg Arg Phe Ala Arg Met Glu Arg Arg
 305 310 315 320
 Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Glu Phe
 325 330 335
 Leu Ala Ala Thr Thr Glu Lys Asp Ile Phe Ala His Leu Gly Leu Glu
 340 345 350
 Tyr Ile Glu Pro Trp Gln Arg Asn Ala
 355 360

<210> SEQ ID NO 65

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: Poecilia reticulata

-continued

<400> SEQUENCE: 65

Lys Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Val Glu Asn Asn
 1 5 10 15
 Asn Arg Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Asn Tyr Glu
 20 25 30
 Phe Asn Glu Ile Glu Gly Arg Cys Leu Ala Phe Arg Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Trp Ala Val Arg Ser Val Gly Ala Thr Leu
 50 55 60
 Asp Leu Pro Cys Leu Gly Glu His Thr Thr Ala Val Met Lys Glu Ile
 65 70 75 80
 Leu Gln Tyr Gly Arg Ser Phe Glu Val Glu Lys Ile Leu Ser Asp Glu
 85 90 95
 Arg Cys Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly Pro
 100 105 110
 Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Phe Ser Asp
 115 120 125
 Val Leu Ala Gln Pro Gly Ile His Leu Asn Arg Met Gln Gln Ser Gly
 130 135 140
 Phe Leu His Tyr Gly Asp Ile Ser Arg Ala Val Ser Lys Ala Glu Ala
 145 150 155 160
 Arg Ala Val Gly Asn Ile Ile Asp Glu Ala Val His Val Ile Thr Pro
 165 170 175
 Asn Ala Ile Leu Ala Leu Thr Gly Gly Phe Arg Arg Gly Lys Asp Phe
 180 185 190
 Gly His Asp Val Asp Phe Ile Val Thr Thr Thr Glu Leu Gly Lys Glu
 195 200 205
 Lys Asn Leu Leu Ile Ser Val Ile Glu Ser Leu Lys Lys Gln Gly Leu
 210 215 220
 Leu Leu Phe Ser Asp Tyr Gln Ala Ser Thr Phe Asp Ile Ser Lys Leu
 225 230 235 240
 Pro Ser His Arg Phe Glu Ala Met Asp His Phe Ala Lys Cys Phe Leu
 245 250 255
 Ile Leu Arg Leu Glu Gly Ser Arg Val Glu Gly Gly Leu Gln Arg Ala
 260 265 270
 Gln Ala Asp Gly Arg Gly Trp Arg Ala Val Arg Val Asp Leu Val Ser
 275 280 285
 Pro Pro Ala Asp Arg Phe Ala Phe Thr Met Leu Gly Trp Thr Gly Ser
 290 295 300
 Arg Met Phe Glu Arg Asp Leu Arg Arg Phe Ala Arg Leu Glu Arg Gln
 305 310 315 320
 Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Glu Phe
 325 330 335
 Leu Thr Ala Ala Thr Glu Lys Asp Ile Phe Asp His Leu Gly Leu Glu
 340 345 350
 Tyr Ile Glu Pro Trp Gln Arg Asn Ala
 355 360

<210> SEQ ID NO 66

<211> LENGTH: 366

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: F57 trunc new (Poecilia reticulata) guppy
[56.66 percent identity to wt calc by NCBI BLASTP]

<400> SEQUENCE: 66

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Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr
1           5           10           15

Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr Glu
20           25           30

Phe Asn Glu Asn Lys Gly Phe Cys Leu Ala Phe Arg Arg Ala Ala Ser
35           40           45

Val Leu Lys Phe Leu Pro Phe Thr Ile Val Arg Met Asn Asp Ile Gln
50           55           60

Gly Leu Pro Trp Met Gly Asp Gln Val Lys Arg Val Ile Glu Glu Ile
65           70           75           80

Leu Glu Glu Gly Glu Ser Ser Lys Val Lys Glu Val Leu Asn Asp Glu
85           90           95

Lys Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg
100          105          110

Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Leu Glu Glu
115          120          125

Val Lys Ala Asp Lys Thr Leu Lys Leu Thr Lys Met Gln Lys Ala Gly
130          135          140

Phe Leu Tyr Tyr Glu Asp Leu Ile Ser Arg Val Ser Lys Ala Glu Ala
145          150          155          160

Asp Ala Val Ser Leu Ile Val Lys Asp Thr Val Trp Lys Phe Leu Pro
165          170          175

Asp Ala Leu Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys Met
180          185          190

Gly His Asp Val Asp Phe Leu Ile Thr Thr Pro Gly Pro Glu Glu Glu
195          200          205

Glu Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly Leu
210          215          220

Leu Leu Tyr Tyr Asp Ile Ile Glu Ser Thr Phe Glu Lys Thr Lys Leu
225          230          235          240

Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe Ala
245          250          255

Ile Leu Lys Leu His Lys Gln Arg Val Asp Asn Gly Asn Ser Asn Gln
260          265          270

Ser Lys Glu Phe Asp Lys Glu Glu Thr Lys Asp Trp Lys Ala Ile Arg
275          280          285

Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ala Tyr Ala Leu Leu
290          295          300

Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala
305          310          315          320

Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys
325          330          335

Thr Lys Arg Ile Phe Leu Lys Ala Ala Ser Glu Glu Glu Ile Phe Ala
340          345          350

His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
355          360          365

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-continued

<210> SEQ ID NO 67

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 67

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn His
 1 5 10 15
 Asn Gln Leu Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Tyr Glu
 20 25 30
 Phe Arg Glu Asn Glu Val Ser Cys Leu Pro Phe Met Arg Ala Ala Ser
 35 40 45
 Val Leu Lys Ser Leu Ser Phe Pro Ile Val Ser Met Lys Asp Ile Glu
 50 55 60
 Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile Glu Gly Ile
 65 70 75 80
 Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
 85 90 95
 Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110
 Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser Lys
 115 120 125
 Ile Lys Ser Asp Lys Ser Leu Arg Phe Thr His Met Gln Lys Ala Gly
 130 135 140
 Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Asn Arg Ala Glu Ala
 145 150 155 160
 Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Ala Phe Leu Pro
 165 170 175
 Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Met Thr
 180 185 190
 Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr Glu Glu
 195 200 205
 Glu Glu Gln Gln Leu Leu His Lys Val Thr Asn Phe Trp Arg Gln Gln
 210 215 220
 Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Glu Lys Phe
 225 230 235 240
 Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys Cys
 245 250 255
 Phe Leu Ile Leu Lys Leu His Arg Gly Leu Val Arg Ser Glu Glu Ser
 260 265 270
 Gly Gln Gln Glu Gly Lys Asp Trp Lys Ala Ile Arg Val Asp Leu Val
 275 280 285
 Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly
 290 295 300
 Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg
 305 310 315 320
 Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val
 325 330 335
 Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu
 340 345 350
 Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

-continued

<210> SEQ ID NO 68
<211> LENGTH: 366
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: N100 trunc new

<400> SEQUENCE: 68

Ser Val Val Gln Lys Ile Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr
1 5 10 15
Leu Asn Asn His Asn Glu Val Phe Thr Arg Ala Phe Asp Ile Leu Ala
20 25 30
Glu Asn Tyr Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg
35 40 45
Arg Ala Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Ile Ser Met
50 55 60
Lys Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Val
65 70 75 80
Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val
85 90 95
Leu Asn Asp Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe
100 105 110
Gly Val Gly Arg Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg
115 120 125
Thr Leu Glu Lys Ile Lys Ser Asp Lys Ser Leu Arg Phe Thr Gln Met
130 135 140
Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Ala Glu Arg Val Asn
145 150 155 160
Arg Ala Glu Ala Glu Ala Ile Glu Met Leu Val Lys Glu Ala Val Val
165 170 175
Ala Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu
180 185 190
Gly Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu
195 200 205
Ala Thr Glu Glu Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe
210 215 220
Trp Arg Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr
225 230 235 240
Phe Glu Lys Phe Lys Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His
245 250 255
Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu His His Gly Arg Val Val
260 265 270
Ser Glu Lys Ser Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg
275 280 285
Val Asp Leu Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu
290 295 300
Gly Trp Thr Gly Ser Pro Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala
305 310 315 320
Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys
325 330 335
Thr Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala
340 345 350

Lys 1	Ile	Ser	Gln 5	Tyr	Ala	Cys	Gln	Arg	Arg 10	Thr	Thr	Leu	Asn 15	Asn	Cys
Asn	Gln	Ile	Phe 20	Thr	Asp	Ala	Phe	Asp 25	Ile	Leu	Ala	Glu	Asn 30	Cys	Glu
Phe	Arg	Glu	Asn 35	Glu	Asp	Ser	Cys 40	Val	Thr	Phe	Met	Arg 45	Ala	Ala	Ser
Val	Leu	Lys	Ser	Leu	Pro	Phe 55	Thr	Ile	Ile	Ser	Met 60	Lys	Asp	Thr	Glu
Gly 65	Ile	Pro	Cys	Leu 70	Gly	Ser	Lys	Val	Lys 75	Cys	Ile	Ile	Glu	Glu	Ile 80
Ile	Glu	Asp	Gly 85	Glu	Ser	Ser	Glu	Val	Lys 90	Ala	Val	Leu	Asn 95	Asp	Glu
Arg	Tyr	Gln	Ser 100	Phe	Lys	Leu	Phe	Thr 105	Ser	Val	Phe	Gly 110	Val	Gly	Leu
Lys	Thr	Ser 115	Glu	Lys	Trp	Phe 120	Arg	Met	Gly	Phe	Arg 125	Thr	Leu	Ser	Lys
Val 130	Arg	Ser	Asp	Glu	Ser	Leu 135	Lys	Phe	Thr	Arg	Met 140	Gln	Arg	Ala	Gly
Phe 145	Leu	Tyr	Tyr	Glu 150	Asp	Leu	Val	Ser	Cys 155	Val	Thr	Arg	Ala	Glu	Ala 160
Glu	Ala	Val	Ser 165	Ala	Leu	Val	Lys	Glu 170	Ala	Val	Trp	Ala	Phe	Leu	Pro
Asp	Ala	Phe 180	Val	Thr	Met	Thr	Gly 185	Gly	Phe	Arg	Arg 190	Gly	Lys	Lys	Met
Gly 195	His	Asp	Val	Asp	Phe	Leu 200	Ile	Thr	Ser	Pro	Gly 205	Ser	Thr	Glu	Asp
Glu 210	Glu	Gln	Gln	Leu 215	Leu	Gln	Lys	Val	Met	Asn 220	Leu	Trp	Glu	Lys	Lys
Gly 225	Leu	Leu	Leu 230	Tyr	Tyr	Asp	Leu 235	Val	Glu	Ser 240	Thr	Phe	Glu	Lys	Leu
Arg	Leu	Pro 245	Ser	Arg	Lys	Val	Asp 250	Ala	Leu	Asp	His 255	Phe	Gln	Lys	Cys
Phe	Leu	Ile 260	Phe	Lys	Leu	Pro	Leu 265	Gln	Arg	Val	Asp 270	Ser	Asp	Gln	Ser
Ser	Trp 275	Gln	Gly	Gly	Lys	Thr	Trp 280	Lys	Ala	Ile	Arg 285	Val	Asp	Leu	Val
Met 290	Cys	Pro	Tyr	Glu	Arg	Arg 295	Ala	Phe	Ala	Leu	Leu 300	Gly	Trp	Thr	Gly
Ser 305	Arg	Gln	Phe	Glu 310	Arg	Asp	Leu	Arg	Arg 315	Tyr	Ala	Thr	His	Glu	Arg 320
Lys	Met	Ile 325	Leu	Asp	Asn	His	Ala 330	Leu	Tyr	Asp 335	Lys	Thr	Lys	Arg	Ile
Phe	Leu	Lys 340	Ala	Glu	Ser	Glu	Glu 345	Glu	Ile	Phe	Ala 350	His	Leu	Gly	Leu

-continued

Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 70

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Sus domesticus

<400> SEQUENCE: 70

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Cys
 1 5 10 15

Asn His Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Asn Tyr Glu
 20 25 30

Phe Arg Glu Asn Glu Thr Phe Cys Leu Ala Phe Met Arg Ala Ala Ser
 35 40 45

Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Glu
 50 55 60

Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile Glu Glu Ile
 65 70 75 80

Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
 85 90 95

Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110

Lys Thr Ser Glu Arg Trp Phe Arg Met Gly Phe Arg Ser Leu Ser Lys
 115 120 125

Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met Gln Lys Ala Gly
 130 135 140

Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu Ala
 145 150 155 160

Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Gln Ala Phe Leu Pro
 165 170 175

Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys Met
 180 185 190

Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Asp Asp
 195 200 205

Glu Glu Gln Gln Leu Leu Pro Lys Val Val Asn Leu Trp Glu Arg Glu
 210 215 220

Gly Leu Leu Leu Tyr Cys Asp Leu Val Glu Ser Thr Leu Glu Lys Ser
 225 230 235 240

Lys Leu Pro Ser Arg Asn Val Asp Ala Leu Asp His Phe Gln Lys Cys
 245 250 255

Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Gly Met Ser
 260 265 270

Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val
 275 280 285

Met Cys Pro Tyr Glu Leu Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly
 290 295 300

Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg
 305 310 315 320

Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Ile
 325 330 335

Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu

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340	345	350
Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala		
355	360	
<210> SEQ ID NO 71		
<211> LENGTH: 362		
<212> TYPE: PRT		
<213> ORGANISM: artificial sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: N27 trunc new [88.13 percent identity to wt calc by NCBI BLASTP] pig		
<400> SEQUENCE: 71		
Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr		
1	5	10 15
Asn Glu Ile Phe Thr Arg Ala Phe Glu Ile Leu Ala Glu Asn Tyr Glu		
	20	25 30
Phe Arg Glu Asn Glu Glu Ser Tyr Val Thr Phe Arg Arg Ala Ala Ser		
	35	40 45
Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Glu		
	50	55 60
Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Val Ile Glu Glu Ile		
	65	70 75 80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu		
	85	90 95
Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg		
	100	105 110
Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu Lys		
	115	120 125
Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met Gln Lys Ala Gly		
	130	135 140
Phe Leu Tyr Tyr Glu Asp Leu Val Ser Arg Val Thr Arg Ala Glu Ala		
	145	150 155 160
Glu Ala Val Ser Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro		
	165	170 175
Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Lys Ile		
	180	185 190
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu Glu		
	195	200 205
Glu Glu Gln Gln Leu Leu His Lys Val Ile Asn Leu Trp Glu Lys Lys		
	210	215 220
Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Leu		
	225	230 235 240
Lys Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys		
	245	250 255
Phe Leu Ile Leu Lys Leu His His Gln Arg Val Gly Ser Gly Lys Ser		
	260	265 270
Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val		
	275	280 285
Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly		
	290	295 300
Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg		
	305	310 315 320

<400> SEQUENCE: 72

Lys 1	Ile	Ser	Gln	Tyr 5	Ala	Cys	Gln	Arg	Arg 10	Thr	Thr	Leu	Asn	Asn 15	Tyr
Asn	Glu	Ile	Phe 20	Thr	Arg	Ala	Phe	Glu	Val 25	Leu	Ala	Glu	Asn 30	Tyr	Glu
Phe	Arg	Glu	Asn	Glu	Glu	Ser	Tyr 40	Val	Thr	Phe	Arg	Arg 45	Ala	Ala	Ser
Val	Leu	Lys	Ser	Leu	Pro	Phe 55	Thr	Ile	Ile	Ser	Met 60	Lys	Asp	Thr	Glu
Gly 65	Ile	Pro	Cys	Leu	Gly 70	Asp	Lys	Val	Lys	Arg 75	Val	Ile	Glu	Glu	Ile 80
Ile	Glu	Asp	Gly	Glu 85	Ser	Ser	Glu	Val	Lys 90	Ala	Val	Leu	Asn	Asp 95	Glu
Arg	Tyr	Gln	Ala 100	Phe	Lys	Leu	Phe	Thr 105	Ser	Val	Phe	Gly	Val 110	Gly	Arg
Lys	Thr	Ser	Glu	Lys	Trp	Phe 120	Arg	Met	Gly	Phe	Arg	Thr 125	Leu	Glu	Lys
Ile	Arg	Ser	Asp	Lys	Thr	Leu 135	Lys	Phe	Thr	Arg	Met 140	Gln	Lys	Ala	Gly
Phe 145	Leu	Tyr	Tyr	Glu	Asp 150	Leu	Ala	Glu	Arg	Val 155	Thr	Arg	Ala	Glu	Ala 160
Glu	Ala	Val	Ser	Val 165	Leu	Val	Lys	Glu	Ala 170	Val	Trp	Ala	Phe	Leu 175	Pro
Asp	Ala	Phe	Val 180	Thr	Met	Thr	Gly	Gly 185	Phe	Arg	Leu	Gly	Lys 190	Lys	Ile
Gly	His	Asp	Val	Asp	Phe	Leu 200	Ile	Thr	Ser	Pro	Gly 205	Ser	Thr	Glu	Glu
Glu 210	Glu	Gln	Gln	Leu	Leu	Pro 215	Lys	Val	Ile	Asn	Leu 220	Trp	Glu	Arg	Lys
Gly 225	Leu	Leu	Leu	Tyr 230	Tyr	Asp	Leu	Val	Glu	Ser 235	Thr	Phe	Glu	Lys	Leu 240
Lys	Leu	Pro	Ser	Arg 245	Lys	Val	Asp	Ala	Leu 250	Asp	His	Phe	Gln	Lys 255	Cys
Phe	Leu	Ile	Leu 260	Lys	Leu	His	His	Gln 265	Arg	Val	Gly	Ser	Gly 270	Lys	Ser
Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp 280	Lys	Ala	Ile	Arg	Val 285	Asp	Leu	Val
Met	Cys	Pro	Tyr	Glu	Arg	Arg 295	Ala	Phe	Ala	Leu	Leu 300	Gly	Trp	Thr	Gly

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Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg
305                      310                      315                      320

Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Ile
                      325                      330                      335

Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu
                      340                      345                      350

Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
                      355                      360

<210> SEQ ID NO 73
<211> LENGTH: 361
<212> TYPE: PRT
<213> ORGANISM: Bubalus bubalis

<400> SEQUENCE: 73

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Lys Thr Thr Leu Asn Asn Tyr
1      5      10      15

Asn His Ile Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Ser Glu
20     25     30

Phe Lys Glu Asn Glu Val Ser Tyr Val Thr Phe Met Arg Ala Ala Ser
35     40     45

Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Gln
50     55     60

Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Cys Val Ile Glu Glu Ile
65     70     75     80

Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
85     90     95

Arg Tyr Gln Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
100    105    110

Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Ser Leu Ser Lys
115    120    125

Ile Thr Ser Asp Lys Thr Leu Lys Phe Thr Lys Met Gln Lys Ala Gly
130    135    140

Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu Ala
145    150    155    160

Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro
165    170    175

Asp Ala Phe Ile Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Lys Ile
180    185    190

Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Ala Glu Asp
195    200    205

Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Lys Lys Gly
210    215    220

Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Phe Lys
225    230    235    240

Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys Cys Phe
245    250    255

Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Gly Arg Ser Asn
260    265    270

Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val Met
275    280    285

Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser
290    295    300

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Arg Gln Phe Glu Arg Asp Ile Arg Arg Tyr Ala Thr His Glu Arg Lys
305 310 315 320

Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val Phe
325 330 335

Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp
340 345 350

Tyr Ile Glu Pro Trp Glu Arg Asn Ala
355 360

<210> SEQ ID NO 74

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: Marmota flaviventris

<400> SEQUENCE: 74

Thr Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Cys
1 5 10 15

Asn Arg Val Phe Thr Asp Ala Phe Asp Val Leu Ala Glu Asn Tyr Glu
20 25 30

Phe Arg Glu Asn Glu Ser Cys Ser Val Val Phe Met Arg Ala Ala Ser
35 40 45

Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Arg Asp Leu Glu
50 55 60

Gly Ile Pro Cys Leu Glu Gly Lys Ala Lys Ser Ile Ile Glu Glu Ile
65 70 75 80

Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
85 90 95

Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
100 105 110

Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser Lys
115 120 125

Ile Arg Ser Asp Lys Ser Leu Lys Phe Thr His Met Gln Lys Ala Gly
130 135 140

Phe Leu Tyr Tyr Glu Asp Leu Val Ser Cys Val Thr Arg Ala Glu Ala
145 150 155 160

Glu Ala Val Ser Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro
165 170 175

Asp Ala Phe Ile Thr Met Thr Gly Gly Phe Arg Arg Gly Lys Asn Ile
180 185 190

Gly His Asp Val Asp Phe Leu Ile Thr Ser Ala Glu Ala Thr Glu Glu
195 200 205

Glu Glu Gln Gln Leu Leu His Lys Val Thr Asn Leu Trp Glu Lys Lys
210 215 220

Gly Leu Leu Leu Tyr Cys Asp Leu Val Glu Ser Thr Phe Glu Lys Leu
225 230 235 240

Lys Thr Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys Cys
245 250 255

Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Asp Lys Ala
260 265 270

Ser Gln Gln Gly Gly Lys Asn Trp Lys Ala Ile Arg Val Asp Leu Val
275 280 285

Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly

-continued

290	295	300
Ser Arg Gln Phe Glu Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg		
305	310	315 320
Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Ile		
	325	330 335
Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu		
	340	345 350
Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala		
	355	360

<210> SEQ ID NO 75
 <211> LENGTH: 362
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Truncated mouse TdT M53

<400> SEQUENCE: 75

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr	
1	5 10 15
Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu Asn Asp Glu	
	20 25 30
Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ser Ser	
	35 40 45
Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys Asp Thr Glu	
	50 55 60
Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu Glu Ile	
65	70 75 80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu	
	85 90 95
Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg	
	100 105 110
Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu Lys	
	115 120 125
Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln Lys Ala Gly	
	130 135 140
Phe Leu Tyr Tyr Glu Asp Leu Ala Glu Arg Val Asn Arg Pro Glu Ala	
145	150 155 160
Glu Ala Ile Arg Met Leu Val Lys Glu Ala Val Val Thr Phe Leu Pro	
	165 170 175
Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Met Thr	
	180 185 190
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr Glu Asp	
	195 200 205
Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe Trp Lys Gln Gln	
	210 215 220
Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe Glu Lys Phe	
225	230 235 240
Lys Gln Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys	
	245 250 255
Phe Leu Ile Leu Lys Leu Asp His Pro Arg Val His Ser Val Lys Ser	
	260 265 270
Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg Val Asp Leu Val	

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275					280					285					
Met	Cys	Pro	Tyr	Asp	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
290						295					300				
Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg
305					310					315					320
Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Val
				325					330					335	
Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu
			340					345					350		
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
	355						360								

<210> SEQ ID NO 76

<211> LENGTH: 379

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: N93or trunc new [95.15 percent identity to wt
(Marmota flaviventris)calc by NCBI BLASTP]

<400> SEQUENCE: 76

Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro	Ala	Val
1				5					10					15	
Gln	Thr	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn
			20					25					30		
Tyr	Asn	Arg	Val	Phe	Thr	Asp	Ala	Phe	Asp	Val	Leu	Ala	Glu	Asn	Tyr
		35					40					45			
Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Ser	Val	Val	Phe	Arg	Arg	Ala	Ala
	50					55					60				
Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Arg	Asp	Leu
65					70					75					80
Glu	Gly	Ile	Pro	Cys	Leu	Glu	Gly	Lys	Ala	Lys	Arg	Ile	Ile	Glu	Glu
			85						90					95	
Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp
			100					105					110		
Glu	Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
		115					120					125			
Arg	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu
		130				135					140				
Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Lys	Phe	Thr	His	Met	Gln	Lys	Ala
145					150					155					160
Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Arg	Val	Thr	Arg	Ala	Glu
			165						170					175	
Ala	Glu	Ala	Val	Ser	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu
			180					185					190		
Pro	Asp	Ala	Phe	Ile	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Asn
		195					200					205			
Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Ala	Glu	Ala	Thr	Glu
	210					215					220				
Glu	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asn	Leu	Trp	Glu	Lys
225					230					235					240
Lys	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys
			245					250						255	

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Leu Lys Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys
 260 265 270
 Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Asp Lys
 275 280 285
 Ala Ser Gln Gln Gly Gly Lys Asn Trp Lys Ala Ile Arg Val Asp Leu
 290 295 300
 Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr
 305 310 315 320
 Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 325 330 335
 Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
 340 345 350
 Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly
 355 360 365
 Leu Asp Tyr Ile Glu Pro Arg Glu Arg Asn Ala
 370 375

<210> SEQ ID NO 77
 <211> LENGTH: 366
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: N72 trunc new [91.76 percent identity to
 wt (Bubalus bubalis) calc by NCBI BLASTP]

<400> SEQUENCE: 77

Ser Val Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr
 1 5 10 15
 Leu Asn Asn Tyr Asn Glu Ile Phe Thr Arg Ala Phe Glu Ile Leu Ala
 20 25 30
 Glu Asn Tyr Glu Phe Lys Glu Asn Glu Glu Ser Tyr Val Thr Phe Arg
 35 40 45
 Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met
 50 55 60
 Lys Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Val
 65 70 75 80
 Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val
 85 90 95
 Leu Asn Asp Glu Arg Tyr Gln Ala Phe Lys Leu Phe Thr Ser Val Phe
 100 105 110
 Gly Val Gly Arg Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg
 115 120 125
 Ser Leu Glu Lys Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met
 130 135 140
 Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Arg Val Thr
 145 150 155 160
 Arg Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp
 165 170 175
 Ala Phe Leu Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu
 180 185 190
 Gly Lys Lys Ile Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly
 195 200 205
 Ser Thr Glu Asp Glu Glu Gln Gln Leu Leu Pro Lys Val Ile Asn Leu
 210 215 220

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Trp Glu Arg Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr
 225 230 235 240
 Phe Glu Lys Phe Lys Leu Pro Ser Arg Gln Val Asp Ala Leu Asp His
 245 250 255
 Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Gly
 260 265 270
 Ser Gly Lys Ser Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg
 275 280 285
 Val Asp Leu Val Met Cys Pro Tyr Glu Gln Arg Ala Phe Ala Leu Leu
 290 295 300
 Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala
 305 310 315 320
 Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys
 325 330 335
 Thr Lys Arg Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala
 340 345 350
 His Leu Gly Leu Asp Tyr Ile Glu Pro Arg Glu Arg Asn Ala
 355 360 365

<210> SEQ ID NO 78
 <211> LENGTH: 379
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: N35 trunc new [89.18 percent identity to wt
 (Panthera tigris altaica) calc by NCBI BLASTP]

<400> SEQUENCE: 78

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Val Val
 1 5 10 15
 Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
 20 25 30
 Tyr Asn Glu Ile Phe Thr Arg Ala Phe Glu Val Leu Ala Glu Asn Tyr
 35 40 45
 Glu Phe Arg Glu Asn Glu Glu Ser Tyr Val Thr Phe Arg Arg Ala Ala
 50 55 60
 Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr
 65 70 75 80
 Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Val Ile Glu Glu
 85 90 95
 Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp
 100 105 110
 Glu Arg Tyr Gln Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
 115 120 125
 Arg Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu
 130 135 140
 Lys Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met Gln Lys Ala
 145 150 155 160
 Gly Phe Leu Tyr Tyr Glu Asp Leu Ala Glu Arg Val Thr Arg Ala Glu
 165 170 175
 Ala Glu Ala Val Ser Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu
 180 185 190
 Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Lys

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195	200	205
Ile Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu		
210	215	220
Glu Glu Glu Gln Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Arg		
225	230	235 240
Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys		
	245	250 255
Leu Lys Leu Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe Gln Lys		
	260	265 270
Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Gly Ser Gly Lys		
	275	280 285
Ser Ser Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu		
	290	295 300
Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr		
305	310	315 320
Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu		
	325	330 335
Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg		
	340	345 350
Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly		
	355	360 365
Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala		
	370	375

<210> SEQ ID NO 79
 <211> LENGTH: 379
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: N27 trunc new [88.13 percent identity to
 wt(Sus scrofa) calc by NCBI BLASTP]

<400> SEQUENCE: 79

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Val Val
1 5 10 15
Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
20 25 30
Tyr Asn Glu Ile Phe Thr Arg Ala Phe Glu Ile Leu Ala Glu Asn Tyr
35 40 45
Glu Phe Arg Glu Asn Glu Glu Ser Tyr Val Thr Phe Arg Arg Ala Ala
50 55 60
Ser Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr
65 70 75 80
Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Val Ile Glu Glu
85 90 95
Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp
100 105 110
Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115 120 125
Arg Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu
130 135 140
Lys Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Arg Met Gln Lys Ala
145 150 155 160

Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Arg	Val	Thr	Arg	Ala	Glu	
				165					170					175		
Ala	Glu	Ala	Val	Ser	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu	
				180					185					190		
Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Lys	
				195					200					205		
Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Thr	Glu	
				210					215					220		
Glu	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys	
				225					230					235		
Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	
				245					250					255		
Leu	Lys	Leu	Pro	Ser	Arg	Thr	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys	
				260					265					270		
Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Gly	Ser	Gly	Lys	
				275					280					285		
Ser	Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	
				290					295					300		
Val	Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	
				305					310					315		
Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	
				325					330					335		
Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	
				340					345					350		
Ile	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	
				355					360					365		
Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
				370					375							
<210> SEQ ID NO 80																
<211> LENGTH: 379																
<212> TYPE: PRT																
<213> ORGANISM: artificial sequence																
<220> FEATURE:																
<223> OTHER INFORMATION: N14 trunc new [92.08 percent identity to wt(Piliocolobus tephrosceles) calc by NCBI BLASTP]																
<400> SEQUENCE: 80																
Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro	Val	Val	
				5					10					15		
Gln	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	
				20					25					30		
Tyr	Asn	Glu	Ile	Phe	Thr	Arg	Ala	Phe	Asp	Ile	Leu	Ala	Glu	Asn	Ser	
				35					40					45		
Glu	Phe	Arg	Glu	Asn	Glu	Asp	Ser	Tyr	Val	Thr	Phe	Arg	Arg	Ala	Ala	
				50					55					60		
Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr	Ile	Ile	Ser	Met	Lys	Asp	Thr	
				65					70					75		
Glu	Gly	Ile	Pro	Cys	Leu	Gly	Ser	Lys	Val	Lys	Arg	Ile	Ile	Glu	Glu	
				85					90					95		
Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	
				100					105					110		
Glu	Arg	Tyr	Gln	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	
				115					120					125		

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Arg Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu
 130 135 140
 Lys Val Arg Ser Asp Glu Ser Leu Lys Phe Thr Arg Met Gln Arg Ala
 145 150 155 160
 Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Arg Val Thr Arg Ala Glu
 165 170 175
 Ala Glu Ala Val Ser Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu
 180 185 190
 Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Lys
 195 200 205
 Met Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Thr Glu
 210 215 220
 Asp Glu Glu Gln Gln Leu Leu Gln Lys Val Met Asn Leu Trp Glu Lys
 225 230 235 240
 Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys
 245 250 255
 Leu Arg Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys
 260 265 270
 Cys Phe Leu Ile Phe Lys Leu Pro Leu Gln Arg Val Gly Ser Asp Gln
 275 280 285
 Ser Ser Trp Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu
 290 295 300
 Val Met Cys Pro Tyr Glu Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr
 305 310 315 320
 Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 325 330 335
 Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
 340 345 350
 Ile Phe Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly
 355 360 365
 Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 370 375

<210> SEQ ID NO 81

<211> LENGTH: 379

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

 <223> OTHER INFORMATION: N100 or trunc new [93.14 percent identity to
 wt(long Rattus norvegicus) calc by NCBI BLASTP]

<400> SEQUENCE: 81

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Val Val
 1 5 10 15
 Gln Lys Ile Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
 20 25 30
 His Asn Glu Leu Phe Thr Arg Ala Phe Asp Ile Leu Ala Glu Asn Tyr
 35 40 45
 Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ala
 50 55 60
 Ser Val Leu Lys Ser Leu Ser Phe Pro Ile Val Ser Met Lys Asp Ile
 65 70 75 80
 Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu Glu

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85				90				95							
Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp
		100						105					110		
Glu	Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly
	115						120					125			
Arg	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu
	130					135					140				
Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	His	Met	Gln	Lys	Ala
145					150					155					160
Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Arg	Val	Asn	Arg	Ala	Glu
			165					170						175	
Ala	Glu	Ala	Val	Ser	Met	Leu	Val	Lys	Glu	Ala	Val	Val	Ala	Phe	Leu
			180					185						190	
Pro	Asp	Ala	Leu	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Met
	195						200					205			
Thr	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Glu	Ala	Thr	Glu
	210					215					220				
Glu	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asn	Phe	Trp	Arg	Gln
225					230					235					240
Gln	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Ile	Ile	Glu	Ser	Thr	Phe	Glu	Lys
			245					250						255	
Phe	Lys	Leu	Pro	Ser	Arg	Thr	Val	Asp	Ala	Leu	Asp	His	Phe	Gln	Lys
		260						265					270		
Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	Arg	Gly	Leu	Val	Arg	Ser	Glu	Glu
		275					280					285			
Ser	Gly	Gln	Gln	Glu	Gly	Lys	Asp	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu
	290					295					300				
Val	Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr
305					310					315					320
Gly	Ser	Pro	Phe	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu
			325					330						335	
Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg
		340						345					350		
Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly
		355					360					365			
Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala					
	370					375									

<210> SEQ ID NO 82

<211> LENGTH: 366

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: N100 trunc new [89.86 percent identity to wt short Rattus norvegicus calc by NCBI BLASTP]

<400> SEQUENCE: 82

Ser	Val	Val	Gln	Lys	Ile	Ser	Glu	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr
1				5					10					15	

Leu	Asn	Asn	His	Asn	Glu	Val	Phe	Thr	Arg	Ala	Phe	Asp	Ile	Leu	Ala
	20							25					30		

Glu	Asn	Tyr	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg	Asp	Ala	Phe	Arg
	35						40					45			

Arg 50	Ala	Ser	Ser	Val	Leu	Lys 55	Ser	Leu	Pro	Phe	Pro 60	Ile	Ile	Ser	Met
Lys 65	Asp	Thr	Glu	Gly	Ile 70	Pro	Cys	Leu	Gly	Asp 75	Lys	Val	Lys	Arg	Val 80
Ile	Glu	Glu	Ile	Ile 85	Glu	Asp	Gly	Glu	Ser 90	Ser	Glu	Val	Lys	Ala 95	Val
Leu	Asn	Asp	Glu	Arg 100	Tyr	Lys	Ala	Phe 105	Lys	Leu	Phe	Thr	Ser 110	Val	Phe
Gly	Val	Gly	Arg	Lys 115	Thr	Ala	Glu	Lys 120	Trp	Phe	Arg	Met 125	Gly	Phe	Arg
Thr	Leu 130	Glu	Lys	Ile	Lys 135	Ser	Asp	Lys	Ser	Leu	Arg 140	Phe	Thr	Gln	Met
Gln 145	Lys	Ala	Gly	Phe 150	Leu	Tyr	Tyr	Glu	Asp 155	Leu	Ala	Glu	Arg	Val	Asn 160
Arg	Ala	Glu	Ala	Glu 165	Ala	Ile	Glu	Met 170	Leu	Val	Lys	Glu	Ala 175	Val	Val
Ala	Phe	Leu	Pro	Asp 180	Ala	Leu	Val	Thr 185	Met	Thr	Gly	Gly	Phe 190	Arg	Leu
Gly	Lys 195	Met	Thr	Gly	His	Asp 200	Val	Asp	Phe	Leu	Ile 205	Thr	Ser	Pro	Glu
Ala 210	Thr	Glu	Glu	Glu	Glu	Gln 215	Gln	Leu	Leu	His	Lys 220	Val	Thr	Asp	Phe
Trp 225	Arg	Gln	Gln	Gly 230	Leu	Leu	Leu	Tyr	Cys	Asp 235	Ile	Ile	Glu	Ser	Thr 240
Phe	Glu	Lys	Phe	Lys 245	Leu	Pro	Ser	Arg	Thr 250	Val	Asp	Ala	Leu	Asp 255	His
Phe	Gln	Lys	Cys	Phe 260	Leu	Ile	Leu	Lys 265	Leu	His	His	Gly	Arg 270	Val	Val
Ser	Glu 275	Lys	Ser	Gly	Gln	Gln	Glu	Gly 280	Lys	Gly	Trp 285	Lys	Ala	Ile	Arg
Val 290	Asp	Leu	Val	Met	Cys	Pro 295	Tyr	Glu	Arg	Arg	Ala 300	Phe	Ala	Leu	Leu
Gly 305	Trp	Thr	Gly	Ser 310	Pro	Phe	Phe	Asn	Arg	Asp 315	Leu	Arg	Arg	Tyr	Ala 320
Thr	His	Glu	Arg	Lys 325	Met	Met	Leu	Asp	Asn 330	His	Ala	Leu	Tyr	Asp 335	Lys
Thr	Lys	Arg	Val	Phe 340	Leu	Glu	Ala	Glu 345	Ser	Glu	Glu	Glu	Ile 350	Phe	Ala
His	Leu 355	Gly	Leu	Asp	Tyr	Ile	Glu	Pro 360	Trp	Glu	Arg	Asn 365	Ala		
<210> SEQ ID NO 83															
<211> LENGTH: 383															
<212> TYPE: PRT															
<213> ORGANISM: artificial sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: F57 trunc new (Poecilia reticulata) [56.66 percent identity to wt calc by NCBI BLASTP]															
<400> SEQUENCE: 83															
Ser 1	Pro	Ser	Pro	Val 5	Pro	Gly	Ser	Gln	Asn 10	Val	Pro	Ala	Pro	Ser	Val 15
Asp	Lys	Ile	Ser	Gln 20	Tyr	Ala	Cys	Gln 25	Arg	Arg	Thr	Thr	Leu 30	Asn	Asn

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Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr
      35              40              45

Glu Phe Asn Glu Asn Lys Gly Phe Cys Leu Ala Phe Arg Arg Ala Ala
      50              55              60

Ser Val Leu Lys Phe Leu Pro Phe Thr Ile Val Arg Met Asn Asp Ile
      65              70              75              80

Gln Gly Leu Pro Trp Met Gly Asp Gln Val Lys Arg Val Ile Glu Glu
      85              90              95

Ile Leu Glu Glu Gly Glu Ser Ser Lys Val Lys Glu Val Leu Asn Asp
      100             105             110

Glu Lys Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
      115             120             125

Arg Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Leu Glu
      130             135             140

Glu Val Lys Ala Asp Lys Thr Leu Lys Leu Thr Lys Met Gln Lys Ala
      145             150             155             160

Gly Phe Leu Tyr Tyr Glu Asp Leu Ile Ser Arg Val Ser Lys Ala Glu
      165             170             175

Ala Asp Ala Val Ser Leu Ile Val Lys Asp Thr Val Trp Lys Phe Leu
      180             185             190

Pro Asp Ala Leu Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys
      195             200             205

Met Gly His Asp Val Asp Phe Leu Ile Thr Thr Pro Gly Pro Glu Glu
      210             215             220

Glu Glu Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly
      225             230             235             240

Leu Leu Leu Tyr Tyr Asp Ile Ile Glu Ser Thr Phe Glu Lys Thr Lys
      245             250             255

Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe
      260             265             270

Ala Ile Leu Lys Leu His Lys Gln Arg Val Asp Asn Gly Asn Ser Asn
      275             280             285

Gln Ser Lys Glu Phe Asp Lys Glu Glu Thr Lys Asp Trp Lys Ala Ile
      290             295             300

Arg Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ala Tyr Ala Leu
      305             310             315             320

Leu Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr
      325             330             335

Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp
      340             345             350

Lys Thr Lys Arg Ile Phe Leu Lys Ala Ala Ser Glu Glu Glu Ile Phe
      355             360             365

Ala His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
      370             375             380

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<210> SEQ ID NO 84

<211> LENGTH: 378

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: F32 trunc new [89.95 percent identity to wt
 Anabas testudineus calc by NCBI BLASTP]

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<400> SEQUENCE: 84

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Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val
1      5      10      15
Ala Thr Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Thr Glu Asn
20      25      30
Asn Asn Lys Ile Phe Thr Asp Ala Phe Glu Val Leu Ala Glu Ser Tyr
35      40      45
Glu Phe Asn Glu Met Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ala
50      55      60
Ser Val Leu Lys Ser Leu Pro Trp Ala Val Gln Ser Leu Gly Ala Thr
65      70      75      80
Gln Asp Leu Pro Cys Leu Gly Glu His Thr Lys Arg Val Met Glu Glu
85      90      95
Ile Leu Gln Tyr Gly Arg Ser Phe Glu Val Glu Lys Ile Leu Ser Asp
100     105     110
Glu Arg Tyr Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115     120     125
Pro Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Phe Glu
130     135     140
Asp Ile Arg Ala Glu Pro Ser Ile His Leu Asn Arg Met Gln Gln Ser
145     150     155     160
Gly Phe Leu His Tyr Gly Asp Ile Ser Arg Ala Val Ser Lys Ala Glu
165     170     175
Ala Arg Ala Leu Gly Asn Ile Ile Asp Glu Ala Val His Ala Ile Thr
180     185     190
Pro Asp Ala Ile Leu Ala Leu Thr Gly Gly Phe Arg Leu Gly Lys Glu
195     200     205
Phe Gly His Asp Val Asp Phe Ile Val Thr Thr Pro Glu Leu Gly Lys
210     215     220
Glu Glu Ser Leu Leu Pro Asn Ile Ile Asp Arg Leu Lys Lys Gln Gly
225     230     235     240
Ile Leu Leu Tyr Ser Asp Tyr Gln Ala Ser Thr Phe Asp Met Ser Lys
245     250     255
Leu Pro Ser His Arg Phe Glu Ala Met Asp His Phe Ala Lys Cys Phe
260     265     270
Leu Ile Leu Arg Leu Glu Gly Ser Gln Val Glu Gly Gly Leu Asn Ser
275     280     285
Ala Glu Gly Asp Ser Arg Gly Trp Arg Ala Val Arg Val Asp Leu Val
290     295     300
Ser Pro Pro Met Asp Arg Tyr Ala Phe Ala Leu Leu Gly Trp Thr Gly
305     310     315     320
Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Phe Ala Arg Met Glu Arg
325     330     335
Arg Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Glu
340     345     350
Phe Leu Ala Ala Thr Thr Glu Lys Asp Ile Phe Ala His Leu Gly Leu
355     360     365
Glu Tyr Ile Glu Pro Trp Gln Arg Asn Ala
370     375

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<210> SEQ ID NO 85

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<211> LENGTH: 377
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: F24 trunc new [83.55 percent identity to wt
      Electric eel calc by NCBI BLASTP]

<400> SEQUENCE: 85

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Leu
1          5          10          15

Pro Thr Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asp Asn
20          25          30

His Asn Lys Ile Phe Thr Asp Ala Leu Glu Val Leu Ala Glu Asn Tyr
35          40          45

Glu Phe Ser Asp Ser Lys Glu Ser Arg Asp Ala Phe Arg Arg Ala Ala
50          55          60

Ser Val Leu Lys Ser Leu Pro Thr Pro Leu Arg Ser Leu Gln Asp Thr
65          70          75          80

Glu Gly Leu Pro Cys Leu Gly Glu Glu Thr Lys Arg Val Ile Glu Glu
85          90          95

Ile Phe Glu Glu Gly Ser Ser Ser Arg Val Glu Asp Ile Leu Ser Asp
100         105         110

Glu Arg Tyr Gln Thr Leu Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115         120         125

Pro Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg Ser Leu Glu
130         135         140

Gln Val Arg Ser Asp Pro Ser Ile His Leu Asn Arg Met Gln Thr Ala
145         150         155         160

Gly Phe Leu Tyr Tyr Glu Asp Ile Ser Lys Pro Val Ser Lys Ala Glu
165         170         175

Ala Lys Ala Leu Gly Asn Ile Ile Glu Glu Thr Ala Ser Ala Phe Ser
180         185         190

Pro Ser Val Thr Ile Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Glu
195         200         205

Phe Gly His Asp Val Asp Phe Leu Leu Thr Val Pro Glu Pro Gly Lys
210         215         220

Glu Asp Gly Leu Leu Pro Ala Val Ile Asp Arg Leu Arg Ser Gln Gly
225         230         235         240

Ile Leu Leu Tyr Ser Asp Phe Gln Glu Ser Thr Phe Asp Leu Ser Lys
245         250         255

Leu Pro Ser Arg Arg Phe Glu Ala Met Asp His Phe Gln Lys Cys Phe
260         265         270

Leu Ile Val Lys Leu Lys Ala Gly Leu Val Glu Gly Glu Gly Ala Asp
275         280         285

Pro Gly Asp Arg Arg Asp Trp Arg Ala Val Arg Val Asp Leu Val Ala
290         295         300

Pro Pro Val Asp Arg Tyr Ala Phe Ala Leu Leu Gly Trp Ser Gly Ser
305         310         315         320

Ala Phe Phe Asn Arg Asp Leu Arg Arg Phe Ala Arg Leu Glu Arg Gly
325         330         335

Met Leu Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Thr Asn Thr Phe
340         345         350

Leu Gln Ala Lys Thr Glu Glu Asp Ile Phe Ala His Leu Gly Leu Asp

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355	360	365
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Tyr Ile Glu Pro Trp Gln Arg Asn Ala
 370 375

<210> SEQ ID NO 86
 <211> LENGTH: 367
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: F15 trunc new [80.38 percent identity to wt
 Salmo truttal calc by NCBI BLASTP]

<400> SEQUENCE: 86

Ser Ser Val Ala Thr Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr	
1 5 10 15	
Met Glu Asn His Asn Lys Ile Phe Thr Asp Ala Phe Glu Val Leu Ala	
20 25 30	
Glu Asn Ser Glu Phe Asn Glu Ser Lys Glu Ser Arg Asp Ala Phe Arg	
35 40 45	
Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Ser Ala Val Arg Ser Leu	
50 55 60	
Gly Asp Thr Glu Gly Leu Pro Cys Leu Gly Glu His Thr Lys Arg Val	
65 70 75 80	
Ile Glu Glu Ile Leu Glu Tyr Gly Arg Ser Ser Lys Val Glu Asp Ile	
85 90 95	
Leu Ser Asp Glu Arg Tyr Gln Thr Met Lys Leu Phe Thr Ser Val Phe	
100 105 110	
Gly Val Gly Pro Lys Thr Ala Glu Lys Trp Tyr Arg Arg Gly Leu Arg	
115 120 125	
Ser Leu Glu Glu Val Arg Ala Glu Pro Ser Ile His Leu Asn Arg Met	
130 135 140	
Gln Arg Ala Gly Phe Leu Tyr Tyr Arg Asp Ile Ser Lys Arg Val Ser	
145 150 155 160	
Lys Ala Glu Ala Lys Ala Leu Gly Asn Ile Ile Glu Glu Thr Val His	
165 170 175	
Ala Ile Ala Pro Asp Ala Ile Val Thr Leu Thr Gly Gly Phe Arg Leu	
180 185 190	
Gly Lys Glu Phe Gly His Asp Val Asp Phe Ile Leu Thr Thr Pro Glu	
195 200 205	
Met Gly Lys Glu Glu Gly Leu Leu Pro Arg Val Ile Asp Arg Leu Arg	
210 215 220	
Asn Gln Gly Ile Leu Leu Tyr Cys Asp Tyr Gln Glu Ser Thr Phe Asp	
225 230 235 240	
Met Ser Lys Leu Pro Ser Arg Thr Phe Glu Ala Met Asp His Phe Gln	
245 250 255	
Lys Cys Phe Leu Ile Ile Lys Leu Lys Glu Gly Leu Val Glu Gly Glu	
260 265 270	
Gly Gly Leu Gln Ser Asp Pro Gly Asp Arg Arg Gly Trp Arg Ala Val	
275 280 285	
Arg Val Asp Leu Val Ala Pro Pro Val Asp Arg Tyr Ala Phe Ala Leu	
290 295 300	
Leu Gly Trp Thr Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Phe	
305 310 315 320	

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Ala Arg Leu Glu Arg Arg Met Leu Leu Asp Asn His Ala Leu Tyr Asp
325 330 335

Lys Thr Lys Lys Ile Phe Leu Pro Ala Lys Thr Glu Glu Asp Ile Phe
340 345 350

Ala His Leu Gly Leu Glu Tyr Ile Glu Pro Trp Gln Arg Asn Ala
355 360 365

<210> SEQ ID NO 87
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: B63 trunc new [85.68 percent identity to wt
Notechis scutatus calc by NCBI BLASTP]

<400> SEQUENCE: 87

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val
1 5 10 15

Asp Ser Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
20 25 30

Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr
35 40 45

Glu Phe Asn Glu Asn Lys Gly Phe Cys Leu Ala Phe Arg Arg Ala Ala
50 55 60

Ser Val Leu Lys Phe Leu Pro Phe Thr Ile Val Arg Val Asn Asp Ile
65 70 75 80

Glu Gly Leu Pro Trp Met Gly Glu Gln Val Lys Arg Ile Ile Glu Asp
85 90 95

Ile Leu Glu Glu Gly Glu Ser Ser Lys Val Lys Ala Val Leu Asn Asn
100 105 110

Glu Asn Tyr Arg Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115 120 125

Arg Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Leu Glu
130 135 140

Glu Val Lys Ala Asp Lys Asn Leu Lys Leu Thr Arg Met Gln Lys Ala
145 150 155 160

Gly Phe Leu His Tyr Glu Asp Leu Ile Ser Arg Val Ser Lys Ala Glu
165 170 175

Ala Asp Ala Ala Ser Leu Ile Val Lys Asp Thr Val Trp Lys Phe Leu
180 185 190

Pro Asn Ala Leu Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys
195 200 205

Met Gly His Asp Val Asp Phe Leu Ile Thr Val Pro Gly Ser Arg Gln
210 215 220

Glu Glu Glu Glu Leu Leu His Thr Val Ile Asp Leu Trp Lys Lys Gln
225 230 235 240

Gly Leu Leu Leu Tyr Tyr Asp Leu Ile Glu Ser Thr Phe Glu Lys Thr
245 250 255

Lys Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys
260 265 270

Phe Ala Ile Leu Lys Leu His Lys Glu Arg Val Asp Lys Gly Asn Ser
275 280 285

Ile Gln Ser Lys Ala Phe Ala Glu Glu Glu Thr Lys Asp Trp Lys Ala
290 295 300

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Ile Arg Val Asp Leu Val Val Thr Pro Phe Glu Gln Tyr Ala Phe Ala
305                310                315                320

Leu Leu Gly Trp Thr Gly Ser Pro Phe Phe Asn Arg Asp Leu Arg Arg
                325                330                335

Tyr Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr
                340                345                350

Asp Lys Thr Lys Lys Ile Phe Leu Lys Ala Ala Ser Glu Glu Glu Ile
                355                360                365

Phe Ala His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
370                375                380

<210> SEQ ID NO 88
<211> LENGTH: 380
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: B59 trunc new [84.74 percent identity to wt
        xenopus calc by NCBI BLASTP]

<400> SEQUENCE: 88

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Glu
1                5                10                15

Val Lys Val Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Gln Asp
20                25                30

Thr Asn Arg Ile Phe Thr Asp Ala Phe Asp Ile Leu Ala Glu Asn Tyr
35                40                45

Glu Phe Asn Glu Asn Lys Gly Pro Cys Val Ala Phe Arg Arg Ala Ser
50                55                60

Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Val Ala Met Lys Asp Leu
65                70                75                80

Glu Gly Leu Pro Trp Leu Gly Asp Gln Met Lys Arg Ile Ile Glu Glu
85                90                95

Ile Leu Glu Glu Gly Lys Ser Ser Lys Val Val Glu Val Met Asn Asp
100               105               110

Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
115               120               125

Arg Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu
130               135               140

Glu Ile Lys Ser Asp Lys Glu Leu Lys Leu Thr Lys Met Gln Lys Ala
145               150               155               160

Gly Leu Leu Tyr Tyr Glu Asp Ile Thr Ser Ala Val Ser Lys Ala Glu
165               170               175

Ala Glu Ala Val Glu Gln Leu Ile Lys Ser Ile Val Trp Lys Phe Val
180               185               190

Pro Asp Ala Ile Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys
195               200               205

Met Gly His Asp Val Asp Ile Leu Ile Thr Thr Pro Arg Lys Gly Glu
210               215               220

Lys Glu Glu Ile Leu His Lys Thr Ile Asn Val Leu Lys Lys Arg Gly
225               230               235               240

Leu Leu Leu Phe Tyr Asn Ile Ile Glu Ser Thr Phe Asp Glu Thr Lys
245               250               255

Leu Pro Ser Arg His Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe

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260	265	270
Thr Ile Leu Lys Leu Gln Lys Gln Gln Leu Asp Asn Gly Asn Ser Asn		
275	280	285
Glu Pro Phe Glu Thr Glu Lys Lys Asn Trp Lys Ala Ile Arg Val Asp		
290	295	300
Leu Val Ile Thr Pro Tyr Glu Gln Tyr Ala Tyr Ala Leu Leu Gly Trp		
305	310	315
Thr Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His		
325	330	335
Glu Lys Arg Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys		
340	345	350
Asn Ile Phe Leu Lys Ala Asn Ser Glu Glu Asp Ile Phe Thr His Leu		
355	360	365
Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala		
370	375	380

<210> SEQ ID NO 89

<211> LENGTH: 383

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: B57 trunc new [83.03 percent identity to wt alligator calc by NCBI BLASTP; 84.07 % id to purple]

<400> SEQUENCE: 89

Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val		
1	5	10
Asp Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn		
20	25	30
Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Leu Ala Glu Asn Tyr		
35	40	45
Glu Phe Asn Glu Asn Lys Gly Phe Cys Leu Ala Phe Arg Arg Ala Ala		
50	55	60
Ser Val Leu Lys Phe Leu Pro Phe Thr Ile Val Arg Met Asn Asp Ile		
65	70	75
Gln Gly Leu Pro Trp Met Gly Asp Gln Val Lys Arg Val Ile Glu Glu		
85	90	95
Ile Leu Glu Glu Gly Glu Ser Ser Lys Val Lys Glu Val Leu Asn Asp		
100	105	110
Glu Lys Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly		
115	120	125
Arg Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Leu Glu		
130	135	140
Glu Val Lys Ala Asp Lys Thr Leu Lys Leu Thr Lys Met Gln Lys Ala		
145	150	155
Gly Phe Leu Tyr Tyr Glu Asp Leu Ile Ser Arg Val Ser Lys Ala Glu		
165	170	175
Ala Asp Ala Val Ser Leu Ile Val Lys Asp Thr Val Trp Lys Phe Leu		
180	185	190
Pro Asp Ala Leu Val Thr Leu Thr Gly Gly Phe Arg Leu Gly Lys Lys		
195	200	205
Met Gly His Asp Val Asp Phe Leu Ile Thr Thr Pro Gly Pro Glu Glu		
210	215	220

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Glu Glu Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly
 225 230 235 240
 Leu Leu Leu Tyr Tyr Asp Ile Ile Glu Ser Thr Phe Glu Lys Thr Lys
 245 250 255
 Leu Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys Phe
 260 265 270
 Ala Ile Leu Lys Leu His Lys Gln Arg Val Asp Asn Gly Asn Ser Asn
 275 280 285
 Gln Ser Lys Glu Phe Asp Lys Glu Glu Thr Lys Asp Trp Lys Ala Ile
 290 295 300
 Arg Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ala Tyr Ala Leu
 305 310 315 320
 Leu Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr
 325 330 335
 Ala Thr His Glu Lys Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp
 340 345 350
 Lys Thr Lys Arg Ile Phe Leu Lys Ala Ala Ser Glu Glu Glu Ile Phe
 355 360 365
 Ala His Leu Gly Leu Asp Tyr Leu Glu Pro Trp Glu Arg Asn Ala
 370 375 380

<210> SEQ ID NO 90
 <211> LENGTH: 370
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: B19 trunc new [92.35 percent identity to wt
 neopelma calc by NCBI BLASTP]

<400> SEQUENCE: 90

Ser Ser Val Ser Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr
 1 5 10 15
 Leu Asn Asn Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Ile Met Ala
 20 25 30
 Glu Asn Tyr Glu Phe Lys Glu Asn Glu Ile Phe Cys Leu Glu Phe Arg
 35 40 45
 Arg Ala Ala Ser Val Leu Lys Phe Leu Pro Phe Pro Val Thr Arg Met
 50 55 60
 Lys Asp Ile Gln Gly Leu Pro Cys Met Gly Asp Arg Val Arg Asp Val
 65 70 75 80
 Ile Glu Glu Ile Ile Glu Glu Gly Glu Ser Ser Arg Val Lys Glu Val
 85 90 95
 Leu Asn Asp Glu Arg Tyr Lys Ala Phe Lys Gln Phe Thr Ser Val Phe
 100 105 110
 Gly Val Gly Arg Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg
 115 120 125
 Thr Leu Glu Glu Val Lys Ala Asp Lys Thr Leu Lys Leu Ser Lys Met
 130 135 140
 Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Ala Glu Arg Val Ser
 145 150 155 160
 Lys Ala Glu Ala Asp Ala Val Ser Leu Ile Val Lys Asn Thr Val Val
 165 170 175
 Thr Phe Leu Pro Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Leu
 180 185 190

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Gly Lys Lys Ile Gly His Asp Ile Asp Phe Leu Ile Thr Asn Pro Gly
 195 200 205
 Pro Arg Glu Asp Glu Glu Leu Leu His Lys Val Val Asp Leu Trp Lys
 210 215 220
 Lys Gln Gly Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Val
 225 230 235 240
 Lys Glu Gln Leu Pro Ser Arg Lys Val Asp Ala Met Asp Asn Phe Gln
 245 250 255
 Lys Cys Phe Ala Ile Leu Lys Leu Tyr Gln Pro Arg Val Asp Asn Ser
 260 265 270
 Ser Tyr Asn Thr Ser Lys Lys Phe Asp Met Ala Glu Val Lys Asp Trp
 275 280 285
 Lys Ala Ile Arg Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ala
 290 295 300
 Tyr Ala Leu Leu Gly Trp Thr Gly Ser Pro Gln Phe Asn Arg Asp Leu
 305 310 315 320
 Arg Arg Tyr Ala Thr His Glu Arg Lys Met Ile Leu Asp Asn His Ala
 325 330 335
 Leu Tyr Asp Arg Arg Lys Arg Ile Phe Leu Lys Ala Gly Ser Glu Glu
 340 345 350
 Glu Ile Phe Ala His Leu Gly Leu Asp Tyr Val Glu Pro Arg Glu Arg
 355 360 365
 Asn Ala
 370

<210> SEQ ID NO 91
 <211> LENGTH: 369
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: specific bovine from Misha
 <400> SEQUENCE: 91

Lys Thr Pro Pro Leu Val Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln
 1 5 10 15
 Arg Lys Thr Thr Leu Asn Asn Tyr Asn Glu Ile Phe Thr Arg Ala Phe
 20 25 30
 Glu Ile Leu Ala Glu Asn Ser Glu Phe Lys Glu Asn Glu Glu Ser Tyr
 35 40 45
 Val Thr Phe Arg Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr
 50 55 60
 Ile Ile Ser Met Lys Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys
 65 70 75 80
 Val Lys Arg Ile Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu
 85 90 95
 Val Lys Ala Val Leu Asn Asp Glu Arg Tyr Gln Ala Phe Lys Leu Phe
 100 105 110
 Thr Ser Val Phe Gly Val Gly Arg Lys Thr Ser Glu Lys Trp Phe Arg
 115 120 125
 Met Gly Phe Arg Ser Leu Glu Lys Ile Arg Ser Asp Lys Thr Leu Lys
 130 135 140
 Phe Thr Lys Met Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val
 145 150 155 160

-continued

Ser Gly Val Thr Arg Ala Glu Ala Glu Ala Val Gly Val Leu Val Lys
 165 170 175
 Glu Ala Val Trp Ala Phe Leu Pro Asp Ala Phe Val Thr Met Thr Gly
 180 185 190
 Gly Phe Arg Leu Gly Lys Lys Ile Gly His Asp Val Asp Phe Leu Ile
 195 200 205
 Thr Ser Pro Gly Ser Ala Glu Asp Glu Glu Gln Leu Leu Pro Lys Val
 210 215 220
 Ile Asn Leu Trp Glu Lys Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val
 225 230 235 240
 Glu Ser Thr Phe Glu Lys Phe Lys Leu Pro Ser Arg Gln Val Asp Thr
 245 250 255
 Leu Asp His Phe Gln Lys Cys Phe Leu Ile Leu Lys Leu His His Gln
 260 265 270
 Arg Val Asp Ser Ser Lys Ser Asn Gln Gln Glu Gly Lys Thr Trp Lys
 275 280 285
 Ala Ile Arg Val Asp Leu Val Met Cys Pro Tyr Glu Asn Arg Ala Phe
 290 295 300
 Ala Leu Leu Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Ile Arg
 305 310 315 320
 Arg Tyr Ala Thr His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu
 325 330 335
 Tyr Asp Lys Thr Lys Arg Val Phe Leu Lys Ala Glu Ser Glu Glu Glu
 340 345 350
 Ile Phe Ala His Leu Gly Leu Asp Tyr Ile Glu Pro Arg Glu Arg Asn
 355 360 365

Ala

<210> SEQ ID NO 92
 <211> LENGTH: 374
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: specific latmeria from Misha

<400> SEQUENCE: 92

Asn Val Pro Ala Pro Ser Val Val Ala Ile Ser Gln Tyr Ala Cys Gln
 1 5 10 15
 Arg Arg Thr Thr Leu Asn Asn His Asn Lys Ile Phe Thr Asp Ala Phe
 20 25 30
 Glu Ile Leu Ala Glu Asn Tyr Glu Phe Asn Glu Asn Glu Gly Pro Cys
 35 40 45
 Leu Ala Phe Arg Arg Ala Ala Ser Leu Leu Lys Ser Leu Pro Tyr Ala
 50 55 60
 Ile Ser Ser Met Lys Asp Leu Glu Gly Leu Pro Cys Leu Gly Asp Gln
 65 70 75 80
 Thr Lys Ala Val Ile Glu Glu Ile Leu Glu Glu Gly Gln Ser Ser Lys
 85 90 95
 Val Gln Asp Val Leu Ser Asp Glu Arg Tyr Lys Ser Ile Lys Leu Phe
 100 105 110
 Thr Ser Val Phe Gly Val Gly Leu Lys Thr Ala Glu Lys Trp Tyr Arg
 115 120 125

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Lys Gly Phe Arg Thr Leu Glu Glu Val Gln Ala Asp Lys Glu Ile Lys
 130 135 140
 Leu Thr Lys Met Gln Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Ile Ser
 145 150 155 160
 Ser Ala Val Thr Lys Ala Glu Ala Glu Ala Ile Gly Gln Ile Ile Glu
 165 170 175
 Asp Thr Val Arg Leu Phe Ala Pro Asp Ala Ile Val Thr Leu Thr Gly
 180 185 190
 Gly Phe Arg Leu Gly Lys Lys Ile Gly His Asp Val Asp Phe Leu Ile
 195 200 205
 Thr Thr Pro Glu Thr Gly Asn Glu Asn Gly Leu Leu His Lys Val Ile
 210 215 220
 Asn Val Leu Gln Asn Gln Gly Ile Leu Leu Tyr Tyr Asp Ile Val Glu
 225 230 235 240
 Ser Thr Phe Asp Lys Thr Arg Leu Pro Ser Arg Lys Val Asp Ala Leu
 245 250 255
 Asp His Phe Gln Lys Cys Phe Ala Ile Leu Lys Leu His Lys Gln Lys
 260 265 270
 Val Asn Thr Ser Asn Ser Glu Glu Ala Glu Glu Pro Ser Asn Thr Glu
 275 280 285
 Thr Lys Asp Trp Lys Ala Ile Arg Val Asp Leu Val Ile Thr Pro Phe
 290 295 300
 Asp Gln Tyr Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser Ala Phe Phe
 305 310 315 320
 Asn Arg Asp Leu Arg Arg Phe Ala Thr His Glu Arg Lys Met Met Leu
 325 330 335
 Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Lys Ile Phe Leu Pro Ala
 340 345 350
 Lys Thr Glu Glu Asp Ile Phe Ala His Leu Gly Leu Asp Tyr Ile Glu
 355 360 365
 Pro Trp Glu Arg Asn Ala
 370

<210> SEQ ID NO 93
 <211> LENGTH: 370
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: specific Puma from Misha

<400> SEQUENCE: 93

Lys Thr Glu Ser Ala Val Val Lys Lys Ile Pro Leu Tyr Ala Cys Gln
 1 5 10 15
 Arg Arg Thr Thr Leu Asn Asn Phe Asn Glu Ile Phe Thr Arg Ala Phe
 20 25 30
 Glu Val Leu Ala Glu Asn Tyr Glu Phe Lys Glu Asn Glu Ile Ser Ser
 35 40 45
 Ala Thr Phe Arg Arg Ala Ala Ser Val Leu Lys Ser Leu Pro Phe Thr
 50 55 60
 Ile Ile Ser Met Lys Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys
 65 70 75 80
 Val Lys Arg Val Ile Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu
 85 90 95

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Val	Lys	Ala	Val	Leu	Asn	Asp	Glu	Arg	Tyr	Gln	Ala	Phe	Lys	Leu	Phe
			100					105					110		
Thr	Ser	Val	Phe	Gly	Val	Gly	Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg
		115					120					125			
Met	Gly	Phe	Arg	Thr	Leu	Glu	Lys	Ile	Lys	Ser	Asp	Lys	Thr	Leu	Lys
	130					135					140				
Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val
145					150					155					160
Ser	Gly	Val	Thr	Arg	Ala	Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys
				165					170					175	
Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly
			180					185					190		
Gly	Phe	Arg	Leu	Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile
		195					200					205			
Thr	Ile	Pro	Gly	Ser	Thr	Asp	Glu	Glu	Glu	Glu	Gln	Leu	Leu	Pro	Lys
	210					215					220				
Val	Ile	Asn	Leu	Trp	Gln	Arg	Lys	Glu	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu
225					230					235					240
Val	Glu	Ser	Thr	Phe	Glu	Lys	Leu	Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp
				245						250				255	
Ala	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His
			260					265					270		
Gln	Arg	Val	Asp	Ser	Gly	Lys	Cys	Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp
		275					280					285			
Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala
	290					295					300				
Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser	Ala	Phe	Phe	Asn	Arg	Asp	Leu
305					310					315					320
Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys	Met	Ile	Leu	Asp	Asn	His	Ala
				325					330					335	
Leu	Tyr	Asp	Lys	Thr	Lys	Lys	Ile	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu
			340					345					350		
Glu	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg
		355					360					365			
Asn	Ala														
															370

<210> SEQ ID NO 94

<211> LENGTH: 370

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: specific N139 from Misha

<400> SEQUENCE: 94

Lys	Thr	Glu	Ser	Ala	Val	Val	Lys	Lys	Ile	Pro	Leu	Tyr	Ala	Cys	Gln
1					5				10					15	
Arg	Arg	Thr	Thr	Leu	Asn	Asn	Phe	Asn	Glu	Ile	Phe	Thr	Arg	Ala	Phe
			20					25					30		
Glu	Val	Leu	Ala	Glu	Asn	Tyr	Glu	Phe	Lys	Glu	Asn	Glu	Ile	Ser	Ser
		35					40					45			
Ala	Thr	Phe	Arg	Arg	Ala	Ala	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Thr
	50					55					60				

-continued

Ile	Ile	Ser	Met	Lys	Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	65	70	75	80
Val	Lys	Arg	Val	Ile	Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	85	90	95	
Val	Lys	Ala	Val	Leu	Asn	Asp	Glu	Arg	Tyr	Gln	Ala	Phe	Lys	Leu	Phe	100	105	110	
Thr	Ser	Val	Phe	Gly	Val	Gly	Leu	Lys	Thr	Ser	Glu	Lys	Trp	Phe	Arg	115	120	125	
Met	Gly	Phe	Arg	Thr	Leu	Glu	Lys	Ile	Lys	Ser	Asp	Lys	Thr	Leu	Lys	130	135	140	
Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	145	150	155	160
Ser	Gly	Val	Thr	Arg	Ala	Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	165	170	175	
Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	180	185	190	
Gly	Phe	Arg	Leu	Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	195	200	205	
Thr	Ile	Pro	Gly	Ser	Thr	Asp	Glu	Glu	Glu	Gln	Leu	Leu	Pro	Lys		210	215	220	
Val	Ile	Asn	Leu	Trp	Gln	Arg	Lys	Glu	Leu	Leu	Tyr	Tyr	Asp	Leu		225	230	235	240
Val	Glu	Ser	Thr	Phe	Glu	Lys	Leu	Lys	Leu	Pro	Ser	Arg	Lys	Val	Asp	245	250	255	
Ala	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	260	265	270	
Gln	Arg	Val	Asp	Ser	Gly	Lys	Cys	Ser	Gln	Gln	Glu	Gly	Lys	Thr	Trp	275	280	285	
Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Arg	Arg	Ala	290	295	300	
Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser	Ala	Phe	Phe	Asn	Arg	Asp	Leu	305	310	315	320
Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys	Met	Ile	Leu	Asp	Asn	His	Ala	325	330	335	
Leu	Tyr	Asp	Lys	Thr	Lys	Lys	Ile	Phe	Leu	Lys	Ala	Glu	Ser	Glu	Glu	340	345	350	
Glu	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	355	360	365	
Asn	Ala															370			

<210> SEQ ID NO 95

<211> LENGTH: 370

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: specific shrew from Misha

<400> SEQUENCE: 95

Asn	Val	Pro	Ala	Pro	Val	Val	Gln	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	1	5	10	15
Arg	Arg	Thr	Thr	Leu	Asn	Asn	His	Asn	His	Ile	Phe	Thr	Asp	Ala	Phe	20	25	30	

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<210> SEQ ID NO 96
<211> LENGTH: 381
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: M27

<400> SEQUENCE: 96
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Asn	Ser	Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro
1				5					10					15	
Val	Val	Lys	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu
			20					25					30		
Asn	Asn	Tyr	Asn	Gln	Leu	Phe	Thr	Asp	Ala	Leu	Asp	Ile	Leu	Ala	Glu
		35					40					45			
Asn	Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Cys	Leu	Ala	Phe	Arg	Arg
	50					55					60				
Ala	Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys
65					70					75					80
Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Ser	Ile	Ile
				85					90					95	
Glu	Gly	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu
			100					105					110		
Asn	Asp	Glu	Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly
		115					120					125			
Val	Gly	Leu	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr
	130					135					140				
Leu	Ser	Lys	Ile	Gln	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln
145					150					155					160
Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Asn	Arg
				165					170					175	
Pro	Glu	Ala	Glu	Ala	Val	Ser	Met	Leu	Val	Lys	Glu	Ala	Val	Val	Thr
			180					185					190		
Phe	Leu	Pro	Asp	Ala	Leu	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly
		195					200					205			
Lys	Met	Thr	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Glu	Ala
	210					215					220				
Thr	Glu	Asp	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asp	Phe	Trp
225					230					235					240
Lys	Gln	Gln	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Ile	Leu	Glu	Ser	Thr	Phe
				245					250					255	
Glu	Lys	Phe	Lys	Gln	Pro	Ser	Arg	Thr	Val	Asp	Ala	Leu	Asp	His	Phe
			260					265					270		
Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	Asp	His	Pro	Arg	Val	His	Ser
		275					280					285			
Val	Lys	Ser	Gly	Gln	Gln	Glu	Gly	Lys	Gly	Trp	Lys	Ala	Ile	Arg	Val
	290					295					300				
Asp	Leu	Val	Met	Cys	Pro	Tyr	Asp	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly
305					310					315					320
Trp	Thr	Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr
				325					330					335	
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
			340					345					350		
Lys	Arg	Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His
		355					360					365			
Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala			
	370					375					380				

<210> SEQ ID NO 97

<211> LENGTH: 383

<212> TYPE: PRT

-continued

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: B10 trunc new [91.15 percent identity to wt
canary calc by NCBI BLASTP]

<400> SEQUENCE: 97

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Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro Ser Val
1          5          10          15

Thr Lys Val Ser Gln Tyr Ser Cys Gln Arg Lys Thr Thr Leu Asn Asn
20          25          30

Tyr Asn Lys Lys Phe Thr Asp Ala Phe Glu Val Met Ala Glu Asn Tyr
35          40          45

Glu Phe Lys Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ala
50          55          60

Ser Leu Leu Lys Ser Leu Pro Phe Pro Val Thr Arg Met Lys Asp Ile
65          70          75          80

Gln Gly Leu Pro Cys Met Gly Asp Gln Val Arg Arg Ile Ile Glu Glu
85          90          95

Ile Ile Glu Glu Gly Glu Ser Ser Arg Val Lys Glu Val Leu Asn Asp
100         105         110

Glu Arg Tyr Lys Ala Phe Lys Gln Phe Thr Ser Val Phe Gly Val Gly
115         120         125

Arg Lys Thr Ser Glu Lys Trp Tyr Arg Met Gly Leu Arg Thr Val Glu
130         135         140

Glu Val Lys Ala Asp Lys Thr Leu Lys Leu Ser Lys Met Gln Lys Ala
145         150         155         160

Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Arg Val Ser Lys Ala Glu
165         170         175

Ala Asp Ala Val Ser Leu Ile Val Lys Asn Thr Val Val Thr Phe Leu
180         185         190

Pro Asp Ala Leu Val Thr Ile Thr Gly Gly Phe Arg Leu Gly Lys Lys
195         200         205

Ile Gly His Asp Ile Asp Phe Leu Ile Thr Asn Pro Gly Pro Arg Glu
210         215         220

Asp Glu Glu Leu Leu His Lys Val Ile Asp Leu Trp Lys Lys Gln Gly
225         230         235         240

Leu Leu Leu Tyr Cys Asp Ile Ile Glu Ser Thr Phe Val Lys Glu Gln
245         250         255

Leu Pro Ser Arg Lys Val Asp Ala Met Asp His Phe Gln Lys Cys Phe
260         265         270

Ala Ile Leu Lys Leu Tyr Gln Pro Arg Val Asp Asn Ser Thr Ser Asn
275         280         285

Thr Ser Lys Lys Leu Glu Met Ala Glu Val Lys Asp Trp Lys Ala Ile
290         295         300

Arg Val Asp Leu Val Ile Thr Pro Phe Glu Gln Tyr Ala Tyr Ala Leu
305         310         315         320

Leu Gly Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr
325         330         335

Ala Ser His Glu Arg Lys Met Ile Leu Asp Asn His Ala Leu Tyr Asp
340         345         350

Arg Arg Lys Arg Ile Phe Leu Lys Ala Gly Ser Glu Glu Glu Ile Phe
355         360         365

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Ala	His	Leu	Gly	Leu	Asp	Tyr	Val	Glu	Pro	Trp	Glu	Arg	Asn	Ala
370						375					380			

<210> SEQ ID NO 98
 <211> LENGTH: 381
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Mouse TdT mutant >TH M27 (A146V, G189E, K394T)

<400> SEQUENCE: 98

Ser	Gly	Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro
1				5				10					15		
Val	Val	Lys	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu
		20						25					30		
Asn	Asn	Tyr	Asn	Gln	Leu	Phe	Thr	Asp	Ala	Leu	Asp	Ile	Leu	Ala	Glu
		35					40					45			
Asn	Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Cys	Leu	Ala	Phe	Arg	Arg
	50					55				60					
Ala	Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys
65				70					75					80	
Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Ser	Ile	Ile
			85					90					95		
Glu	Gly	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu
		100					105					110			
Asn	Asp	Glu	Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly
	115					120					125				
Val	Gly	Leu	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr
	130					135					140				
Leu	Ser	Lys	Ile	Gln	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln
145				150					155					160	
Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Asn	Arg
		165					170					175			
Pro	Glu	Ala	Glu	Ala	Val	Ser	Met	Leu	Val	Lys	Glu	Ala	Val	Val	Thr
		180					185					190			
Phe	Leu	Pro	Asp	Ala	Leu	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly
	195					200					205				
Lys	Met	Thr	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Glu	Ala
	210				215					220					
Thr	Glu	Asp	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asp	Phe	Trp
225				230					235					240	
Lys	Gln	Gln	Gly	Leu	Leu	Leu	Tyr	Cys	Asp	Ile	Leu	Glu	Ser	Thr	Phe
		245					250						255		
Glu	Lys	Phe	Lys	Gln	Pro	Ser	Arg	Thr	Val	Asp	Ala	Leu	Asp	His	Phe
		260					265					270			
Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	Asp	His	Pro	Arg	Val	His	Ser
		275				280					285				
Val	Lys	Ser	Gly	Gln	Gln	Glu	Gly	Lys	Gly	Trp	Lys	Ala	Ile	Arg	Val
	290					295				300					
Asp	Leu	Val	Met	Cys	Pro	Tyr	Asp	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly
305				310					315					320	
Trp	Thr	Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr
			325					330					335		

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His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr
 340 345 350

Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His
 355 360 365

Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 370 375 380

<210> SEQ ID NO 99
 <211> LENGTH: 381
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Mouse TdT mutant (MS 13-34 (A146V, G186E,
 K394T, Q455F, W506R))

<400> SEQUENCE: 99

Ser Gly Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro
 1 5 10 15

Val Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu
 20 25 30

Asn Asn Tyr Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu
 35 40 45

Asn Asp Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg
 50 55 60

Ala Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys
 65 70 75 80

Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile
 85 90 95

Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu
 100 105 110

Asn Asp Glu Arg Tyr Gln Ala Phe Lys Leu Phe Thr Ser Val Phe Gly
 115 120 125

Val Gly Arg Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr
 130 135 140

Leu Glu Lys Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln
 145 150 155 160

Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Asn Arg
 165 170 175

Pro Glu Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Thr
 180 185 190

Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu Gly
 195 200 205

Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala
 210 215 220

Thr Glu Asp Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe Trp
 225 230 235 240

Lys Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe
 245 250 255

Glu Lys Phe Lys Gln Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe
 260 265 270

Gln Lys Cys Phe Leu Ile Leu Lys Leu Asp His Pro Arg Val His Ser
 275 280 285

Val Lys Ser Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg Val
 290 295 300

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Asp Leu Val Met Cys Pro Tyr Asp Arg Arg Ala Phe Ala Leu Leu Gly
305 310 315 320

Trp Thr Gly Ser Ala Phe Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr
325 330 335

His Glu Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr
340 345 350

Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His
355 360 365

Leu Gly Leu Asp Tyr Ile Glu Pro Arg Glu Arg Asn Ala
370 375 380

<210> SEQ ID NO 100

<211> LENGTH: 381

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Mouse TdT mutant (MS 34-1 (A146V, G186E, K394T, Q455F))

<400> SEQUENCE: 100

Ser Gly Ser Pro Ser Pro Val Pro Gly Ser Gln Asn Val Pro Ala Pro
1 5 10 15

Val Val Lys Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu
20 25 30

Asn Asn Tyr Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu
35 40 45

Asn Asp Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg
50 55 60

Ala Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys
65 70 75 80

Asp Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile
85 90 95

Glu Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu
100 105 110

Asn Asp Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly
115 120 125

Val Gly Arg Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr
130 135 140

Leu Glu Lys Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln
145 150 155 160

Lys Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Asn Arg
165 170 175

Pro Glu Ala Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Thr
180 185 190

Phe Leu Pro Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu Gly
195 200 205

Lys Met Thr Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala
210 215 220

Thr Glu Asp Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe Trp
225 230 235 240

Lys Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe
245 250 255

Glu Lys Phe Lys Gln Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe

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260					265					270					
Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	Asp	His	Pro	Arg	Val	His	Ser
	275						280					285			
Val	Lys	Ser	Gly	Gln	Gln	Glu	Gly	Lys	Gly	Trp	Lys	Ala	Ile	Arg	Val
	290					295					300				
Asp	Leu	Val	Met	Cys	Pro	Tyr	Asp	Arg	Arg	Ala	Phe	Ala	Leu	Leu	Gly
305						310					315				320
Trp	Thr	Gly	Ser	Ala	Phe	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr
			325						330					335	
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
			340					345					350		
Lys	Arg	Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His
	355						360					365			
Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala			
	370					375					380				

<210> SEQ ID NO 101
 <211> LENGTH: 381
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Mouse TdT mutant (TH c2_5 (A146V, G189E))

<400> SEQUENCE: 101

Ser	Gly	Ser	Pro	Ser	Pro	Val	Pro	Gly	Ser	Gln	Asn	Val	Pro	Ala	Pro
1				5					10					15	
Val	Val	Lys	Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu
		20						25					30		
Asn	Asn	Tyr	Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu	Asp	Ile	Leu	Ala	Glu
		35					40					45			
Asn	Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Cys	Leu	Ala	Phe	Arg	Arg
	50					55					60				
Ala	Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys
65					70					75					80
Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile	Ile
			85					90						95	
Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu
		100					105						110		
Asn	Asp	Glu	Arg	Tyr	Lys	Ser	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly
		115					120					125			
Val	Gly	Leu	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr
	130					135					140				
Leu	Glu	Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln
145				150						155					160
Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Asn	Arg
			165					170						175	
Pro	Glu	Ala	Glu	Ala	Val	Ser	Thr	Leu	Val	Lys	Glu	Ala	Val	Val	Thr
		180						185					190		
Phe	Leu	Pro	Asp	Ala	Leu	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly
	195					200						205			
His	Met	Thr	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Glu	Ala
	210				215						220				
Thr	Glu	Asp	Glu	Glu	Gln	Gln	Leu	Leu	His	Lys	Val	Thr	Asp	Phe	Trp

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225	230	235	240
Lys Gln Gln Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe	245	250	255
Glu Lys Phe Lys Gln Pro Ser Arg Lys Val Asp Ala Leu Asp His Phe	260	265	270
Gln Lys Cys Phe Leu Ile Leu Lys Leu Asp His Leu Arg Val His Ser	275	280	285
Ala Lys Ser Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg Val	290	295	300
Asp Leu Val Met Cys Pro Tyr Asp Arg Arg Ala Phe Ala Leu Leu Gly	305	310	315
Trp Thr Gly Ser Val Gln Phe Lys Arg Asp Leu Arg Arg Tyr Ala Thr	325	330	335
His Glu Arg Lys Met Met Leu Asp Glu His Ala Leu Tyr Asp Lys Thr	340	345	350
Lys Arg Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His	355	360	365
Leu Gly Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala	370	375	380

<210> SEQ ID NO 102
 <211> LENGTH: 361
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: (M57 without His tag & linker & with Q4E mutation) bovine-mouse chimera

<400> SEQUENCE: 102

Lys Ile Ser Gln Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr	1	5	10	15
Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu Asn Asp Glu	20	25	30	
Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ser Ser	35	40	45	
Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys Asp Thr Glu	50	55	60	
Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu Glu Ile	65	70	75	80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu	85	90	95	
Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg	100	105	110	
Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Glu Lys	115	120	125	
Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln Lys Ala Gly	130	135	140	
Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Thr Arg Ala Glu Ala	145	150	155	160
Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro	165	170	175	
Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Lys Ile	180	185	190	

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Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Ala	Glu	Asp
	195						200					205			
Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys	Lys	Gly
	210					215					220				
Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Phe	Lys
225					230					235					240
Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe
			245						250					255	
Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Ser	Ser	Lys	Ser	Asn
		260					265						270		
Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Met
		275					280					285			
Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser
	290					295					300				
Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys
305					310					315					320
Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Val	Phe
				325					330					335	
Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp
		340						345					350		
Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala							
	355						360								

<210> SEQ ID NO 103
 <211> LENGTH: 361
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: (M59 without His Tag and linker & with Q4E mutation) bovine-mouse chimera

<400> SEQUENCE: 103

Lys	Ile	Ser	Gln	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr
1			5						10					15	
Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu	Asp	Ile	Leu	Ala	Glu	Asn	Asp	Glu
		20						25					30		
Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg	Asp	Ala	Phe	Arg	Arg	Ala	Ser	Ser
		35					40					45			
Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys	Asp	Thr	Glu
	50					55					60				
Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile	Ile	Glu	Glu	Ile
65				70						75					80
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	Glu
			85						90					95	
Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Arg
			100					105					110		
Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu	Lys
		115					120					125			
Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Ala	Glu	Arg	Val	Asn	Arg	Pro	Glu	Ala
145					150					155					160
Glu	Ala	Ile	Arg	Met	Leu	Val	Lys	Glu	Ala	Val	Val	Thr	Phe	Leu	Pro
				165					170					175	

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Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Met Thr
 180 185 190
 Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Ala Glu Asp
 195 200 205
 Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Lys Lys Gly
 210 215 220
 Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Phe Lys
 225 230 235 240
 Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys Cys Phe
 245 250 255
 Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Ser Lys Ser Asn
 260 265 270
 Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val Met
 275 280 285
 Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser
 290 295 300
 Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg Lys
 305 310 315 320
 Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val Phe
 325 330 335
 Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp
 340 345 350
 Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 104

<211> LENGTH: 362

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: (M27 with maximal N-terminal deletion plus Q4E mutation) mouse

<400> SEQUENCE: 104

Lys Ile Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn Tyr
 1 5 10 15
 Asn Gln Leu Phe Thr Asp Ala Leu Asp Ile Leu Ala Glu Asn Asp Glu
 20 25 30
 Phe Arg Glu Asn Glu Glu Ser Cys Leu Ala Phe Arg Arg Ala Ser Ser
 35 40 45
 Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys Asp Thr Glu
 50 55 60
 Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Ser Ile Ile Glu Gly Ile
 65 70 75 80
 Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu
 85 90 95
 Arg Tyr Lys Ser Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Leu
 100 105 110
 Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu Ser Lys
 115 120 125
 Ile Gln Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln Lys Ala Gly
 130 135 140
 Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Asn Arg Pro Glu Ala

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145	150	155	160
Glu Ala Val Ser Met Leu Val Lys Glu Ala Val Val Thr Phe Leu Pro	165	170	175
Asp Ala Leu Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Met Thr	180	185	190
Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Glu Ala Thr Glu Asp	195	200	205
Glu Glu Gln Gln Leu Leu His Lys Val Thr Asp Phe Trp Lys Gln Gln	210	215	220
Gly Leu Leu Leu Tyr Cys Asp Ile Leu Glu Ser Thr Phe Glu Lys Phe	225	230	235
Lys Gln Pro Ser Arg Thr Val Asp Ala Leu Asp His Phe Gln Lys Cys	245	250	255
Phe Leu Ile Leu Lys Leu Asp His Pro Arg Val His Ser Val Lys Ser	260	265	270
Gly Gln Gln Glu Gly Lys Gly Trp Lys Ala Ile Arg Val Asp Leu Val	275	280	285
Met Cys Pro Tyr Asp Arg Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly	290	295	300
Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu Arg	305	310	315
Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val	325	330	335
Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu	340	345	350
Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala	355	360	

<210> SEQ ID NO 105

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: M54 with maximal N-terminal deletion plus Q4E mutation

<400> SEQUENCE: 105

Lys Ile Ser Glu Tyr Ala Cys Gln Arg Lys Thr Thr Leu Asn Asn Tyr	1	5	10	15
Asn Glu Ile Phe Thr Arg Ala Phe Glu Ile Leu Ala Glu Asn Ser Glu	20	25	30	
Phe Lys Glu Asn Glu Glu Ser Tyr Val Thr Phe Arg Arg Ala Ala Ser	35	40	45	
Val Leu Lys Ser Leu Pro Phe Thr Ile Ile Ser Met Lys Asp Thr Glu	50	55	60	
Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu Glu Ile	65	70	75	80
Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp Glu	85	90	95	
Arg Tyr Gln Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly Arg	100	105	110	
Lys Thr Ser Glu Lys Trp Phe Arg Met Gly Phe Arg Ser Leu Glu Lys	115	120	125	

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Ile Arg Ser Asp Lys Thr Leu Lys Phe Thr Lys Met Gln Lys Ala Gly
 130          135          140

Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Thr Arg Ala Glu Ala
145          150          155          160

Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe Leu Pro
          165          170          175

Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys Lys Ile
          180          185          190

Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Ala Glu Asp
          195          200          205

Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Lys Lys Gly
210          215          220

Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys Phe Lys
225          230          235          240

Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys Cys Phe
          245          250          255

Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Ser Lys Ser Asn
          260          265          270

Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu Val Met
          275          280          285

Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr Gly Ser
290          295          300

Ala Phe Phe Asn Arg Asp Ile Arg Arg Tyr Ala Thr His Glu Arg Lys
305          310          315          320

Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg Val Phe
          325          330          335

Leu Lys Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly Leu Asp
          340          345          350

Tyr Ile Glu Pro Arg Glu Arg Asn Ala
          355          360

<210> SEQ ID NO 106
<211> LENGTH: 362
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: M77 without His tag

<400> SEQUENCE: 106

Lys Lys Ile Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn Asn
 1          5          10          15

Tyr Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu Asn Asp
          20          25          30

Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala Ser
          35          40          45

Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys Asp Thr
          50          55          60

Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu Glu
65          70          75          80

Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn Asp
          85          90          95

Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val Gly
          100          105          110

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Arg	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu
		115					120					125			
Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln	Lys	Ala
	130					135					140				
Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Thr	Arg	Ala	Glu
145					150					155					160
Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu
				165					170					175	
Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Lys
			180					185					190		
Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Ala	Glu
		195					200					205			
Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys	Lys
	210					215					220				
Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Phe
225					230					235					240
Lys	Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe	Gln	Lys	Cys
				245					250					255	
Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Ser	Ser	Lys	Ser
			260					265					270		
Asn	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val
		275					280					285			
Met	Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly
	290					295					300				
Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg
305					310					315					320
Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Val
				325					330					335	
Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu
			340					345					350		
Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala						
		355					360								

<210> SEQ ID NO 107

<211> LENGTH: 361

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: C1-40 without His tag

<400> SEQUENCE: 107

Lys	Ile	Ser	Glu	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr
1				5					10					15	
Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu	Asp	Ile	Leu	Ala	Glu	Asn	Asp	Glu
			20					25					30		
Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg	Asp	Ala	Phe	Arg	Arg	Ala	Ser	Ser
		35					40					45			
Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys	Asp	Thr	Glu
		50				55					60				
Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile	Ile	Glu	Glu	Ile
65					70					75					80
Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn	Asp	Glu
				85					90					95	

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Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val	Gly	Arg
			100					105					110		
Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu	Glu	Lys
		115					120					125			
Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly
	130					135					140				
Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Thr	Arg	Ala	Glu	Ala
145					150					155					160
Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro
			165						170					175	
Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys	Lys	Ile
			180					185					190		
Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Ala	Glu	Asp
		195					200					205			
Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys	Lys	Gly
	210					215					220				
Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys	Phe	Lys
225					230					235					240
Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe	Gln	Lys	Cys	Phe
			245						250					255	
Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Ser	Ser	Lys	Ser	Asn
		260					265						270		
Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu	Val	Met
		275					280					285			
Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser
	290					295					300				
Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys
305					310					315					320
Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg	Val	Phe
			325						330					335	
Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly	Leu	Asp
		340					345						350		
Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala							
		355					360								

<210> SEQ ID NO 108

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: A2-20 without His tag

<400> SEQUENCE: 108

Val	Thr	Thr	Val	Ser	Glu	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr	Leu	Asn
1				5					10					15	
Asn	Tyr	Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu	Asp	Ile	Leu	Ala	Glu	Asn
		20					25						30		
Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg	Asp	Ala	Phe	Arg	Arg	Ala
		35				40					45				
Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys	Asp
	50					55					60				
Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile	Ile	Glu
65					70					75					80

-continued

Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn
 85 90 95
 Asp Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val
 100 105 110
 Gly Arg Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu
 115 120 125
 Glu Lys Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln Lys
 130 135 140
 Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Thr Arg Ala
 145 150 155 160
 Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe
 165 170 175
 Leu Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys
 180 185 190
 Lys Ile Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Ala
 195 200 205
 Glu Asp Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Lys
 210 215 220
 Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys
 225 230 235 240
 Phe Lys Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys
 245 250 255
 Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Ser Lys
 260 265 270
 Ser Asn Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu
 275 280 285
 Val Met Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr
 290 295 300
 Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 305 310 315 320
 Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
 325 330 335
 Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly
 340 345 350
 Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 109

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: ii-ell-40 without His tag

<400> SEQUENCE: 109

Val Ala Thr Val Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn
 1 5 10 15
 Asn Tyr Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu Asn
 20 25 30
 Asp Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala
 35 40 45
 Ser Ser Val Leu Lys Ser Leu Pro Phe Pro Ile Thr Ser Met Lys Asp
 50 55 60

-continued

Thr Glu Gly Ile Pro Cys Leu Gly Asp Lys Val Lys Arg Ile Ile Glu
 65 70 75 80
 Glu Ile Ile Glu Asp Gly Glu Ser Ser Glu Val Lys Ala Val Leu Asn
 85 90 95
 Asp Glu Arg Tyr Lys Ala Phe Lys Leu Phe Thr Ser Val Phe Gly Val
 100 105 110
 Gly Arg Lys Thr Ala Glu Lys Trp Phe Arg Met Gly Phe Arg Thr Leu
 115 120 125
 Glu Lys Ile Arg Ser Asp Lys Ser Leu Arg Phe Thr Gln Met Gln Lys
 130 135 140
 Ala Gly Phe Leu Tyr Tyr Glu Asp Leu Val Ser Gly Val Thr Arg Ala
 145 150 155 160
 Glu Ala Glu Ala Val Gly Val Leu Val Lys Glu Ala Val Trp Ala Phe
 165 170 175
 Leu Pro Asp Ala Phe Val Thr Met Thr Gly Gly Phe Arg Leu Gly Lys
 180 185 190
 Lys Ile Gly His Asp Val Asp Phe Leu Ile Thr Ser Pro Gly Ser Ala
 195 200 205
 Glu Asp Glu Glu Gln Leu Leu Pro Lys Val Ile Asn Leu Trp Glu Lys
 210 215 220
 Lys Gly Leu Leu Leu Tyr Tyr Asp Leu Val Glu Ser Thr Phe Glu Lys
 225 230 235 240
 Phe Lys Leu Pro Ser Arg Gln Val Asp Thr Leu Asp His Phe Gln Lys
 245 250 255
 Cys Phe Leu Ile Leu Lys Leu His His Gln Arg Val Asp Ser Ser Lys
 260 265 270
 Ser Asn Gln Gln Glu Gly Lys Thr Trp Lys Ala Ile Arg Val Asp Leu
 275 280 285
 Val Met Cys Pro Tyr Glu Asn Arg Ala Phe Ala Leu Leu Gly Trp Thr
 290 295 300
 Gly Ser Pro Gln Phe Asn Arg Asp Leu Arg Arg Tyr Ala Thr His Glu
 305 310 315 320
 Arg Lys Met Met Leu Asp Asn His Ala Leu Tyr Asp Lys Thr Lys Arg
 325 330 335
 Val Phe Leu Glu Ala Glu Ser Glu Glu Glu Ile Phe Ala His Leu Gly
 340 345 350
 Leu Asp Tyr Ile Glu Pro Trp Glu Arg Asn Ala
 355 360

<210> SEQ ID NO 110

<211> LENGTH: 363

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: 9-40 without His tag

<400> SEQUENCE: 110

Val Ala Thr Val Ser Glu Tyr Ala Cys Gln Arg Arg Thr Thr Leu Asn
 1 5 10 15
 Asn Tyr Asn Glu Leu Phe Thr Arg Ala Leu Asp Ile Leu Ala Glu Asn
 20 25 30
 Asp Glu Phe Arg Glu Asn Glu Glu Ser Arg Asp Ala Phe Arg Arg Ala
 35 40 45

-continued

Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met	Lys	Asp
50						55					60				
Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile	Ile	Glu
65					70					75					80
Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val	Leu	Asn
				85					90					95	
Asp	Glu	Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe	Gly	Val
			100					105					110		
Gly	Arg	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg	Thr	Leu
		115					120					125			
Glu	Lys	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met	Gln	Lys
	130					135					140				
Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Thr	Arg	Ala
145					150					155					160
Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp	Ala	Phe
				165					170					175	
Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu	Gly	Lys
			180					185					190		
Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly	Ser	Ala
		195					200					205			
Glu	Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp	Glu	Lys
	210					215					220				
Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Phe	Glu	Lys
225					230					235					240
Phe	Lys	Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe	Gln	Lys
				245					250					255	
Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	His	Gln	Arg	Val	Asp	Ser	Ser	Lys
			260					265					270		
Ser	Asn	Gln	Gln	Glu	Gly	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val	Asp	Leu
		275					280					285			
Val	Met	Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly	Trp	Thr
	290					295					300				
Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr	His	Glu
305					310					315					320
Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr	Lys	Arg
				325					330					335	
Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His	Leu	Gly
			340					345					350		
Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala					
		355					360								

<210> SEQ ID NO 111

<211> LENGTH: 365

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: (45-751)(internal His tag)

<400> SEQUENCE: 111

Met	Ala	Glu	Lys	Lys	Ile	Ser	Glu	Tyr	Ala	Cys	Gln	Arg	Arg	Thr	Thr
1				5					10					15	

Leu	Asn	Asn	Tyr	Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu	Asp	Ile	Leu	Ala
		20						25					30		

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Glu	Asn	Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg	Asp	Ala	Phe	Arg
	35						40					45			
Arg	Ala	Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro	Ile	Thr	Ser	Met
	50					55					60				
Lys	Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys	Val	Lys	Arg	Ile
65					70					75					80
Ile	Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu	Val	Lys	Ala	Val
			85					90						95	
Leu	Asn	Asp	Glu	Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe	Thr	Ser	Val	Phe
			100					105					110		
Gly	Val	Gly	Arg	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg	Met	Gly	Phe	Arg
		115					120					125			
Thr	Leu	Glu	Glu	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg	Phe	Thr	Gln	Met
	130					135					140				
Gln	Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val	Ser	Gly	Val	Thr
145					150					155					160
Arg	Ala	Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys	Glu	Ala	Val	Trp
			165					170						175	
Ala	Phe	Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly	Gly	Phe	Arg	Leu
			180					185					190		
Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile	Thr	Ser	Pro	Gly
		195					200					205			
Ser	Ala	Glu	Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val	Ile	Asn	Leu	Trp
	210					215					220				
Glu	Lys	Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val	Glu	Ser	Thr	Leu
225					230					235					240
Glu	Lys	Phe	Lys	Leu	Pro	Ser	Arg	Gln	Val	Asp	Thr	Leu	Asp	His	Phe
			245					250						255	
Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	Arg	Gln	Arg	Val	Asp	Ser
			260					265					270		
Ser	Lys	His	His	His	His	His	His	Lys	Thr	Trp	Lys	Ala	Ile	Arg	Val
		275						280				285			
Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe	Ala	Leu	Leu	Gly
	290					295					300				
Trp	Thr	Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg	Arg	Tyr	Ala	Thr
305					310					315					320
His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu	Tyr	Asp	Lys	Thr
			325					330						335	
Lys	Arg	Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu	Ile	Phe	Ala	His
			340					345					350		
Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn	Ala			
	355						360					365			

<210> SEQ ID NO 112

<211> LENGTH: 369

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: (46-737) (internal His tag)

<400> SEQUENCE: 112

Met	Ala	His	His	Gly	Ser	Glu	Lys	Lys	Ile	Ser	Glu	Tyr	Ala	Cys	Gln
1				5					10					15	

-continued

Arg	Arg	Thr	Thr	Leu	Asn	Asn	Tyr	Asn	Glu	Leu	Phe	Thr	Arg	Ala	Leu
			20					25					30		
Asp	Ile	Leu	Ala	Glu	Asn	Asp	Glu	Phe	Arg	Glu	Asn	Glu	Glu	Ser	Arg
	35					40					45				
Asp	Ala	Phe	Arg	Arg	Ala	Ser	Ser	Val	Leu	Lys	Ser	Leu	Pro	Phe	Pro
	50					55					60				
Ile	Thr	Ser	Met	Lys	Asp	Thr	Glu	Gly	Ile	Pro	Cys	Leu	Gly	Asp	Lys
65				70					75					80	
Val	Lys	Arg	Ile	Ile	Glu	Glu	Ile	Ile	Glu	Asp	Gly	Glu	Ser	Ser	Glu
			85					90						95	
Val	Lys	Ala	Val	Leu	Asn	Asp	Glu	Arg	Tyr	Lys	Ala	Phe	Lys	Leu	Phe
			100					105					110		
Thr	Ser	Val	Phe	Gly	Val	Gly	Arg	Lys	Thr	Ala	Glu	Lys	Trp	Phe	Arg
		115					120					125			
Met	Gly	Phe	Arg	Thr	Leu	Glu	Glu	Ile	Arg	Ser	Asp	Lys	Ser	Leu	Arg
	130					135					140				
Phe	Thr	Gln	Met	Gln	Lys	Ala	Gly	Phe	Leu	Tyr	Tyr	Glu	Asp	Leu	Val
145				150					155					160	
Ser	Gly	Val	Thr	Arg	Ala	Glu	Ala	Glu	Ala	Val	Gly	Val	Leu	Val	Lys
			165					170					175		
Glu	Ala	Val	Trp	Ala	Phe	Leu	Pro	Asp	Ala	Phe	Val	Thr	Met	Thr	Gly
			180					185					190		
Gly	Phe	Arg	Leu	Gly	Lys	Lys	Ile	Gly	His	Asp	Val	Asp	Phe	Leu	Ile
	195					200						205			
Thr	Ser	Pro	Gly	Ser	Ala	Glu	Asp	Glu	Glu	Gln	Leu	Leu	Pro	Lys	Val
	210					215					220				
Ile	Asn	Leu	Trp	Glu	Lys	Lys	Gly	Leu	Leu	Leu	Tyr	Tyr	Asp	Leu	Val
225				230					235					240	
Glu	Ser	Thr	Leu	Glu	Lys	Phe	Lys	Arg	Pro	Ser	Arg	Gln	Val	Asp	Thr
			245					250					255		
Leu	Asp	His	Phe	Gln	Lys	Cys	Phe	Leu	Ile	Leu	Lys	Leu	His	Arg	Gln
		260					265						270		
Arg	Val	Asp	Ser	Ser	Lys	His	His	His	His	His	His	Glu	Thr	Trp	Lys
	275					280						285			
Ala	Ile	Arg	Val	Asp	Leu	Val	Met	Cys	Pro	Tyr	Glu	Asn	Arg	Ala	Phe
	290				295						300				
Ala	Leu	Leu	Gly	Trp	Thr	Gly	Ser	Pro	Gln	Phe	Asn	Arg	Asp	Leu	Arg
305				310					315					320	
Arg	Tyr	Ala	Thr	His	Glu	Arg	Lys	Met	Met	Leu	Asp	Asn	His	Ala	Leu
			325					330					335		
Tyr	Asp	Lys	Thr	Lys	Arg	Val	Phe	Leu	Glu	Ala	Glu	Ser	Glu	Glu	Glu
			340				345					350			
Ile	Phe	Ala	His	Leu	Gly	Leu	Asp	Tyr	Ile	Glu	Pro	Trp	Glu	Arg	Asn
	355					360					365				

Ala

<210> SEQ ID NO 113

<211> LENGTH: 552

<212> TYPE: PRT

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 113

-continued

Met	Ala	Ser	Gln	Lys	Val	Phe	Gly	Ile	Thr	Gly	Pro	Val	Ser	Thr	Val
1				5					10					15	
Gly	Ala	Thr	Ala	Ala	Glu	Asn	Lys	Leu	Asn	Asp	Ser	Leu	Ile	Gln	Glu
			20					25					30		
Leu	Lys	Lys	Glu	Gly	Ser	Phe	Glu	Thr	Glu	Gln	Glu	Thr	Ala	Asn	Arg
		35					40					45			
Val	Gln	Val	Leu	Lys	Ile	Leu	Gln	Glu	Leu	Ala	Gln	Arg	Phe	Val	Tyr
	50					55					60				
Glu	Val	Ser	Lys	Lys	Lys	Asn	Met	Ser	Asp	Gly	Met	Ala	Arg	Asp	Ala
65					70					75					80
Gly	Gly	Lys	Ile	Phe	Thr	Tyr	Gly	Ser	Tyr	Arg	Leu	Gly	Val	His	Gly
			85						90					95	
Pro	Gly	Ser	Asp	Ile	Asp	Thr	Leu	Val	Val	Val	Pro	Lys	His	Val	Thr
			100					105					110		
Arg	Glu	Asp	Phe	Phe	Thr	Val	Phe	Asp	Ser	Leu	Leu	Arg	Glu	Arg	Lys
		115					120					125			
Glu	Leu	Asp	Glu	Ile	Ala	Pro	Val	Pro	Asp	Ala	Phe	Val	Pro	Ile	Ile
	130					135					140				
Lys	Ile	Lys	Phe	Ser	Gly	Ile	Ser	Ile	Asp	Leu	Ile	Cys	Ala	Arg	Leu
145					150					155					160
Asp	Gln	Pro	Gln	Val	Pro	Leu	Ser	Leu	Thr	Leu	Ser	Asp	Lys	Asn	Leu
			165						170					175	
Leu	Arg	Asn	Leu	Asp	Glu	Lys	Asp	Leu	Arg	Ala	Leu	Asn	Gly	Thr	Arg
		180						185					190		
Val	Thr	Asp	Glu	Ile	Leu	Glu	Leu	Val	Pro	Lys	Pro	Asn	Val	Phe	Arg
		195					200					205			
Ile	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala	Gln	Arg	Arg	Ala	Val	Tyr
	210					215					220				
Ala	Asn	Ile	Phe	Gly	Phe	Pro	Gly	Gly	Val	Ala	Trp	Ala	Met	Leu	Val
225					230					235					240
Ala	Arg	Ile	Cys	Gln	Leu	Tyr	Pro	Asn	Ala	Cys	Ser	Ala	Val	Ile	Leu
			245						250					255	
Asn	Arg	Phe	Phe	Ile	Ile	Leu	Ser	Glu	Trp	Asn	Trp	Pro	Gln	Pro	Val
		260						265					270		
Ile	Leu	Lys	Pro	Ile	Glu	Asp	Gly	Pro	Leu	Gln	Val	Arg	Val	Trp	Asn
	275						280					285			
Pro	Lys	Ile	Tyr	Ala	Gln	Asp	Arg	Ser	His	Arg	Met	Pro	Val	Ile	Thr
	290					295					300				
Pro	Ala	Tyr	Pro	Ser	Met	Cys	Ala	Thr	His	Asn	Ile	Thr	Glu	Ser	Thr
305					310					315					320
Lys	Lys	Val	Ile	Leu	Gln	Glu	Phe	Val	Arg	Gly	Val	Gln	Ile	Thr	Asn
			325						330				335		
Asp	Ile	Phe	Ser	Asn	Lys	Lys	Ser	Trp	Ala	Asn	Leu	Phe	Glu	Lys	Asn
		340						345					350		
Asp	Phe	Phe	Phe	Arg	Tyr	Lys	Phe	Tyr	Leu	Glu	Ile	Thr	Ala	Tyr	Thr
		355					360					365			
Arg	Gly	Ser	Asp	Glu	Gln	His	Leu	Lys	Trp	Ser	Gly	Leu	Val	Glu	Ser
	370					375					380				
Lys	Val	Arg	Leu	Leu	Val	Met	Lys	Leu	Glu	Val	Leu	Ala	Gly	Ile	Lys
385					390					395					400
Ile	Ala	His	Pro	Phe	Thr	Lys	Pro	Phe	Glu	Ser	Ser	Tyr	Cys	Cys	Pro

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405	410	415
Thr Glu Asp Asp Tyr Glu Met Ile Gln Asp Lys Tyr Gly Ser His Lys		
420	425	430
Thr Glu Thr Ala Leu Asn Ala Leu Lys Leu Val Thr Asp Glu Asn Lys		
435	440	445
Glu Glu Glu Ser Ile Lys Asp Ala Pro Lys Ala Tyr Leu Ser Thr Met		
450	455	460
Tyr Ile Gly Leu Asp Phe Asn Ile Glu Asn Lys Lys Glu Lys Val Asp		
465	470	475
Ile His Ile Pro Cys Thr Glu Phe Val Asn Leu Cys Arg Ser Phe Asn		
485	490	495
Glu Asp Tyr Gly Asp His Lys Val Phe Asn Leu Ala Leu Arg Phe Val		
500	505	510
Lys Gly Tyr Asp Leu Pro Asp Glu Val Phe Asp Glu Asn Glu Lys Arg		
515	520	525
Pro Ser Lys Lys Ser Lys Arg Lys Asn Leu Ser Ser Gly Glu Asn Leu		
530	535	540
Tyr Phe Gln Gly Ser Ser Gly Ser		
545	550	

<210> SEQ ID NO 114

<211> LENGTH: 674

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Myceliophthora sp.

<400> SEQUENCE: 114

Met Ala Glu Arg Ile Tyr Gly Val Thr Pro Pro Ile Ser Thr Ala Leu		
1	5	10
Pro Thr Glu Glu Glu Lys Arg Leu Asn Asn Ala Leu His Gln Glu Leu		
20	25	30
Arg Ala Gln Gly Thr Phe Glu Ser Pro Ala Glu Thr Glu Lys Arg Lys		
35	40	45
Glu Val Leu Arg Gln Leu Glu Lys Ile Thr Asn Val Phe Val Gln Arg		
50	55	60
Ala Ala Ala Glu Lys Glu Pro Lys Asn Thr Ile Leu Ile Arg Asp Ala		
65	70	75
Ile Gly Arg Val Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val Tyr Gly		
85	90	95
Pro Gly Ser Asp Met Asp Thr Leu Val Val Ala Pro Lys Tyr Val Thr		
100	105	110
Val Glu Gln Tyr Phe Arg Ile Phe Pro Glu Val Leu Val Glu Met Ala		
115	120	125
Pro Pro Gly Ala Ile Thr Asp Leu Thr Pro Val Pro Glu Ala Phe Val		
130	135	140
Pro Ile Ile Lys Phe Glu Phe Ser Gly Ile Ser Ile Asp Leu Ile Phe		
145	150	155
Cys Ser Ile Gln Thr Leu Lys Gln Leu Pro Ala Asp Lys Asn Trp Ser		
165	170	175
Leu Ala Asp Asn Asn Leu Leu Arg Gly Leu Ser Glu Asn Glu Val Arg		
180	185	190
Ser Leu Asn Gly Thr Arg Val Thr Asp Asp Ile Leu Asn Leu Val Pro		

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195					200					205					
Glu	Pro	Ala	Thr	Phe	Lys	Leu	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala
210						215					220				
Gln	Arg	Lys	Ala	Ile	Tyr	Ala	Asn	Ile	Met	Gly	Tyr	Pro	Gly	Gly	Val
225					230					235					240
Ala	Trp	Ala	Met	Leu	Val	Ala	Arg	Val	Cys	Gln	Leu	Tyr	Pro	Lys	Ala
				245					250					255	
Thr	Ser	Ala	Val	Ile	Val	Asn	Lys	Phe	Phe	His	Ile	Met	Leu	Lys	Trp
			260					265					270		
Pro	Trp	Pro	Leu	Pro	Val	Leu	Leu	Lys	Asp	Ile	Glu	Tyr	Gly	Cys	Pro
		275					280					285			
Val	Thr	Arg	Val	Ala	Val	Trp	Asn	Pro	Lys	Ile	Tyr	Ala	Ser	Asp	Arg
290						295					300				
Asn	His	Arg	Met	Pro	Ile	Ile	Thr	Pro	Ser	Tyr	Pro	Ser	Met	Cys	Ala
305					310					315					320
Thr	His	Asn	Val	Gly	Arg	Ser	Ser	Met	Ala	Val	Ile	Lys	Asp	Glu	Leu
			325						330					335	
Glu	Lys	Gly	Val	Gln	Val	Thr	Glu	Asp	Ile	Met	Arg	Gly	Lys	Arg	Pro
			340					345					350		
Trp	Lys	Asp	Leu	Phe	Thr	Lys	His	Thr	Phe	Phe	Thr	Ser	Gly	Phe	Arg
		355					360					365			
Tyr	Tyr	Leu	Thr	Val	Ile	Ser	Ser	Ser	Arg	Thr	Lys	Lys	Ala	Gln	Asn
370						375					380				
Val	Trp	Ser	Gly	Phe	Val	Glu	Ser	Arg	Val	Arg	Leu	Leu	Val	Asn	Lys
385					390					395					400
Leu	Glu	Met	His	Pro	Ser	Ile	Ser	Leu	Ala	Arg	Pro	Phe	Asn	Lys	Gly
			405						410					415	
Tyr	Asp	Arg	Glu	His	Arg	Cys	Lys	Asn	Asn	Ala	Gln	Leu	Glu	Glu	Val
			420					425					430		
Val	Ser	Leu	Gly	Ser	Leu	Ala	Tyr	Met	Tyr	Lys	Pro	Ala	Thr	Gly	Glu
	435						440					445			
Ala	Lys	Ala	Glu	Ala	Lys	Lys	Glu	Val	Lys	Thr	Glu	Val	Lys	His	Glu
	450					455					460				
Ser	Gly	Ala	Ser	Val	Lys	Asn	Glu	Ala	Pro	Glu	Ala	Lys	Gly	Glu	Asp
465					470					475					480
Gly	Val	Arg	Ala	Lys	Arg	Glu	Asn	Gly	Asp	Glu	Ala	Gln	Leu	Pro	Pro
			485						490					495	
Ala	Thr	Gly	Ile	Lys	Pro	Glu	Pro	Thr	Asp	Gly	Ala	Gly	Ala	Asp	Val
			500					505					510		
Lys	Leu	Glu	Asp	Ile	Pro	Val	Lys	Lys	Asp	Glu	Pro	Glu	Glu	Met	Thr
	515						520					525			
Ile	Tyr	Thr	Thr	Asn	His	Tyr	Ile	Gly	Leu	Gln	Leu	Val	Glu	Gly	Ala
530						535						540			
Lys	Thr	Leu	Asp	Leu	Ser	Arg	Glu	Val	Asn	Asp	Trp	Lys	Ala	Met	Cys
545					550					555					560
Thr	Ser	Asn	Glu	Leu	Tyr	Glu	Glu	Gly	Thr	Met	Phe	Leu	Ser	Ile	Gln
			565						570					575	
His	Val	Arg	Asn	Thr	Ala	Leu	Pro	Asp	Asp	Val	Phe	Glu	Pro	Gly	Glu
			580					585					590		
Thr	Arg	Pro	Arg	Pro	Ala	Lys	Lys	Ser	Leu	Lys	Arg	Val	Ala	Ser	Glu
	595						600					605			

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Asp Pro Gly Asn Lys Gln Thr Gln Pro Pro Ala Lys Lys Gln Val Gln
 610 615 620
 Asp Lys Ala Pro Pro Ala Ala Gln Gln Gln Gln Gln Gln His Gln
 625 630 635 640
 Gln Gln Gln Gln Pro Ala Ser Thr Ala Ala Ala Ala Ala Gly Ser Ser
 645 650 655
 Gly Glu Asn Leu Tyr Phe Gln Gly Ser Ser Gly Ser His His His His
 660 665 670
 His His

 <210> SEQ ID NO 115
 <211> LENGTH: 672
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Thielavia sp.

 <400> SEQUENCE: 115
 Met Ala Glu Arg Thr Tyr Gly Val Thr Pro Pro Ile Ser Thr Ala Leu
 1 5 10 15
 Pro Thr Glu Gln Glu Lys Ala Leu Asn Lys Ala Leu His Asp Glu Leu
 20 25 30
 Arg Ala Gln Gly Thr Phe Glu Ser Arg Ala Glu Thr Glu Lys Arg Lys
 35 40 45
 Glu Val Leu Ala Gln Leu Glu Lys Ile Thr Asn Ala Phe Val Gln Arg
 50 55 60
 Ala Ala Arg Glu Lys His Gly Lys Asn Ala Ile Leu Ile Arg Asp Ala
 65 70 75 80
 Ile Gly Arg Val Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val Tyr Gly
 85 90 95
 Pro Gly Ser Asp Met Asp Thr Leu Val Val Ala Pro Lys Tyr Val Thr
 100 105 110
 Val Glu Gln Tyr Phe Arg Ile Phe Pro Glu Val Leu Val Glu Met Ala
 115 120 125
 Pro Pro Ala Ala Ile Thr Asp Leu Thr Pro Val Pro Glu Ala Phe Val
 130 135 140
 Pro Ile Ile Lys Phe Glu Phe Ser Gly Ile Ser Ile Asp Leu Ile Phe
 145 150 155 160
 Cys Ser Ile Gln Thr Leu Ile Gln Leu Pro Ala Asp Lys Ser Trp Ser
 165 170 175
 Leu Ala Asp Asn Asn Leu Leu Arg Gly Leu Ser Glu Asn Ala Val Arg
 180 185 190
 Ser Leu Asn Gly Thr Arg Val Thr Asp Glu Ile Leu His Leu Val Pro
 195 200 205
 Glu Pro Ala Thr Phe Lys Leu Ala Leu Arg Ala Ile Lys Leu Trp Ala
 210 215 220
 Gln Arg Lys Ala Ile Tyr Ala Asn Ile Met Gly Tyr Pro Gly Gly Val
 225 230 235 240
 Ala Trp Ala Met Leu Val Ala Arg Val Cys Gln Leu Tyr Pro Lys Ala
 245 250 255
 Thr Ser Ala Val Ile Val Asn Lys Phe Phe Asn Ile Met Leu Lys Trp
 260 265 270

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Pro	Trp	Pro	Leu	Pro	Val	Leu	Leu	Lys	Asp	Ile	Glu	Tyr	Asn	Gly	Pro	275	280	285
Val	Thr	Arg	Val	Pro	Val	Trp	Asn	Pro	Lys	Leu	Tyr	Ala	Ser	Asp	Arg	290	295	300
Asn	His	Lys	Met	Pro	Ile	Ile	Thr	Pro	Ala	Tyr	Pro	Ser	Met	Cys	Ala	305	310	315
Thr	His	Asn	Val	Gly	Arg	Ser	Ser	Met	Val	Val	Ile	Gln	Gln	Glu	Leu	325	330	335
Lys	Lys	Gly	Ala	Glu	Val	Thr	Glu	Glu	Ile	Met	Leu	Gly	Arg	Arg	Pro	340	345	350
Trp	Lys	Asp	Leu	Phe	Thr	Lys	His	Thr	Phe	Phe	Thr	Ser	Gly	Phe	Lys	355	360	365
Tyr	Tyr	Leu	Thr	Val	Ile	Ser	Ser	Ser	Arg	Thr	Lys	Lys	Ala	Gln	Asn	370	375	380
Val	Trp	Ser	Gly	Phe	Ile	Glu	Ser	Arg	Val	Arg	Leu	Leu	Val	Asn	Lys	385	390	395
Ile	Glu	Met	His	Pro	Ser	Ile	Ala	Leu	Ala	Arg	Pro	Phe	Asn	Lys	Gly	405	410	415
Tyr	Asp	Arg	Met	His	Arg	Cys	Lys	Asn	Asp	Ala	Gln	Val	Glu	Glu	Val	420	425	430
Val	Ser	Ala	Gly	Ser	Leu	Ala	Tyr	Val	Tyr	Thr	Pro	Pro	Ala	Phe	Gly	435	440	445
Asp	Glu	Lys	Val	Lys	Ser	Glu	Thr	Lys	Ser	Glu	Val	Lys	Gln	Glu	Val	450	455	460
Lys	Gln	Glu	Val	Arg	Gln	Asp	Asp	Val	Ile	Gln	Asp	Gly	Val	Pro	Val	465	470	475
Lys	Gln	Glu	Lys	Ala	Glu	Val	Arg	Ala	Glu	Asp	Gly	Val	Arg	Ile	Lys	485	490	495
Arg	Glu	Leu	Ser	Glu	Glu	Val	Gln	Leu	Pro	Pro	Pro	Thr	Asn	Val	Lys	500	505	510
Pro	Glu	Pro	His	Asp	Asp	Glu	Ala	Lys	Val	Lys	Leu	Glu	Asp	Ile	Pro	515	520	525
Glu	Lys	Glu	Ala	Gln	Pro	Glu	Asp	Met	Glu	Ile	Tyr	Thr	Thr	Asn	His	530	535	540
Tyr	Ile	Gly	Leu	Gln	Leu	Val	Glu	Gly	Ala	Lys	Ser	Leu	Asp	Leu	Ser	545	550	555
Arg	Glu	Val	Asn	Asp	Trp	Lys	Ala	Met	Cys	Met	Ser	Asn	Glu	Leu	Tyr	565	570	575
Glu	Glu	Gly	Leu	Met	Phe	Leu	Ser	Ile	Gln	His	Leu	Lys	Asn	Thr	Ala	580	585	590
Leu	Pro	Asp	Asp	Val	Phe	Glu	Pro	Gly	Glu	Val	Lys	Pro	Arg	Pro	Gly	595	600	605
Lys	Lys	Val	Leu	Lys	Arg	Gly	Ala	Ser	Glu	Glu	Pro	Ser	Lys	Gln	Gln	610	615	620
Pro	Pro	Ala	Lys	Arg	Gln	Ala	His	Val	Gln	Pro	Arg	Ala	Pro	Ala	Ala	625	630	635
Gln	Gln	Pro	Ser	Ser	Thr	Ala	Thr	Ala	Ala	Ala	Gly	Ser	Ser	Gly	Glu	645	650	655
Asn	Leu	Tyr	Phe	Gln	Gly	Ser	Ser	Gly	Ser	His	His	His	His	His	His	660	665	670

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<210> SEQ ID NO 116
<211> LENGTH: 566
<212> TYPE: PRT
<213> ORGANISM: Pyronema omphalodes

<400> SEQUENCE: 116

Met Ser Ala Pro Pro Lys Gln Tyr Gly Thr Thr Pro Pro Ile Asn Thr
 1          5          10          15
Asn Pro Pro Thr Lys Glu Asp Met Glu Arg Asn Ser Glu Leu Val Gln
 20          25          30
Glu Leu Lys Asn Gln Asn Cys Phe Glu Glu Lys Ala Glu Ser Asp Lys
 35          40          45
Arg Val Lys Val Leu Glu Ile Leu Gln Thr Leu Gly Glu Ala Phe Val
 50          55          60
Lys Lys Ala Cys Ala Ala Lys Gly Leu Pro Asp His Leu Val Asn Asn
 65          70          75          80
Ser Gly Gly Lys Ile Phe Thr Phe Gly Ser Tyr Arg Leu Gly Ala Tyr
 85          90          95
Gly Pro Gly Ser Asp Ile Asp Thr Leu Leu Val Val Pro Ala His Ile
100          105          110
Glu Arg Ala Asp Phe Phe Gln His Val Pro Lys Met Leu Ser Glu Leu
115          120          125
Asn Pro Pro Ala Glu Glu Val Ser Pro Val Pro Asp Ala Phe Val Pro
130          135          140
Ile Ile Lys Phe Glu Leu Gln Ser Ile Ser Ile Asp Leu Ile Phe Ala
145          150          155          160
Lys Val Pro Ser Val Asn Ser Ile Pro Arg Asp Met Thr Leu Glu Asn
165          170          175
Lys Asp Ile Leu Lys Gly Cys Asp Glu Ala Asn Leu Arg Gly Leu Asn
180          185          190
Gly Val Arg Leu Thr Asp Glu Leu Leu Gly Leu Val Pro His Pro Val
195          200          205
Asn Phe Arg Met Ala Leu Arg Ala Ile Lys Leu Trp Ala Lys Ser Arg
210          215          220
Ala Ile Tyr Ala Asn Val Met Gly Phe Pro Gly Gly Ile Ala Trp Ala
225          230          235          240
Met Leu Val Ala Lys Ile Cys Gln Leu Tyr Pro Met Ala Val Ser Ala
245          250          255
Val Ile Val Cys Lys Phe Phe Thr Ile Tyr Thr Lys Trp Lys Trp Pro
260          265          270
Gln Pro Val Leu Leu Lys Asp Ile Leu Asp Glu Gly Gly Gln Gly Ala
275          280          285
Gly Leu Arg Val Trp Asn Pro Lys Ile Tyr Ala Ser Asp Arg Gly His
290          295          300
Leu Met Pro Val Ile Thr Pro Asn Tyr Pro Cys Met Cys Ser Thr His
305          310          315          320
Asn Ile Thr Lys Ser Thr Lys Ala Val Ile Leu Arg Glu Met Asp Arg
325          330          335
Ser Leu Asn Ile Ile His Glu Ile Met Asp Gly Lys Ser Pro Trp Ser
340          345          350
Lys Leu Phe Ala Pro His Thr Phe Phe Thr Gln Asp Tyr Arg Tyr Tyr
355          360          365

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Leu Arg Val Ile Ser Ala Ala Arg Thr Lys Asp Glu Gln Leu Leu Trp
 370 375 380
 Ser Gly Leu Val Glu Ser Lys Leu Arg His Leu Val Ser Lys Leu Glu
 385 390 395 400
 Met Leu Asp Asn Ile Gln Leu Ala His Pro Phe Asn Lys Gly Phe Glu
 405 410 415
 Tyr Glu Val Glu Cys Asn Asn Glu Asp Glu Val Met Arg Val Ile Arg
 420 425 430
 Gly Glu Asn Ile Asp Gln Ser Glu Ala Asn Gly Thr Ala Val Asp Ser
 435 440 445
 Asn Gly Lys Ser Ser Lys Val Lys Ala Tyr Thr Thr Ser Phe Tyr Val
 450 455 460
 Gly Leu Val Leu Asp Thr Ser Lys Ser Lys Gln Phe Asp Ile Ser Trp
 465 470 475 480
 Ala Cys His Glu Phe Tyr Asp Ile Cys Lys Thr Trp Asn Leu Tyr Asn
 485 490 495
 Asp Asp Met His Ser Ile His Val Val Asn Thr Arg Asn Phe Glu Leu
 500 505 510
 Pro Asp Asn Val Phe Lys Pro Gly Glu Thr Lys Pro Thr Lys Lys Thr
 515 520 525
 Lys Thr Ile Lys Arg Lys Ile Pro Asn Ala Ala Gly Lys Lys Arg Gly
 530 535 540
 Ala Asp Glu Ile Ser Ala Pro Asp Ser Lys Arg Ala His Ile Asn Glu
 545 550 555 560
 Val Pro Pro Asn Ala Asn
 565

<210> SEQ ID NO 117

<211> LENGTH: 598

<212> TYPE: PRT

<213> ORGANISM: Tilletia indica

<400> SEQUENCE: 117

Met Thr Lys His Leu Gly Val Thr Pro Ile Val Ser Asp Ala Ser Pro
 1 5 10 15
 Thr Ala His Glu Ile Ala Leu Glu Thr Gln Leu Val Thr Glu Leu Thr
 20 25 30
 Thr Gln Asn Cys Phe Glu Ser Pro Gln Asp Ser Arg Leu Arg Glu Val
 35 40 45
 Val Leu Gly Lys Ala Asp Lys Leu Val Lys Glu Phe Val Tyr Arg Ala
 50 55 60
 Ala Ile Ala Arg Gly Leu Pro Glu Ser Ala Ala Arg Glu Leu Gly Gly
 65 70 75 80
 Lys Ile Phe Thr Phe Gly Ser Tyr Arg Leu Gly Val His Gly Pro Gly
 85 90 95
 Ser Asp Ile Asp Thr Leu Cys Val Val Pro Lys His Val Gln Arg Glu
 100 105 110
 Asp Phe Phe Thr Ile Phe Glu Gln Met Leu Arg Gln Arg Glu Asp Val
 115 120 125
 Thr Asp Val Ala Pro Val Pro Glu Ala Tyr Val Pro Leu Ile Ala Val
 130 135 140
 Asn Phe Met Gly Ile Ala Ile Asp Phe Leu Phe Ala Arg Leu Ser Leu
 145 150 155 160

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Ser	Arg	Ile	Asp	Asp	Ser	Leu	Asp	Leu	Ser	Asp	Asn	Thr	Ile	Leu	Lys
				165						170				175	
Asn	Met	Asp	Asp	Gln	Asp	Ile	Arg	Ser	Val	Gly	Gly	Ser	Arg	Val	Thr
				180					185					190	
Asp	Glu	Ile	Leu	Arg	Leu	Val	Pro	Asn	Val	Glu	Thr	Phe	His	Thr	Ala
				195				200					205		
Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala	Gln	Arg	Arg	Ala	Ile	Tyr	Lys	Asn
				210				215				220			
Met	Val	Gly	Phe	Pro	Gly	Gly	Val	Ala	Trp	Ala	Met	Leu	Val	Ala	Arg
						230					235				240
Ile	Cys	Gln	Leu	Tyr	Pro	Asn	Gly	Asn	Ala	Ala	Val	Ile	Ile	Glu	Arg
				245					250					255	
Phe	Phe	Ile	Ile	Met	Phe	Gln	Trp	Asn	Trp	Pro	Gln	Pro	Val	Met	Leu
				260				265						270	
Lys	His	His	Ile	Tyr	Glu	Ala	Asp	Pro	Ile	Leu	Lys	Val	Trp	Asn	Pro
				275				280					285		
Arg	Leu	Tyr	Ala	Ala	Asp	Arg	Ser	His	Arg	Met	Pro	Ile	Ile	Thr	Pro
				290				295				300			
Ala	Tyr	Pro	Gly	Met	Cys	Ala	Thr	His	Asn	Ile	Thr	Ala	Ser	Thr	Gln
						310					315				320
Ala	Val	Leu	Thr	Leu	Glu	Phe	Lys	Arg	Gly	Ile	Glu	Ile	Met	Ser	Ala
						325				330				335	
Met	Ser	Ala	Ser	Ile	Thr	Asn	Pro	Ala	Leu	Pro	Gln	Thr	Thr	Trp	Ala
						340			345					350	
Thr	Leu	Phe	Glu	Ser	Arg	Arg	Phe	Phe	Glu	Glu	Tyr	Lys	Tyr	Tyr	Leu
								360					365		
Gln	Ile	Ile	Ala	Ser	Ser	Gly	Ser	Ala	Asp	Leu	Gln	Leu	Lys	Trp	Ala
						375						380			
Gly	Thr	Val	Glu	Ser	Lys	Leu	Arg	His	Leu	Val	Leu	Ser	Thr	Glu	Ala
						390					395				400
Gln	Pro	Gly	Val	Lys	Ile	Ala	His	Pro	Tyr	Thr	Arg	Glu	Ile	Asp	Leu
						405				410				415	
Val	Ser	Glu	Cys	His	Thr	Asp	Glu	Glu	Val	Arg	Arg	Val	Ala	Thr	Gly
						420			425				430		
Asp	Val	Pro	Pro	Glu	Val	Ala	Ala	Arg	Ala	Arg	Thr	Val	Glu	Gly	Glu
							440						445		
Gly	Val	Leu	Pro	Glu	Ser	Lys	Glu	Val	Asp	Lys	Leu	Lys	Glu	Ile	Glu
						455					460				
Ser	Lys	Glu	Glu	Glu	Ala	Arg	Lys	Lys	Gly	Arg	Arg	Thr	Ile	Trp	Thr
						470				475					480
Thr	Thr	Phe	Tyr	Ile	Gly	Leu	Ala	Leu	His	Asp	Gly	Ser	Asp	Asp	Lys
						485				490				495	
Thr	Gly	Gln	Ser	Ala	Pro	Lys	Arg	Leu	Asp	Leu	Val	Gly	Pro	Ile	Gln
						500			505				510		
Asp	Phe	Lys	Gly	Arg	Cys	Thr	Glu	Trp	Ala	Thr	Tyr	Asp	Glu	Ser	Thr
								520					525		
Met	Gly	Val	Val	Val	Arg	His	Val	Lys	Arg	Ser	Gln	Leu	Pro	Asp	Ser
						535						540			
Val	Phe	Pro	Asp	Gly	Lys	Arg	Pro	Ala	Ala	Pro	Val	Lys	Lys	Val	Val
						550				555					560

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Lys Lys Arg Lys Arg Asn Ala Gly Met Ala Glu Ala Ala Ala Glu Thr
565 570 575

Lys Glu Glu Glu Gly Ala Asp Glu Gln Glu Ala Lys Arg Ala Arg Thr
580 585 590

Asn Gly Asp Ser Ala Ala
595

<210> SEQ ID NO 118

<211> LENGTH: 574

<212> TYPE: PRT

<213> ORGANISM: Clathrospora elynae

<400> SEQUENCE: 118

Met Asn Gly Pro Pro Lys Thr Gln Tyr Gly Val Thr Ser Ala Ile Ser
1 5 10 15

Glu Ala Ala Pro Thr Glu Glu Asp Arg Leu Leu Asn Asp Lys Leu Ile
20 25 30

Glu Thr Leu Lys Arg Glu Asn Val Phe Glu Thr Pro Glu Gly Asn Ala
35 40 45

Lys Arg Glu Glu Val Ile Leu His Leu Gln Lys Val Val Glu Glu Phe
50 55 60

Val Arg Arg Val Gly Lys Gln Lys Gly Val Pro Gln Ser Thr Ile Asp
65 70 75 80

Ala Ala Gly Gly Lys Val Phe Thr Phe Gly Ser Tyr Ala Leu Gly Val
85 90 95

His Gly Pro Ser Ser Asp Ile Asp Thr Leu Val Val Ala Pro Lys Phe
100 105 110

Val Thr Ile Asp Glu Phe Phe Gln Thr Phe Pro Pro Thr Phe Lys Glu
115 120 125

Met Ser Arg Val Glu Asp Ile Lys Glu Phe Val Pro Val Glu Asp Ala
130 135 140

Phe Val Pro Ile Ile Lys Met Glu Tyr Arg Gly Val Ser Ile Asp Leu
145 150 155 160

Leu Phe Ala Ser Leu Pro Arg Met Ala Ser Ile Pro Lys Asp Met Asp
165 170 175

Thr Ile Asp Lys Lys Asn Leu Glu Gly Leu Gly Glu Ser Ala Thr Arg
180 185 190

Ser Val Asn Gly Thr Arg Val Thr Lys Glu Leu Leu Ala Ala Val Pro
195 200 205

Gln Asn Asn Ser Phe Arg His Ala Leu Arg Ala Ile Lys Leu Trp Ser
210 215 220

Ser Arg Arg Gly Ile Tyr Gly Ala Val Phe Gly Tyr Pro Gly Gly Val
225 230 235 240

Ala Trp Ala Ile Met Val Ala Arg Ile Cys Gln Leu Tyr Pro Phe Ala
245 250 255

Asn Gly Ala Thr Ile Val Ser Lys Phe Phe Ser Leu Met Tyr Lys Trp
260 265 270

Thr Trp Pro Arg Pro Val Met Leu Lys His Ile Glu Glu Gly Ala Met
275 280 285

Gly Leu Arg Val Trp Asn Pro Gln Ile Tyr Gly Gly Asp Arg Ala His
290 295 300

Leu Met Pro Ile Ile Thr Pro Ala Phe Pro Ser Met Cys Ala Thr His
305 310 315 320

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Thr Val Met Pro Ser Thr Leu Arg Ile Met Lys Glu Glu Phe Gly Arg
      325                      330                      335
Ala Asp Lys Ile Leu Gln His Val Phe Ala Gly Thr Lys Lys Trp Asp
      340                      345                      350
Ala Leu Phe Glu Arg His Ser Phe Phe Thr Lys Asp His Lys Tyr Tyr
      355                      360                      365
Leu Ser Val Val Ala Ala Ser Arg Thr Lys Glu Ala Asn Ser Thr Phe
      370                      375                      380
Ser Gly Leu Val Gln Ser Lys Ile Arg His Ile Val Lys Gly Ile Asp
      385                      390                      395                      400
Asp Gly Gln Thr Gly Val Asp Ile Ala Arg Pro Tyr Ile Asp Tyr Phe
      405                      410                      415
Glu Arg Tyr His Arg Cys Lys Asp Glu Asp Gln Ile Phe Gln Val Cys
      420                      425                      430
Gln Gly Lys Leu Asp His Met Ile Pro Ala Ser Glu Leu Pro Ala Glu
      435                      440                      445
Gly Thr Ala Pro Ala Asn Gly Asp Ser His Ile Met Tyr Thr Thr Thr
      450                      455                      460
Phe Tyr Ile Gly Leu Thr Leu Pro Leu Gly Thr Ala Thr Glu Gly Thr
      465                      470                      475                      480
Lys Ser Leu Asp Ile Ser Tyr Pro Val Ser Gln Phe Lys Asn Phe Ile
      485                      490                      495
Thr Asp Ala Asp Lys Phe Asp Lys Glu Thr Met Ser Val Lys Val Val
      500                      505                      510
His Thr Arg Ser Ser Ala Leu Pro Asp Asp Val Phe Val Gln Gly Glu
      515                      520                      525
Thr Arg Pro Lys Lys Pro Val Lys Glu Lys Lys Lys Lys Glu Ala Lys
      530                      535                      540
Ser Ala Lys Arg Gln Phe Ala Asp Thr Gly Leu Asp Val Arg Asp Ser
      545                      550                      555                      560
Arg Leu Thr Lys Pro Ser Gln Arg Ser Ser Ala Ser Arg Glu
      565                      570

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<210> SEQ ID NO 119

<211> LENGTH: 546

<212> TYPE: PRT

<213> ORGANISM: Drechslereella brochopaga

<400> SEQUENCE: 119

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Met Pro Ala Ser Val Asp Val Leu Lys Arg Leu Gln Lys Ile Thr Glu
1      5      10      15
Glu Phe Val Arg Gln Val Tyr Thr Ser Lys Asn Gln Asn Glu Leu Thr
20     25     30
Thr Asn Ser Ala Gly Gly Lys Val Phe Thr Tyr Gly Ser Tyr Arg Leu
35     40     45
Gly Val Val Gly Pro Gly Ser Asp Ile Asp Thr Leu Val Val Ala Pro
50     55     60
Lys Leu Val Thr Arg Glu Asp Phe Phe Lys Phe Tyr Pro Pro Leu Leu
65     70     75     80
Lys Ala Leu Asn Lys Ala Asn Glu Pro Pro Val Ile Glu Glu Leu Ala
85     90     95
Pro Val Pro Asp Ala Phe Val Pro Ile Ile Lys Phe Val Met Ser Gly

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100							105					110				
Ile	Ser	Ile	Asp	Leu	Ile	Phe	Cys	Arg	Leu	Gly	Val	Ala	Gln	Val	Pro	
	115						120					125				
Ala	Asp	Met	Thr	Leu	Glu	Asp	Lys	Asn	Leu	Leu	Arg	Gly	Leu	Asp	Glu	
	130					135					140					
Lys	Glu	Leu	Arg	Ser	Leu	Asn	Gly	Thr	Arg	Val	Thr	Asp	Glu	Ile	Leu	
145					150					155					160	
Thr	Leu	Val	Pro	Val	Pro	Ala	Val	Phe	Lys	His	Ala	Leu	Arg	Ala	Ile	
			165					170						175		
Lys	Ile	Trp	Ala	Lys	Cys	Arg	Ala	Ile	Tyr	Gly	Asn	Val	Tyr	Gly	Phe	
		180						185					190			
Pro	Gly	Gly	Val	Ala	Trp	Ala	Met	Leu	Val	Ala	Arg	Ile	Cys	Gln	Leu	
		195					200					205				
Tyr	Pro	Ser	Ala	Val	Ser	Ala	Val	Ile	Val	Ser	Lys	Phe	Phe	Arg	Ile	
	210					215					220					
Leu	Gly	Gln	Trp	Asn	Trp	Pro	Gln	Pro	Val	Leu	Leu	Lys	Pro	Ile	Glu	
225				230						235					240	
Asp	Gly	Pro	Leu	Asn	Val	Arg	Val	Trp	Asn	Pro	Lys	Leu	Tyr	Pro	Ser	
			245					250					255			
Asp	Arg	Asn	His	Leu	Met	Pro	Ile	Ile	Thr	Pro	Ala	Tyr	Pro	Ser	Met	
		260					265					270				
Cys	Ser	Thr	His	Asn	Ile	Thr	Pro	Ser	Thr	Lys	Ala	Val	Ile	Leu	Gly	
		275					280					285				
Glu	Met	Ala	Lys	Ala	Ala	Asp	Ile	Val	Asp	Arg	Ile	Ile	Thr	Gly	Asn	
	290					295					300					
Gly	Asn	Trp	Asn	Gln	Leu	Phe	Gln	Arg	His	Thr	Phe	Phe	Thr	Lys	Asp	
305				310						315					320	
Tyr	Lys	Tyr	Tyr	Leu	Thr	Val	Lys	Ala	Ser	Ala	Arg	Asp	Gln	Glu	Gln	
			325					330					335			
Ser	Val	Lys	Trp	Ser	Gly	Leu	Val	Glu	Ser	Lys	Ile	Arg	His	Leu	Val	
		340					345					350				
Met	Lys	Leu	Glu	Met	Leu	Ser	Asp	Val	Ile	Ala	Ser	Ala	Arg	Pro	Tyr	
	355					360						365				
Val	Lys	Pro	Phe	Glu	Lys	Val	Gln	Tyr	Cys	Arg	Asn	Gln	Glu	Glu	Ala	
	370				375						380					
His	Lys	Ile	Ala	Met	Gly	Gln	His	Val	Pro	Asp	Ser	Pro	Pro	Ala	Glu	
385				390						395					400	
Ala	Ala	Asn	Ala	Ala	Thr	Gly	Glu	Thr	Ala	Gly	Glu	Val	Ser	Glu	Gln	
			405					410						415		
Glu	Lys	Glu	Gly	Gly	Val	Pro	Val	Trp	Ile	Thr	Thr	Phe	His	Ile	Gly	
		420						425					430			
Ile	Glu	Leu	Thr	Pro	Gly	Met	Val	Gly	Gln	Val	Asn	Ile	Ser	Trp	Pro	
	435					440						445				
Ala	Gln	Glu	Phe	Tyr	Ile	Met	Cys	His	Asn	Trp	Glu	Gly	Tyr	Glu	Glu	
	450					455					460					
Asn	Met	His	Cys	Val	Lys	Leu	Glu	Leu	Leu	Arg	Asn	Trp	Asn	Leu	Pro	
465				470						475				480		
Asp	Glu	Cys	Phe	Asp	Phe	Ser	Lys	Gly	Asp	Ala	Lys	Pro	Ser	Arg	Pro	
			485						490					495		
Ile	Val	Lys	Arg	Lys	Val	Ile	Arg	Lys	Val	Val	Ser	Ser	Ser	Glu	Val	
		500						505					510			

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Ser Gln Ser Gly Met Lys Arg Thr His Ala Glu Glu Asn Pro Asn Ala
 515 520 525
 Asp Glu Gly Asp Val Lys Arg Thr Lys Val Asn Ala Pro Val Thr Val
 530 535 540
 Gln Gly
 545
 <210> SEQ ID NO 120
 <211> LENGTH: 632
 <212> TYPE: PRT
 <213> ORGANISM: Magnaporthiopsis poae
 <400> SEQUENCE: 120
 Met Ala Glu Arg Gln Leu Gly Val Thr Pro Pro Val Ser Val Ser Leu
 1 5 10 15
 Pro Ser Glu Phe Glu Leu Gln Ala Thr Asp Ala Leu Leu Ala Glu Leu
 20 25 30
 Arg Ala Gln Asn Ser Phe Glu Ser Pro Leu Glu Thr Gln Lys Arg His
 35 40 45
 Arg Val Leu Ala Ser Leu Gln Thr Ile Ala Asp Ala Phe Val Lys Val
 50 55 60
 Val Ala Gln Glu Arg Glu Pro His Asn Ala Val Leu Ile Lys Asp Ala
 65 70 75 80
 Arg Ala Lys Val Phe Ala Tyr Gly Ser Leu Arg Leu Gly Val Trp Gly
 85 90 95
 Pro Gly Ser Asp Ile Asp Thr Leu Val Val Gly Pro Arg Tyr Val Thr
 100 105 110
 Arg Glu Asp Tyr Phe Lys His Phe Pro Asp Leu Leu Val Lys Met Ala
 115 120 125
 Pro Pro Gly Ala Ile Thr Asp Leu Ala Val Val Gln Glu Ala Phe Val
 130 135 140
 Pro Ile Ile Lys Phe Glu Tyr His Gly Ile Ser Ile Asp Leu Ile Phe
 145 150 155 160
 Ser Arg Ile Ala Thr Leu Lys Gln Leu Pro Ser Asp Pro Ala Trp Asp
 165 170 175
 Leu Lys Asp Asn Asn Leu Leu Arg Gly Leu Asp Glu Lys Glu Leu Arg
 180 185 190
 Ser Val Asn Gly Thr Arg Val Thr Asp Glu Ile Leu Ser Leu Val Pro
 195 200 205
 Glu Gln Asn Ile Phe Arg Thr Ala Leu Arg Ala Ile Lys Leu Trp Ala
 210 215 220
 Gln Arg Arg Ala Val Tyr Gly Asn Ile Met Gly Phe Pro Gly Gly Val
 225 230 235 240
 Ala Trp Ala Met Leu Val Ala Arg Val Cys Gln Leu Tyr Pro Lys Ala
 245 250 255
 Thr Ser Ser Val Ile Val Asn Lys Phe Tyr Ser Ile Leu Leu Gln Trp
 260 265 270
 Pro Trp Pro Gln Pro Val Leu Leu Lys Pro Ile Gly Asp Gly Pro Leu
 275 280 285
 Gln Val Arg Val Trp Asn Pro Arg Leu Tyr Lys Gly Asp Ser Tyr His
 290 295 300
 Leu Met Pro Val Ile Thr Pro Ala Tyr Pro Ser Met Cys Ala Thr Phe

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<210> SEQ ID NO 121
<211> LENGTH: 631
<212> TYPE: PRT
<213> ORGANISM: Cryptococcus depauperatus

<400> SEQUENCE: 121
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Met Leu Gly Val Thr Pro Pro Ile Ser Ile Glu Ala Pro Lys Pro Gly
1 5 10 15
Asp Ile His Ser Ser Glu Ala Leu Met Thr Asp Leu Ile Ala Met Asn
20 25 30

-continued

Gln	Phe	Glu	Ser	Asp	Ser	Glu	Arg	Lys	Ile	Arg	Glu	Lys	Leu	Leu	Ser
		35					40					45			
Asn	Ile	Ala	Gln	Leu	Val	Val	Lys	Phe	Val	His	Asp	Val	Ser	Leu	Lys
	50					55					60				
Ile	Gly	Met	Ser	Glu	Lys	Met	Ala	Ser	Glu	Ser	Gly	Gly	Arg	Ile	Tyr
65					70					75				80	
Thr	Ser	Gly	Ser	Tyr	Arg	Leu	Gly	Val	His	Gly	Pro	Gly	Ser	Asp	Ile
			85						90					95	
Asp	Thr	Ile	Cys	Val	Cys	Pro	Arg	His	Ile	Tyr	Arg	Glu	His	Phe	Phe
			100					105					110		
Gly	Glu	Phe	Gln	Glu	Met	Leu	Arg	Ala	Trp	Pro	Glu	Val	Thr	Glu	Ile
		115					120					125			
Thr	Ala	Val	Thr	Gly	Ala	Phe	Val	Pro	Val	Met	Lys	Thr	Val	Ile	Ser
	130					135					140				
Gly	Val	Glu	Val	Asp	Leu	Leu	Phe	Ala	Arg	Val	Asn	Leu	Pro	Glu	Ala
145					150					155					160
Gly	Asp	Ser	Leu	Asp	Ile	Glu	Lys	Asp	Glu	Ile	Leu	Arg	Gly	Val	Asp
			165					170						175	
Asp	Ala	Ser	Gln	Arg	Ser	Leu	Asn	Gly	Pro	Arg	Val	Thr	Asp	Met	Ile
			180					185					190		
Leu	Asn	Leu	Val	Pro	Asp	Val	Thr	Thr	Phe	Arg	Thr	Ala	Leu	Arg	Ser
	195						200					205			
Ile	Arg	Leu	Trp	Ala	Arg	Arg	Arg	Gly	Ile	Tyr	Ser	Asn	Val	Leu	Gly
	210				215						220				
Phe	Pro	Gly	Gly	Val	Ala	Trp	Ala	Leu	Leu	Thr	Ala	Arg	Ile	Cys	Gln
225					230					235					240
Leu	Tyr	Pro	Asn	Ala	Thr	Pro	Ser	Ile	Ile	Val	Gly	Lys	Phe	Phe	Pro
			245					250						255	
Ile	Tyr	Tyr	Gln	Trp	Asn	Trp	Pro	Gln	Pro	Val	Ile	Leu	Lys	Lys	Ile
			260					265					270		
Glu	Asn	Gly	Pro	Pro	Asn	Met	Gln	His	Ala	Val	Trp	Asn	Pro	Lys	Leu
	275						280					285			
Asp	Arg	Arg	Asp	Met	Ala	His	Arg	Met	Pro	Val	Ile	Thr	Pro	Ala	Tyr
	290					295					300				
Pro	Ser	Met	Cys	Ser	Thr	His	Asn	Ile	Thr	Ala	Ser	Thr	Met	Ser	Ile
305					310					315					320
Ile	Ser	Lys	Glu	Met	Leu	Arg	Ala	Met	Gln	Ile	Thr	Asp	Asp	Ile	Leu
			325					330						335	
Arg	Glu	Pro	Gly	Ala	Ser	Trp	Ala	Pro	Leu	Phe	Glu	Lys	Val	Asp	Phe
			340				345						350		
Phe	Ser	Met	Tyr	Lys	Thr	Tyr	Val	Gln	Val	Val	Ala	Ser	Ala	Ser	Thr
		355					360					365			
Ala	Glu	Gly	Ile	Lys	Asp	Trp	Ser	Gly	Thr	Val	Glu	Ala	Arg	Ile	Arg
	370					375					380				
Thr	Leu	Val	Gln	Asp	Leu	Glu	Asn	Thr	Asp	Ser	Ile	Leu	Thr	Ala	His
385					390					395					400
Pro	Leu	Val	Gly	Gly	Val	Ser	Arg	Val	Phe	Tyr	Cys	Thr	Thr	Glu	Glu
			405					410						415	
Glu	Gln	Ala	Ala	Ala	Ser	Gln	Gly	Glu	Leu	Thr	Ser	Glu	Met	Ile	Asp
		420					425					430			
Arg	Thr	Glu	Glu	Glu	Val	Thr	Asp	Lys	Glu	His	Arg	Asn	Ile	Phe	Thr

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435					440					445					
Lys	Ser	Phe	Phe	Ile	Gly	Leu	Glu	Ile	Glu	Lys	Lys	Thr	Lys	Glu	Gly
450						455					460				
Gly	Gly	Arg	Val	Leu	Asn	Leu	Phe	Tyr	Pro	Ser	Lys	Lys	Phe	Cys	Ser
465					470					475					480
Met	Cys	Gln	Ser	Trp	Asp	Arg	Tyr	Asn	Glu	Met	Glu	Met	Ser	Val	Ile
				485					490						495
Leu	Arg	Pro	Ala	Lys	Arg	Ser	Asp	Leu	Pro	Ser	Phe	Val	Phe	Pro	Asp
			500					505					510		
Gly	Ile	Pro	Ile	Ser	Lys	Lys	Lys	Ala	Lys	Arg	Gln	Gln	Gln	Asn	Gly
		515					520					525			
Ser	Ala	Asp	Ala	Ser	Leu	Ser	Asp	Ser	Val	Glu	Gly	Gln	Gly	Pro	Ser
530						535					540				
Lys	Arg	Thr	Lys	Ala	Ala	Pro	Pro	Ser	Pro	Asn	Gln	Ala	Ser	Asn	Pro
545					550					555					560
Arg	Cys	Asn	Gly	Leu	Pro	Val	Gln	Asn	Gly	Ser	Asp	Pro	Val	Ser	Ser
				565					570						575
Lys	Ile	Gly	Glu	Glu	Val	Asp	Ile	Lys	Pro	Pro	Pro	Gly	Ile	Glu	Asn
			580					585					590		
Leu	Pro	Pro	Leu	Ser	Asn	Ala	Ala	Met	Ser	Ser	Phe	Ala	Thr	Ala	Ala
		595				600						605			
Lys	Gly	Val	Ala	Thr	Thr	Gln	Asp	Phe	Asn	Lys	Glu	Gly	Leu	Val	Val
610						615					620				
Leu	Asn	Gln	Gln	Ser	Thr	Leu									
625					630										

<210> SEQ ID NO 122
 <211> LENGTH: 571
 <212> TYPE: PRT
 <213> ORGANISM: Golovinomyces cichoracearum
 <400> SEQUENCE: 122

Met	Ala	Glu	Lys	Ser	Phe	Gly	Val	Thr	Ala	Pro	Leu	Ser	Val	Thr	Pro
1				5					10					15	
Pro	Thr	Glu	Ser	Glu	Asn	Gln	Ala	Ser	Ser	Ala	Leu	Ile	Glu	Glu	Leu
			20					25					30		
Lys	Arg	Gln	Asn	Asn	Tyr	Glu	Asn	Ala	Ile	Glu	Thr	Ala	Asn	Arg	Gln
		35					40					45			
Lys	Val	Leu	Asn	Ser	Leu	Gln	Leu	Ile	Thr	Glu	Glu	Phe	Val	Arg	Gln
		50				55				60					
Val	Ser	Arg	Ala	Gln	Gly	Leu	Pro	Ala	Asn	Leu	Ile	Lys	Ser	Ala	Gly
65					70				75					80	
Gly	Met	Val	Val	Thr	Phe	Gly	Ser	Tyr	Lys	Leu	Gly	Val	Ile	Gly	Pro
				85				90					95		
Gly	Ser	Asp	Ile	Asp	Thr	Leu	Ile	Val	Ala	Pro	Gln	Asn	Val	Thr	Lys
		100						105					110		
Glu	Asp	Phe	Phe	Ser	Arg	Phe	Pro	Asp	Leu	Leu	Arg	Ser	Met	Ala	Thr
		115					120					125			
Glu	Gly	Asn	Ile	Thr	Glu	Leu	Thr	Ala	Lys	Pro	Asp	Ala	Phe	Ala	Pro
		130				135					140				
Cys	Ile	Thr	Leu	Lys	Tyr	Ala	Gly	Ile	Asp	Ile	Asp	Leu	Leu	Phe	Gly
145					150				155						160

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Arg	Val	Lys	Leu	Ser	Gln	Val	Pro	Arg	Asn	Leu	Ser	Leu	Leu	Asn	Gln
				165					170					175	
Asn	Met	Leu	Arg	Gly	Leu	Asn	Asn	Glu	Glu	Val	Arg	Ser	Leu	Asn	Gly
		180						185					190		
Val	Arg	Val	Ala	Asp	Glu	Ile	Leu	Asn	Leu	Val	Pro	Glu	Pro	Ala	Ile
		195					200					205			
Phe	Arg	Thr	Ala	Leu	Arg	Thr	Ile	Lys	Leu	Trp	Gly	Thr	Arg	Arg	Ala
	210					215					220				
Ile	Ala	Gly	Asn	Ile	Tyr	Gly	Phe	Pro	Gly	Gly	Val	Thr	Trp	Ala	Ile
225					230					235					240
Leu	Val	Ala	Arg	Ile	Cys	Gln	Leu	Tyr	Pro	Lys	Ala	Thr	Ser	Ser	Thr
			245						250					255	
Ile	Val	Phe	Lys	Phe	Phe	Arg	Ile	Met	Glu	Lys	Trp	Arg	Trp	Pro	Met
		260						265					270		
Pro	Val	Leu	Leu	Lys	Glu	Ile	Asp	Asn	Leu	Asn	Ala	Leu	Gly	Leu	Lys
		275					280					285			
Val	Trp	Asn	Pro	Lys	Ile	Tyr	Gly	Ser	Asp	Lys	Asn	His	Ile	Met	Pro
	290					295					300				
Ile	Ile	Thr	Pro	Ala	Tyr	Pro	Glu	Met	Cys	Thr	Thr	His	Asn	Phe	Ser
305					310					315					320
Leu	Ser	Thr	Lys	Ala	Ile	Leu	Glu	Lys	Glu	Leu	Lys	Arg	Gly	Gly	Asp
			325						330					335	
Ile	Thr	Asp	Arg	Val	Met	Ser	Gly	Lys	Ala	Ser	Trp	Lys	Asp	Leu	Phe
		340						345					350		
Ala	Lys	His	Thr	Phe	Phe	Thr	Asn	Gly	Tyr	Lys	Tyr	Tyr	Leu	Ser	Val
		355					360					365			
Val	Ser	Ala	Ser	Arg	Asn	Lys	Glu	Ser	Gln	Leu	Phe	Trp	Ser	Gly	Phe
	370					375					380				
Val	Glu	Ser	Lys	Val	Arg	Leu	Leu	Val	Asn	Lys	Leu	Glu	Tyr	His	Pro
385					390					395					400
Ser	Ile	Ala	Leu	Ala	His	Pro	Phe	Asn	Lys	Gly	Phe	Thr	Arg	Val	His
			405						410					415	
Arg	Cys	Arg	Thr	Glu	Glu	Glu	Val	Asp	Leu	Val	Lys	Asn	Gly	Ser	Leu
			420					425					430		
Lys	Phe	His	Ala	Thr	Asp	Ile	Ala	Thr	Ser	Thr	Thr	Gly	His	Gly	Leu
		435					440					445			
Ser	Val	Glu	Thr	Thr	Ser	Lys	Glu	Lys	Leu	Asn	Glu	Thr	Lys	Asn	Val
	450					455					460				
Glu	Gln	Asp	Glu	Asn	Leu	Ile	Met	Val	Tyr	Thr	Ala	Thr	His	Tyr	Ile
465					470					475					480
Gly	Leu	Glu	Leu	Ala	Lys	Gly	Ala	Lys	Ser	Leu	Asp	Leu	Ser	Tyr	Glu
			485						490					495	
Val	Glu	Glu	Phe	Lys	Ser	Ile	Cys	Thr	Ser	Ser	Glu	Ala	Tyr	Asn	Gln
			500					505					510		
Glu	Glu	Asn	Ser	Leu	Gly	Val	Ala	His	Thr	Lys	Asn	Cys	Asp	Leu	Pro
		515					520					525			
Asp	Asp	Val	Phe	Thr	Glu	Gly	Glu	Leu	Lys	Pro	Thr	Arg	Pro	Leu	Lys
	530					535					540				
Gln	Lys	Lys	Lys	Arg	Pro	Ala	Thr	Glu	Asp	Ser	Asn	Pro	Val	Pro	Ser
545					550					555					560
Lys	Arg	Gln	Gln	Thr	Ser	Leu	Ile	Ala	Val	Gly					

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565	570
<210> SEQ ID NO 123	
<211> LENGTH: 600	
<212> TYPE: PRT	
<213> ORGANISM: Hortaea werneckii	
<400> SEQUENCE: 123	
Met Ala Glu Lys Gln Tyr Gly Val Thr Pro Ala Phe Ser Leu Glu Pro	
1 5 10 15	
Pro Ser Pro Lys Asp Leu Lys Leu Asn Asp Ala Leu Leu Ala Glu Phe	
20 25 30	
Lys Ala Gln Asn Asn Phe Ala Pro Gln Ser Asp Thr Glu Lys Arg Glu	
35 40 45	
Ala Ile Leu Lys Lys Leu Glu Gly Leu Leu Gln Arg Met Val Gln Glu	
50 55 60	
Val Gly Arg Lys Lys Gly Leu Pro Gln Ser Ile Leu Glu Val Ala Gly	
65 70 75 80	
Gly Lys Val Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val Tyr Gly Pro	
85 90 95	
Asn Ser Asp Val Asp Thr Leu Met Val Gly Pro Lys His Val Thr Arg	
100 105 110	
Glu Asp Phe Phe Glu His Met Pro Pro Leu Ile Arg Ser Ala Trp Ala	
115 120 125	
Glu Asp Gln Ile Gly Gly Leu Val Pro Val Pro Gly Ile Gly Thr Pro	
130 135 140	
Ile Ile Lys Leu Glu Leu Glu Gly Val Asp Ile Asp Leu Ile Tyr Ser	
145 150 155 160	
Ser Leu Gln Leu Ser Ser Ile Pro Lys Asp Ile Glu Leu Lys Asp Asp	
165 170 175	
Asn Leu Leu Arg Gly Leu Asp Asp Thr Asp Arg Arg Cys Val Asn Gly	
180 185 190	
Thr Arg Val Thr Asn Arg Ile Leu Glu Leu Val Pro Gln Thr Lys Thr	
195 200 205	
Phe Arg Leu Ala Leu Arg Ala Ile Lys Leu Trp Ser Ser Gln Arg Ala	
210 215 220	
Ile Tyr Gly Asn Ile Val Gly Phe Pro Gly Gly Val Ala Trp Ala Ile	
225 230 235 240	
Leu Val Ala Arg Val Cys Gln Leu Tyr Pro Lys Ala Ala Ala Pro Leu	
245 250 255	
Leu Ile Ser Lys Phe Phe Phe Ile Met Lys Arg Trp Asn Trp Pro Lys	
260 265 270	
Pro Val Phe Leu Gln His Lys Glu Glu Thr Ser Leu Gln Leu Arg Glu	
275 280 285	
Trp Asp Pro Ile Gln Tyr Arg Gly Asp Gly Phe His Leu Met Pro Ile	
290 295 300	
Leu Thr Pro Ala Tyr Pro Ser Met Asn Thr Ala His Thr Val Gly Pro	
305 310 315 320	
Ser Thr Lys Met Ile Ile Met Arg Glu Leu Glu Arg Gly Glu Asn Ile	
325 330 335	
Val Asn Asp Ile Tyr Ala Asn Lys Arg Pro Trp Lys Asp Leu Phe Gln	
340 345 350	

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Arg	His	Thr	Phe	Phe	Thr	Asn	Ala	Tyr	Lys	His	Tyr	Ile	Cys	Val	Val
		355					360					365			
Val	Ala	Ala	Lys	Asn	Lys	Asp	Ala	His	Asp	Asn	Trp	Ser	Gly	Leu	Val
	370					375					380				
Asn	Ser	Lys	Leu	Lys	Phe	Leu	Val	Lys	Gly	Ile	Glu	Asp	Ser	Gly	Gly
385					390					395					400
Ser	Ser	Val	Glu	Leu	Val	Gln	Pro	Phe	Asn	Lys	Gly	Phe	Ser	Arg	Val
			405						410					415	
His	Glu	Cys	Gln	Thr	Asn	Glu	Gln	Val	Glu	Lys	Val	Leu	Asp	Gly	Ser
			420					425					430		
Leu	Glu	Cys	Gln	Val	Lys	Glu	Thr	Lys	Thr	Thr	Glu	Glu	Gly	Asn	Asp
	435					440						445			
Thr	Ala	Ile	Ala	Asn	Gly	Val	Ala	Gln	Thr	Asp	Thr	Glu	Gly	Leu	Glu
	450					455					460				
Val	Pro	Lys	Ala	Asp	Ser	Glu	Ala	Arg	Ala	Glu	Gly	Ser	Gly	Thr	Thr
465					470					475					480
Lys	Leu	Trp	Thr	Thr	Thr	Phe	Tyr	Leu	Gly	Ile	Gly	Leu	Lys	Lys	Gly
			485						490					495	
Ala	Thr	Asp	Leu	Asp	Ile	Ser	Val	Pro	Val	Arg	Asn	Leu	Gln	Gly	Asp
			500					505					510		
Cys	Thr	Ala	Trp	Ala	Asp	Tyr	Asn	Pro	Asp	Leu	His	Ser	Ile	Lys	Ile
		515					520					525			
Lys	His	Ile	Arg	Asn	Phe	Asn	Leu	Pro	Asp	Asp	Val	Phe	Lys	Glu	Gly
	530					535					540				
Glu	Val	Lys	Pro	Gln	Arg	Pro	Lys	Lys	Lys	Thr	Asn	Ala	Ala	Lys	Ala
545					550					555					560
Ala	Glu	Ala	Thr	Ser	Asn	Lys	Arg	Ser	Phe	Ser	Asp	Thr	Gly	Leu	Asp
			565						570					575	
Pro	Asn	Ala	Asp	Pro	Ala	Lys	Arg	Arg	Gln	Ser	Gly	Asn	Val	Pro	Thr
		580						585					590		
Pro	Val	Ala	Asn	Gly	Ser	Val	Gly								
		595				600									

<210> SEQ ID NO 124

<211> LENGTH: 589

<212> TYPE: PRT

<213> ORGANISM: Valsa sordida

<400> SEQUENCE: 124

Met	Ala	Glu	Ile	Gln	Tyr	Gly	Met	Thr	Pro	Pro	Leu	Ser	Thr	Glu	Leu
1			5					10					15		
Pro	Lys	Asp	Pro	Glu	Lys	Arg	Ala	Asn	Asp	Ala	Leu	Phe	Ala	Glu	Leu
	20						25					30			
Lys	Ala	Gln	Asn	Thr	Tyr	Glu	His	Ala	Thr	Glu	Thr	Gln	Lys	Arg	Glu
	35					40						45			
Lys	Val	Leu	Lys	Ser	Leu	Gln	Ser	Ile	Ala	Asp	Val	Phe	Val	Gln	Lys
	50					55				60					
Val	Ala	Gln	Arg	Val	His	Lys	Glu	Ser	Pro	Ala	Ile	Gln	Lys	Ser	Ala
65					70				75					80	
Arg	Gly	Arg	Val	Phe	Thr	Tyr	Gly	Ser	Tyr	Arg	Leu	Gly	Val	Phe	Gly
			85					90						95	
Pro	Gly	Ser	Asp	Ile	Asp	Thr	Leu	Val	Val	Ala	Pro	Lys	Tyr	Val	Thr
		100					105						110		

Arg	Glu	Asp	Phe	Phe	Glu	Leu	Phe	Pro	Gly	Leu	Leu	Lys	Glu	Leu	Ala
		115					120					125			
Pro	Ala	Gly	Ser	Ile	Thr	Asp	Leu	Thr	Ala	Val	Thr	Asp	Ala	Phe	Val
		130					135					140			
Pro	Ile	Ile	Lys	Phe	Glu	Tyr	Ser	Asp	Ile	Ser	Ile	Asp	Leu	Ile	Phe
		145					150					155			160
Ser	Arg	Ile	Ala	Thr	Leu	Thr	Glu	Ile	Pro	Ala	Leu	Thr	Thr	Lys	Trp
				165					170					175	
Asp	Leu	Leu	Asp	Asn	Asn	Leu	Leu	Arg	Gly	Leu	Asp	Asp	Ala	Glu	Leu
				180					185					190	
Arg	Ser	Leu	Asn	Gly	Thr	Arg	Val	Thr	Asp	Asp	Ile	Leu	Lys	Leu	Val
				195					200					205	
Pro	Glu	Gln	Thr	Ser	Phe	Arg	Leu	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp
				210					215					220	
Ala	Gln	Arg	Arg	Ala	Ile	Tyr	Ala	Asn	Ile	Met	Gly	Phe	Pro	Gly	Gly
				225							235				
Val	Ala	Trp	Ala	Met	Leu	Val	Ala	Arg	Ile	Cys	Gln	Leu	Tyr	Pro	Lys
				245							250				
Ala	Asn	Gly	Ala	Val	Ile	Val	Asp	Lys	Phe	Phe	His	Ile	Ile	Arg	Arg
				260											270
Trp	Pro	Trp	Pro	Gln	Pro	Val	Leu	Leu	Lys	Asn	Pro	Glu	Asp	Gly	Pro
				275											285
Leu	Gln	Val	Arg	Val	Trp	Asn	Pro	Lys	Val	Tyr	Lys	Gly	Asp	Ser	Tyr
				290											300
His	Leu	Met	Pro	Val	Ile	Thr	Pro	Ala	Tyr	Pro	Ser	Met	Cys	Ser	Thr
				305											320
Phe	Asn	Val	Thr	His	Ser	Asn	Lys	Ala	Val	Ile	Gln	Lys	Glu	Leu	Asp
				325											335
His	Phe	Ala	Asp	Val	Val	Gln	Gln	Ile	Met	Met	Gly	Lys	Leu	Pro	Trp
				340											350
Lys	Ser	Leu	Phe	Val	Lys	His	Thr	Phe	Phe	Thr	Lys	Asp	Tyr	Lys	Tyr
				355											365
Tyr	Ile	Ala	Val	Ile	Ala	Ala	Ser	Thr	Thr	Lys	Glu	Asn	Ser	Lys	Ile
				370											380
Trp	Gly	Gly	Phe	Val	Glu	Ser	Lys	Ile	Arg	Leu	Leu	Val	Gln	Gly	Leu
				385											400
Glu	Arg	His	Ser	Ser	Ile	Arg	Leu	Ala	Arg	Pro	Phe	Asn	Lys	Gly	Tyr
				405											415
Glu	Arg	Lys	His	Arg	Thr	Val	Ser	Gly	Met	Asn	Glu	Trp	His	Asp	Ile
				420											430
Leu	Asp	Gly	Ser	Leu	Asp	Tyr	Val	Val	Lys	Asp	Glu	Gln	Val	Ala	Asp
				435											445
Glu	Lys	Ala	Lys	Thr	Glu	Pro	Gly	Asn	Asp	Ala	Val	Lys	Thr	Glu	Pro
				450											460
Thr	Ser	Pro	Ala	Lys	Val	Lys	Gln	Glu	Glu	Gly	Ile	Glu	Ser	Leu	Glu
				465											480
Ile	Lys	Asp	Asp	Ala	Lys	Glu	Thr	Asp							

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Asp Val Tyr Thr Ser Thr His Tyr Ile Gly Ile Glu Leu His Asp Glu
    515                      520                      525

His Tyr Ser Trp Ser Leu Pro Asp Asp Val Phe Glu Ala Gly Glu Thr
    530                      535                      540

Lys Pro Thr Arg Pro Leu Lys Lys Lys Ala Ala Gln Pro Pro Asn Gly
    545                      550                      555                      560

Ala Lys Arg Lys Thr Pro Thr Glu Gly Ala Leu Asn Ser Ala Ile Asp
    565                      570                      575

Ala Ala Lys Arg Gln Lys Thr Ser Thr Val Ser Ala Gly
    580                      585

<210> SEQ ID NO 125
<211> LENGTH: 588
<212> TYPE: PRT
<213> ORGANISM: Wallemia mellicola

<400> SEQUENCE: 125

Met Ala Glu Lys Gln Trp Gly Ile Thr Pro Pro Ile Ser Leu Ala Ser
 1          5          10          15

Pro Thr Asp Leu Glu Lys Gln Val Asp Ala Gly Leu Ile Gln Glu Leu
 20          25          30

Lys Asp Gln Asn Thr Ile Glu Ser Glu Gln Glu Ser Leu Asn Arg Glu
 35          40          45

Lys Val Leu Ala Leu Val Gln Ser Leu Val Lys Glu Phe Val Lys Arg
 50          55          60

Val Ser Ile Arg Asn Gly Met Ser Glu Ser Leu Ala Asp Gln Ala Gly
 65          70          75          80

Gly Lys Ile Phe Thr Phe Gly Ser Tyr Arg Leu Gly Val Asn Gln Pro
 85          90          95

Gly Ala Asp Ile Asp Thr Leu Cys Val Val Pro Lys His Val Ser Arg
100         105         110

Glu Asp Phe Phe Asp Val Phe Glu Pro Leu Leu Lys Thr Arg Glu Glu
115         120         125

Val Ser Val Cys Ala Gly Val Pro Asp Ala Tyr Val Pro Ile Ile Lys
130         135         140

Thr Thr Ile Ser Gly Ile Glu Ile Asp Phe Leu Val Ala Arg Leu Ala
145         150         155         160

Leu Ala Thr Ile Pro Asp Asp Leu Glu Leu Ala Asp Asp Asn Leu Leu
165         170         175

Lys Asn Leu Asp Glu Arg Cys Val Arg Ser Leu Asn Gly Ser Arg Val
180         185         190

Thr Asp Glu Ile Leu Arg Val Val Pro Asn Val Gln Val Phe Arg Glu
195         200         205

Ser Leu Arg Cys Ile Lys Leu Trp Ala Gln Arg Arg Ala Ile Tyr Ser
210         215         220

Asn Val Asn Gly Phe Leu Gly Gly Val Ala Trp Ala Met Leu Val Ala
225         230         235         240

Arg Ile Cys Gln Leu Tyr Pro Asn Glu Asn Ala Gly Ala Val Ile Ala
245         250         255

Lys Phe Phe Thr Ile Leu Tyr Gln Trp Lys Trp Pro His Pro Val Leu
260         265         270

Leu Lys Glu Ile Glu Asp Gly Pro Leu Lys Val Arg Val Trp Asn Pro
275         280         285

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Lys Val Gly Ser Ser Thr Leu Leu Ile Tyr Pro Ser Asp Arg Ala His
 290 295 300
 Arg Met Pro Ile Ile Thr Pro Ala Tyr Pro Ser Met Cys Ser Thr His
 305 310 315 320
 Asn Val Gly Pro Ser Thr Gln Glu Val Met Arg Gln Glu Phe Lys Arg
 325 330 335
 Gly Met Asp Ile Leu Asp Gly Ile His Lys Gly Thr Thr Thr Trp Ser
 340 345 350
 Gln Leu Phe Asn Lys His Glu Phe Phe Ser Lys Tyr Lys His Tyr Leu
 355 360 365
 Gln Ile Ile Ala Ser Gly Gln Thr Ala Glu Ser Gln Lys Lys Trp Ser
 370 375 380
 Gly Ala Val Glu Ser Lys Val Arg Gln Leu Val Ser Lys Leu Glu Leu
 385 390 395 400
 Val Asp Gly Ile Glu Leu Ala His Pro Phe Val Lys Gly Phe Ser Glu
 405 410 415
 Val Phe Ile Cys Leu Asn Asp Glu Glu Val Gln His Ala Ala Glu Tyr
 420 425 430
 Asn Pro Ser Glu Glu Val Lys Glu Arg Ser Lys Lys Pro Glu Glu Tyr
 435 440 445
 Glu Asn Lys Glu Ala Glu Asp Gly Ile His Arg Val Tyr Ile Ser Thr
 450 455 460
 Phe Tyr Ile Gly Leu Ala Ile Gln Pro Arg Asn Pro Glu Thr Asn Glu
 465 470 475 480
 Lys Arg Thr Leu Asn Leu Thr Tyr Pro Thr Asn Glu Phe Met Lys Leu
 485 490 495
 Thr Lys Leu Trp Asp Gln Phe Val Glu Gly Glu Met Lys Ile Phe Val
 500 505 510
 Lys Asn Ile Lys Ala Ser Ala Leu Pro Asp Val Val Phe Glu Gly Gly
 515 520 525
 Ile Arg Thr Ser Ser Ser Lys Ser Ser Lys Lys Lys Ser Lys Arg Pro
 530 535 540
 Ser Asn Val Met Asp Glu Ser Asn Gly Glu Asn Glu Gln Asn Lys Arg
 545 550 555 560
 Leu Lys Pro Asp Asn Thr Ser Asn Asn Glu Gln Lys Ser Glu Ala Thr
 565 570 575
 Ala Leu Ala Pro Ser Ala Val Pro Thr Glu Ser Thr
 580 585

<210> SEQ ID NO 126

<211> LENGTH: 587

<212> TYPE: PRT

<213> ORGANISM: Xylaria flabelliformis

<400> SEQUENCE: 126

Met Ala Glu Lys Thr Trp Gly Ile Thr Lys Pro Ile Ser Ser Ala Pro
 1 5 10 15
 Pro Thr Pro Ala Glu Ser His Ala Thr Ala Leu Leu Leu Glu Glu Leu
 20 25 30
 Arg Arg Gln Asn Thr Phe Glu Thr Ala Ser Glu Ile Lys Lys Arg Glu
 35 40 45
 Lys Val Val Asn Asp Leu Gln Arg Ile Ala Asp Glu Phe Val Arg Lys

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50					55					60					
Val 65	Ala	Arg	Ala	Lys 70	Glu	Pro	His	Asn	Glu	Val 75	Leu	Ile	Arg	Asp	Ala 80
Arg	Gly	Glu	Val	Phe 85	Thr	Tyr	Gly	Ser	Tyr 90	Cys	Leu	Gly	Val	Tyr 95	Gly
Pro	Gly	Ser	Asp 100	Ile	Asp	Thr	Leu	Val 105	Thr	Ala	Pro	Arg	Tyr 110	Val	Thr
Arg	Asp	Asp 115	Tyr	Phe	Lys	Tyr	Phe 120	Pro	Asp	Leu	Leu	Thr	Glu	Met	Ala
Pro	Pro 130	Gly	Ala	Ile	Thr	Asn 135	Leu	Thr	Ala	Val	Glu 140	Asp	Ala	Phe	Val
Pro 145	Ile	Ile	Lys	Phe	Glu 150	Tyr	Trp	Gly	Ile	Ser 155	Ile	Asp	Leu	Ile	Phe 160
Ser	Arg	Ile	Ala	Thr 165	Leu	Thr	Gln	Phe	Pro 170	Pro	His	Lys	Gln	Leu 175	Glu
Leu	Thr	Ser	Asn 180	Glu	His	Leu	Arg	Gly 185	Leu	Asp	Asp	Arg	Glu 190	Leu	Arg
Ser	Leu 195	Asn	Gly	Thr	Arg	Val	Thr 200	Lys	Glu	Ile	Leu	Asn 205	Leu	Val	Pro
Glu 210	Gln	Ser	Thr	Phe	Arg	Thr 215	Ala	Leu	Arg	Ala	Ile 220	Lys	Leu	Trp	Ala
Gln 225	Arg	Arg	Ala	Ile 230	Tyr	Ala	Asn	Ile	Ile	Gly 235	Phe	Pro	Gly	Gly	Val 240
Val	Trp	Ala	Met 245	Met	Val	Ala	Arg	Val	Cys 250	Gln	Leu	Tyr	Pro	Lys 255	Ala
Thr	Ser	Ala	Thr 260	Ile	Val	Gly	Lys	Phe 265	Phe	Leu	Val	Met	Lys 270	Gly	Trp
Pro	Trp 275	Pro	Ile	Pro	Val	Gln	Leu 280	Lys	His	Met	Glu	Asp 285	Gly	Pro	Leu
Asn 290	Val	Arg	Val	Trp	Asn	Pro 295	Lys	Ile	Tyr	Lys	Ser 300	Asp	Ser	Phe	His
Leu 305	Met	Pro	Val	Ile 310	Thr	Pro	Ala	Tyr	Pro	Gln 315	Met	Cys	Ala	Thr	Phe 320
Asn	Ile	Thr	Lys 325	Ser	Thr	Lys	Thr	Ile	Ile 330	Gln	Arg	Glu	Leu	Glu 335	Arg
Gly	Val	Glu	Leu 340	Thr	Asp	Lys	Ile	Leu 345	Gly	Ser	Glu	Arg	Pro 350	Trp	Lys
Asp	Leu 355	Phe	Val	Lys	His	Thr	Phe 360	Phe	Thr	Gln	Asp	His 365	Lys	Tyr	Tyr
Leu 370	Ala	Val	Ile	Ala	Thr	Ser 375	Thr	Thr	Lys	Glu	Ala 380	His	Lys	Ile	Trp
Ser 385	Gly	Phe	Val	Glu	Ser	Lys 390	Val	Arg	Ile	Leu	Val 395	Gly	Glu	Leu	Glu 400
Arg	His	Ser	Ser 405	Ile	Ala	Leu	Ala	Arg	Pro 410	Phe	Asn	Lys	Gly	Phe 415	Glu
Arg	Gln	His	Leu 420	Thr	Lys	Thr	Asp	Gln 425	Gln	Ala	Ala	Glu	Val 430	Gln	Ser
Gly	Ser 435	Leu	Ala	Tyr	Val	Thr	Glu 440	Asp	Glu	Asp	Asn 445	Ser	Asp	Asp	Ala
Thr 450	Val	Thr	Lys	Ala	Glu	Asn 455	Gly	Thr	Asp	Gly	Asn 460	Ala	Asp	Ser	Ser

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Asp Gln Lys Val Thr Ala Glu Asn Thr Thr Gln Pro Arg Arg Thr Tyr
 465 470 475 480
 Thr Thr Thr His Tyr Ile Gly Leu Glu Leu Ala Glu Gly Ser Lys Ser
 485 490 495
 Leu Asp Leu Ser Tyr Gln Val Asn Ala Phe Lys Gln Leu Cys Ser Glu
 500 505 510
 Trp Glu Lys Tyr Ser Ala Glu Phe Asn Ser Leu Ser Val Gln His Val
 515 520 525
 Lys Asn Ile Asn Leu Pro Glu Asp Val Phe Glu Pro Gly Glu Thr Lys
 530 535 540
 Pro Ala Arg Pro Leu Lys Lys Pro Thr Ala Asn Gly Thr Ala Pro Thr
 545 550 555 560
 Thr Gln Arg Lys Arg Asn Ala Pro Thr Glu Gly His Pro Pro Pro Ala
 565 570 575
 Lys Arg Gln Gln Ser Ser Ala Ala Ala Ala Gly
 580 585

<210> SEQ ID NO 127

<211> LENGTH: 592

<212> TYPE: PRT

<213> ORGANISM: Chaetomium globosum

<400> SEQUENCE: 127

Met Ser Glu Ile Ile Tyr Gly Val Thr Pro Pro Ile Ser Thr Thr Leu
 1 5 10 15
 Pro Thr Glu Pro Glu Lys Arg Leu Asn His Ala Leu His Gln Glu Leu
 20 25 30
 Arg Ala Gln Gly Thr Phe Glu Ser Pro Gln Glu Thr Glu Lys Arg Lys
 35 40 45
 Glu Val Leu Arg Gln Leu Glu Lys Ile Thr Thr Val Phe Val Gln Arg
 50 55 60
 Ala Ala Ala Gln Lys Glu Pro Lys Asn Thr Phe Met Ile Arg Asp Ala
 65 70 75 80
 Ile Gly Arg Val Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val Tyr Gly
 85 90 95
 Pro Gly Ser Asp Met Asp Thr Leu Val Val Ala Pro Lys Tyr Val Thr
 100 105 110
 Val Glu Gln Tyr Phe Glu Ile Phe Pro Glu Val Leu Val Glu Met Ala
 115 120 125
 Pro Pro Gly Ala Ile Thr Asp Leu Thr Pro Val Pro Glu Ala Phe Val
 130 135 140
 Pro Ile Ile Lys Phe Glu Phe Ser Gly Ile Ser Ile Asp Leu Ile Phe
 145 150 155 160
 Cys Ser Ile Gln Thr Leu Lys Gln Leu Pro Asp Glu Lys Asn Trp Ser
 165 170 175
 Leu Ala Asp Asn Asn Leu Leu Arg Gly Leu Ser Glu Asn Glu Val Arg
 180 185 190
 Ser Leu Asn Gly Thr Arg Val Thr Asp Asp Ile Leu His Leu Val Pro
 195 200 205
 Glu Pro Ala Thr Phe Lys Leu Ala Leu Arg Ala Ile Lys Leu Trp Ala
 210 215 220
 Gln Arg Lys Ala Ile Tyr Ala Asn Ile Met Gly Tyr Pro Gly Gly Val

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225	230	235	240
Ala Trp Ala Met Leu Val Ala Arg Val Cys Gln Leu Tyr Pro Lys Ala	245	250	255
Thr Ser Ala Val Val Val Asn Lys Phe Phe His Ile Met Phe Lys Trp	260	265	270
Pro Trp Pro Leu Pro Val Leu Leu Lys Asp Ile Glu Tyr Gly Ser Pro	275	280	285
Val Thr Arg Val Pro Val Trp Asn Pro Lys Ser Ser Arg Arg Ala Gly	290	295	300
Pro Lys Arg His Arg Thr Ser Met Arg Ala Ser Ser Asn Arg Gly Cys	305	310	315
Gly Cys Trp Leu Thr Asp Leu Arg Gly Thr Ile Leu Leu Pro Trp Pro	325	330	335
Gly Pro Ser Thr Arg Val Thr Thr Glu Asn Thr Asp Ala Arg Thr Ile	340	345	350
Gly Ser Trp Glu Glu Val Val Ser Val Gly Ser Leu Ala Tyr Val Tyr	355	360	365
Lys Pro Pro Ala Asp Ser Glu Glu Lys Ala Lys Ala Glu Pro Lys Val	370	375	380
Glu Pro Lys Arg Glu Val Lys Gln Glu Ile Lys Ala Glu Asn Arg Glu	385	390	395
Pro Ala Ala Ala Glu Glu Lys Leu Glu Thr Arg Gly Glu Asp Gly Met	405	410	415
Arg Ile Lys Gln Glu His Ser Glu Ser Thr Gln Pro Pro Pro Ser	420	425	430
Asn Val Lys Pro Glu Pro His His Asp Thr Asn Ser Glu Val Lys Leu	435	440	445
Glu Asp Ile Pro Gln Lys Lys Lys Asp Glu Pro Asp Glu Met Glu Ile	450	455	460
Tyr Thr Thr Asn His Tyr Ile Gly Leu Gln Leu Val Glu Gly Ala Lys	465	470	475
Ser Leu Asp Leu Ser Arg Glu Val Asn Asp Trp Lys Ser Met Cys Thr	485	490	495
Ser Asn Glu Leu Phe Asp Glu Gly Leu Met Phe Leu Ser Ile Gln His	500	505	510
Leu Lys Asn Thr Asn Leu Pro Asp Asp Val Phe Glu Pro Gly Glu Thr	515	520	525
Lys Pro Arg Pro Val Lys Lys Asn Leu Lys Arg Ala Ala Ser Glu Gly	530	535	540
Pro Gly Asn Met Gln Pro Gln Pro Pro Val Lys Arg His Ala Pro Gly	545	550	555
Lys Pro Ser Ser His Ser Ala Gln Gln Gln Gln Gln Gln Gln Lys	565	570	575
Gln Arg Gln Gln Arg Pro Ala Ser Thr Ala Ala Ala Ala Val Ala Gly	580	585	590

<210> SEQ ID NO 128

<211> LENGTH: 591

<212> TYPE: PRT

<213> ORGANISM: Lachancea thermotolerans

<400> SEQUENCE: 128

-continued

Met	Asn	Gln	Gln	Lys	Thr	Tyr	Gly	Val	Thr	Gly	Pro	Ile	Ser	Thr	Ala
1				5					10					15	
Gly	Pro	Thr	Ala	Ala	Glu	Asn	Gln	Leu	Asn	Asp	Ala	Leu	Ile	Gln	Glu
			20					25					30		
Leu	Lys	Lys	Glu	Lys	Ser	Phe	Glu	Ser	Glu	Glu	Asp	Thr	Lys	Lys	Arg
		35					40					45			
Val	Glu	Val	Leu	Arg	Ile	Leu	Gln	Asn	Leu	Ala	Gln	Glu	Phe	Val	Tyr
	50					55					60				
Gln	Val	Ser	Lys	Lys	Arg	Asn	Met	Ser	Asp	Gly	Met	Ala	Lys	Asp	Ala
65					70					75					80
Gly	Gly	Lys	Ile	Phe	Thr	Tyr	Gly	Ser	Tyr	Arg	Leu	Gly	Val	His	Gly
			85						90					95	
Pro	Gly	Ser	Asp	Ile	Asp	Thr	Leu	Val	Val	Val	Pro	Lys	His	Val	Thr
			100					105					110		
Arg	Glu	Asp	Phe	Phe	Thr	Val	Phe	Asp	Gln	Ile	Leu	Arg	Thr	Arg	Ser
		115						120				125			
Glu	Leu	Glu	Glu	Ile	Ala	Pro	Val	Pro	Asp	Ala	Phe	Val	Pro	Ile	Ile
	130					135					140				
Lys	Ile	Lys	Phe	Ser	Gly	Ile	Ser	Ile	Asp	Leu	Ile	Cys	Ala	Arg	Leu
145					150					155					160
Asp	Ile	Ala	Gln	Val	Pro	Val	Asn	Leu	Thr	Leu	Ala	Asp	Lys	Asn	Leu
			165					170						175	
Leu	Arg	Asn	Leu	Asp	Glu	Lys	Asp	Leu	Arg	Ala	Leu	Asn	Gly	Thr	Arg
		180						185					190		
Val	Thr	Asp	Glu	Ile	Leu	Gln	Leu	Val	Pro	Lys	Pro	Thr	Ser	Phe	Lys
		195					200					205			
Ile	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala	Gln	Arg	Arg	Ala	Val	Tyr
	210					215					220				
Ala	Asn	Ile	Phe	Gly	Phe	Pro	Gly	Gly	Val	Ala	Trp	Ala	Met	Leu	Val
225					230					235					240
Ala	Arg	Ile	Cys	Gln	Leu	Tyr	Pro	Asn	Ala	Cys	Ser	Ala	Val	Ile	Leu
			245						250					255	
Thr	Arg	Phe	Phe	His	Ile	Leu	Thr	Lys	Trp	Asn	Trp	Pro	Gln	Pro	Val
		260						265					270		
Leu	Leu	Lys	Pro	Ile	Glu	Asp	Gly	Pro	Leu	Gln	Val	Arg	Val	Trp	Asn
		275					280					285			
Pro	Arg	Ile	Tyr	Ala	Gln	Asp	Arg	Ser	His	Lys	Met	Pro	Val	Ile	Thr
	290					295					300				
Pro	Ala	Tyr	Pro	Ser	Met	Cys	Ala	Thr	His	Asn	Ile	Ser	Glu	Ser	Thr
305					310					315					320
Lys	Lys	Val	Ile	Leu	Ala	Glu	Leu	Glu	Arg	Gly	Ala	Gln	Ile	Ser	Ser
			325						330					335	
Glu	Ile	Phe	Ser	Asn	Lys	Lys	Thr	Trp	Ser	Asp	Leu	Phe	Gln	Lys	His
		340						345					350		
Asp	Phe	Phe	Tyr	Lys	Tyr	Lys	Phe	Tyr	Leu	Thr	Val	Met	Ala	Ser	Thr
		355					360					365			
Ser	Gly	Ser	Ser	Glu	Gln	His	Leu	Lys	Trp	Ser	Gly	Leu	Val	Glu	Ser
	370					375					380				
Lys	Leu	Arg	Leu	Leu	Val	Gln	Lys	Leu	Glu	Thr	Leu	Gly	Gly	Ile	Asn
385					390					395					400
Leu	Ala	His	Pro	Phe	Thr	Lys	Pro	Phe	Glu	Ala	Ser	Tyr	Val	Tyr	Asp

-continued

405	410	415
Asn Glu Ala Gln Cys Lys Asp Ile Ile Asp Asn Tyr Gly Thr His Lys		
420	425	430
Ala Gln Asp Ile Leu Ala Gln Tyr Thr Glu Val Thr Asp Asp Asn Lys		
435	440	445
Asp Thr Asp Gly Val Lys Asp Lys Ala Gln Val His Ile Thr Thr Met		
450	455	460
Tyr Ile Gly Leu Asp Val Ala Leu Asp Gly Lys Lys Gln Val Asp Ile		
465	470	475
His Val Pro Cys Ser Asp Phe Phe Asn Leu Cys Arg Ser Phe Ser Glu		
485	490	495
Tyr Asp Asp Thr Asp Met Phe Ser Leu Met Ile Lys Tyr Val Lys Leu		
500	505	510
Tyr Asp Leu Pro Asp Asn Val Tyr Val Glu Gly Glu Glu Arg Pro Val		
515	520	525
Lys His Ser Lys Arg Lys Lys Leu Gly Lys Asp Ser Lys Lys Ser Lys		
530	535	540
Arg Pro Lys Ser Asn Ser Leu Lys Ala Glu Glu Pro Thr Ser Gly Thr		
545	550	555
Thr Asn Ser Glu Lys Asn Asn Ser Gln Gly Pro Asn Lys Pro Val Lys		
565	570	575
His Glu Pro Pro Met Gln Thr Ser Pro Asn Pro Leu Ala Gly Ser		
580	585	590

<210> SEQ ID NO 129
 <211> LENGTH: 558
 <212> TYPE: PRT
 <213> ORGANISM: Schizosaccharomyces octosporus

<400> SEQUENCE: 129

Met Ser Val Lys Gln Trp Gly Ile Thr Pro Pro Ile Ser Thr Ala Pro		
1	5	10
Ala Thr Glu Gln Glu Asn Ala Leu Asn Thr Ala Leu Ile Asp Glu Leu		
20	25	30
Lys Arg Gln Asn Leu Phe Glu Ser Ser Ala Glu Ser Glu Arg Arg Ile		
35	40	45
Lys Val Leu Asp Asp Leu Gln Arg Ile Ala Thr Glu Phe Val Lys Lys		
50	55	60
Val Ser Leu Ala Lys His Met Asn Glu Lys Met Ala Asn Glu Ala Gly		
65	70	75
Gly Lys Ile Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val Tyr Gly Pro		
85	90	95
Gly Ser Asp Ile Asp Thr Leu Leu Val Val Pro Lys His Val Ser Arg		
100	105	110
Glu Asn Phe Phe Gln Asp Leu Glu Pro Met Leu Arg Glu Arg Glu Glu		
115	120	125
Ile Thr Glu Leu Ala Ser Val Pro Asp Ala Tyr Val Pro Ile Ile Lys		
130	135	140
Phe Lys Tyr Phe Asp Ile Ser Ile Asp Leu Ile Phe Ala Arg Leu Ser		
145	150	155
Val Pro Lys Val Pro Arg Thr Leu Glu Leu Ser Asp Asn Asn Leu Leu		
165	170	175

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Lys	Gly	Val	Glu	Glu	Arg	Cys	Ile	Leu	Ser	Leu	Asn	Gly	Thr	Arg	Val
			180					185					190		
Thr	Asp	Gln	Ile	Leu	Gln	Leu	Val	Pro	Asn	Arg	Ala	Val	Phe	Lys	His
		195					200					205			
Ala	Leu	Arg	Ala	Val	Lys	Phe	Trp	Ala	Gln	Arg	Arg	Ala	Ile	Tyr	Ala
	210					215					220				
Asn	Val	Ile	Gly	Phe	Pro	Gly	Gly	Val	Ala	Trp	Ala	Met	Met	Val	Ala
225					230					235					240
Arg	Ile	Cys	Gln	Leu	Tyr	Pro	Asn	Ala	Val	Ser	Ser	Val	Ile	Val	Ala
			245						250					255	
Lys	Phe	Phe	Arg	Ile	Leu	His	Gln	Trp	Asn	Trp	Pro	Gln	Pro	Ile	Leu
		260						265					270		
Leu	Lys	Pro	Ile	Glu	Asp	Gly	Pro	Leu	Gln	Val	Arg	Ile	Trp	Asn	Pro
		275					280					285			
Lys	Leu	Tyr	Pro	Ser	Asp	Lys	Ala	His	Arg	Met	Pro	Ile	Ile	Thr	Pro
	290					295					300				
Ala	Tyr	Pro	Ser	Met	Cys	Ala	Thr	His	Asn	Ile	Thr	Pro	Ser	Thr	Gln
305					310					315					320
Ala	Ile	Ile	Leu	Ser	Glu	Met	Val	Arg	Ala	Gly	Glu	Ile	Ala	Asp	Gln
			325						330					335	
Ile	Met	Val	Gly	Ser	Val	Gly	Trp	Ser	Ala	Leu	Phe	Gln	Lys	His	Asp
		340						345					350		
Phe	Phe	His	Arg	Tyr	Lys	His	Tyr	Leu	Met	Ile	Thr	Ala	Ala	Ser	Lys
		355					360					365			
Thr	Ala	Glu	Gly	Gln	Leu	Lys	Trp	Ser	Gly	Leu	Val	Glu	Ser	Lys	Leu
	370					375					380				
Arg	His	Leu	Val	Thr	Arg	Leu	Glu	Leu	Val	Glu	Ala	Ile	Ala	Leu	Ala
385					390					395					400
His	Pro	Phe	Asn	Lys	Gly	Phe	Asp	Lys	Ile	His	Asn	Cys	Lys	Ser	Glu
			405						410					415	
Glu	Glu	Ala	His	Gln	Leu	Ala	Ser	Gly	Ile	Ser	Leu	Asp	Val	Ala	Ser
		420						425					430		
Gln	Thr	Ala	Asp	Leu	Glu	Arg	Leu	Ala	Ala	Asp	Ala	Asn	Gly	Gln	Asn
	435						440					445			
Glu	Asn	Glu	Asp	Glu	Lys	Glu	Glu	Lys	Lys	Ile	Tyr	Pro	Val	Tyr	Thr
	450					455					460				
Thr	Thr	Phe	Tyr	Ile	Gly	Leu	Glu	Leu	Glu	Lys	Lys	Lys	Gly	Gln	Pro
465					470					475					480
Ile	Lys	Lys	Leu	Asp	Ile	Ser	Trp	Pro	Ala	Gln	Glu	Phe	Tyr	Glu	Leu
			485						490					495	
Cys	Lys	Lys	Trp	Asp	Arg	Tyr	Asp	Glu	Thr	Gln	Val	Asn	Val	Phe	Ile
			500					505					510		
Lys	Asn	Thr	Arg	Asn	Val	Asn	Leu	Pro	Asp	Glu	Val	Phe	Glu	Pro	Gly
		515					520					525			
Glu	Glu	Lys	Pro	Lys	Ser	Ser	Lys	Lys	Arg	Ser	Tyr	Asn	Asp	Ala	Gly
	530						535				540				
Asn	Asn	Asn	Glu	Thr	Met	Lys	Arg	Gln	Lys	Ile	Ser	Thr	Ala		
545					550					555					

<210> SEQ ID NO 130

<211> LENGTH: 608

<212> TYPE: PRT

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<213> ORGANISM: *Exophiala spinifera*

<400> SEQUENCE: 130

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Met Ala Glu Lys Gln Trp Gly Met Thr Pro Pro Met Ser Thr Ala Leu
 1           5           10           15

Pro Glu Pro Ile Asp Thr Glu Lys Thr Ala Asp Leu Ile Glu Glu Leu
 20           25           30

Lys Lys Glu Asn Asn Tyr Glu Pro Leu Glu Ala Thr Gln Lys Arg Met
 35           40           45

Ala Thr Leu Gly Leu Leu Asn Arg Val Thr Gln Glu Phe Val Arg Glu
 50           55           60

Val Ser Arg Arg Arg Arg Met Pro Pro Ser Gln Ile Glu Gln Phe Gly
 65           70           75           80

Gly Lys Ile Phe Pro Tyr Gly Ser Tyr Arg Leu Gly Val Phe Gly Pro
 85           90           95

Gly Ser Asp Ile Asp Thr Leu Ala Val Ala Pro Arg His Val Thr Arg
 100          105          110

Glu Asp Phe Phe Glu Tyr Phe Pro Thr Val Leu Lys Arg Met Thr Ala
 115          120          125

Glu Gly Asp Ile Ser Ser Leu Thr Pro Val Pro Asp Ser Tyr Val Pro
 130          135          140

Ile Ile Lys Leu Val Leu Asn Asp Ile Glu Ile Asp Leu Ile Phe Ala
 145          150          155          160

Ser Ile Ala Ser Leu Gln Thr Ile Pro Lys Asn Leu Thr Leu Asn Asp
 165          170          175

Asn Asn Leu Leu Thr Gly Leu Asp Gln Ala Thr Ile Arg Ala Val Thr
 180          185          190

Gly Pro Arg Val Thr Asp Glu Ile Leu Ser Leu Val Pro Glu Gln Lys
 195          200          205

Thr Phe Arg Thr Ala Leu Arg Ala Ile Lys Leu Trp Ala Gln Arg Arg
 210          215          220

Ala Val Tyr Ala Asn Ile Val Gly Tyr Pro Gly Gly Val Ala Trp Ala
 225          230          235          240

Met Leu Val Ala Arg Val Cys Gln Leu Tyr Pro His Ala Val Gly Ala
 245          250          255

Thr Leu Val Asp Lys Phe Phe Phe Val Met Lys Asn Trp Asp Trp Pro
 260          265          270

Thr Pro Val Met Leu Lys Asp Ile Glu Gln Pro Lys Pro Ser Gln Ala
 275          280          285

Asn Glu Phe Lys Val Trp Asn Pro Ala Leu Tyr Lys Gly Asp Lys Lys
 290          295          300

Met Leu Met Pro Ile Ile Thr Pro Ala Phe Pro Ser Met Ser Ala Thr
 305          310          315          320

Tyr Asn Ile Ser Lys Ser Gly Lys Thr Val Ile Leu Arg Glu Leu Glu
 325          330          335

Arg Ala Ser Gln Ile Thr Asn Asn Ile Phe Ala Gly Lys Ala Arg Trp
 340          345          350

Ser Asp Leu Phe Lys Lys His Ser Phe Phe Thr Ala Asp His Lys Tyr
 355          360          365

Tyr Leu Gly Val Thr Ala Ser Thr Leu Asn Ala Asp Ser Ala Lys Gln
 370          375          380

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Trp Ala Gly Leu Val Glu Ser Lys Val Arg Ile Phe Val Met Leu Leu
 385 390 395 400
 Glu Gly Ile Pro Asp Ile Thr Leu Ala Arg Pro Phe Thr Lys Gly Phe
 405 410 415
 Lys Arg Val His Lys Cys Ala Asp Glu Thr Gln Ile Arg Glu Val Gln
 420 425 430
 Lys Gly Ser Met Lys Tyr Lys Phe Glu Glu Thr Lys Thr Val Glu Thr
 435 440 445
 Thr Asp Pro Glu Leu Val Thr Ser Asn Gly Asp Gly Ala Ala Val Pro
 450 455 460
 Ala Ala Asn Gly Ala Lys Pro Glu Thr Asp Gln Gly Ala His Thr Val
 465 470 475 480
 Tyr Thr Tyr Thr Phe Tyr Ile Gly Ile Asp Thr Thr Ala Lys Gly Ser
 485 490 495
 Leu Asn Ile Ala Pro Ser Phe Gln Asn Phe Lys Asp Ile Cys Glu Gly
 500 505 510
 Trp Ala Ser Phe Asn Arg Asp Ser His Phe Leu Thr Leu Ala Ser Val
 515 520 525
 Lys Cys Trp Asp Leu Pro Asp Asp Val Phe Asp Ala Lys Ala Gly Glu
 530 535 540
 Val Lys Pro Ser Arg Pro Val Lys Lys Val Ala Lys Gln Val Lys Ala
 545 550 555 560
 Glu Ala Lys Ser Gly Val Arg Arg Ser Ile Asn Glu Val Glu Ala Thr
 565 570 575
 Glu Thr Asn Gly Asp Ala Ala Lys Arg Gln Lys Leu Met Thr Pro Thr
 580 585 590
 Pro Thr Pro Thr Pro Thr Pro Ala Pro Lys Pro Thr Ala Ala Pro Ala
 595 600 605

 <210> SEQ ID NO 131
 <211> LENGTH: 626
 <212> TYPE: PRT
 <213> ORGANISM: *Scedosporium apiospermum*

 <400> SEQUENCE: 131

 Met Ala Asp Arg Gln Leu Gly Val Thr Pro Pro Ile Ser Met Ala Leu
 1 5 10 15
 Pro Thr Pro Ala Glu Ile Glu Ala Asn Asn Ala Met Val Glu Glu Leu
 20 25 30
 Arg Lys Gln Gly Ile Phe Glu Ser Lys Glu Glu Thr Asp Lys Arg Asn
 35 40 45
 Leu Val Leu Glu Ser Leu Gln Lys Met Cys Asp Glu Phe Val Thr Arg
 50 55 60
 Val Ala Gln Glu Lys Asp Gln Gly Leu Ala Arg Asp Ala Arg Gly Lys
 65 70 75 80
 Ile Phe Thr Tyr Gly Gly Phe Arg Leu Gly Val Phe Gly Pro Gly Ser
 85 90 95
 Asp Ile Asp Thr Leu Val Val Ala Pro Lys Tyr Val Thr Arg Asp Asp
 100 105 110
 Tyr Phe Ala Ile Phe Pro Asp Leu Leu Leu Glu Met Ala Pro Lys Gly
 115 120 125
 Ala Ile Thr Gly Leu Ala Val Val Thr Asp Ala Phe Val Pro Ile Ile
 130 135 140

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Lys	Phe	Glu	Tyr	Trp	Gly	Ile	Ser	Ile	Asp	Met	Ile	Phe	Ser	Arg	Ile	145	150	155	160
Ala	Met	Leu	Lys	Lys	Leu	Pro	Thr	Asn	Thr	Ser	Gln	Phe	Asn	Leu	Thr	165	170	175	
Asp	Thr	Ala	Leu	Leu	Arg	Gly	Leu	Asp	Glu	Thr	Glu	Ile	Arg	Ser	Leu	180	185	190	
Asn	Gly	Thr	Arg	Val	Ala	His	Glu	Ile	Leu	Asp	Leu	Val	Pro	Glu	Gln	195	200	205	
Ser	Thr	Phe	Gln	Met	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala	Gln	Arg	210	215	220	
Arg	Ala	Ile	Tyr	Ala	Asn	Val	Met	Gly	Tyr	Pro	Gly	Gly	Val	Ala	Trp	225	230	235	240
Ala	Met	Leu	Val	Ala	Arg	Val	Cys	Gln	Leu	Tyr	Pro	Arg	Ala	Thr	Ala	245	250	255	
Ala	Thr	Ile	Val	Ser	Lys	Phe	Phe	Cys	Ile	Met	Arg	Gln	Trp	Pro	Trp	260	265	270	
Pro	Gln	Pro	Val	Leu	Leu	Lys	His	Ile	Glu	Arg	Ala	Pro	Leu	Gly	Tyr	275	280	285	
Arg	Ile	Trp	Asn	Pro	Val	Val	Tyr	Pro	Ser	Asp	Lys	His	His	Leu	Met	290	295	300	
Pro	Ile	Ile	Thr	Pro	Ala	Tyr	Pro	Ser	Met	Asn	Ala	Ala	Tyr	Asn	Ile	305	310	315	320
Asn	Arg	Ser	Ser	Met	Ser	Ile	Ile	Gln	Thr	Glu	Leu	Ser	Arg	Ala	Asp	325	330	335	
Gly	Ile	Thr	Asp	Gly	Ile	Val	Val	Gly	Gln	Arg	Pro	Trp	Ser	Asp	Leu	340	345	350	
Phe	Glu	Lys	His	Thr	Phe	Phe	Thr	Ala	Asp	Tyr	Lys	Tyr	Tyr	Leu	Ala	355	360	365	
Ile	Val	Ser	Ser	Gly	Ile	Thr	Lys	Asp	Ala	His	Lys	Lys	Trp	Ser	Gly	370	375	380	
Phe	Val	Glu	Ser	Lys	Val	Arg	Met	Leu	Val	Leu	Ala	Leu	Asp	Arg	His	385	390	395	400
Asp	Thr	Ile	Ala	Leu	Ala	Gln	Ala	Phe	Val	Lys	Gly	Tyr	Asp	Arg	Val	405	410	415	
His	Lys	Cys	Asn	Ala	Asp	Ser	Glu	Ile	Gln	Lys	Val	Gln	Gln	Gly	Asp	420	425	430	
Leu	Ala	Tyr	Met	Ile	Lys	Glu	Glu	Asp	Ala	Pro	Lys	Glu	Glu	Gln	Thr	435	440	445	
Ser	Pro	Val	Lys	Ser	Glu	Thr	Lys	Pro	Asp	Pro	Gly	Ala	Asn	Glu	Lys	450	455	460	
Asn	Ser	Ala	Gly	Gly	Ser	Asn	Ser	Asn	Gly	Asp	Val	Ala	Gly	Ser	Ala	465	470	475	480
Asp	Ala	Val	Lys	Asn	Glu	Glu	Gly	Ala	Asn	Gly	Val	Lys	Ala	Gly	Asp	485	490	495	
Glu	Asn	Val	Tyr	Ile	Tyr	Thr	Thr	Thr	His	Tyr	Ile	Gly	Leu	Ala	Leu	500	505	510	
Arg	Pro	Gly	Gln	Lys	Ser	Ile	Asn	Leu	Ala	Asn	Glu	Val	Gly	Glu	Phe	515	520	525	
Lys	Lys	Leu	Cys	Lys	Gly	Trp	Glu	Lys	Phe	Asp	His	Gln	Leu	Asn	Ala	530	535	540	

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Ile Thr Val Gln His Leu Arg Asn Ser Asp Leu Pro Asp Asp Val Phe
545                    550                    555                    560

Ala Glu Gly Glu Val Lys Pro Arg Lys Arg Val Val Arg Lys Ala Asn
                    565                    570                    575

Gly Thr Gly Pro Ser Pro Pro Val Asp Ala Lys Gln Ala Thr Asn Gly
                    580                    585                    590

Thr Thr Ser Gln Val Asp Ser Lys Lys Arg Ala Ala Ala Glu Thr Gly
                    595                    600                    605

Ala Ala Ala Pro Ala Ala Lys Arg Arg Gln Pro Ala Ser Val Val Ala
610                    615                    620

Ala Gly
625

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<210> SEQ ID NO 132
<211> LENGTH: 613
<212> TYPE: PRT
<213> ORGANISM: Trichoderma citrinoviride

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<400> SEQUENCE: 132

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Met Ala Glu Gln Asn Tyr Gly Val Thr Pro Pro Ile Ser Val Thr Leu
1      5      10      15

Pro Thr Glu Ala Glu Asn Arg Ala Ser Asp Ala Leu Leu Glu Glu Leu
20     25     30

Arg Arg Gln Lys Thr Phe Glu Ser Pro Ser Asp Thr Glu Lys Arg His
35     40     45

Glu Ile Cys Asn Glu Phe Val Arg Lys Val Ala Arg Glu Arg Glu Pro
50     55     60

Lys Asn Glu Ile Leu Ile Lys Asn Ala Arg Gly Lys Val Phe Thr Tyr
65     70     75     80

Gly Ser Phe Arg Leu Gly Val Phe Gly Pro Gly Ser Asp Ile Asp Thr
85     90     95

Leu Ile Val Ala Pro Lys Tyr Val Thr Arg Glu Asp Tyr Phe Asn Tyr
100    105    110

Phe Pro Asp Leu Leu Val Ser Met Ala Pro Pro Gly Ala Ile Thr Asp
115    120    125

Leu Thr Val Val Lys Asp Ala Phe Val Pro Ile Ile Lys Phe Glu Tyr
130    135    140

Ser Gly Ile Ser Ile Asp Leu Ile Phe Ser Arg Ile Ile Gln Lys Gln
145    150    155    160

Ile Ala Pro Asp Phe His Ser Leu Lys Asp Ser Ser Leu Leu Arg Gly
165    170    175

Leu Asp Glu Ala Glu Leu Arg Ser Leu Asn Gly Thr Arg Val Thr Asp
180    185    190

Glu Ile Leu Glu Leu Val Pro Glu Lys Ser Thr Phe Lys Leu Ala Leu
195    200    205

Arg Ala Ile Lys Leu Trp Ala Gln Arg Arg Ala Val Tyr Ala Asn Ile
210    215    220

Met Gly Phe Pro Gly Gly Val Ala Trp Ala Met Leu Val Ala Arg Val
225    230    235    240

Cys Gln Leu Tyr Pro Lys Ala Thr Met Ser Val Ile Val Asn Lys Phe
245    250    255

Phe Leu Val Ile Gly Gln Trp Arg Trp Pro Gln Pro Val Leu Leu Lys
260    265    270

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Pro Ile Glu Gly Gly Pro Leu Pro Val Arg Val Trp Asn Pro Lys Val
    275                      280                      285

Tyr Lys Gly Asp Ser Phe His Leu Met Pro Val Ile Thr Pro Ala Tyr
    290                      295                      300

Pro Ser Met Cys Ala Thr Phe Asn Ile Thr Arg Ser Ser Met Thr Ile
    305                      310                      315                      320

Ile Gln Arg Glu Leu Arg Arg Gly Leu Glu Ile Ser Glu Gln Ile Met
                      325                      330                      335

Val Gly Lys Arg Pro Trp Ser Asp Leu Phe Val Lys His Thr Phe Phe
                      340                      345                      350

Thr Ser Gly Tyr Arg Tyr Tyr Ile Ser Val Val Ser Ala Ser Lys Asp
    355                      360                      365

Lys Glu Ala His Lys Val Trp Ser Gly Tyr Val Glu Ser Lys Ile Arg
    370                      375                      380

Met Leu Val Gln Lys Leu Glu Gln His Pro Ser Ile Ala Leu Ala His
    385                      390                      395                      400

Ala Phe Asn Lys Gly Tyr Asp Arg Arg His Leu Cys Arg Asn Glu Gln
                      405                      410                      415

Glu Ile Gly Gln Val Gln Glu Gly Ser Leu Glu Phe Met Ile Lys Asp
    420                      425                      430

Trp Asp Asp Asn Lys Leu Asn Gly Ala Ala Ile Lys Pro Glu Lys Val
    435                      440                      445

Glu Gly Asp Ser Lys Pro Asn Gly Ala Asp Ile Ile Gln Pro Glu Lys
    450                      455                      460

Val Glu Gly Glu Ser Lys Leu Ser Asp Ile Pro Val Lys Pro Glu Met
    465                      470                      475                      480

Asp Glu Ala Glu Ser Lys Leu Ser Glu Val Ala Ile Lys Thr Glu Thr
                      485                      490                      495

Lys Asp Gly Ser Thr Glu Ser Val Phe Pro Val Glu Val Tyr Thr Thr
    500                      505                      510

Thr His Tyr Ile Gly Leu Glu Leu Glu Glu Gly Ala Lys Ser Leu Asp
    515                      520                      525

Leu Ser Tyr Gln Val Asp Glu Phe Lys Val Leu Cys Thr Ser Trp Lys
    530                      535                      540

Lys Tyr Gln Glu Asp Leu Glu His Leu Val Ser Leu Gly Val Gln His
    545                      550                      555                      560

Val Arg Asn Phe Asn Leu Pro Asp Asp Val Phe Glu Pro Gly Glu Gln
                      565                      570                      575

Lys Pro Gln Lys Lys Ser Ser Ala Lys Ser Leu Ala Asn Lys Lys Arg
    580                      585                      590

Gly Ala Pro Glu Asp Asn Thr Pro Pro Ala Lys Arg Gln Gln Ala Ser
    595                      600                      605

Val Ala Ala Ala Gly
    610

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<210> SEQ ID NO 133

<211> LENGTH: 598

<212> TYPE: PRT

<213> ORGANISM: Aspergillus thermomutatus

<400> SEQUENCE: 133

Met Ser Ala Pro Pro Val Arg Gln Trp Gly Val Thr Pro Pro Ile Ser

-continued

1	5	10	15
Thr Ala Leu Pro Thr Pro Asp Glu Leu Ala Ala Asn Asp Asp Leu Ile	20	25	30
Ser Glu Leu Lys Ala Gln Asn Asn Phe Glu Ser Pro Ala Glu Thr Glu	35	40	45
Arg Arg Lys Gln Val Leu Gln Leu Ile Gln Arg Val Thr His Glu Phe	50	55	60
Val Lys Val Val Ser Arg Lys Lys Gly Leu Ser Pro Ala Ala Val Glu	65	70	75
Ala Ala Gly Gly Lys Ile Phe Thr Tyr Gly Ser Tyr Arg Leu Gly Val	85	90	95
Tyr Gly Pro Gly Ser Asp Ile Asp Thr Leu Val Val Gly Pro Lys His	100	105	110
Val Leu Ile Asp Asp Phe Phe Ser Asp Phe Pro Pro Val Leu Glu Lys	115	120	125
Met Ala Pro Pro Gly Ala Ile Glu Lys Met Thr Pro Val Pro Asp Ala	130	135	140
Phe Val Pro Ile Ile Lys Leu Glu Leu Ser Gly Ile Ser Ile Asp Leu	145	150	155
Ile Phe Ala Arg Leu Ile Val Ser Ser Val Pro Leu Asn Leu Asp Leu	165	170	175
Lys Asn Asn Asp Tyr Leu Arg Gly Leu Asp Glu Lys Glu Val Arg Ser	180	185	190
Leu Asn Gly Thr Arg Val Thr Asp Glu Ile Leu Glu Leu Val Pro Gln	195	200	205
Gln Lys Thr Phe Arg Leu Ala Leu Arg Ala Ile Lys Leu Trp Ala Gln	210	215	220
Arg Arg Ala Ile Tyr Ser Asn Ile Val Gly Phe Pro Gly Gly Val Ala	225	230	235
Trp Ala Met Leu Val Ala Arg Val Cys Gln Leu Tyr Pro His Ala Thr	245	250	255
Gly Ser Val Ile Val Gly Lys Phe Phe Arg Ile Met Asn Lys Trp Ala	260	265	270
Trp Pro Gln Pro Val Leu Leu Lys Pro Ile Glu Asp Gly Pro Leu Gln	275	280	285
Ile Lys Val Trp Asn Pro Lys Ile Tyr His Gly Asp Arg Phe His Leu	290	295	300
Met Pro Ile Ile Thr Pro Ala Tyr Pro Ser Met Cys Ala Thr His Asn	305	310	315
Ile Ser Met Ser Thr Lys Ala Val Ile Leu Arg Glu Leu Gln Arg Gly	325	330	335
Gly Asp Ile Val Asp Lys Ile Phe Leu Lys Gln Leu Thr Trp Asn Asp	340	345	350
Leu Phe Ala Arg His Ser Phe Phe Thr His Asp Tyr Lys Tyr Tyr Leu	355	360	365
Ser Ile Thr Ala Ser Ser Arg Thr Lys Glu Ala Glu Ser Val Trp Ser	370	375	380
Gly Leu Val Glu Ser Lys Ile Arg His Leu Val Gly Ala Leu Asp Arg	385	390	395
Lys Pro Thr Ile Ala Val Ala His Pro Phe Pro Lys Gly Phe Glu Arg	405	410	415

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Val His Ile Ile Ser Asn Glu Glu Glu Ala Glu Ala Val Lys Asn Gly
420 425 430

Ser Thr Lys Tyr Gln Asp Lys Gly Thr Lys Thr Glu Thr Thr Asp Glu
435 440 445

Thr Lys Asp Ala Ala His Gln Ala Ala Ala Gln Ser Gly Val Glu Asn
450 455 460

Ala Glu Val Glu Pro Val Gly Glu Asn Ala Asn Gly Asp Ser Arg Ile
465 470 475 480

Ile Tyr Thr Thr Thr Tyr Tyr Ile Gly Leu Glu Leu Lys Pro Leu Glu
485 490 495

Pro Gly Ala Ser Arg Ser Leu Asp Ile Ser Thr Asp Ala Gln Ile Phe
500 505 510

Lys Ser Thr Cys Thr Ser Trp Ala Gly Tyr Gln Pro Gly Ile Asn Asp
515 520 525

Leu Ser Ile Thr His Val Arg Asn Phe Asp Leu Pro Glu Asp Val Phe
530 535 540

Gln Pro Gly Glu Ala Arg Pro Thr Arg Pro Lys Lys Lys Val Ile Lys
545 550 555 560

Lys Pro Glu Ala Gly Ala Gln Lys Arg Gly Ile Asp Ser Leu Asp Asp
565 570 575

Ala Ser Leu Pro Ala Ala Lys Arg Gln Val Thr Ser Asn Gly Ile Ser
580 585 590

Ser Thr Pro Thr Pro Ala
595

<210> SEQ ID NO 134
<211> LENGTH: 579
<212> TYPE: PRT
<213> ORGANISM: Sodiomyces alkalinus

<400> SEQUENCE: 134

Met Gln Asp Arg Pro Leu Gly Val Thr Pro Pro Ile Ser Thr Ser Leu
1 5 10 15

Pro Thr Asp Ala Glu His Ala Val Asn Glu Ser Leu Leu Glu Glu Leu
20 25 30

Lys Arg Gln Asn Thr Phe Glu Ser Pro Ala Glu Thr Ala Lys Arg Glu
35 40 45

Lys Val Leu Glu Gln Leu Gln Leu Ile Cys Asn Glu Phe Val Arg Arg
50 55 60

Val Ala Arg Glu Arg Glu Ala Gly Ser Glu Ala Leu Ile Lys Glu Ala
65 70 75 80

Arg Gly Arg Ile Phe Thr Tyr Gly Ser Phe Arg Leu Gly Val Tyr Gly
85 90 95

Pro Gly Ser Asp Ile Asp Ala Leu Val Val Ala Pro Lys Tyr Val Thr
100 105 110

Arg Asp Asp Tyr Phe Lys Leu Phe Pro Gly Leu Leu Gln Glu Ile Pro
115 120 125

Pro Thr Gly Ala Ile Thr Asp Leu Ala Val Val Ser Asp Ala Phe Val
130 135 140

Pro Ile Ile Lys Phe Asp Phe Leu Gly Ile Ser Ile Asp Leu Ile Phe
145 150 155 160

Ser Arg Ile Ala Ser Leu Lys Gln Leu Pro Lys Asp Lys Asp Trp Asn

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165					170					175					
Leu	Lys	Asp	Ser	Asn	Ile	Leu	Arg	Gly	Leu	Asp	Glu	Ala	Glu	Leu	Arg
			180					185					190		
Ser	Leu	Asn	Gly	Thr	Arg	Val	Thr	Asp	Glu	Ile	Ile	Ser	Leu	Val	Pro
		195					200					205			
Glu	Pro	Ser	Thr	Phe	Arg	Leu	Ala	Leu	Arg	Ala	Ile	Lys	Leu	Trp	Ala
	210					215					220				
Gln	Arg	Arg	Ala	Val	Tyr	Ala	Asn	Ile	Met	Gly	Phe	Pro	Gly	Gly	Val
225					230					235					240
Ala	Trp	Ala	Met	Leu	Val	Ala	Arg	Val	Cys	Gln	Leu	Tyr	Pro	Lys	Ala
			245						250					255	
Thr	Ser	Ser	Val	Val	Val	Asn	Lys	Phe	Phe	His	Ile	Met	Arg	Arg	Trp
			260					265					270		
Pro	Trp	Pro	Gln	Pro	Val	Leu	Leu	Lys	Ala	Val	Glu	Ser	Gly	Pro	Leu
		275					280					285			
Gln	Val	Arg	Val	Trp	Asn	Pro	Lys	Leu	Tyr	Lys	Gly	Asp	Gln	Phe	His
	290					295					300				
Leu	Met	Pro	Ile	Ile	Thr	Pro	Ala	Tyr	Pro	Ser	Met	Cys	Ala	Thr	Tyr
305					310					315					320
Asn	Ile	Thr	Lys	Ser	Ala	Met	Thr	Val	Ile	Gln	Arg	Glu	Leu	Gln	Arg
				325					330					335	
Gly	Cys	Glu	Ile	Thr	Asp	Ser	Val	Met	Met	Ser	Lys	Gln	Pro	Trp	Ser
		340						345					350		
Asp	Leu	Phe	Val	Lys	His	Ala	Phe	Phe	Thr	Ser	Asp	Tyr	Lys	His	Tyr
		355					360					365			
Ile	Ser	Val	Ile	Thr	Thr	Ser	Thr	Thr	Lys	Glu	Ala	His	Lys	Ile	Trp
	370					375					380				
Ser	Gly	Tyr	Val	Glu	Ser	Lys	Val	Arg	Val	Leu	Val	Gln	Gly	Leu	Glu
385					390					395					400
Gln	His	Pro	Ser	Ile	Ala	Leu	Ala	His	Ala	Phe	Asn	Lys	Gly	Tyr	Asp
				405					410					415	
Arg	Arg	His	Lys	Cys	Ser	Ser	Glu	Gln	Glu	Ile	Asn	Gln	Val	Gln	Glu
			420					425					430		
Gly	Ser	Leu	Asp	Tyr	Leu	Leu	Lys	Ala	Gly	Asp	Ser	Ser	Ala	Gly	Asp
		435					440					445			
Pro	Ala	Glu	Gly	Asn	Lys	Ser	Pro	Arg	Arg	Pro	Ala	Asp	Gly	Glu	His
	450					455						460			
Pro	Glu	Ile	Thr	Pro	Glu	Ala	Thr	Glu	Thr	Ile	Val	Phe	Thr	Thr	Thr
465					470					475					480
His	Tyr	Ile	Gly	Leu	Glu	Leu	Val	Glu	Gly	Ala	Lys	Ser	Leu	Asp	Leu
				485					490					495	
Ser	Tyr	Gln	Val	Asp	Ser	Phe	Lys	Gln	Leu	Cys	Gly	Gln	Trp	Glu	Lys
			500					505					510		
Phe	Asn	Ser	Asn	Leu	Asn	Tyr	Leu	Ser	Val	Gln	His	Val	Arg	Asn	Val
		515					520					525			
Lys	Leu	Pro	Asp	Asp	Val	Phe	Glu	Pro	Cys	Glu	Thr	Arg	Pro	Gln	Lys
	530					535						540			
Lys	Ser	Ala	Thr	Asn	Gly	Ser	Leu	Gln	Lys	Lys	Lys	Arg	Gly	Ala	Ser
545					550					555					560
Glu	Val	Asp	Met	His	Pro	Pro	Ala	Lys	Arg	Gln	Gln	Ser	Ser	Val	Thr
				565					570					575	

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Ala Ala Gly

<210> SEQ ID NO 135

<211> LENGTH: 606

<212> TYPE: PRT

<213> ORGANISM: Neohortaea acidophila

<400> SEQUENCE: 135

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Met Asp Pro Lys Leu Gly Val Ser Gly Thr Met Ser Met Glu Pro Pro
1      5      10      15

Ser Pro Lys Asp Ile Lys Leu Asp Asp Ala Leu Leu Gln Glu Leu Lys
20     25     30

Ser Arg Asn Glu Phe Glu Ala Pro Glu Glu Thr Gln Arg Arg His Ser
35     40     45

Val Leu Asp Arg Leu Glu Ser Val Leu Lys Arg Leu Val Ala Met Val
50     55     60

Gly Lys Gln Gln Gly Leu Pro Pro Gly Ile Leu Asn Glu Ala Gly Gly
65     70     75     80

Lys Ile Phe Thr Phe Gly Ser Phe Glu Leu Gly Val Tyr Gly Pro Lys
85     90     95

Ser Asp Met Asp Thr Leu Met Ala Ala Pro Lys His Val Ser Arg Glu
100    105    110

Asp Phe Phe Gln Tyr Met Pro Asp Leu Leu Arg Lys Glu Phe Lys Pro
115    120    125

Glu Glu Ile Ala Glu Leu Thr Pro Val Pro Gly Ile Ser Val Pro Ile
130    135    140

Ile Lys Leu Glu Leu Cys Gly Val Ser Val Asp Leu Ile Phe Cys Arg
145    150    155    160

Leu His Leu Gln Ser Val Pro Lys Ser Gln Glu Leu Ser Asn Leu Asp
165    170    175

Leu Leu Arg Gly Leu Asp Asp Thr Asp Leu Lys Cys Val Asn Gly Thr
180    185    190

Arg Val Thr Arg Arg Ile Leu Glu Leu Val Pro Gln Thr Lys Val Phe
195    200    205

Arg Met Ala Leu Arg Ala Val Lys Leu Trp Ala Lys Gln Arg Ala Leu
210    215    220

Tyr Gly Asn Ile Val Gly Tyr Pro Gly Gly Val Ala Tyr Ala Met Met
225    230    235    240

Val Ala Arg Ile Cys Gln Leu Tyr Pro Arg Ala Ala Ala Pro Leu Val
245    250    255

Ile Trp Lys Phe Phe Tyr Leu Met Arg Lys Trp Asn Trp Pro Ser Pro
260    265    270

Val Leu Leu Gln Asn His Glu Glu Gly Ser Ile Asn Leu Arg Glu Trp
275    280    285

Asp Pro Ser Ile Tyr Pro Gly Asp Lys Arg His Leu Met Pro Ile Ile
290    295    300

Thr Pro Ala Phe Pro Arg Met Asn Ala Cys His Thr Ile Gly Pro Ser
305    310    315    320

Thr Lys Lys Val Leu Leu Gln Glu Met Glu Arg Ala Glu Gly Ile Val
325    330    335

Arg Ser Ile Tyr Glu Ser Gly Arg Pro Trp Arg Asp Leu Phe Gln Arg
340    345    350

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His Ser Phe Phe Thr Asp Ala Tyr Arg His Tyr Ile Cys Val Ile Thr
 355 360 365
 Ala Gly Arg Thr Lys Glu Ala Gln Gln Ala Trp Ser Gly Leu Val Glu
 370 375 380
 Ser Lys Val Lys Trp Leu Ile Val Gly Ile Glu His Ser Asp Ala Lys
 385 390 395 400
 Ser Val Glu Leu Val Gln Pro Tyr Asn Lys Gly Phe Asn Arg Val His
 405 410 415
 Glu Cys Lys Gly Asp Ala Asp Ile Asp Lys Thr Leu Asp Gly Asn Leu
 420 425 430
 Asp Cys Gln Val Lys Glu Ile Lys Thr Val Thr Thr Glu Gln Ala Gly
 435 440 445
 Asp Val Gln Phe Gln Ala Ala Ala Gln Thr Asp Thr Asp Gly Gln Glu
 450 455 460
 Val Pro Ala Pro Asn Gly Glu Ala Glu Val Pro Pro Gln Thr Asp Gly
 465 470 475 480
 Pro Gln Thr Ile Trp Thr Thr Thr Phe Tyr Leu Gly Ile Gly Leu Thr
 485 490 495
 Lys Gly Ala Asn Ser Leu Asp Ile Ser Ser Pro Ile Arg Asp Phe Thr
 500 505 510
 Gln Gln Cys Lys Glu Trp Gln Asn Tyr Asp Glu Asn Leu His Ser Ile
 515 520 525
 Arg Val Lys His Met Arg Asn Tyr Asp Leu Pro Ala Asp Val Phe Ala
 530 535 540
 Glu Gly Glu Thr Arg Pro Thr Arg Thr Lys Lys Lys Ser Ala Pro Lys
 545 550 555 560
 Thr Thr Asp Pro Thr Ala Met Asp Ala Ala Asn Lys Lys Arg Ser Phe
 565 570 575
 Asn Asn Ser Gly Leu Asp Thr Leu Asp Asp Pro Ala Lys Arg Arg Ala
 580 585 590
 Ser Ala Asn Gly Thr Ala Thr Pro Asn Gly Val Pro Pro Arg
 595 600 605

<210> SEQ ID NO 136

<211> LENGTH: 405

<212> TYPE: PRT

<213> ORGANISM: Schizosaccharomyces pombe

<400> SEQUENCE: 136

Met Asn Ile Ser Ser Ala Gln Phe Ile Pro Gly Val His Thr Val Glu
 1 5 10 15
 Glu Ile Glu Ala Glu Ile His Lys Asn Leu His Ile Ser Lys Ser Cys
 20 25 30
 Ser Tyr Gln Lys Val Pro Asn Ser His Lys Glu Phe Thr Lys Phe Cys
 35 40 45
 Tyr Glu Val Tyr Asn Glu Ile Lys Ile Ser Asp Lys Glu Phe Lys Glu
 50 55 60
 Lys Arg Ala Ala Leu Asp Thr Leu Arg Leu Cys Leu Lys Arg Ile Ser
 65 70 75 80
 Pro Asp Ala Glu Leu Val Ala Phe Gly Ser Leu Glu Ser Gly Leu Ala
 85 90 95
 Leu Lys Asn Ser Asp Met Asp Leu Cys Val Leu Met Asp Ser Arg Val

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100				105				110							
Gln	Ser	Asp	Thr	Ile	Ala	Leu	Gln	Phe	Tyr	Glu	Glu	Leu	Ile	Ala	Glu
	115						120					125			
Gly	Phe	Glu	Gly	Lys	Phe	Leu	Gln	Arg	Ala	Arg	Ile	Pro	Ile	Ile	Lys
	130					135					140				
Leu	Thr	Ser	Asp	Thr	Lys	Asn	Gly	Phe	Gly	Ala	Ser	Phe	Gln	Cys	Asp
145					150					155					160
Ile	Gly	Phe	Asn	Asn	Arg	Leu	Ala	Ile	His	Asn	Thr	Leu	Leu	Leu	Ser
			165						170						175
Ser	Tyr	Thr	Lys	Leu	Asp	Ala	Arg	Leu	Lys	Pro	Met	Val	Leu	Leu	Val
			180						185				190		
Lys	His	Trp	Ala	Lys	Arg	Lys	Gln	Ile	Asn	Ser	Pro	Tyr	Phe	Gly	Thr
	195						200						205		
Leu	Ser	Ser	Tyr	Gly	Tyr	Val	Leu	Met	Val	Leu	Tyr	Tyr	Leu	Ile	His
	210					215					220				
Val	Ile	Lys	Pro	Pro	Val	Phe	Pro	Asn	Leu	Leu	Leu	Ser	Pro	Leu	Lys
225					230					235					240
Gln	Glu	Lys	Ile	Val	Asp	Gly	Phe	Asp	Val	Gly	Phe	Asp	Asp	Lys	Leu
			245						250					255	
Glu	Asp	Ile	Pro	Pro	Ser	Gln	Asn	Tyr	Ser	Ser	Leu	Gly	Ser	Leu	Leu
			260				265						270		
His	Gly	Phe	Phe	Arg	Phe	Tyr	Ala	Tyr	Lys	Phe	Glu	Pro	Arg	Glu	Lys
		275					280						285		
Val	Val	Thr	Phe	Arg	Arg	Pro	Asp	Gly	Tyr	Leu	Thr	Lys	Gln	Glu	Lys
	290					295					300				
Gly	Trp	Thr	Ser	Ala	Thr	Glu	His	Thr	Gly	Ser	Ala	Asp	Gln	Ile	Ile
305					310					315					320
Lys	Asp	Arg	Tyr	Ile	Leu	Ala	Ile	Glu	Asp	Pro	Phe	Glu	Ile	Ser	His
			325						330					335	
Asn	Val	Gly	Arg	Thr	Val	Ser	Ser	Ser	Gly	Leu	Tyr	Arg	Ile	Arg	Gly
			340				345						350		
Glu	Phe	Met	Ala	Ala	Ser	Arg	Leu	Leu	Asn	Ser	Arg	Ser	Tyr	Pro	Ile
		355					360						365		
Pro	Tyr	Asp	Ser	Leu	Phe	Glu	Glu	Ala	Pro	Ile	Pro	Pro	Arg	Arg	Gln
	370					375				380					
Lys	Lys	Thr	Asp	Glu	Gln	Ser	Asn	Lys	Lys	Leu	Leu	Asn	Glu	Thr	Asp
385					390					395					400
Gly	Asp	Asn	Ser	Glu											
			405												

<210> SEQ ID NO 137

<211> LENGTH: 333

<212> TYPE: PRT

<213> ORGANISM: T. brucei

<400> SEQUENCE: 137

Met	Pro	Pro	Ser	Pro	Ala	Val	Val	Gly	Arg	Ser	Leu	Val	Asn	Ser	Phe
1				5					10					15	

Lys	Gln	Phe	Val	Ser	Lys	Asp	Leu	His	Thr	Arg	His	Val	Asp	Ala	Thr
		20						25					30		

Tyr	Arg	Leu	Val	Leu	Asp	Cys	Val	Ala	Ala	Val	Asp	Pro	Leu	Met	Arg
		35					40					45			

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Leu	Tyr	Thr	Phe	Gly	Ser	Thr	Val	Val	Tyr	Gly	Val	His	Glu	Lys	Gly
50						55					60				
Ser	Asp	Val	Asp	Phe	Val	Val	Leu	Asn	Lys	Thr	Asp	Val	Glu	Asp	Gly
65					70					75					80
Lys	Gly	Gly	Asp	Ala	Ala	Thr	Gln	Val	Ala	Lys	Gly	Leu	Gln	Ala	Asp
			85						90					95	
Ile	Leu	Ala	Lys	Leu	Ala	Arg	Val	Ile	Arg	Gln	Lys	His	Leu	Ser	Trp
			100					105					110		
Asn	Val	Glu	Glu	Val	Arg	Arg	Thr	Arg	Val	Pro	Val	Val	Arg	Val	Lys
		115					120					125			
Gly	Gly	Gly	Ala	Val	Asp	Phe	Asp	Ile	Thr	Ala	Tyr	Arg	Arg	Asn	Gly
	130					135					140				
Val	Arg	Asn	Ser	Ala	Leu	Leu	Arg	Ala	Tyr	Phe	Glu	Gln	Asn	Pro	Pro
145					150					155					160
Cys	Arg	Trp	Leu	Ser	Met	Ser	Ile	Lys	Arg	Trp	Ser	Lys	Gln	Thr	Gly
			165					170						175	
Leu	Asn	Ala	Ser	Val	Ile	Gly	Gly	Ser	Ile	Thr	Ser	Tyr	Gly	Phe	Asn
		180						185					190		
Leu	Met	Val	Val	Tyr	Tyr	Leu	Leu	Gln	Arg	Asn	His	Leu	Gln	Phe	Val
	195						200					205			
Pro	Pro	Ser	Thr	Ile	Asp	Val	Ser	Arg	Val	Glu	Pro	Leu	Pro	Pro	His
	210					215					220				
Leu	Pro	Leu	Glu	Glu	Pro	Ala	Asp	Glu	Gly	Leu	Glu	Leu	Gly	Thr	Gln
225					230					235					240
Val	Leu	Asp	Phe	Leu	His	Phe	Phe	Leu	His	Glu	Phe	Asp	Ser	Asp	Lys
			245					250						255	
Gln	Val	Ile	Ser	Leu	Asn	Arg	Pro	Gly	Ile	Thr	Thr	Lys	Glu	Glu	Leu
			260					265					270		
Asp	Trp	Thr	Lys	Ser	Ala	Glu	Asp	Phe	Ala	Arg	Met	Asn	Gly	Glu	Lys
		275					280					285			
Val	His	Tyr	Gln	Trp	Cys	Ile	Glu	Asp	Pro	Tyr	Glu	Leu	Asn	Leu	Asn
	290					295					300				
Val	Gly	Arg	Asn	Val	Thr	Pro	Leu	Lys	Arg	Asp	Phe	Leu	Arg	Arg	His
305				310						315					320
Leu	Glu	Lys	Ala	Arg	Asp	Thr	Ala	Leu	Leu	Thr	Ile	Val			
			325					330							

<210> SEQ ID NO 138

<211> LENGTH: 349

<212> TYPE: PRT

<213> ORGANISM: S. pombe

<400> SEQUENCE: 138

Gly	Ser	His	Met	Ser	Tyr	Gln	Lys	Val	Pro	Asn	Ser	His	Lys	Glu	Phe
1				5					10					15	
Thr	Lys	Phe	Cys	Tyr	Glu	Val	Tyr	Asn	Glu	Ile	Lys	Ile	Ser	Asp	Lys
		20						25				30			
Glu	Phe	Lys	Glu	Lys	Arg	Ala	Ala	Leu	Asp	Thr	Leu	Arg	Leu	Cys	Leu
		35				40					45				
Lys	Arg	Ile	Ser	Pro	Asp	Ala	Glu	Leu	Val	Ala	Phe	Gly	Ser	Leu	Glu
	50					55					60				
Ser	Gly	Leu	Ala	Leu	Lys	Asn	Ser	Asp	Met	Asp	Leu	Cys	Val	Leu	Met
65				70						75					80

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Asp Ser Arg Val Gln Ser Asp Thr Ile Ala Leu Gln Phe Tyr Glu Glu
 85 90 95
 Leu Ile Ala Glu Gly Phe Glu Gly Lys Phe Leu Gln Arg Ala Arg Ile
 100 105 110
 Pro Ile Ile Lys Leu Thr Ser Asp Thr Lys Asn Gly Phe Gly Ala Ser
 115 120 125
 Phe Gln Cys Asp Ile Gly Phe Asn Asn Arg Leu Ala Ile His Asn Thr
 130 135 140
 Leu Leu Leu Ser Ser Tyr Thr Lys Leu Asp Ala Arg Leu Lys Pro Met
 145 150 155 160
 Val Leu Leu Val Lys His Trp Ala Lys Arg Lys Gln Ile Asn Ser Pro
 165 170 175
 Tyr Phe Gly Thr Leu Ser Ser Tyr Gly Tyr Val Leu Met Val Leu Tyr
 180 185 190
 Tyr Leu Ile His Val Ile Lys Pro Pro Val Phe Pro Asn Leu Leu Leu
 195 200 205
 Ser Pro Leu Lys Gln Glu Lys Ile Val Asp Gly Phe Asp Val Gly Phe
 210 215 220
 Asp Asp Lys Leu Glu Asp Ile Pro Pro Ser Gln Asn Tyr Ser Ser Leu
 225 230 235 240
 Gly Ser Leu Leu His Gly Phe Phe Arg Phe Tyr Ala Tyr Lys Phe Glu
 245 250 255
 Pro Arg Glu Lys Val Val Thr Phe Arg Arg Pro Asp Gly Tyr Leu Thr
 260 265 270
 Lys Gln Glu Lys Gly Trp Thr Ser Ala Thr Glu His Thr Gly Ser Ala
 275 280 285
 Asp Gln Ile Ile Lys Asp Arg Tyr Ile Leu Ala Ile Glu Asp Pro Phe
 290 295 300
 Glu Ile Ser His Asn Val Gly Arg Thr Val Ser Ser Ser Gly Leu Tyr
 305 310 315 320
 Arg Ile Arg Gly Glu Phe Met Ala Ala Ser Arg Leu Leu Asn Ser Arg
 325 330 335
 Ser Tyr Pro Ile Pro Tyr Asp Ser Leu Phe Glu Glu Ala
 340 345

<210> SEQ ID NO 139

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: T. boudieri

<400> SEQUENCE: 139

Met Ser Gly Pro Pro Ser Gly Pro Pro Ser Gly Pro Arg Tyr Arg Gly
 1 5 10 15
 Gly His Arg Asn Phe Gly Pro Arg His Leu Gln Ser Gln Glu Arg Gln
 20 25 30
 Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Pro Tyr Gln Pro
 35 40 45
 Ser Pro Gln Glu Leu Ala Val Leu Ala Glu Val Val Ile Glu Gln Ile
 50 55 60
 Ala Leu Ala Thr Pro Ser Ala Glu Glu Leu Glu Arg Lys Thr Lys Leu
 65 70 75 80
 Arg Glu Arg Leu Arg Lys Leu Cys Thr Asn Val Ser Pro Thr Ala Glu

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85					90					95					
Leu	Glu	Ala	Phe	Gly	Ser	Leu	Val	Ser	Gly	Phe	Ala	Thr	Val	Gly	Ser
		100						105					110		
Asp	Leu	Asp	Leu	Val	Leu	Val	Asp	Lys	Gly	Gly	Ser	Thr	Asn	Arg	Pro
	115						120					125			
Phe	Phe	Glu	Val	Pro	Leu	Leu	Leu	Glu	Gln	Glu	Leu	Lys	Asp	Asn	Gly
	130					135					140				
Leu	Glu	Ala	Gln	Leu	Leu	Ala	Lys	Thr	Arg	Val	Pro	Ile	Leu	Lys	Ile
145						150					155				160
Lys	Gln	Pro	Ala	Thr	Glu	Glu	Tyr	Pro	Lys	Glu	Val	Ala	Ala	Asp	Ile
			165						170						175
Gly	Phe	Ala	Asn	Pro	Leu	Ala	Ile	Arg	Asn	Thr	Asp	Met	Leu	Ser	Met
			180					185					190		
Tyr	Ser	Lys	Cys	Asp	Pro	Arg	Val	Ile	Asp	Met	Val	Arg	Phe	Ile	Lys
	195						200					205			
Arg	Trp	Ala	Lys	Arg	Arg	Lys	Ile	Asn	Asn	Pro	Tyr	Lys	Gly	Thr	Leu
	210					215					220				
Ser	Ser	Tyr	Gly	Tyr	Val	Leu	Met	Val	Leu	His	Tyr	Leu	Ile	Asn	Ile
225						230					235				240
Val	Glu	Pro	Pro	Val	Leu	Pro	Asn	Leu	Gln	Leu	Tyr	Pro	Ile	Pro	Ala
			245						250					255	
Glu	Thr	Pro	Lys	Asp	Glu	Ile	Thr	Thr	Glu	Glu	Gly	His	Asn	Val	Trp
			260					265					270		
Tyr	Tyr	Lys	Asp	Val	Ala	Glu	Ile	Gln	Arg	Arg	Val	Ala	Glu	Gly	Thr
	275						280					285			
Met	Thr	Thr	Asn	Thr	Met	Asp	Leu	Pro	Leu	Leu	Leu	Gly	Phe	Phe	
	290					295					300				
Glu	Phe	Tyr	Ala	Tyr	Arg	Phe	Gly	Trp	Val	Lys	Asp	Ile	Ile	Ser	Ile
305						310					315				320
Arg	Thr	Lys	Gly	Gly	Leu	Ile	Ser	Lys	Val	Asp	Lys	Gly	Trp	Thr	Val
			325						330					335	
Val	Ala	Val	Arg	Val	Gly	Lys	Cys	Glu	Ala	Glu	Tyr	Lys	Asp	Arg	Tyr
			340					345					350		
Leu	Phe	Ala	Ile	Glu	Asp	Pro	Phe	Glu	Thr	Asn	His	Asn	Val	Ser	Arg
	355						360					365			
Thr	Cys	Asn	Ile	Tyr	Gly	Val	Arg	Lys	Ile	Arg	Asp	Glu	Phe	Lys	Arg
	370					375					380				
Ala	Asn	Arg	Ile	Met	Arg	Met	Arg	Glu	Gly	Val	Gly	Ala	Leu	Arg	Tyr
385						390					395				400
Lys	Leu	Phe	Glu	Glu	Ala	Pro	Glu	Asp	Val	Arg	Pro	Gln	His	His	Gln
			405						410					415	
Lys	Gln	Ser	Ile	Ala	Gly	Glu	Glu	His	Val	Asn	Gly	Ala	Pro	Glu	Ser
			420					425					430		
Gly	Arg	Glu	Asp	Arg	Glu	Gly	Asn	Gln	Leu	Val	Glu	Glu	Leu	Glu	Gly
		435					440					445			
His	Ser	Asn	Gly	Asn	Gly	Ile	His	Pro	Ser	Glu	Gly	Tyr	Pro	Gly	Pro
	450					455					460				
Gly	Leu	Ala	Ala	Val	Val	Glu	Gly	Val	Gln	Arg	Leu	Ser	Val	Asp	Ala
465						470					475				480

-continued

<211> LENGTH: 761

<212> TYPE: PRT

<213> ORGANISM: D. stenobrocha

<400> SEQUENCE: 140

Met Glu Leu Thr Thr Gly Asp Phe Gly Gly Pro Pro Gly Ala Ser Asn
1 5 10 15

Ala Thr Gly Pro Gly Gln Lys Pro Glu Asp Pro Arg Thr Met Asp Pro
20 25 30

Val Gly Glu Gln Thr Leu Glu Ser Arg Leu Arg Gly Met Leu Leu His
35 40 45

Gly Gly Pro Pro Pro Val Asp Pro Arg Thr Gln Leu Arg His Pro Ser
50 55 60

Gly Leu Ala Gln Gln Ala Gln Gln Ala Gln Gln Gln Pro Ile Gly Ile
65 70 75 80

Asp Pro Ala Val Pro Gly His Leu His Gln Leu Gln Thr Ala Ala Ala
85 90 95

Pro Pro Gln Gln Ser Arg Leu Pro Pro Ser Pro Asn Ile His His Gln
100 105 110

His Leu Gln Pro Arg Leu Leu Gln Arg Ser Asn Thr Leu Pro Pro Pro
115 120 125

Gly Gly Leu Thr Gln Pro Pro Gly Leu Glu Arg Ala Gly Thr Phe
130 135 140

His Gly Pro Val Pro Val Ala Gln Phe Gly Tyr His Gly Ser Tyr Pro
145 150 155 160

Thr Ala His Ser Gln Arg Gly Arg Tyr Gly Asn Arg Arg Asn Gly Phe
165 170 175

Gln Asn Ala Pro Val Pro Gln Met Asn Asn Asn Leu His Phe Pro Pro
180 185 190

Leu Gly Thr Val Pro Ser Gln Pro Pro Pro Pro Leu Ser Glu Arg Ser
195 200 205

Ser Leu Ser Asn Met Asn Asn Pro Arg Arg His Ser Gly Pro Gly His
210 215 220

Tyr His Thr Asp Asn Asn His Ser Pro Pro Tyr Lys Arg Tyr Glu Ala
225 230 235 240

Phe His Gly Ala Pro Gly Ser Pro Gln Gly Tyr Ser Gly Ser Pro Pro
245 250 255

Gln Asn Arg Gln Gly Arg Pro Tyr His Ser Ser Arg Gly Ser His Gly
260 265 270

Gly Arg Asn Asn Arg Asn Phe Trp Pro Leu Asp Tyr Asp Gly Leu Thr
275 280 285

Asn Tyr Ala Lys Tyr Ile Val Glu Ser Val Ser Pro Thr Pro Glu Glu
290 295 300

Ile Ala Met Lys Asp Asn Met Leu Arg Arg Ile Ser Glu Ile Cys Asp
305 310 315 320

Lys Leu Val Pro Gly Ser Arg Ile Ile Pro Phe Gly Ser Leu Val Ser
325 330 335

Gly Phe Ala Thr Lys Gly Ala Asp Met Asp Val Ile Phe Ala His Asp
340 345 350

Thr Ile Asp Pro Ala Pro Ser Ser His Glu Ser Asn Ile Pro Val Arg
355 360 365

Leu Ala Asn Glu Phe Leu Lys Arg Gly Phe Glu Val Asp Leu Leu Ile

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370					375					380					
Lys 385	Thr	Arg	Val	Pro	Ile 390	Leu	Lys	Leu	Lys	Thr 395	Pro	Gly	Glu	Pro	Val 400
Ser	Arg	Pro	Gly	Ser 405	Pro	Val	Ser	Glu	Gly 410	Asp	Gly	Asp	Ser	Ser	Glu 415
Glu	Pro	Trp	Pro	Glu 420	Asn	Val	Ser	Cys 425	Asp	Ile	Gly	Phe	Lys	Ala	His 430
Leu	Gly	Ile	Thr	Asn 435	Ser	His	Phe 440	Phe	Arg	Thr	Tyr	Ser 445	His	Cys	Asp
His	Arg 450	Phe	Arg	Glu	Met 455	Val	Leu 455	Phe	Val	Lys	Gln 460	Trp	Ser	Lys	Asn
Arg 465	Asp	Leu	Asn	Ser	Pro 470	Tyr	Phe	Gly	Thr 475	Leu	Ser	Ser	Tyr	Gly	Tyr 480
Val	Leu	Met	Val	Ala 485	His	Phe	Leu	Ile	Asn 490	Val	Val	Gln	Pro	Pro	Val 495
Leu	Pro	Asn	Leu	Gln 500	Leu	Ile	Pro	Pro 505	Ala	Ala	Glu	Thr	Ala	Glu	Ser
Glu	Leu	Met	Gln	Glu 515	Gly	Phe	Asn 520	Ile	Trp	Tyr	Phe	Lys 525	Asp	Leu	Glu
Arg	Ile 530	Ala	Ser	Gly	Glu	Phe 535	Leu	Pro	Gly	Gly	Arg 540	Asn	Glu	Met	Ser
Leu 545	Gly	Gln	Leu	Val	His 550	Glu	Phe	Phe	Gln 555	Tyr	Tyr	Thr	Thr	Asn	Phe 560
Asn	Phe	Val	Ser	Glu 565	Cys	Val	Thr	Ile	Arg 570	Thr	Pro	Gly	Gly	Val	Met 575
Tyr	Lys	Gln	Glu 580	Lys	Gly	Trp	Thr	Ser 585	Ala	Arg	Glu	Arg	Val	Gly	Glu
Met	Asn 595	Asn	Thr	Tyr	Gln	Asp	Arg 600	Tyr	Leu	Leu	Ala	Leu 605	Glu	Asp	Pro
Phe 610	Glu	Val	Ser	His	Asn	Val 615	Gly	Arg	Thr	Cys	Gly 620	Gly	Ala	Gly	Val
Arg 625	Arg	Ile	Arg	Ala	Glu 630	Met	Gln	Arg	Ala	Ala 635	His	Ile	Ile	Arg	Lys 640
Val	Ser	Ala	Lys	Glu 645	Gly	Gly	Thr	Pro	Arg 650	Ser	Ala	Gly	Trp	Glu	Met 655
Pro	Leu	Val	Val 660	Glu	Asp	Leu	Met	Thr 665	Thr	Val	Arg	Glu	Thr	Asn	Arg 670
Gly	Phe 675	Arg	Asn	Arg	Arg	Val	Asn 680	Gln	Lys	Asn	Leu	Met 685	Glu	Trp	Ile
Arg 690	Asn	Asp	Trp	Thr	Val	Gly 695	Lys	Cys	Leu	Val	Gln 700	Ile	Asp	Ala	Ala
Ala 705	Glu	Glu	Leu	Arg	Lys 710	Lys	Arg	Glu	Leu	Gly 715	Leu	Pro	Asp	Asp	Ala 720
Glu	Leu	Pro	Pro	Glu 725	Ser	Glu	Glu	Asp	Ser 730	Glu	Gly	Asp	Glu	Asp	Ala 735
Ser	Glu	Glu	Glu 740	Glu	Glu	Ser	Ser	Ser 745	Asp	Asp	Gly	Ala	Val	Arg	Val 750
Val	Gly 755	Gly	Gly	Ser	Arg	Gly	Gly	Val							

<210> SEQ ID NO 141

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<211> LENGTH: 337
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Phytomonas sp.

<400> SEQUENCE: 141

Met Thr Ser Leu Ser Ala Val Asn Val Gly Lys Leu Val Cys Asp Ser
1          5          10          15
Phe Gly Lys Leu Leu Ser Lys Asn Leu Asp Ile Lys Pro Ile Val Gln
          20          25          30
Thr His Glu Ile Val Ser Gln Ser Met Gln Thr Val Asp Pro Ser Met
          35          40          45
Ala Leu Tyr Val Phe Gly Ser Thr Ala Val Tyr Gly Phe His Glu Ser
          50          55          60
Gly Cys Asp Val Asp Phe Val Ala Leu Asn Gln Lys Asp Val Ser Asp
          65          70          75          80
Gly Lys Ala Ala Asp Pro Ser Ser Glu Ile Ala Lys Gly Leu Gln Val
          85          90          95
Asp Phe Leu Ser Arg Leu Glu Ala Ser Leu Arg Glu Met His Asn Leu
          100         105         110
Ala Trp Lys Met Asp Leu Val Arg Arg Thr Arg Val Pro Val Leu Arg
          115         120         125
Val Lys Gly Asp Pro Cys Gly Ile Asp Phe Asp Val Thr Ala Arg Arg
          130         135         140
Arg Asn Gly Val Arg Asn Ser Ala Leu Leu Ser Ala Tyr Phe Lys Gln
          145         150         155         160
Lys Pro Glu Thr Arg Trp Leu Ser Met Ala Ile Lys Gln Trp Ser Lys
          165         170         175
Arg Ala Gly Phe Asn Met Ser Val Asp Gly Gly Cys Leu Thr Ser Tyr
          180         185         190
Gly Cys Asn Leu Met Val Val Tyr Tyr Leu Leu Gln Arg Gln Leu Val
          195         200         205
Lys Phe Val Asp Pro Glu Arg Cys Asp Val Ala His Ile Pro Ser Leu
          210         215         220
Pro Ser Tyr Leu Pro Leu Glu His Pro Ala Gln Asn Gly Ser Glu Leu
          225         230         235         240
Gly Asp Met Val Leu Asp Phe Leu Asn Tyr Tyr Leu His Glu Phe Asn
          245         250         255
Pro Glu Thr Glu Val Ile Ser Leu Ser Arg Ala Asp Lys Thr Thr Lys
          260         265         270
Glu Met Ile Tyr Trp Thr Lys Gln Ala Glu Asp Met Ala Arg Ile Ser
          275         280         285
Gly Glu Lys Val Ser Tyr Arg Trp Cys Ile Glu Asp Pro Phe Glu His
          290         295         300
Asn Leu Asn Val Gly Arg Tyr Val Thr Pro Phe Lys Leu Thr Leu Leu
          305         310         315         320
Arg Lys His Met Glu Arg Ala Lys Glu Thr Ala Leu Leu Leu Ser Ile
          325         330         335

Ser

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<210> SEQ ID NO 142

<211> LENGTH: 355

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<212> TYPE: PRT

<213> ORGANISM: B. saltans

<400> SEQUENCE: 142

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Met Leu Ser Pro Ala Ala Ile Gly Lys Ala Ile Leu Asp His His Ala
1      5      10      15
Ser Leu Ile Gly Ala Asp Leu Ser Val Lys Arg Val His Asp Ala His
20     25     30
Ser Leu Val Lys Asn Ser Leu Asp Ser Val Ala Pro Asp Leu Lys Leu
35     40     45
Tyr Thr Phe Gly Ser Ser Thr Val Phe Gly Phe His Glu Pro Lys Ser
50     55     60
Asp Val Asp Phe Val Ala Leu Arg Gln Glu Asp Ile Val Asp Gly Lys
65     70     75     80
Gly Gly Asp Ser Thr Ser Gln Leu Ala Lys Gly Leu Gln Thr Gln Val
85     90     95
Leu Ala Lys Leu Ala Ala Ser Val Arg Gln Lys Asn Val Gln Trp Ala
100    105    110
Val Glu Glu Val Arg Arg Ala Arg Val Pro Val Val Lys Val Lys Ala
115    120    125
Pro His Ile Asp Phe Asp Ile Thr Ala His Arg Arg Asn Gly Val Arg
130    135    140
Asn Ser Ala Leu Leu Arg His Tyr Leu Thr Gln Val Pro Glu Asn Arg
145    150    155    160
Trp Leu Ser Ile Ala Ile Lys Ser Trp Ser Lys Arg Val Gly Met Asn
165    170    175
Gly Pro Val Gly Gly Tyr Leu Thr Ser Tyr Gly Phe Asn Ile Leu Val
180    185    190
Val Tyr Tyr Leu Leu His Arg Arg Arg His Glu Gln Pro Glu Val Glu
195    200    205
Gln Lys Glu Gly Gln Asp Gly Ala Ala Ser Ala Ser Val Asp Gly Lys
210    215    220
Gln Gln Lys Glu Leu Thr Phe Ile Asp Lys Asp Thr Leu Asp Val Ser
225    230    235    240
Leu Ile Pro Pro Ile Pro Glu Tyr Leu Ala Leu Glu Pro Pro Asn Pro
245    250    255
Glu Thr Leu Gly Glu Gln Val Leu Asp Phe Phe Asp Phe Tyr Leu Ser
260    265    270
Arg Phe Pro Met Glu Ser His Val Ile Ser Leu Ser His Lys Glu Pro
275    280    285
Ile Thr Lys Gln Ser Leu Asn Trp Thr Lys Thr Ala Glu Asp Met Lys
290    295    300
Asn Asn Thr Ser Met Glu Lys Val Phe Tyr Arg Leu Cys Ile Glu Asp
305    310    315    320
Pro Tyr Glu Val Asn Leu Asn Val Gly Arg Asn Val Ser Pro Phe Lys
325    330    335
Phe Asp Leu Met Lys Lys His Phe Val Lys Gly Arg Ala Thr Ala Leu
340    345    350
Gly Leu Leu
355

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<210> SEQ ID NO 143

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<211> LENGTH: 425

<212> TYPE: PRT

<213> ORGANISM: A. deanei

<400> SEQUENCE: 143

Met Pro Pro Asn Pro Arg Gln Leu Gly Ala Ala Met Ile Glu Gln Phe
1 5 10 15

Arg Pro Leu Leu Gln Ala Pro Gln Gln Ser Gln Leu Thr Val Leu Ser
20 25 30

Glu Ala Pro Leu Phe Gln Arg Ile Gln Arg Glu Val His Glu Leu Asp
35 40 45

Pro Gly Ala Arg Val Tyr Ile Phe Gly Ser Thr Arg Val Tyr Gly Phe
50 55 60

His Asp Ser Gly Ala Tyr Gln Gln Gln Ser Ala Gln Val Pro Pro Ser
65 70 75 80

Asp Val Asp Met Ala Val Leu Arg Arg Glu Asp Leu Leu Asp Gly Ser
85 90 95

Gly Val Asp Pro Ser Ser Glu Leu Thr Arg Asn Val Gln Ala Glu Phe
100 105 110

Leu Glu Lys Leu Leu Asn Arg Leu Gln Gln Ser Glu Gly Ser Ala Ser
115 120 125

Ile Arg Ser Asn Cys Ser Ser Leu Val Lys Arg Thr Arg Val Pro Val
130 135 140

Leu Arg Val Lys His Leu Val Ser Asn His Ser Pro Pro His Asp Lys
145 150 155 160

Glu Asp Asn Ile Ser Val Gln Val Pro Phe Asp Phe Asp Ile Thr Cys
165 170 175

Gly Arg Arg Cys Gly Ile Arg Asn Ser Ala Leu Phe Phe His Tyr Val
180 185 190

Gln Gln Met Pro Met Ile Arg Phe Leu Leu Leu Ser Val Lys Lys Trp
195 200 205

Ser Lys Gln Thr Gly Leu Asn Ser Ala Ile Lys Ala Pro Tyr Ser Asn
210 215 220

Asn Asn Asn Gln Asn His Asn Leu Thr Ser Val Met Gly Gly Ser Leu
225 230 235 240

Thr Ser Tyr Gly Phe His Val Leu Leu Leu Tyr Tyr Leu Leu Arg Arg
245 250 255

Gly Val Val Gln Tyr Ile Pro Met Thr Val Ser Ser Ala Ala Gly Ser
260 265 270

Thr Asp Arg Ile Pro Asn Pro Gln Leu Asp Val Asn Ala Ile Pro Pro
275 280 285

Val Pro Thr Phe Leu Pro Leu Leu Asp Pro Ala Leu Glu Pro His Gly
290 295 300

Pro Gln Gly Asp Pro Val Arg Tyr Leu Gly Glu Leu Ala Leu Asp Trp
305 310 315 320

Cys His Phe Tyr Leu His Glu Phe Asp Val Glu Lys Glu Val Val Ser
325 330 335

Leu Ser Arg Phe Asp Asp Tyr Phe Ala Asp Ser Pro Pro Ser Gly Pro
340 345 350

Phe Pro Val Ile Thr Lys Asp Val Leu Gly Trp Thr Lys Gln Arg Glu
355 360 365

Asp Gln Gly Arg Leu Arg Gly Glu Lys Val His Tyr Thr Leu Cys Ile

-continued

370	375	380
Glu Asp Pro Tyr Glu Val Asn Leu Asn Val Gly Arg Asn Val Thr Ser		
385	390	395 400
Leu Lys Trp Met Leu Phe Arg Lys His Leu Glu Lys Ala Ile Gln Thr		
	405	410 415
Gly Leu Leu Leu Cys Pro Pro Ser Lys		
	420	425

<210> SEQ ID NO 144
 <211> LENGTH: 432
 <212> TYPE: PRT
 <213> ORGANISM: P. lactucaedebilis

<400> SEQUENCE: 144

Met Pro Ser Ser Ser Pro Leu Gly Ala Thr Leu Pro Asn Ser Ser Ser		
1	5	10 15
Leu Ala Val Ser Ala Ala Ala Val Val Pro Asp Pro Leu Phe Ala		
	20	25 30
Phe Thr Gln Phe Ala Val Asp Ala Val Arg Asn Asn Thr Val Ser His		
	35	40 45
Ala Glu Leu Val Glu Lys Glu Arg Leu Arg Gln Glu Leu Glu Ala Val		
	50	55 60
Leu Gln Ala Leu Gln Pro Thr Ala Arg Leu Val Val Tyr Gly Ser Leu		
	65	70 75 80
Ala Asn Asn Leu Ala Ile Ala Asn Cys Asp Met Asp Leu Met Asp Ala		
	85	90 95
Leu His Asp Val Ser Val Asp Leu Pro Ala Leu Tyr Thr Glu Ala Leu		
	100	105 110
Asn Ser Gln Gly Tyr Ser Val Lys Leu Leu Ser Lys Thr Arg Thr Pro		
	115	120 125
Ile Val Lys Val Ser Leu Glu His Met Ala Val Pro Tyr Cys Val Asp		
	130	135 140
Ile Ser Phe Asp Asn Ala Leu Ala Leu His Asn Thr Lys Leu Ile Ala		
	145	150 155 160
Thr Tyr Ala Ala Cys Asp Ser Arg Val Pro Ile Leu Leu Met Phe Val		
	165	170 175
Lys Leu Trp Thr Ala Val Arg Arg Ile Asn Asp Pro His Ser Gly Thr		
	180	185 190
Leu Ser Ser Tyr Gly Tyr Ala Leu Leu Leu Ile Phe Tyr Leu Gln Asn		
	195	200 205
Arg Cys Asn Ser Pro Pro Val Leu Pro Asn Leu Gln Leu Ile Ser Ala		
	210	215 220
Met Gly Thr Arg Ser Ala Asp Glu Leu Glu Cys Ser Gly Tyr Asp Ile		
	225	230 235 240
Trp Phe Phe Lys Asp Val Glu Lys Ile Gln Gln Ala Gln Val Val Lys		
	245	250 255
Asn Thr Ser Ser Ile Gly Thr Leu Leu Glu Gly Phe Phe Ser Tyr Phe		
	260	265 270
Ala Tyr Glu Phe Asp Phe Arg Asp Leu Cys Ile Ser Ile Arg Thr Pro		
	275	280 285
Gly Gly Ile Val Thr Lys Thr Ala Lys Thr Trp Thr Gln Met Val Glu		
	290	295 300

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His Leu Asn Asp Arg Gly Asp Ala Lys Val Lys Asp Arg Tyr Ile Leu
305                      310                      315                      320

Ser Ile Glu Asp Pro Phe Glu Ile Val His Asn Val Gly Arg Ser Val
                      325                      330                      335

Thr Arg Ala Gly Leu Tyr Glu Ile Arg Gly Glu Phe Met Ala Ala Thr
                      340                      345                      350

Arg Tyr Val Arg Gly Asn Lys Leu Asp Glu Ile Leu Asn Pro His Arg
                      355                      360                      365

Arg Glu Ile Val Ala Pro Thr Ser Ser Thr Ala Met His Ser Pro Gln
                      370                      375                      380

Gln Gln His Gly Thr Gln Gln Val Ser Pro Gly Glu Gly Ser Ser Pro
385                      390                      395                      400

Thr Pro Pro Ala Val His Ala Glu Ala Arg Pro Ser Thr Gly Thr Ala
                      405                      410                      415

Thr Ser Thr Lys Thr Ser Gln Gln Ala Glu Thr Val Val Lys Ala Glu
                      420                      425                      430

<210> SEQ ID NO 145
<211> LENGTH: 437
<212> TYPE: PRT
<213> ORGANISM: S. culicis

<400> SEQUENCE: 145

Met Pro Pro Pro Thr Pro Met Gln Leu Gly Arg Ala Ile Leu Arg His
1      5      10      15

Tyr Ala Pro Ser Ile Val Val Pro Thr Ala Ala Thr Ser Gly Pro Ile
20     25     30

Ser Ser Ala Ala Ala Ser Thr Ala Ser Leu Ala Ser Ile His Pro Leu
35     40     45

Phe Glu Phe Ile Gln Gly Cys Ala Arg Arg Val Asp Pro Gln Thr His
50     55     60

Leu Tyr Val Phe Gly Ser Thr Arg Val Tyr Gly Phe Ser Glu Val Ser
65     70     75     80

Ala Pro Thr Ser Val Gly Pro Pro Ala Ala Leu Ala Arg Pro Val Lys
85     90     95

Asn Asp Val Asp Ile Ala Ala Leu Ser Ala Ala Asp Val Ala Thr Thr
100    105    110

Pro Ser Pro His Ser Ala Asp Ala Val Val Asp Lys Gly Ser Glu Leu
115    120    125

Ala Lys Ser Leu Gln Ile Asp Phe Leu Glu Lys Leu Lys Met Gln Leu
130    135    140

Gln Thr Gln Arg Gln His Gln Gly Asp Gly Ala Pro Pro Ala Ala Ser
145    150    155    160

Leu Ser Trp Glu Met Glu Val Val Lys Arg Ala Arg Val Pro Val Leu
165    170    175

Arg Leu Gln Pro Arg Pro Ala Tyr Thr Thr Ser Ala Ala Gly Gly Val
180    185    190

Pro Thr Tyr Asn Val Asp Val Thr Tyr Gly Arg Arg Cys Gly Val Leu
195    200    205

Asn Ser Ala Leu Leu Arg Arg Tyr Met Asp Gln Gln Pro Asp Leu Arg
210    215    220

Tyr Leu Cys Leu Ala Val Lys Arg Trp Ser Lys Leu Thr Gly Leu Asn
225    230    235    240

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<210> SEQ ID NO 146
<211> LENGTH: 467
<212> TYPE: PRT
<213> ORGANISM: B. meristosporus

<400> SEQUENCE: 146

Met Ser Leu Glu Thr Asp Arg Ile Val Ser Glu Phe Lys Asp Leu Ala
1          5          10          15
Ile Ser Gln Thr Ser Glu Thr Asn Gln Pro Val Pro Asp Val Lys Val
          20          25          30
Ser Ser Val Glu Asp Ser Gln His His Ser Lys Ala His Pro Pro Lys
          35          40          45
Asn Glu Asn Gly Asn Asn Gln Gly His Arg His Gly Lys Lys Pro Phe
          50          55          60
His Arg Gly Ser Thr Phe Gly Glu Gln Gly Lys Ser Asp Arg Ser Tyr
65          70          75          80
Gly Ala Arg Ser Phe His Pro Asn Arg Gly Thr Gly Gln Tyr Glu Gly
          85          90          95
Lys Arg Val Leu Asp Lys Ser Asp Pro Arg Tyr Leu Lys Ala Arg Ala
          100          105          110
Arg Phe Ile Ser Val Leu Thr Glu Gln Val Ser Glu Leu Tyr Leu Lys
          115          120          125
Leu Thr Pro Ser Val Glu Glu Val Ser Arg Arg Glu Gln Leu Tyr Phe
          130          135          140
Arg Ile His Ser Ile Cys Glu Glu Leu Phe Pro Asp Ala Gln Leu Phe

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145	150	155	160
Gln Phe Gly Ser Thr Ala Asn Gly Phe Gly Leu Lys Asn Ala Asp Val			
	165	170	175
Asp Phe Cys Leu Cys Thr Ala Glu Asn Ser Ser Leu Thr Ala Val Ala			
	180	185	190
Phe Val Glu Gln Leu Gly Ser Val Leu Lys Glu Arg Gly Met Glu Asn			
	195	200	205
Val Met Leu Leu Thr Arg Thr Arg Ile Pro Ile Ile Lys Leu Lys Asp			
	210	215	220
Pro Val Ser Glu Leu Asn Cys Asp Ile Gly Phe Asn Asn Arg Leu Ala			
	225	230	235
Val Tyr Asn Thr Arg Leu Leu Arg Met Tyr Ala Glu Val Asp Ser Arg			
	245	250	255
Val Lys Glu Ile Val Ala Ile Val Lys His Trp Ala Lys Ser Arg Gln			
	260	265	270
Ile Asn Glu Pro Tyr Leu Gly Thr Leu Ser Ser Tyr Ala Tyr Val Leu			
	275	280	285
Met Ile Ile Phe Val Leu Gln Gln Arg Gly Val Leu Pro Cys Leu Gln			
	290	295	300
Gln Ile Cys Glu Gly Glu Lys Lys Glu Val Leu Val Asp Asn Tyr Asp			
	305	310	315
Ala Tyr Phe Phe Asp Asn Ile Thr Asp Leu Pro Lys Tyr Trp Lys Ser			
	325	330	335
Gln Asn Asn Glu Ser Leu Gly Glu Leu Leu Tyr Glu Phe Phe Arg Phe			
	340	345	350
Tyr Ala Ser Asp Phe Ser Tyr Tyr Gly Ser Val Val Ser Val Arg Thr			
	355	360	365
Gly Gln Val Met Ser Lys Glu Glu Lys Gly Trp Thr Ala Glu Ala Thr			
	370	375	380
Arg Glu Ser Gly Ala Ser His Val Gln Asp Arg Tyr Trp Val Cys Ile			
	385	390	395
Glu Asp Pro Phe Glu Val Thr His Asn Leu Gly Arg Pro Val Asn Lys			
	405	410	415
Asn Ser Leu Tyr Thr Ile Arg Gly Glu Phe Met Arg Ala Ala Asn Ile			
	420	425	430
Leu Gly Gly Thr Arg Thr Ser Thr Ile Leu Glu Arg Leu Cys Gln Val			
	435	440	445
Tyr Val Pro Glu Glu Glu Pro Pro Arg Ala Pro Pro Ala Pro His Glu			
	450	455	460
Arg Ser Asn			
465			

<210> SEQ ID NO 147

<211> LENGTH: 354

<212> TYPE: PRT

<213> ORGANISM: N. californiae

<400> SEQUENCE: 147

Met Lys Thr Thr Glu Arg Thr Cys Thr Phe Glu Phe Tyr Asp Glu Leu			
1	5	10	15
Thr Glu Glu Ile Asn Lys Ile Val Asn Lys Ile Lys Ala Lys Pro Asn			
20	25	30	


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<210> SEQ ID NO 148
<211> LENGTH: 344
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Perkinsela sp.

<400> SEQUENCE: 148

Met Leu Asp Pro Ser Glu Thr Cys Met Gly Arg Ile Lys Asn Ile His
1             5             10             15

Ala Ile Ile Thr Ser Lys Leu Met Gly Leu Leu Val Glu Lys Asn Ala

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-continued

20					25					30					
Gln	Ile	Tyr	Cys	Phe	Gly	Ser	Ser	Met	Val	Asn	Gly	Val	Val	Glu	Lys
	35						40					45			
Lys	Ser	Asp	Val	Asp	Val	Val	Tyr	Leu	Thr	Arg	Glu	Asp	Leu	Gln	Lys
	50					55					60				
Pro	Leu	Ser	Ile	Gly	Asp	Val	Cys	Asn	Pro	Gln	Ser	Arg	Ser	Glu	Gln
65					70					75				80	
Thr	Ser	Ile	Leu	Ser	Ala	Ile	Ser	Lys	Ile	Leu	Met	Lys	Asp	Ser	Glu
			85						90					95	
Leu	Phe	Ser	Thr	Val	Gln	Glu	Lys	Pro	Arg	Ala	Arg	Val	Pro	Tyr	Val
			100					105					110		
Arg	Gly	Val	Leu	Lys	Asn	Gly	Leu	Glu	Ile	Asp	Ile	Ser	Ala	His	Arg
	115					120						125			
Arg	Asn	Gly	Val	Arg	Asn	Ser	Leu	Leu	Leu	Arg	Ser	Tyr	Phe	Ser	Gln
	130					135						140			
Ile	Pro	Pro	Thr	Arg	Pro	Ser	Leu	Pro	Asn	Ala	Thr	Thr	Ser	Val	Tyr
145					150					155					160
Arg	Met	Leu	Ser	Leu	Ala	Leu	Lys	Phe	Trp	Gly	Lys	Arg	Thr	Gly	Leu
			165						170					175	
Val	Asp	Pro	Val	Gln	Thr	Phe	Leu	Thr	Ser	Tyr	Ala	Phe	Asn	Val	Leu
			180					185					190		
Ile	Cys	Tyr	Tyr	Leu	Gln	Gln	Arg	Gly	Gly	Met	Asp	Phe	Ile	His	Pro
	195						200					205			
Glu	Ser	Ile	Leu	Ile	Pro	Lys	Gly	His	Pro	Thr	Val	Pro	Asp	Tyr	Arg
	210					215					220				
Glu	Ile	Ala	Leu	Ala	Gly	Thr	Gln	Gly	His	Thr	Cys	Leu	Gly	Trp	Tyr
225					230					235					240
Met	Arg	Asp	Phe	Leu	Lys	Phe	Tyr	Asn	His	Glu	Phe	Asp	Tyr	Asn	Asn
			245						250					255	
Thr	Val	Val	Ser	Leu	Ser	Arg	Lys	Gly	Ile	Thr	Thr	Lys	Glu	Tyr	Leu
			260					265					270		
Gly	Trp	Gly	Leu	Arg	Asp	Glu	Glu	Arg	Met	His	Gly	Thr	Asp	Gly	Asn
	275						280					285			
Ala	Phe	Phe	Tyr	Arg	Phe	Cys	Val	Glu	Asp	Pro	Tyr	Glu	Asn	Arg	Leu
	290					295					300				
Asn	Leu	Gly	Arg	Phe	Val	Thr	Pro	Leu	Arg	Tyr	Ser	Met	Phe	Arg	Met
305					310					315					320
Ala	Leu	His	Gln	Ala	Gln	Leu	Asn	Gly	Phe	Gly	Tyr	Leu	Asn	Leu	Lys
			325						330					335	
Asn	Tyr	Gly	Ala	Lys	Ile	Val	Asp								
			340												

<210> SEQ ID NO 149

<211> LENGTH: 352

<212> TYPE: PRT

<213> ORGANISM: S. complicate

<400> SEQUENCE: 149

Met	Met	Ala	His	Pro	Ser	Gln	Gly	Gly	Ala	His	Pro	Leu	Ser	Ala	Gly
1				5					10					15	
Pro	Thr	Asn	Pro	Gln	Pro	Tyr	Gln	Asp	Ala	Tyr	Ser	Gly	Phe	Thr	Glu
			20				25					30			

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Tyr Val Leu Ser Ile Asn Arg Pro Gln Tyr Pro Ser Ala Asn Glu Met
    35              40              45

Tyr Leu Lys Glu Gln Leu Arg Ser Leu Ile Asp Val Glu Ile Val Gln
    50              55              60

Arg Ile Gln Pro Ser Ala Arg Val Ile Pro Phe Gly Ser Leu Val Asn
    65              70              75              80

Gly Phe Ala Thr Ala Asn Ser Asp Leu Asp Leu Cys Ile Leu Asp Asp
              85              90              95

Ser Pro Asn Pro Arg Tyr Lys Ile Lys Thr Glu Leu Pro Glu Val Leu
    100              105              110

Ala Ala Glu Phe Glu Arg Tyr Gly Phe Glu Val Lys Leu Leu Thr Lys
    115              120              125

Thr Arg Ile Pro Ile Ile Lys Leu Val Gln Gly Pro Thr Ala Gln Phe
    130              135              140

Pro Leu Gly Leu Ala Met Asp Ile Gly Phe Glu Asn Arg Leu Ala Leu
    145              150              155              160

His Asn Thr Arg Leu Leu Ala Thr Tyr Ser Arg Ile Asp Ser Arg Leu
              165              170              175

Arg Glu Met Val Leu Phe Val Lys His Trp Ala Lys Val Arg Gly Ile
    180              185              190

Asn Ser Ser Tyr His Gly Thr Leu Ser Ser Tyr Gly Tyr Val Leu Thr
    195              200              205

Ile Leu His Phe Leu Ile Asn Val Ala Ser Pro Ser Cys Leu Pro Asn
    210              215              220

Leu Gln His Ile Gly Ala Gln Val Pro Val Pro Phe Glu Glu Leu Glu
    225              230              235              240

Cys Glu Gly Tyr Asn Ile Trp Phe Phe Lys Asp Leu Thr Asn Ile Pro
              245              250              255

Pro Ser Leu Asn Arg Arg Ser Ile Gly Glu Leu Met Ala Glu Tyr Phe
    260              265              270

Ser Tyr Tyr Ala Gln Phe Asp Phe Arg Asn Met Val Ile Ser Ile Arg
    275              280              285

Thr Pro Gly Gly Leu Leu Thr Lys Thr Ser Lys Gly Trp Thr Gln Ala
    290              295              300

Leu Asp Arg Val Gly Pro Glu Asp Gln Lys Val Arg Ser Arg Asn Phe
    305              310              315              320

Leu Cys Ile Glu Asp Pro Phe Glu Ile Thr His Asn Val Gly Arg Thr
              325              330              335

Val Gly Lys Asn Gly Leu Tyr Asp Val Trp Pro Phe Val Ser Phe Ala
    340              345              350

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<210> SEQ ID NO 150

<211> LENGTH: 388

<212> TYPE: PRT

<213> ORGANISM: *S. ochraceum*

<400> SEQUENCE: 150

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Met Ala Thr Val Phe Ala Gln Ser His Ser Leu Leu Asn Thr Arg Gln
  1              5              10              15

Pro His Gln His Val Gln Gln Pro Ala Gln Gln Pro Pro Gln Gln Gln
    20              25              30

Gln Gln Gln Gln Ser Thr Arg Pro Lys Gln Leu Ser Lys Pro Arg Phe
    35              40              45

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Tyr Ala Glu Phe Ser Gln Cys Leu Phe Asp Phe Val Ile Gln Leu Leu
 50 55 60
 Pro Thr Pro Glu Glu Leu Ala Ile Lys Glu Asp Val Arg Lys Leu Leu
 65 70 75 80
 Glu Arg Leu Ile Arg Thr Leu Glu Pro Asp Ser Arg Leu Leu Ser Phe
 85 90 95
 Gly Ser Thr Ala Asn Gly Phe Ser Leu Arg Asn Ser Asp Met Asp Leu
 100 105 110
 Cys Cys Leu Ile Asp Ser Asp Asp Arg Leu Ala Ala Ser Asp Leu Val
 115 120 125
 Thr Met Leu Gly Asp Leu Leu Glu Arg Glu Thr Lys Phe His Val Lys
 130 135 140
 Pro Leu Pro Tyr Ala Arg Ile Pro Ile Val Lys Leu Ser Leu Asp Pro
 145 150 155 160
 Ser Pro Gly Leu Pro Phe Gly Ile Ala Cys Asp Ile Gly Phe Glu Asn
 165 170 175
 Arg Leu Ala Leu Glu Asn Thr Arg Leu Leu Met Cys Tyr Ala Met Ile
 180 185 190
 Asp Pro Ala Arg Val Arg Thr Met Val Leu Phe Leu Lys Val Trp Cys
 195 200 205
 Lys Arg Arg Lys Ile Asn Ser Pro Tyr Lys Gly Thr Leu Ser Ser Tyr
 210 215 220
 Gly Tyr Val Leu Leu Val Ile Phe Phe Leu Val His Val Lys Asn Pro
 225 230 235 240
 Pro Val Leu Pro Asn Leu Gln Ser Met Pro Pro Leu Arg Pro Met Asp
 245 250 255
 Lys Glu Glu Ser Thr Leu Asn Gly His Asn Val Trp Phe Phe Asp Asp
 260 265 270
 Ile Asn Leu Leu Arg Gln Arg Trp Gln Ser Thr Asn Thr Glu Ser Val
 275 280 285
 Ala Glu Leu Leu Val Asp Phe Phe Lys Phe Tyr Ser Arg Glu Phe Pro
 290 295 300
 Tyr Asn Ser Gly Val Val Ser Ile Arg Ala Gly Gly Ile Lys Lys Asp
 305 310 315 320
 Ser Lys Gly Trp Phe Ser Glu Ala Glu Arg Gly Ser Ala Arg Glu Arg
 325 330 335
 Asn Arg Leu Cys Ile Glu Asp Pro Phe Glu Thr Asp Phe Asn Val Ala
 340 345 350
 Arg Cys Val Thr Arg Asp Gly Leu Tyr Thr Ile Arg Gly Glu Leu Met
 355 360 365
 Arg Ala Ser Arg Ile Leu Ala Ala Arg Pro Glu Arg Pro Ile Val Ala
 370 375 380
 Leu Ala Gln Leu
 385

<210> SEQ ID NO 151

<211> LENGTH: 421

<212> TYPE: PRT

<213> ORGANISM: G. androsaceus

<400> SEQUENCE: 151

Met Ala Ala Tyr Ser Pro Thr Met Ser Ser Asn Gln His Tyr His Glu

-continued

1	5	10	15
Leu Asn Ser	Ala Ser Pro Pro	Ala Arg Arg Lys Trp	Leu Ala Asp Phe
	20	25	30
Ser Glu Cys	Leu Phe Ser Phe	Val Ile Gln Leu Leu	Pro Thr Gln Glu
	35	40	45
Glu Leu Ala	Val Lys Glu Asp	Val Arg Lys Leu Leu	Glu Arg Leu Ile
	50	55	60
Arg Thr Ile	Glu Pro Glu Ser	Arg Leu Leu Ser	Phe Gly Ser Thr Ala
	65	70	75
Asn Gly Tyr	Gly Leu Arg Asn	Ser Asp Met Asp	Leu Cys Cys Leu Ile
	85	90	95
Asp Ser Glu	Glu Arg Leu Ala	Ala Ser Asp Leu	Val Thr Met Leu Gly
	100	105	110
Asp Leu Leu	Glu Arg Glu Thr	Lys Phe His Val	Lys Pro Leu Pro His
	115	120	125
Ala Arg Ile	Pro Ile Val Lys	Leu Ser Leu Asp	Pro Ser Pro Gly Leu
	130	135	140
Pro Leu Gly	Ile Ala Cys Asp	Ile Gly Phe Glu	Asn Arg Leu Ala Leu
	145	150	155
Glu Asn Thr	Arg Leu Leu Met	Cys Tyr Ala Met	Ile Asp Pro Thr Arg
	165	170	175
Val Arg Thr	Met Val Leu Phe	Leu Lys Val Trp	Ser Lys Arg Arg Lys
	180	185	190
Ile Asn Ser	Pro Tyr Lys Gly	Thr Leu Ser Ser	Tyr Gly Tyr Val Leu
	195	200	205
Leu Val Ile	Tyr Phe Leu Val	His Val Lys Asn	Pro Pro Val Leu Pro
	210	215	220
Asn Leu Gln	Gln Met Pro Pro	Leu Arg Pro Ile	Pro Lys Glu Glu Thr
	225	230	235
Tyr Leu Gly	Gly Phe Asn Thr	Trp Phe Phe Asp	Asp Ile Glu Leu Leu
	245	250	255
Arg Gln Arg	Trp His Ser Glu	Asn Thr Glu Thr	Val Ala Glu Leu Leu
	260	265	270
Ile Asp Phe	Phe Arg Tyr Phe	Ser Arg Asp Phe	Pro Tyr Gly Val Gly
	275	280	285
Val Ala Ser	Ile Arg Ala Gly	Leu Leu Lys Lys	Asp Ser Lys Gly Trp
	290	295	300
Gln Asn Asp	Leu Ser Ala Ser	Arg Tyr Asn Asp	Ala Arg Glu Arg Asn
	305	310	315
Arg Phe Cys	Ile Glu Asp Pro	Phe Glu Thr Asp	Tyr Asn Val Ala Arg
	325	330	335
Cys Val Thr	Lys Asp Gly Leu	Tyr Thr Ile Arg	Gly Glu Phe Met Arg
	340	345	350
Ala Ser Arg	Val Leu Ala Ala	Arg Pro Glu Arg	Ala Ile Leu Ala Leu
	355	360	365
Ala Asp Leu	Cys Asp Glu Arg	Lys Asp Glu Asp	Pro Cys Arg Arg Ala
	370	375	380
Ser Leu Tyr	Ser Phe Ser Ser	Ser Ala Pro Thr	Asn Thr Val Phe Gly
	385	390	395
Gly Lys Ser	Ile Asn Ala Phe	Gln Arg Asn Thr	Asp Tyr Ser Leu Ser
	405	410	415

-continued

Phe Asp Gln Thr Pro
420

<210> SEQ ID NO 152

<211> LENGTH: 512

<212> TYPE: PRT

<213> ORGANISM: T. equiperdum

<400> SEQUENCE: 152

Met Gly Val Arg Leu Tyr Ser Cys Asp Ala Cys Pro His Ala Val Phe
1 5 10 15
Thr Thr His Ala Ala Leu Leu Ala His Ala Glu Glu His His Ala Asp
20 25 30
Leu Leu Pro Asp His Ala Arg Leu Arg Arg Ile Ala Gln Lys Leu Asn
35 40 45
Pro Val Trp Asn Arg Ala Leu Asn Ala Arg Arg Asn Thr Ile Thr Ser
50 55 60
Trp Gly Lys Lys Ile Phe His Val Ala Ala Gln Arg Asp Ala Gly Glu
65 70 75 80
Ser Lys Met Gln Glu Ala His Arg Ala Arg Ala Gln Leu Glu Cys Val
85 90 95
Val Arg Arg Trp His Asp Lys Ala Arg Val Phe Ile Phe Gly Ser Ser
100 105 110
Val Ala Met Gly Val Trp Asp Gly Thr Ala Asp Ile Asp Phe Ala Val
115 120 125
Val Asp Val Asp Ala Met Glu Arg Gly Ser Trp Pro Pro Leu Glu Lys
130 135 140
Asn Ala Val Arg Ser Ile Thr Glu Leu Leu Arg Arg Val Gly Phe Ser
145 150 155 160
Phe Val Asn Leu Glu Pro Ile Ser His Ala Arg Val Pro Ile Ile Lys
165 170 175
His His Ala Ser Ser Pro Ile Leu Thr Val Ala Arg Arg Asp Ala Glu
180 185 190
Asp Val Val Ala Arg Ser Ile Arg Phe Ile Leu Asn Gly Pro Ala Thr
195 200 205
Arg Glu Asp Arg Leu Leu Leu Glu Gly Ser Val Arg Asp Ala Val Gly
210 215 220
Pro Thr Gly Val Gln Gln Val Trp Trp Asn Arg Thr Ser Asp Met Met
225 230 235 240
Ser Ala Thr Leu Glu Ser Thr Thr Ala Ala Val Arg Ala Ala Met Cys
245 250 255
Ser Pro Ala Leu Ala Ser Ala Ser Leu Arg Thr Lys Val Gln Pro Ala
260 265 270
His Asp Glu Cys Arg Pro Glu Leu Tyr Asn Ile Asp Phe Ala Leu Ser
275 280 285
Phe Arg Ala Phe Gly Ile Arg Asn Ser Thr Leu Leu Arg Lys Tyr Leu
290 295 300
Leu Ser His Pro Cys Ala Arg Pro Gly Ala Ile Val Leu Lys Asp Trp
305 310 315 320
Ser Lys Thr Ser Gly Val Asn Asn Ser Val Asn Gly Tyr Phe Thr Ser
325 330 335
Tyr Ala Ile Asn Ile Met Trp Ile Tyr Tyr Leu Val Gln Lys Gly Tyr

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340				345				350							
Val	Pro	Tyr	Val	Asp	Pro	Leu	Glu	Ile	Pro	Glu	Ser	Leu	Val	Asn	Tyr
	355						360					365			
Thr	Asp	Phe	Asp	Pro	Arg	Tyr	Thr	Pro	Met	Ile	Asp	Pro	Glu	Ile	Thr
	370					375					380				
Asn	Thr	Glu	Arg	Glu	Glu	Leu	Tyr	Lys	Ala	Ala	Gly	Asp	Met	Leu	Val
	385				390					395					400
Gly	Phe	Phe	Tyr	Phe	Tyr	Ser	Phe	Glu	Phe	Asp	Trp	Gly	His	Asn	Val
			405						410					415	
Ile	Ser	Leu	Asn	Arg	Pro	Gly	Ile	Thr	Thr	Lys	Arg	Met	Leu	Gly	Trp
			420				425						430		
His	Val	Glu	Asp	Val	Val	Pro	Val	Ala	Ser	Thr	Ser	Val	Ser	Ser	Gly
	435						440					445			
Gly	Gly	Gly	Ser	Asn	Val	Lys	Arg	His	Pro	Thr	Arg	Tyr	Glu	Leu	Cys
	450				455						460				
Ile	Glu	Asp	Pro	Tyr	Glu	Glu	Asn	Leu	Asn	Leu	Gly	Arg	His	Ile	Gly
	465				470					475					480
Val	Thr	Lys	Ser	Leu	Arg	Val	Arg	Thr	Glu	Leu	Tyr	Arg	Gly	Leu	Leu
			485						490					495	
Ser	Leu	Leu	Lys	Glu	Gly	Glu	Thr	Arg	Ser	Cys	Val	Phe	Ala	Ala	Ala
			500						505				510		

<210> SEQ ID NO 153

<211> LENGTH: 531

<212> TYPE: PRT

<213> ORGANISM: M. conica

<400> SEQUENCE: 153

Met	Thr	Ala	Gly	Arg	Asp	Ile	Arg	Ser	Pro	Pro	Arg	Gly	Gln	Arg	Gly
1				5					10					15	
Gln	Gly	Phe	Asn	Lys	Ser	Gln	Asn	Ser	Pro	Arg	Ala	Leu	Pro	Leu	Asn
		20						25					30		
Ala	Leu	Asp	His	Phe	Pro	Pro	Leu	Gly	Thr	Gln	Pro	Thr	Lys	Ala	Lys
		35					40					45			
Thr	Pro	Pro	Pro	Asn	Tyr	Ser	Gln	Gln	Gln	Gln	Ser	Thr	Thr	Ala	Lys
	50					55					60				
Pro	Gln	Gln	Arg	Thr	Gln	Tyr	Gln	Pro	Leu	Pro	Ile	Asp	Thr	Arg	Asn
	65				70				75						80
Thr	Asn	Pro	His	His	Ala	Arg	Gly	Glu	Val	His	Gly	Arg	Arg	Asn	Phe
			85					90						95	
Arg	Gly	Asp	Tyr	Asn	Gln	Arg	Gln	Leu	Arg	Arg	Asp	Ile	Gly	Pro	Glu
		100						105					110		
Val	Cys	Gln	Tyr	Leu	Ser	Glu	Leu	Ser	Lys	Lys	Val	Ile	Ala	Glu	Ala
		115					120					125			
Ala	Pro	Pro	Glu	Ser	Glu	Ile	Leu	Val	Lys	Arg	Ala	Leu	Leu	Glu	Arg
	130					135					140				
Leu	Glu	Ala	Ile	Ser	Lys	Thr	Ile	Val	Pro	Asp	Ala	Lys	Leu	Ile	Ala
	145				150					155					160
Phe	Gly	Ser	Leu	Val	Thr	Gly	Phe	Ala	Thr	Ala	Asn	Ser	Asp	Leu	Asp
			165					170						175	
Val	Ile	Phe	Thr	Gly	Gly	Asp	Arg	Val	Asp	Ile	Leu	Asn	Glu	Ser	Ser
			180					185					190		

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Asp	Asp	Pro	Ser	Ser	Asn	Phe	Arg	Ile	Pro	Met	Leu	Leu	Glu	Ser	Lys
	195						200					205			
Leu	Gln	Glu	Glu	Gly	Phe	Gly	Thr	Thr	Leu	Leu	Thr	Lys	Thr	Arg	Val
	210					215					220				
Pro	Ile	Leu	Lys	Leu	Val	Gln	Lys	Ala	Thr	Glu	Gln	Ser	Pro	Tyr	Glu
225					230					235					240
Leu	Gln	Cys	Asp	Ile	Gly	Phe	Ser	Asn	His	Leu	Ala	Leu	Tyr	Asn	Thr
			245					250						255	
Gln	Met	Leu	Leu	Thr	Tyr	Ser	Lys	Cys	Asp	Pro	Arg	Val	Lys	Glu	Met
		260						265					270		
Met	Ile	Phe	Ile	Lys	Trp	Trp	Ala	Lys	Arg	Arg	His	Ile	Asn	Asn	Pro
	275						280					285			
Tyr	Arg	Gly	Thr	Leu	Ser	Ser	Tyr	Gly	Tyr	Ala	Leu	Ile	Val	Leu	His
	290					295					300				
Tyr	Leu	Ile	Asn	Val	Val	Lys	Pro	Pro	Val	Leu	Pro	Asn	Leu	Gln	Thr
305					310					315					320
Phe	Pro	Val	Pro	Asp	Ser	Ala	Pro	Thr	Asn	Glu	Ile	Ile	Phe	Glu	Gly
			325						330					335	
Asp	Ser	Asp	Asp	Thr	Phe	Glu	Ile	Trp	Phe	Tyr	Lys	Asp	Ile	Glu	Lys
		340						345					350		
Leu	Pro	Lys	Ser	Asp	Asn	Ala	Met	Asp	Ile	Gly	Glu	Leu	Leu	Lys	Gly
		355					360					365			
Phe	Phe	Glu	Tyr	Tyr	Ala	His	Asn	Phe	Gln	Trp	Gly	Arg	Glu	Val	Val
	370					375					380				
Ser	Ile	Arg	Thr	Lys	Gly	Gly	Leu	Met	Thr	Lys	Gln	Glu	Lys	Gly	Trp
385					390					395					400
Val	Ala	Ala	Val	Ile	Lys	Pro	Gly	Arg	Thr	Glu	Asn	Ser	Glu	Val	Lys
			405						410					415	
Asn	Arg	Tyr	Leu	Phe	Ala	Val	Glu	Asp	Pro	Phe	Glu	Thr	Glu	His	Asn
		420						425					430		
Val	Ser	Arg	Thr	Cys	Asn	Gly	Pro	Gly	Val	Asn	Arg	Ile	Lys	Asp	Glu
		435					440					445			
Phe	Lys	Arg	Ala	Val	Trp	Leu	Ile	Arg	Val	Arg	Asp	Gly	Gly	Lys	Thr
	450					455					460				
Leu	Phe	Gln	Asn	Leu	Cys	Met	Glu	Ala	Pro	Pro	Glu	Arg	Val	Trp	Val
465					470					475					480
Arg	Arg	Glu	Asp	Arg	Asp	Arg	Arg	Gly	Ile	Glu	Glu	Ala	Gly	Gly	His
			485					490						495	
Gly	Ser	Lys	Asp	His	Gln	Gly	Ala	Glu	Gly	Glu	Ala	Gly	Asp	Gly	Val
		500						505					510		
Lys	His	Glu	Ser	Val	Val	Leu	Ser	Glu	Asp	Gly	Arg	Asn	Ala	Leu	Glu
		515					520					525			
Ser	Leu	Val													
	530														

<210> SEQ ID NO 154

<211> LENGTH: 474

<212> TYPE: PRT

<213> ORGANISM: P. murina

<400> SEQUENCE: 154

Met	Asp	Ser	Val	Lys	Gly	Phe	Glu	Gly	Val	Glu	Ser	Val	Val	Phe	Gln
1				5					10					15	

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Lys	Glu	Met	Lys	Lys	Leu	Thr	Lys	Ile	Ser	Asp	Phe	Ser	Gly	Lys	Asn	
			20					25					30			
Tyr	Gly	Ser	Gly	Asn	Glu	Ile	Glu	Asp	Leu	Lys	Lys	Ser	Leu	Glu	Lys	
		35					40					45				
Met	Met	Leu	Glu	Ser	Val	Asn	Ile	Gln	Glu	Lys	Gly	Val	Phe	Leu	Arg	
		50				55					60					
Glu	Lys	Gly	Ser	Glu	Gly	Ser	Tyr	Glu	Ala	Asn	Leu	Asn	Asp	Lys	Asp	
65					70					75					80	
Val	Leu	Lys	Asn	Asn	Lys	Lys	Asp	Ser	Asn	Phe	Arg	Asn	Ile	Lys	Tyr	
			85					90						95		
Tyr	Glu	Pro	Lys	Asp	Tyr	Gly	Leu	Tyr	Val	Asn	Gly	Lys	Ala	Val	Trp	
			100					105					110			
Arg	Lys	Asn	Tyr	Asp	Glu	Tyr	Phe	Val	Leu	Ser	Thr	Phe	Ile	Ser	His	
		115					120					125				
Thr	Leu	Lys	His	Ile	Met	Pro	Thr	Asn	Glu	Glu	Ile	Ser	Gln	Lys	Glu	
	130					135					140					
Arg	Phe	Arg	Phe	Phe	Leu	Ser	Glu	Ile	Leu	Asn	Thr	Cys	Arg	Pro	Ser	
145					150					155					160	
Ala	Lys	Leu	Val	Leu	Phe	Gly	Ser	Val	Ala	Ser	Gly	Leu	Ala	Ile	Val	
			165						170					175		
Asn	Ser	Asp	Met	Asp	Phe	Cys	Val	Met	Asp	Asp	Ser	Leu	Asp	Leu	His	
		180						185					190			
Thr	Asp	Glu	Phe	Leu	Lys	Ile	Phe	Ser	Glu	Glu	Ile	Lys	Lys	Tyr	Gly	
	195						200					205				
Met	Glu	Thr	Thr	Leu	Leu	Phe	Arg	Thr	Arg	Val	Ser	Ile	Ile	Lys	Val	
	210					215				220						
Asn	Ser	Lys	Gly	Ser	Phe	Asn	Phe	Pro	Gln	Gly	Ile	Ser	Cys	Asp	Ile	
225					230					235					240	
Gly	Phe	Asn	Asn	Lys	Leu	Ala	Ile	Tyr	Asn	Thr	Lys	Leu	Leu	Ala	Thr	
			245						250					255		
Tyr	Ser	Lys	Cys	Asp	His	Arg	Val	Arg	Lys	Ile	Ile	Leu	Phe	Val	Lys	
		260						265					270			
Tyr	Trp	Ala	Lys	Arg	Arg	Lys	Ile	Asn	Asp	Pro	Tyr	His	Gly	Thr	Leu	
	275					280						285				
Ser	Ser	Tyr	Gly	Tyr	Val	Leu	Leu	Ile	Leu	His	Tyr	Leu	Ile	Asn	Ile	
	290				295						300					
Val	Pro	Ile	Pro	Leu	Leu	Pro	Asn	Leu	Gln	Gln	Met	Lys	Val	Ala	Ser	
305					310					315					320	
Trp	Arg	Ser	Ile	Pro	Gln	Ser	Glu	Ile	Glu	Cys	Asp	Gly	Tyr	Asn	Val	
			325						330					335		
Trp	Phe	Tyr	Lys	Asp	Ile	Asp	Ile	Ser	Cys	Ser	Phe	Ser	Asn	Asn	Thr	
		340						345					350			
Asp	Ser	Leu	Gly	Lys	Leu	Val	Tyr	Gly	Phe	Phe	Tyr	Tyr	Tyr	Ala	Tyr	
		355					360					365				
Gln	Phe	Asn	Trp	Lys	Asp	His	Val	Val	Ser	Ile	Arg	Thr	Gln	Thr	Gly	
	370					375					380					
Leu	Leu	Thr	Lys	Gln	Glu	Lys	Gly	Trp	Thr	Gln	Ala	Met	Glu	Arg	Val	
385					390					395					400	
Phe	Val	Ser	Asp	Lys	Ile	Phe	Lys	Asp	Arg	Tyr	Ile	Leu	Ala	Ile	Glu	
			405						410					415		

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Asp Pro Phe Glu Ile Thr His Asn Val Gly Arg Thr Val Asn Lys Gln
420 425 430

Ser Ser His Ile Ile Arg Gly Glu Phe Phe Arg Ala Ser Lys Leu Ser
435 440 445

Ser Ser Lys Leu Arg Lys Lys Ile Phe Asn Glu Leu Cys Glu Glu Arg
450 455 460

Ile Ile Ile Asn Asn Phe Ser Asn Val Ile
465 470

<210> SEQ ID NO 155

<211> LENGTH: 445

<212> TYPE: PRT

<213> ORGANISM: *S. japonicus*

<400> SEQUENCE: 155

Met Ser Gly Asn Gly Asn Gly His Phe Ile Pro Gly Val His Thr Val
1 5 10 15

Glu Glu Ile Glu Arg Gln Phe Val Asn Leu Ala Leu Gln Lys Arg Asn
20 25 30

Gln Ala Ala Ala Ala Ala Ala Ala Ala Ala Glu Arg Glu Leu Asp Pro
35 40 45

Val Thr Cys Phe Leu Leu Ser Thr Tyr Asp Asp Val Arg Val Ser Asp
50 55 60

Asp Glu Leu Arg Glu Lys Asp Ala Ile Met Asn Leu Leu Lys His Val
65 70 75 80

Val His Ser Val Arg Pro Glu Ala Asp Ile Val Ala Phe Gly Ser Ile
85 90 95

Gln Ser Gly Leu Ala Leu Lys Asn Ser Asp Ile Asp Ala Cys Ile Leu
100 105 110

Leu Pro Asp Ile Gly Glu Glu Met Glu Glu Phe Ala Ser Glu Cys Phe
115 120 125

Glu Arg Phe Thr Ala Leu Gly Phe Glu Gly Lys Tyr Leu Arg Lys Ala
130 135 140

Arg Ile Pro Ile Ile Lys Leu Leu Ser Asp Thr Lys Asn Arg Tyr Tyr
145 150 155 160

Tyr Gly Phe Gln Cys Asp Ile Gly Phe Asn Asn Gln Leu Ala Ile Tyr
165 170 175

Asn Thr Ser Leu Leu His Gln Tyr Ser Leu Ile Asp Pro Arg Cys Lys
180 185 190

Gln Leu Ala Ile Leu Val Lys Tyr Trp Ala Lys Gln Lys Arg Ile Asn
195 200 205

Ser Pro Tyr Tyr Gly Thr Leu Ser Ser Tyr Gly Tyr Val Leu Met Val
210 215 220

Leu Phe Tyr Leu Ile His Val Val Arg Pro Ala Val Leu Pro Asn Leu
225 230 235 240

Gln Asp Ser Pro His Lys Gln Asp Leu Tyr Val Glu Gly Phe Asn Val
245 250 255

Gly Phe Val Arg Gly Thr Thr Val Ala Arg Arg Asn Thr Glu Ser Leu
260 265 270

Pro Gln Leu Leu Ala Gly Phe Tyr Gly Phe Phe Ala His Glu Phe Asn
275 280 285

Tyr Arg Glu Ser Val Ile Ser Ile Arg Gln Pro Gly Gly Leu Leu Lys
290 295 300

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Lys Val Asp Lys Asp Trp Thr Leu Ala Lys Glu His Thr Gly Ser Ala
 305 310 315 320
 Asp Gln Val Ile Lys Asp Arg Tyr Val Leu Ala Ile Glu Asp Pro Phe
 325 330 335
 Glu Ile Thr His Asn Val Gly Arg Thr Val Ser Lys Ala Gly Leu Phe
 340 345 350
 Glu Ile Arg Gly Glu Phe Met Gln Ala Thr Arg Leu Leu Asn Ala Ala
 355 360 365
 Thr Leu His Ser Thr Ser Ala Ile Arg Lys Leu Arg Ala Ser Leu Phe
 370 375 380
 Lys Glu Arg Leu Glu Pro Lys Ser His Leu Lys Tyr Gln Lys Ala Leu
 385 390 395 400
 Arg Gln Lys Arg Met Gly Thr Glu Gly Lys Ala Ala Ala Glu Gly Lys
 405 410 415
 Thr Asn Asn Pro Thr Ala Ala Ser Gly Ser Asp Ser Ala Pro Ser Glu
 420 425 430
 Ser Ala Ser Arg Thr Ala Thr Val Glu Ser Arg Glu Pro
 435 440 445

<210> SEQ ID NO 156

<211> LENGTH: 620

<212> TYPE: PRT

<213> ORGANISM: A. nigricans

<400> SEQUENCE: 156

Met Asn Pro Pro Gln Gln Thr Thr Gly Leu Glu Asp His Leu Arg Ser
 1 5 10 15
 Leu Ile Ile Asn Asn Gln Thr Pro Pro Asn His Ser Ile Ser Pro Ala
 20 25 30
 Ala Lys Ile Pro His Lys His His Pro Arg Thr Ser Thr Asp Ser Arg
 35 40 45
 Gly Phe Val Arg Ser Gln Thr Ile His Arg Pro Gly Pro Ala Ala Ser
 50 55 60
 Gln Pro Pro Ser Asn Gly Pro Asn Arg Arg Phe Pro Gly Glu Ala Gln
 65 70 75 80
 Arg Pro Asp His Arg Pro Pro His Leu Arg Gly Pro Ser Gln Asn Arg
 85 90 95
 Asn Tyr Trp His Gly Ala Pro Pro Pro Pro Arg Gln Val Val Pro Gln
 100 105 110
 Gly Leu Ser Pro Pro Ala Ala Thr Asn Leu His His Phe Pro Pro Leu
 115 120 125
 Gly Ala Lys Val Ala Pro Pro Pro Pro Ser Leu Pro Ser Tyr His Ser
 130 135 140
 Val Pro Thr Ala Pro Cys His Arg Pro Glu Asn Gln Arg Pro Val Tyr
 145 150 155 160
 Gln Asn Asp Asn Arg Asp Ser Tyr His Tyr Lys Gln Arg Gly Leu Tyr
 165 170 175
 Asp Gly Tyr Leu Ala Glu Gln Trp Arg Glu Leu Asp Lys Met Ala Lys
 180 185 190
 Glu Val Leu Leu Thr Ala Thr Pro Val Asp Gly Glu Lys Glu Ala Lys
 195 200 205
 Glu Lys Phe Met Gln Asp Leu Glu Ala Val Cys Gln Lys Val Glu Pro

-continued

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Thr Ala Lys Leu Ile Pro Phe Gly Ser Ile Val Thr Gly Phe Ala Thr		
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Arg Gly Ser Asp Ile Asp Cys Val Phe Thr Ser Asn Thr Asp Ser Ala		
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Leu Gln Asn Asp Lys Ile Thr Glu Pro Glu Thr Pro Phe Asp Arg His		
	260	265 270
Asp Glu Leu Val Thr Gln Leu Gln Glu Ser Gly Tyr Asn Ala Gln Leu		
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Leu Thr Arg Thr Arg Val Pro Ile Ile Lys Leu Val Arg Pro Ala Thr		
	290	295 300
Pro Asp Val Pro Asp Ser Val Ser Cys Asp Ile Gly Phe Asn Asn Phe		
305	310	315 320
Leu Ala Ile His Asn Thr Arg Leu Leu Arg Thr Tyr Ala Ala Cys Asp		
	325	330 335
Glu Arg Leu Val Gln Met Val Leu Phe Ile Lys Trp Trp Ala Lys Arg		
	340	345 350
Arg His Ile Asn Ser Pro Tyr Arg Gly Thr Leu Ser Ser Tyr Gly Tyr		
	355	360 365
Ala Leu Leu Ile Val His Tyr Leu Ile Asn Ile Ala Gln Pro Pro Val		
	370	375 380
Leu Pro Asn Leu Gln Leu Phe His Pro Pro Ala Thr Gln Thr Ser Thr		
385	390	395 400
Glu Leu Gln Tyr Glu His Asn Gly Asn Val Cys Asn Ile Trp Tyr Leu		
	405	410 415
Lys Asp Thr Ser Thr Leu Pro Arg Ser Ala Asn Lys Ala Ser Ile Gly		
	420	425 430
Glu Leu Leu Arg Gly Phe Phe Glu Tyr Tyr Ala Phe Thr Phe Arg Trp		
	435	440 445
Gln Asn Asp Val Val Ser Ile Arg Thr Ala Gly Gly Ile Leu Thr Lys		
	450	455 460
Met Glu Lys Asn Trp Cys Ala Ala Arg Ser Arg Pro Gly Gly Arg Glu		
465	470	475 480
Glu Gly Gln Val Trp Glu Val Lys Asp Arg Val Asp Val Gln His Arg		
	485	490 495
Tyr Leu Leu Ala Ile Glu Asp Pro Phe Glu Thr Thr His Asn Val Ala		
	500	505 510
Arg Thr Cys Thr Phe Asn Gly Val Gly Arg Ile Lys Ser Glu Leu Lys		
	515	520 525
Arg Ala Met Tyr Leu Ile Lys Ser Arg Asp Pro Gly His His Arg Leu		
	530	535 540
Arg Thr Glu Leu Phe Glu Glu Ala Pro Pro Glu Arg Pro Phe Val Arg		
545	550	555 560
Asn Gly Lys Glu Val Val Asp Lys Leu Glu Lys Val Glu Ile Val Glu		
	565	570 575
Glu Arg Ala Gly Val Ala Val Glu Ile Val Glu Glu Arg Ala Gly Val		
	580	585 590
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<220> FEATURE:
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<223> OTHER INFORMATION: n is deoxyinosine

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1: A method of synthesizing a polynucleotide having a predetermined sequence capable of forming a G4 structure, the method comprising the steps of:

- (a) providing, attached to a synthesis support, initiators each with a free 3'-hydroxyl; and
- (b) repeating in a reaction mixture including the synthesis support, until the polynucleotide is formed, cycles of (i) contacting under elongation conditions the initiators or elongated fragments having free 3'-O-hydroxyls with a 3'-O-blocked nucleoside triphosphate and a template-independent polymerase so that the initiators or elongated fragments are elongated by incorporation of a 3'-O-blocked nucleoside triphosphate to form 3'-O-blocked elongated fragments, and (ii) deblocking the

elongated fragments to form elongated fragments having free 3'-hydroxyls, wherein the reaction mixture for elongating the initiators or elongated fragments comprise polyC oligonucleotides capable of forming duplexes with regions of the polynucleotide.

2: The method according to claim 1 wherein said initiators comprise said polyC oligonucleotides.

3: The method according to claim 1, wherein said synthesis support further comprises said polyC oligonucleotides attached thereto.

4: The method according to claim 1, wherein said reaction mixture comprises polyC oligonucleotides in solution whenever said elongated fragments are G4-prone polynucleotides.

5: The method according to claim 1, wherein said polynucleotide is an RNA and wherein said template-independent polymerase is a poly(A) polymerase or a poly(U) polymerase or variant thereof.

6: The method according to claim 5, wherein said 3'-O-blocked nucleoside triphosphate is a 3'-O-azidomethyl-ribonucleoside triphosphate.

7: The method according to claim 1, wherein said polynucleotide is a DNA and wherein said template-independent polymerase is a terminal deoxynucleotidyltransferase (TdT) or variant thereof.

8: The method according to claim 7, wherein said polyC oligonucleotide has a length in the range of from 2 to 20 nucleotides.

9: The method according to claim 7, further including a step of cleaving said polynucleotide from said synthesis support.

10: The method according to claim 7, wherein said 3'-O-blocked nucleoside triphosphate is selected from the group consisting of 3'-O-(2-nitrobenzyl) nucleoside triphosphate, 3'-O-allyl nucleoside triphosphate, 3'-O-amine nucleoside triphosphate, 3'-O-azidomethyl nucleoside triphosphate, 3'-O-(2-cyanoethyl) nucleoside triphosphate, and 3'-O-propargyl nucleoside triphosphate.

11: The method of claim 10, wherein said 3'-O-blocked nucleoside triphosphate is a 3'-O-azidomethyl nucleoside triphosphate.

12: The method of claim 10, wherein said 3'-O-blocked nucleoside triphosphate is a 3'-O-amine nucleoside triphosphate.

13: A kit for synthesizing a polynucleotide of a predetermined sequence using a template-free polymerase comprising a synthesis support having attached thereto initiators comprising polyC oligonucleotides.

14: The kit according to claim 13, wherein said polynucleotide is a polydeoxyribonucleotide and wherein said template-free polymerase is a terminal deoxynucleotidyltransferase or variant thereof.

15: The kit according to claim 13, wherein said polynucleotide is a polyribonucleotide and wherein said template-free polymerase is a poly(A) polymerase or a poly(U) polymerase or variant thereof.

16: The kit according to claim 13, wherein said synthesis support comprises a population of microparticles.

17: The method according to claim 2, wherein said synthesis support further comprises said polyC oligonucleotides attached thereto.

18: The method according to claim 2, wherein said polynucleotide is a DNA and wherein said template-independent polymerase is a terminal deoxynucleotidyltransferase (TdT) or variant thereof.

19: The method according to claim 3, wherein said polynucleotide is a DNA and wherein said template-independent polymerase is a terminal deoxynucleotidyltransferase (TdT) or variant thereof.

20: The kit according to claim 14, wherein said synthesis support comprises a population of microparticles.

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